

Reed–Solomon Codes Beyond Johnson: Efficient Decoding and Smaller Cryptographic Proofs

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September 2026

Abstract

List decoding Reed–Solomon codes is a central problem in coding theory and, together with mutual correlated agreement (MCA), underpins the soundness of many succinct cryptographic proofs. In this work, we give precise quantitative bounds on Reed–Solomon list decoding and MCA up to capacity, provide deterministic decoding algorithms that remain efficient over cryptographically large fields, and specialize our bounds to reduce proof sizes in implemented proof systems. Our bounds apply to arbitrary prescribed evaluation domains; beyond Johnson, they require sufficiently large field characteristic.

Refining the hidden-derivative method of Brakensiek, Chen, Putterman, Zhang, and Zheng, we obtain an explicit agreement threshold $a_1(\rho)$ strictly below the Johnson bound $\sqrt{\rho}$, using only the first derivative, without requiring its evaluations in the received word. At agreement $a_1(\rho) + \eta_1$, where $\eta_1 > 0$, our sharper interpolation and candidate counts give list size $O_\rho(n/\eta_1^2)$ and MCA error $O_\rho(n^2/(q\eta_1^4))$ for codes of length n over $\mathbb{F}_q$. Higher derivatives give explicit quantitative bounds up to capacity, uniformly over all rates, sharpening prior MCA results in Zheng’s unpublished manuscript and Jeronimo’s work, concurrent with ours. At fixed rate, for agreement gap $\eta_0 > 0$ above Johnson, we improve the MCA error bound of Ben-Sasson et al. (BCHKS) from $O_\rho(n/(q\eta_0^5))$ to $O_\rho(n/(q\eta_0^3))$, in every characteristic.

Our decoders use agreement constraints to recover close messages without enumerating initial field values. With fast explicit field arithmetic, at fixed rate and a fixed positive agreement margin above the respective threshold, deterministic decoding takes $\tilde{O}(n \log q)$ bit operations above Johnson in every characteristic and $\tilde{O}(n^2 \log q)$ above our first-order curve in sufficiently large characteristic. Decoding remains polynomial in n and $\log q$ at every fixed positive gap from capacity, under the same requirement of sufficiently large characteristic.

Our first-derivative bounds give the first proof-size reductions in existing proof-system implementations from provable Reed–Solomon proximity-gap bounds beyond the Johnson radius. At unchanged security targets, we save 79.4 KB (11.1%) for ProveKit passport proofs, 13.2 KB (4.63%) for ZisK compressed final proofs, and 59.8 KB (4.85%) for LambdaVM CPU subproofs. We have formally verified the list-decoding and MCA bounds and concrete parameter certificates in ArkLib, a Lean library for verified cryptographic proofs, and proved correctness of a simplified version of our list-decoder.

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1 Introduction

For decades, a central question in Reed–Solomon list decoding was how to close the gap between the Johnson radius and capacity. A Reed–Solomon code of length n and rate $\rho = k/n$ encodes a degree- $< k$ polynomial by its evaluations at n points [RS60]. Given possibly corrupted evaluations, list decoding returns every such polynomial agreeing with the received word in sufficiently many positions. A lower required agreement fraction therefore permits more errors. The Guruswami–Sudan algorithm recovers every polynomial agreeing on more than roughly a $\sqrt{\rho}$ fraction of positions [Sud97, GS99]. This agreement threshold is known as the Johnson bound; its exact finite form is $\sqrt{(k-1)/n}$. Capacity asks for agreement only $\rho + \delta$, for any fixed $\delta > 0$, with list size polynomial in n [GRS25, Section 7.4]. The challenge was to achieve this on arbitrary prescribed evaluation sets, with an efficient algorithm to recover the list.

These bounds also affect the size of succinct cryptographic proofs. Many proof systems use FRI [BBHR18], STIR [ACFY24a], or WHIR [ACFY24b] to check Reed–Solomon encodings at a few random positions; sharper correlated-agreement bounds can justify fewer queries at the same soundness level [BCI⁺23, BGKS19, GMW25]. Because these protocols combine words using random coefficients, their analyses need more than a bound on each word’s list: they need to relate candidates before and after a combination. Mutual correlated agreement (MCA) provides this connection: except for a small set of challenges, agreements in the combined word can be traced back to shared agreements in its two inputs. For $f, g \in \mathbb{F}_q^n$ and uniformly random $z \in \mathbb{F}_q$, MCA requires that, except with small probability, every sufficiently close polynomial P for $f + zg$ has the form

$$P = F + zG,$$

where F, G satisfy the same degree bound and agree with f, g , respectively, wherever P agrees with $f + zg$. Thus P ’s full agreement set equals the witnesses’ common agreement set; we refer to this as *full agreement-set line MCA*. An exceptional challenge can create extra agreements through cancellation of the two inputs’ discrepancies from their witnesses. Work on correlated agreement and its mutual form established these guarantees in the Johnson regime [BCI⁺23, BCH⁺26, Hab25, BCGM25]. The agreement threshold, list size, and probability of an exceptional combination all enter the resulting soundness bounds.

In a recent breakthrough, Brakensiek, Chen, Putterman, Zhang, and Zheng (BCPZZ) achieved Reed–Solomon list decoding up to capacity over prime fields [BCP⁺26a]. Their hidden-derivative method builds on Naskręcki’s submission to the Proximity Prize on better.codes [Nas26]: derivatives constrain candidate polynomials even though their evaluations are absent from the received word. Corresponding MCA guarantees now also reach capacity [rs-26, Jer26].

These results establish capacity guarantees, but leave sharp quantitative bounds and efficient large-field decoding unresolved. To reduce concrete proof sizes, we need list-size and MCA-error bounds that justify fewer queries at the field sizes and block lengths actually used. How much improvement beyond Johnson can we obtain with just the first derivative, and does it suffice to reduce proof sizes in implemented cryptographic proof systems? Knowledge soundness also requires efficient algorithms: when an extractor uses list decoding, its running time includes the cost of decoding. This requirement appears in analyses of FRI-based proof systems [Ben21, BGK⁺23] and in the compilers accompanying STIR [ACFY24a, Lemma 7.3] and WHIR [ACFY24b, Theorem 7.5]. For a cryptographic field of size $q = 2^{\Theta(\lambda)}$, where λ is the security parameter, this calls for decoding time polynomial in n and $\log q$. This brings us to our goal:

**Can we decode beyond Johnson efficiently over cryptographically large fields,
and obtain agreement bounds strong enough to reduce proof sizes in existing systems?**

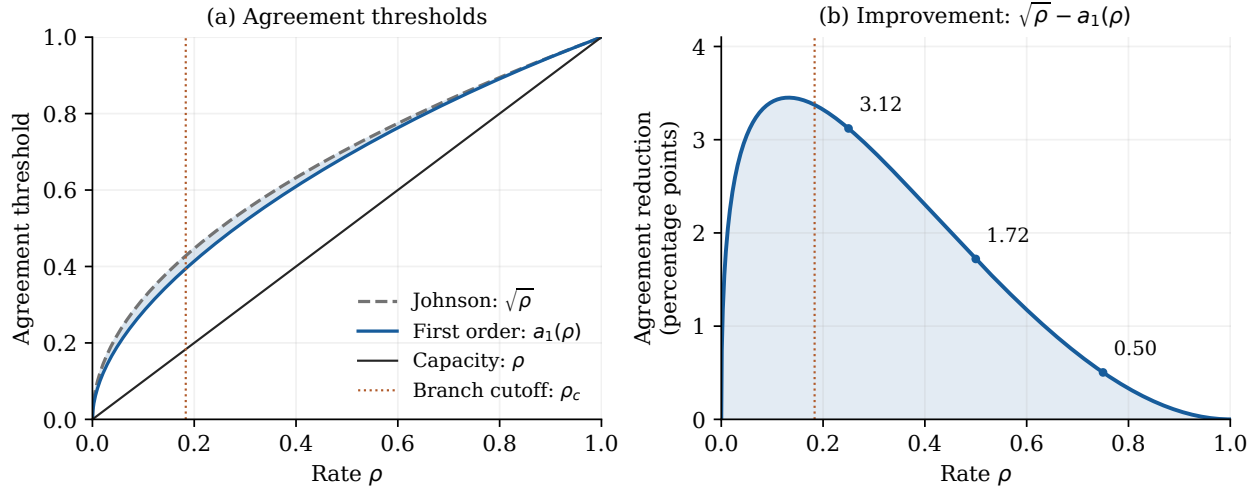


Figure 1: The sufficient first-order threshold obtained by optimizing our support with the rectangle rank bound lies below Johnson. Left: $a_1(\rho)$, Johnson's $\sqrt{\rho}$, and capacity ρ ; the shaded region shows the improvement. Right: the same improvement $\sqrt{\rho} - a_1(\rho)$ in percentage points. The dotted line marks $\rho_c = 11 - 3\sqrt{13}$, where the formula changes branches. The main bounds apply at positive gap η_1 above the curve.

1.1 Our Results

We make four main contributions in this paper. First, we give sharper quantitative bounds for Reed–Solomon list decoding and MCA, from the Johnson regime to capacity, with particular attention to what can already be achieved using just one hidden derivative. Second, we develop deterministic decoding algorithms that remain efficient over cryptographically large fields, where searching over field elements is prohibitively expensive. Third, we use these quantitative bounds to reduce query counts and proof sizes in ProveKit, ZisK, and LambdaVM at their existing security targets. Finally, we formally verify the mathematical bounds, concrete parameter certificates, and correctness of a simplified decoder in Lean.

In the bounds below, subscripts on $O(\cdot)$ indicate the parameters on which its hidden constants may depend, and $\tilde{O}(\cdot)$ suppresses logarithmic factors.

One derivative beyond Johnson. Write $\rho = k/n$ for the rate and a for the required agreement fraction. We first ask how far one hidden derivative takes us below Johnson, without using the higher derivative orders needed to approach capacity. Using one hidden derivative, we lower the sufficient agreement threshold from 0.5 to approximately 0.4688 at rate $1/4$, and from 0.7071 to approximately 0.6899 at rate $1/2$. Optimizing our monomial support with the rectangle rank bound gives this explicit sufficient threshold, denoted $a_1(\rho)$. It lies strictly between ρ and $\sqrt{\rho}$ and is asymptotic to $\sqrt{\rho/2}$ as $\rho \rightarrow 0$, a factor of $1/\sqrt{2}$ times Johnson's threshold. Figure 1 shows the improvement; Section 4.3 derives the resulting two-branch formula.

A lower agreement threshold makes each random query more likely to detect an error, so even a small improvement can reduce the number of queries needed at a fixed security target. Section 8 quantifies the resulting query and proof-size savings in existing implementations.

Write $\eta_1 = a - a_1(\rho) > 0$ for the agreement margin above this curve. To obtain quantitative bounds, we choose a bound $B_\partial = O_\rho((\eta_1 + 1/n)^{-1})$ on the exponent of the derivative variable in the interpolation polynomial.

Regime	Characteristic for counting	Agreement and list size	MCA exceptional count	Decoder bit complexity
Zeroth order (derivative-free)	Unrestricted	<i>Agreement</i> $a = \sqrt{(k-1)/n} + \eta_0$ <i>List size (classical)</i> $O_\rho(\eta_0^{-1})$	$O_\rho(n\eta_0^{-3})$	$\tilde{O}_{\rho,\eta_0}(n \log q)$
First derivative	$\text{char } F = 0$ or $p > \max\{k-1, B_\partial\}$	<i>Agreement</i> $a = a_1(\rho) + \eta_1$ <i>List size</i> $O_\rho(n\eta_1^{-2})$	$O_\rho(n^2\eta_1^{-4})$	$\tilde{O}_{\rho,a}(n^2 \log q)$
Gap δ from capacity	$\text{char } F = 0$ or $p > k-1$	<i>Agreement</i> $a \geq \rho + \delta$ <i>List size</i> $C_\delta n^{d_\delta}$	$C'_\delta n^{d_\delta+1}$	$C''_\delta n^{O(d_\delta+1)} \cdot \tilde{O}(\log q)$

Table 1: Our quantitative guarantees for prescribed Reed–Solomon evaluation sets. The zeroth-order list bound is classical [GS99]; all decoder bounds are deterministic. Here $\eta_0 = a - \sqrt{(k-1)/n} > 0$ and $\eta_1 = a - a_1(\rho) > 0$; d_δ is defined in Theorem 1.2. An exceptional count E gives line-MCA error at most E/q . The constants depend on the rate and agreement in the first two rows, and only on δ in the last row, which requires $n \geq N_\delta$. Here B_∂ is the derivative-degree bound introduced before Theorem 1.1. Decoder bounds use the finite-field presentations explained below, with $q \geq n$. First-order decoding needs only $p > k-1$, whereas its displayed inverse-gap counting bounds use the stronger hypothesis in the field column. The decoder handles the remaining characteristic cases separately. The precise algorithmic hypotheses appear in Section 9.

Theorem 1.1 (First-order list decoding and MCA). *Fix $0 < \rho < 1$ and $a = a_1(\rho) + \eta_1 < 1$, with $\eta_1 > 0$. Let $n \geq 2$, $2 \leq k \leq pn$, and $an \leq A \leq n$ be integers, where A is the required number of agreements. There is an explicit nonnegative integer $B_\partial = O_\rho((\eta_1 + 1/n)^{-1})$ such that the following holds. Over every field of characteristic zero or $p > \max\{k-1, B_\partial\}$, and every set of n distinct evaluation points, the agreement list has size $O_\rho(n/\eta_1^2)$ and full agreement-set line MCA has exceptional count $O_\rho(n^2/\eta_1^4)$. Over $\mathbb{F}_q$, divide the exceptional count by q for the soundness error.*

The characteristic restriction comes from reconstruction, rather than local interpolation: in characteristic p , both 0 and $(X-c)^p$ solve $P' = 0$ and have the same value at c , so the equation and initial value need not determine the later coefficients. The condition $p > k-1$ excludes this obstruction among degree- $< k$ messages; $p > B_\partial$ ensures separability in the derivative variable after repeated factors are removed.

Section 4.3 gives the explicit choice of B_∂ and the remaining interpolation parameters.

Higher derivatives approaching capacity. To approach capacity, we allow more derivatives in the interpolation equation. We give the explicit bound $\lceil \exp(1.5/\delta) \rceil$ uniformly over all rates for $0 < \delta < 0.24$; first order suffices at larger gaps from capacity (Section 10.2). The order matters because it controls the exponents in our list-size, MCA, and decoding bounds.

Theorem 1.2 (Uniform capacity bounds, informal). *Fix $0 < \delta < 1$, and put $d_\delta = \lceil \exp(1.5/\delta) \rceil$ for $\delta < 0.24$ and $d_\delta = 1$ otherwise. There are explicit constants N_δ, C_δ , depending only on δ and controlling the minimum length and counting bounds, such that for every $n \geq N_\delta$, $1 \leq k \leq n$, and integer $k + \delta n \leq A \leq n$, the following holds. Over any field of characteristic zero or $p > k-1$, and any n distinct evaluation points, every received*

word has at most $C_\delta n^{d_\delta}$ degree- $< k$ polynomials agreeing in at least A positions, and full agreement-set line MCA has exceptional count at most $C_\delta n^{d_\delta+1}$. Over $\mathbb{F}_q$, the corresponding error is at most $C_\delta n^{d_\delta+1}/q$.

Theorem 6.8 gives the explicit constants and proves this statement. The same derivative order, constants, and minimum length work for every rate: none deteriorates as the rate tends to zero.

Table 1 compares the Johnson, first-order, and capacity guarantees. Its counting bounds also apply in characteristic zero; the runtime column concerns finite fields with fast arithmetic.

Quantitative MCA at Johnson. We also improve MCA without using derivatives or changing the Johnson agreement threshold. At agreement $a = \sqrt{(k-1)/n} + \eta_0$, with $\eta_0 > 0$, we obtain an exceptional count of $O_\rho(n/\eta_0^3)$ in every characteristic. Over $\mathbb{F}_q$, dividing by q gives the soundness error. This bound remains available in small characteristic, where our first-order theorem need not apply (Section 5.7).

Efficient decoding over large fields. We now turn from bounding the lists to computing them. Above Johnson, we obtain deterministic near-linear dependence on n when the rate and the agreement margin above Johnson are fixed constants. The bit bound requires an explicit representation supporting fast field arithmetic. For the informal statement, we use an arbitrary polynomial basis: write $q = p^e$ and supply an irreducible degree- e polynomial $f(t) \in \mathbb{F}_p[t]$, representing $\mathbb{F}_q$ as $\mathbb{F}_p[t]/(f(t))$.

Theorem 1.3 (Efficient decoding, informal). *Fix $0 < \rho < 1$, $q \geq n$, distinct evaluation points in $\mathbb{F}_q$, and $1 \leq k \leq \rho n$. For $\sqrt{\rho} < a \leq 1$, the complete degree- $< k$ agreement list at threshold a can be found deterministically in $\tilde{O}_{\rho,a}(n \log q)$ bit operations over every finite field supplied in such a polynomial basis. For $a_1(\rho) < a < 1$ and $p > k-1$, the corresponding bound is $\tilde{O}_{\rho,a}(n^2 \log q)$.*

The detailed decoder theorems also allow other representations with the operations specified in Definition 9.1, including relative polynomial bases and fixed-height towers.

Beyond Johnson, the interpolation equation can have many more solutions than the agreement list. We use the received agreements to narrow the candidates before recovering their coefficients. This avoids searching over field elements that never lead to a close message. It gives the quadratic first-order bound in Corollary 9.12 and the higher-order bound $\tilde{O}_{\rho,a,d}(n^{O(d+1)} \log q)$ at fixed rate, agreement, and sufficient derivative order d (Theorem 9.15). We explain the recovery principle in Section 2.4. The proof-size savings below use our counting bounds; efficient decoding separately supports extraction in knowledge-soundness arguments.

Families of received words. The proof systems we consider often combine many received words into one, rather than just two. Some also reuse the same random challenge across several combinations, so the corresponding agreement guarantees must hold together. We extend the common-agreement guarantee to these settings (Section 7). Combinations whose coefficients are powers of one challenge need a separate polynomial-curve argument. A depth- h multilinear fold repeatedly combines pairs of words by $(1-r)u + rv$, using one random coefficient r for all pairs at a level and a fresh coefficient at the next level. Interleaving is precisely the setting in which several combinations share their random coefficients. Its guarantee lets us charge at most E/q per level, not per pair, giving total error at most hE/q when each level has at most E exceptional challenges (Corollary 7.8).

Finite parameters and proof-size reductions. We obtain the first proof-size reductions in existing proof-system implementations from provable Reed–Solomon proximity-gap bounds beyond the Johnson radius. Our first-derivative bounds permit fewer queries at existing security targets, saving field values and authentication hashes.

These gains are practical in systems that already use sufficiently large challenge fields with sufficiently large characteristics. ProveKit, ZisK, and LambdaVM use BN254 or cubic Goldilocks, with field sizes of approximately 254 and 192 bits, respectively. Our first-order line-MCA error scales as $O_p(n^2/(q\eta_1^4))$. At these systems’ code parameters, the large field cardinalities keep the explicit finite bounds within the existing security budgets, including the batching and folding contributions. Higher derivatives lower the agreement threshold further but enlarge the list and exceptional-count bounds, making first order the useful tradeoff here.

Against the existing implementations, we measure mean reductions of 79.4 KB (11.1%) for ProveKit passport proofs and 16.7 KB (10.3%) for its lookup proofs, together with reductions of 13.2 KB (4.63%) for ZisK’s compressed final proofs and 59.8 KB (4.85%) for LambdaVM’s CPU subproofs. [Section 8](#) gives the measurements, finite security certificates, and comparisons with optimized Johnson parameters.

Our sharper Johnson-regime MCA bound may also reduce proof sizes over binary fields, where the first-derivative theorem does not apply. We leave that evaluation to future work.

Formal verification in Lean. We have formally verified our quantitative list-decoding and MCA bounds in ArkLib, a Lean library for verified cryptographic proofs [\[DA26a\]](#). The proofs cover both branches of the first-order curve, the uniform capacity bounds, and their affine, polynomial-curve, and interleaved extensions. We also verify the finite parameter inequalities and proof-size accounting underlying our ProveKit, ZisK, and LambdaVM reductions. We also formalize a simplified executable decoder and prove that it returns every polynomial meeting the requested degree and agreement bounds, and no others, without duplicates [\[DA26b\]](#). The bounded-input list-recovery extension, fast reconstruction algorithms, bit-complexity analysis, and complete implemented proof systems remain outside the formalization scope. [Appendix I](#) maps the main quantitative results to their Lean declarations and explains a representative statement.

AI assistance and author responsibility. GPT-5.6 Sol, GPT-6 Astra, and Claude Fable 5.1 discovered and refined the main results, including the list-decoding and MCA bounds and the concrete specializations, with our close prompting and guidance. Sol and Astra carried out the Lean formalization and wrote most of the manuscript under our direction.¹ We take full responsibility for the results, proofs, and exposition. [Section 10](#) discusses prior and contemporaneous work, including what those works disclose about their use of AI.

1.2 Related Works

We compare the main guarantees here, separating the choice of code, the strength of the agreement bounds, and the cost of computing a list. [Section 10](#) gives the detailed parameter comparisons and algorithmic qualifications.

Capacity on prescribed evaluation sets. Brakensiek, Gopi, and Makam proved capacity list bounds with high probability over random evaluation sets in sufficiently large fields [\[BGM23\]](#); subsequent work reduced the required field size from exponential to polynomial and then linear in n [\[GZ23, AGL24, AGG⁺25\]](#). These results neither give efficient capacity decoders nor establish the same bounds for evaluation domains already fixed by a protocol. Folded Reed–Solomon and multiplicity codes offer another route to capacity, but change the received symbols: multiplicity codes, for example, transmit derivative evaluations [\[GR08, GW13, Kop15\]](#). Our input consists only of ordinary Reed–Solomon values on any prescribed set of distinct evaluation points.

For this input, BCPZZ’s hidden-derivative method builds on Naskręcki’s Proximity Prize submission [\[Nas26, BCP⁺26a\]](#). Their initial low-rate result was extended to all constant rates through a padding observation of Alrabiah, Goyal, and Guruswami [\[BCP⁺26a, Section 5\]](#). Zheng’s unpublished manuscript

¹The authors estimate total agent usage across all stages and authors at the equivalent of ten to twelve weekly usage allowances of an OpenAI Pro subscription, with roughly half devoted to Lean formalization.

establishes MCA at low rates, while Jeronimo’s work, concurrent with ours, reaches all rates and also treats polynomial combinations and list recovery [rs-26, Jer26]. Our higher-order contribution is quantitative: for agreement $\rho + \delta$, instantiating BCPZZ’s parameters gives sufficient order $\exp(O_\rho(\delta^{-1} \log(1/\delta)))$ at fixed positive rate, whereas our explicit $\lceil \exp(1.5/\delta) \rceil$ bound works uniformly over all rates for $0 < \delta < 0.24$. The derivative order controls the exponents in the list-size, MCA, and decoding bounds.

First-order bounds and MCA. To obtain our first-order curve, we revisit the interpolation of Naskręcki and BCPZZ, restricting the allowed monomials and counting the independent conditions they must satisfy. Optimizing the derivative degree gives the two branches of the curve. For list bounds and MCA, we use Kopparty’s Taylor reconstruction and Zheng’s symbolic solution-family and incidence argument [Kop15, rs-26], sharpening the counts by treating each factor of the interpolation polynomial separately. At Johnson, our exceptional count $O_\rho(n/\eta_0^3)$ improves BCHKS’s $O_\rho(n/\eta_0^5)$ bound in every characteristic [BCH⁺26, full version, Theorem 4.6]. After division by the field size, this improves the MCA-error bound by a factor of η_0^2 . Earlier Proximity Prize submissions already exhibit the shorter Taylor reconstruction and shared degree accounting behind these savings at the fixed challenge instance [Bit26a, jie26a]. Section 10.5 distinguishes these antecedents and later multiple-equation methods from our general bounds. For affine and interleaved families, we apply the reductions of Bordage, Chiesa, Guan, and Manzur and of Jo [BCGM25, Jo26a]; coefficients that are powers of one challenge require the separate polynomial-curve argument.

Deterministic decoding over large fields. Fast Guruswami–Sudan interpolation and power-series lifting already have near-linear arithmetic cost [JNSV17, NRS17]. Augot–Pecquet and Chatterjee, Harsha, and Kumar (CHK) use received values to avoid general factorization, with CHK obtaining deterministic polynomial time over every finite field at every agreement strictly above Johnson [AP00, CHK26]. Goyal, Harsha, Kumar, and Shankar also state a deterministic near-linear Johnson decoder, but their reconstruction invokes a polynomial root-finding routine whose base case uses univariate factorization [GHKS24, Ale05]. Our zeroth-order decoder combines fast arithmetic with agreement-based recovery to obtain a deterministic near-linear *bit* bound, without a finite-field root-finding oracle, at fixed rate and positive Johnson gap over the supplied polynomial bases in Theorem 1.3. The distinction matters when the field is cryptographically large.

Beyond Johnson, the stated BCPZZ and Jeronimo decoders enumerate initial derivative values, giving polynomial dependence on q [BCP⁺26a, Jer26]. We retain Kopparty’s Taylor reconstruction and the symbolic solution-family viewpoint of Zheng and Jeronimo, but impose received agreements before extracting individual candidates. This makes the cost polynomial in n and $\log q$, with a quadratic dependence on n at first order. Section 2.4 explains how the three decoders achieve this.

1.3 Organization

Section 2 explains the counting and decoding arguments. The formal development begins with common interpolation in Section 3 and the first-order support and agreement curve in Section 4. Section 5 turns these equations into list and MCA bounds, including the Johnson specialization. Higher derivatives approach capacity in Section 6, and Section 7 treats larger families of received words. The applications in Section 8 use these finite bounds to reduce proof sizes; Section 9 then develops the algorithms that compute the lists. Detailed comparisons with prior work appear in Section 10. The core counting, transfer, and decoder-coverage proofs accompany their statements. The appendices supply general algebraic tools, finite certificates, decoder arithmetic, experimental details, and the Lean map.

2 Technical Overview

We explain the main ideas behind our list-decoding and MCA bounds, starting with the first derivative and then turning to higher orders. We then show how agreement constraints lead to efficient decoding algorithms.

Fix a field F , distinct evaluation points $\alpha_1, \dots, \alpha_n \in F$, a received word $y \in F^n$, and an agreement threshold $k \leq A \leq n$. A *message* is a polynomial of degree at most $D = k - 1$, and it is *close* if it agrees with y in at least A positions. We use ρ as an upper bound on the rate k/n and a as a lower bound on the agreement fraction A/n , with $0 < \rho < a < 1$. The positive-order reconstruction arguments assume characteristic zero or sufficiently large positive characteristic; we specify the degree-dependent bound below. The zeroth-order decoder works in every characteristic. We describe the first-order construction for $k \geq 2$; constant codes have an elementary separate treatment.

2.1 First-order interpolation and list bounds

2.1.1 Agreement forces a hidden-derivative equation

Our first task is to construct one equation satisfied by every close message, although we do not know where it agrees. Guruswami–Sudan interpolation chooses a nonzero $Q(X, Y)$ with multiplicity m at every received pair [GS99]. If its allowed monomials ensure that $\deg_X Q(X, P(X)) < mA$, then A agreements force $Q(X, P(X)) = 0$: a nonzero polynomial cannot have more roots, counted with multiplicity, than its degree. The conditions are linear in the coefficients of Q , so more coefficients than independent conditions guarantee a solution.

Hidden derivatives preserve this argument but change the local conditions [Nas26, BCP⁺26a]. Fix a received pair $(\alpha, b) = (\alpha_i, y_i)$ at which $P(\alpha) = b$; backward Taylor expansion gives

$$P(\alpha + T) - b - TP'(\alpha + T) \equiv 0 \pmod{T^2}.$$

Thus the secant slope $(P(\alpha + T) - b)/T$ and the derivative differ by a multiple of T . We use this backward expansion because P and P' are both evaluated at the same point $X = \alpha + T$, allowing us to substitute them into one global differential equation. Introduce variables Y_0, Y_1 for the value-derivative pair (P, P') , called the first-order jet. We now seek a polynomial $Q(X, Y_0, Y_1)$ and impose the local identity

$$Q(\alpha + T, b + TY_1 + T^2V, Y_1) \equiv 0 \pmod{T^m}$$

in formal variables V, Y_1 . Substituting $Y_1 = P'(\alpha + T)$ and the polynomial $V = (P(\alpha + T) - b - TP'(\alpha + T))/T^2$ forces multiplicity m at each agreement. We restrict the monomials so that $\deg_X Q(X, P, P') < mA$, and the same root-counting argument applies. No derivative evaluations are received. The resulting nonzero Q satisfies $Q(X, P, P') = 0$ for every close message; we call it a *hidden-derivative explainer*.

The local dependencies exploited in Naskręcki’s construction are also central to our count [Nas26]. We derive a rank bound for a family of supports with separate derivative and total-degree bounds, then optimize these bounds to obtain the rate curve. Adding Y_1 creates more coefficients, but also more local conditions. These conditions are not independent: the identity above relates the possible value and derivative at each position. We count those dependencies while restricting both $\deg_{Y_1} Q \leq B_\partial$ and $\deg_{Y_0, Y_1} Q \leq B_{\text{jet}}$. Here B_∂ bounds the exponent of the derivative variable, whereas B_{jet} bounds total degree in the two message variables; neither is a derivative order.

Concretely, we allow the monomials

$$X^x Y_0^{j_0} Y_1^{j_1}, \quad j_1 \leq B_\partial, \quad j_0 + j_1 \leq B_{\text{jet}}, \quad x + D(j_0 + j_1) < mA.$$

The last inequality ensures the required degree bound after substituting any message. At total message degree t , there are $\min\{t, B_\partial\} + 1$ choices of (j_0, j_1) and $(mA - Dt)_+$ choices of x , where $w_+ = \max\{w, 0\}$. Thus the number of coefficients we can choose is

$$N_{\text{coef}} = \sum_{t=0}^{B_{\text{jet}}} (\min\{t, B_\partial\} + 1)(mA - Dt)_+.$$

To see why we must count independent conditions rather than formal equations, take $m = 4$ at one received pair and write $T = X - \alpha$. Under the local substitution $Y_0 = b + TY_1 + T^2V$,

$$T^2(Y_0 - b) - T^3Y_1 \mapsto T^4V \equiv 0 \pmod{T^4}.$$

The two terms on the left have the same local image, so they do not impose independent conditions. Accounting for these dependencies gives a local rank bound R_{loc} ; when $B_\partial \leq m/2$, it is

$$R_{\text{loc}} = (B_\partial + 1) \left(\frac{m(m+1-B_\partial)}{2} + \frac{B_\partial(2B_\partial+1)}{6} \right).$$

We can therefore interpolate a nonzero explainer whenever $N_{\text{coef}} > nR_{\text{loc}}$. For example, take $(n, k, A) = (1000, 250, 480)$ and $(m, B_\partial, B_{\text{jet}}) = (24, 10, 46)$. There are 2,402,466 available coefficients and at most $1000 \cdot 2365 = 2,365,000$ independent conditions. The surplus is positive at agreement 0.48, below the exact Johnson threshold $\sqrt{249/1000} \approx 0.499$. This certifies the interpolation step; the list and MCA bounds still require the counting arguments below.

Optimizing these degree bounds gives a positive coefficient surplus whenever $a > a_1(\rho)$, so a sufficiently large finite multiplicity m gives the required nonzero explainer. Sections 4.2 and 4.3 derive the counts and the two branches of the curve. Local interpolation works in every characteristic. For the following first-order counting argument, we assume characteristic zero or $p > \max\{D, B_\partial\}$.

2.1.2 From differential solutions to a candidate cover

A derivative-free polynomial equation has only boundedly many polynomial roots, since each root gives a factor $Y - P(X)$. A differential equation can have far more solutions. For example, $Q = Y_1$ gives $P' = 0$; in characteristic zero or $p > D$, every degree- $\leq D$ solution is constant. Over a field of characteristic greater than 8, the word $(3, 3, 8)$ has only one constant message agreeing twice, namely $P = 3$, even though every field constant solves $P' = 0$. The differential equation becomes useful only when we retain the received agreements.

Kopparty's recurrence reconstructs a solution from its initial Taylor coefficients [Kop15]: writing $P(c + T) = u_0 + u_1T + u_2T^2 + \dots$ at a center c , we first obtain the constraint $Q(c, u_0, u_1) = 0$ on the initial pair. To determine the next coefficient, differentiate $Q(X, P, P') = 0$ and evaluate at $X = c$. Using subscripts on Q for partial derivatives gives, when $k \geq 3$,

$$2Q_{Y_1}(c, u_0, u_1)u_2 = -Q_X(c, u_0, u_1) - u_1Q_{Y_0}(c, u_0, u_1).$$

Thus, once the initial value u_0 and derivative u_1 are chosen with $Q_{Y_1}(c, u_0, u_1) \neq 0$, the equation determines u_2 and then each later coefficient. We call these *regular initial data at c* ; the characteristic hypothesis makes the integers $1, \dots, D$ appearing in the recurrence invertible.

Regularity at a center must be distinguished from the condition that $Q_{Y_1}(X, P, P')$ is not identically zero. Such a polynomial can still vanish at a particular center. For counting, we work over an auxiliary extension and take c transcendental over the algebraic closure $\bar{F}$. No nonzero polynomial over $\bar{F}$ vanishes at c , so this

one center works for every solution over $\overline{F}$ with this nonzero specialization of Q_{Y_1} . Finding effective centers for decoding is a separate step.

Following Zheng's use of symbolic solution families [rs-26], we keep u_0, u_1 as variables on the curve $Q(c, u_0, u_1) = 0$ and express the later coefficients by Kopparty's rational recurrence. On the regular locus, the resulting coefficient map is

$$(u_0, u_1) \mapsto (u_0, u_1, u_2(u_0, u_1), \dots, u_D(u_0, u_1)).$$

After converting Taylor coefficients to coefficients in X , each image point represents a polynomial through its coefficient vector. We take the algebraic closure of the image, meaning the smallest set defined by polynomial equations that contains it. This set is a union of curves, each a family with one free parameter, and possibly isolated points. Below we add the solutions missed by regular reconstruction. The resulting *candidate cover* contains every differential solution, hence every close message, but may contain more: a Taylor truncation need not solve the equation globally, let alone agree with the received word. Different parameters can also give the same message.

Call the completed cover C , identifying polynomials with their coefficient vectors over an algebraically closed auxiliary field. The desired list is the following subset of this cover:

$$\mathcal{L}(y, A) = \{P \in C \cap F[X]_{<k} : |\{i : P(\alpha_i) = y_i\}| \geq A\}.$$

Because $A \geq k$ and the received pairs lie in F , agreement also enforces the field condition. If a degree- $< k$ polynomial over the auxiliary field agrees at any k received positions, interpolating those pairs over F gives a polynomial with the same values. Their difference has degree less than k and at least k roots, so it is zero. Thus every candidate with enough agreements already has its coefficients in F , even if its initial parameters lie in a larger field.

We still have to identify the candidates with those agreements. The displayed set describes the list; it does not tell us how to enumerate a curve or find its qualifying points. For the counting theorem, we bound their number by intersecting the cover with agreement equations. In [Section 2.4.1](#), we turn the same observation into an algorithm: polynomial gcds separate agreeing candidates, and k agreements recover a message by interpolation. First we need the curves' degrees, since dimension alone cannot bound how many points meet the agreement threshold.

2.1.3 Many agreements leave few points on the curve

Zheng's incidence argument uses the received positions to count points in the solution families [rs-26]. We apply it with sharper degree bounds. Identify a candidate polynomial with its coefficient vector. Agreement at position i is the linear equation $P(\alpha_i) = y_i$, an *agreement hyperplane* in this space. Any k evaluations determine a polynomial. Hence a curve of distinct candidates cannot satisfy k agreement equations identically: their intersection would be one point.

The degree of a curve component controls how many points a linear equation can select from it, provided the equation does not hold on the whole component. Suppose a curve component of degree Δ lies in exactly $h \leq D$ agreement hyperplanes. Every close point must lie on at least $A - h$ of the remaining $n - h$ hyperplanes. Each of these cuts the curve in at most Δ points. Counting incidences between close points and these proper sections therefore bounds the number of close points by

$$\Delta \frac{n - h}{A - h} \leq \Delta \frac{n - D}{A - D}.$$

The second inequality uses $h \leq D < A$ and the fact that $(n - h)/(A - h)$ increases with h . This is why agreement gives a small list. The higher-dimensional version repeats the argument on hyperplane sections ([Lemma 5.1](#)).

Accounting for the solutions missed by reconstruction. To use this incidence bound, we need a cover of controlled degree that includes the solutions where the Taylor recurrence fails. We use the regular–singular separation already present in Naskręcki’s construction [Nas26], retaining the individual factor degrees in the count. Remove repeated factors from the explainer and write its squarefree part as $Q_{\text{free}}Q_{\text{der}}$: the first factor is independent of Y_1 , and every irreducible factor of the second involves Y_1 . If $Q_{\text{der}} = 1$, all solutions already satisfy $Q_{\text{free}} = 0$. Otherwise, the bound $p > B_\partial$ in positive characteristic makes Q_{der} separable in Y_1 . Applying the Taylor map at the transcendental center covers all its solutions for which $(Q_{\text{der}})_{Y_1}(X, P, P')$ is not identically zero. A solution missed by this map makes both $Q_{\text{der}}(X, P, P')$ and $(Q_{\text{der}})_{Y_1}(X, P, P')$ identically zero. We need a consequence of these two identities that no longer involves P' .

The *resultant* supplies this consequence. For two univariate polynomials, $\text{Res}_Y(f, g)$ is a polynomial expression in their coefficients that vanishes when they have a common root. It can be computed as a determinant, without finding the roots. When f is monic, its meaning is particularly simple:

$$\text{Res}_Y(f, g) = \prod_{f(\theta)=0} g(\theta),$$

with roots counted according to multiplicity. If the coefficients themselves depend on other variables, the resultant is a polynomial in those variables; every common solution of the two original equations makes it zero.

Apply this operation in Y_1 to Q_{der} and its Y_1 derivative. Separability makes their resultant a nonzero polynomial in X, Y_0 . For each missed solution, substituting $Y_0 = P(X)$ makes this polynomial identically zero. Including the derivative-free factors gives one ordinary equation for all remaining messages:

$$Q_{\text{free}} \text{Res}_{Y_1}(Q_{\text{der}}, (Q_{\text{der}})_{Y_1}) = 0.$$

Each polynomial solution of this nonzero equation gives a factor $Y_0 - P(X)$, so their number is bounded by its Y_0 -degree. This is how we count the solutions whose separant vanishes identically, without applying the failed Taylor rule to them.

Thus one regular family and one derivative-free equation cover all solutions. Our degree estimate counts the regular family once, rather than charging a separate Taylor family for every successive derivative of Q , and retains the separate Y_1 -degree throughout the image-degree estimate. The resulting sum of regular component degrees is $O(DB_{\text{jet}}B_\partial)$. At fixed ρ and positive gap $\eta_1 = a - a_1(\rho)$, B_{jet} and B_∂ are constants, so this total degree is $O(D) = O(n)$. Also $(n - D)/(A - D) = O(1)$. The summed incidence bound is therefore $O(n)$; adding the derivative-free part preserves this linear list bound. The successive-stage alternative remains available when it gives a better finite certificate (Lemmas 5.6 and 5.9).

2.2 MCA: follow candidates as the received word varies

For MCA, we need one exceptional set that works for every close candidate, including a candidate chosen after seeing the challenge. Zheng’s persistent-line argument supplies the geometric way to organize this requirement; Jeronimo independently uses the same type of symbolic solution family [rs-26, Jer26]. We retain that organization and sharpen the degrees and agreement thresholds that control its exceptional set.

2.2.1 Common witnesses and accidental agreements

Fix received words $f, g \in F^n$ and follow the received line $f + zg$. For degree- $< k$ polynomials F_0, G_0 , define their common agreement set by

$$S(F_0, G_0) = \{i : F_0(\alpha_i) = f_i \text{ and } G_0(\alpha_i) = g_i\}.$$

The polynomial $F_0 + zG_0$ agrees with $f + zg$ at these positions for every challenge z . The MCA conclusion is a converse: outside one small set $\mathcal{E}$, every close degree- $< k$ polynomial P has witnesses F_0, G_0 over F such that

$$P = F_0 + zG_0, \quad \{i : P(\alpha_i) = f_i + zg_i\} = S(F_0, G_0).$$

The equality matters. Expressing P as a linear combination of witnesses does not by itself rule out extra agreements created by cancellation.

Any k agreeing positions already determine a possible witness pair: interpolate f and g separately on those positions to obtain F_0, G_0 . The polynomials P and $F_0 + zG_0$ then agree at k points, so they are equal. As z varies, these pairs $(z, F_0 + zG_0)$ trace a *witness line* in the joint space of challenges and message coefficients. At another position, the agreement residual on this line is

$$(F_0(\alpha_i) - f_i) + z(G_0(\alpha_i) - g_i).$$

It vanishes for every z exactly when both witnesses agree there. Otherwise it is a nonzero polynomial of degree at most one, so the position can add an agreement at at most one challenge.

For example, with constant messages and received line $(z, z, 1)$, the candidate $P = z$ has witnesses $(0, 1)$ on the first two positions. At $z = 1$ it also agrees at the third, solely by cancellation. That challenge must be excluded to obtain equality of the full agreement sets.

For one fixed witness line, at most n challenges can cause such accidental agreements. We cannot simply multiply this bound by the number of choices of k positions: there can be $\binom{n}{k}$ such choices. Instead, we place all close candidates in one algebraic family and use its degree to control both the relevant witness lines and the candidates away from them.

2.2.2 One bounded-degree family for all challenges

We first need an equation that follows the challenge without acquiring a large degree in it. Following BCIKS's function-field viewpoint and Zheng's symbolic interpolation [BCI⁺23, rs-26], replace z by an indeterminate Z and interpolate $Q(X, Z, Y_0, Y_1)$ over the rational-function field $F(Z)$. The fixed-word coefficient surplus ensures a nonzero solution over this field. To obtain a useful degree bound for the joint family, we also bound the degree of its coefficients in Z .

Suppose the interpolation matrix has N columns and rank $R < N$ over $F(Z)$, with entries in $F[Z]$ of degree at most b_Z . Choose R original rows spanning the row space. If each unknown coefficient is a polynomial of degree at most H in Z , there are $N(H + 1)$ scalar unknowns. Each chosen row becomes a polynomial identity of degree at most $H + b_Z$, hence imposes at most $H + b_Z + 1$ scalar equations. Consequently,

$$N(H + 1) > R(H + b_Z + 1) \iff (N - R)(H + 1) > Rb_Z$$

guarantees a nonzero polynomial solution. Taking $H = \lfloor Rb_Z / (N - R) \rfloor$ suffices. At fixed interpolation parameters and positive gap, b_Z is bounded and $N - R$ is a fixed fraction of N , so H is independent of n . This is the quantitative role of retaining the interpolation surplus rather than merely proving it positive.

We also prevent the equation itself from disappearing at a challenge. Divide its coefficient polynomials in Z by their gcd. The quotients still satisfy the homogeneous interpolation identities, and their gcd is now one. A polynomial combination of them therefore equals 1, so they cannot all vanish at the same z , even over an extension field. Thus $Q(X, z, Y_0, Y_1)$ is nonzero at every challenge, while substituting a close message still makes it vanish. The graded version of this argument uses the individual row and column degrees to improve finite parameters (Lemma 3.12 and Proposition 5.10).

We now repeat the Taylor construction while allowing the challenge to vary, following Zheng's symbolic-family construction. As above, let Q_{der} be the product of the distinct factors involving Y_1 , and use the

transcendental Taylor center c . The regular initial data satisfy

$$Q_{\text{der}}(c, Z, u_0, u_1) = 0, \quad (Q_{\text{der}})_{Y_1}(c, Z, u_0, u_1) \neq 0.$$

Kopparty's recurrence expresses each later Taylor coefficient as a rational function of Z, u_0, u_1 . After converting back to coefficients in X , it gives the map

$$(Z, u_0, u_1) \mapsto (Z, P_{Z, u_0, u_1}).$$

Every regular solution is recovered from its own initial values $u_0 = P(c)$ and $u_1 = P'(c)$, so taking the algebraic closure of this image covers all regular pairs (z, P) . Keeping Z in the output records which challenge belongs to each candidate. For a fixed symbolic challenge, the initial pair lies on a curve; allowing the challenge to vary adds one parameter. Thus the joint image has dimension at most two, even though the message itself has k coefficients. Solutions where this recurrence fails are handled separately below.

To turn this description into a bound on exceptional challenges, we need the degrees of these families, not only their dimensions. Write J for the total degree of the joint image, and B_{fib} for the degree of its candidate curve when the challenge is an indeterminate. The Taylor map also controls these degrees: at fixed interpolation parameters, its formulas have degree $O(n)$. A linear section of the coefficient image pulls back to an equation of that degree on the initial data. Counting the image of a curve uses one such section; counting the joint surface uses two. The resulting bounds are

$$B_{\text{fib}} = O(n), \quad J = O(n^2).$$

Tracking the degree in the challenge separately gives the finite versions of these bounds.

It remains to account for solutions outside the regular family. As in the fixed-word construction, these satisfy the derivative-free equation obtained from the resultant. The MCA argument for equations without derivatives bounds their exceptional challenges, including those where the equation vanishes identically. This contribution also fits within $O(n^2)$; the Johnson specialization below explains that argument. For the regular family, we now use J and B_{fib} to bound the two ways in which MCA can fail.

2.2.3 Two counts for the regular family

Choose an integer L with $k \leq L < A$. Within the joint cover, retain every witness line with at least L common agreements. To see what remains to prove, write $T_z(P)$ for the full agreement set of a close candidate P with $f + zg$. If (z, P) lies on a retained line with witnesses F_0, G_0 , then

$$P = F_0 + zG_0, \quad S(F_0, G_0) \subseteq T_z(P).$$

The inclusion holds because every common agreement is an agreement at every challenge. Equality would give exactly the MCA conclusion; the only possible failure on this line is an extra agreement in $T_z(P) \setminus S(F_0, G_0)$. We therefore count two classes of pairs: close candidates outside all retained lines, and candidates on retained lines with an extra agreement. Every MCA failure in the regular family belongs to one of these classes. The counts may include harmless pairs too; an upper bound on the exceptional set is all we need.

The first class is finite. Any k agreements place a candidate on a witness line. If the line is not contained in the cover, it meets the cover in only finitely many points. If it is contained but has fewer than L common agreements, reaching A agreements requires at least one additional position whose residual is not identically zero on the line. Each such residual permits at most one challenge. There are finitely many choices of positions and lines, so these observations establish finiteness. We obtain the useful bound by counting proper hyperplane sections of the cover, not by enumerating those choices.

The second class consists of accidental agreements on retained lines. To bound the number of these lines, regard Z as an indeterminate: each line gives the single polynomial $F_0 + ZG_0$ over $F(Z)$. Distinct witness

pairs give distinct polynomials, and each has at least L agreements with $f + Zg$. We can therefore apply the fixed-word list count to this one symbolic received word. This is the reason for keeping a separate degree bound B_{fib} .

A component confined to one challenge value does not appear over $F(Z)$. It is not discarded: its high-agreement points are included in the joint-cover count for the first class. Thus passing to a symbolic challenge does not omit exceptional fibers.

2.2.4 From the two degree bounds to quadratic MCA

The same incidence argument used for lists now gives, for the regular family, the bounds

$$\begin{aligned} \#\{\text{close points off retained lines}\} &\leq J \frac{n-D}{A-D} \frac{n-L+1}{A-L+1}, \\ \#\{\text{retained witness lines}\} &\leq B_{\text{fib}} \frac{n-D}{L-D}. \end{aligned}$$

For the first inequality, intersecting any k agreement hyperplanes leaves only a witness line. A surface therefore has at most $D = k - 1$ universal agreements, giving the first incidence factor. A proper agreement section cuts it to curves; curves not retained as witness lines have at most $L - 1$ universal agreements. The two incidence factors account for these two successive cuts. The second inequality is the earlier curve count, now over $F(Z)$ and with agreement threshold L .

This also explains the choice of L . Increasing it makes the retained lines easier to count, through the denominator $L - D$, but leaves fewer agreements above the retention threshold, through $A - L + 1$. When $A - D$ is a fixed positive fraction of n , choose L between D and A so that both margins are linear in n . All displayed ratios are then bounded. Since $J = O(n^2)$ and $B_{\text{fib}} = O(n)$, the first class contributes $O(n^2)$ points, hence at most that many challenges under projection $(z, P) \mapsto z$. Each retained line has at most $n - L$ positions outside its common set, so the second class contributes another $O(n) \cdot O(n) = O(n^2)$ challenges.

Take $\mathcal{E}$ to contain these challenges and the exclusions from the ordinary part of the cover. Outside $\mathcal{E}$, a close candidate lies on a retained line and has no accidental agreements. Interpolation on any k common positions supplies its witnesses over F , proving the full agreement-set equality. For a uniform challenge in $\mathbb{F}_q$, dividing the exceptional count by q gives the MCA error. Lemma 5.2 states the finite transfer; our first-order degree bounds make its exceptional count quadratic in n .

2.2.5 The Johnson specialization

Without hidden derivatives, the same argument starts from BCHKS's symbolic Guruswami–Sudan interpolant and factor aggregation [BCH⁺26]. The relevant families have smaller dimension, and we sharpen the degree charged to them. At fixed rate and Johnson agreement gap η_0 , BCHKS's exceptional bound has scale n/η_0^5 , while ours has scale n/η_0^3 .

We save one inverse-gap factor by reconstructing only the coefficients of the actual message, instead of continuing to the larger cutoff allowed by the interpolation bound. We save another by bounding the dependence on the challenge separately from the dependence on the initial message value. These two savings have finite-instance antecedents in the Proximity Prize submissions of BitWonka and jieyilong [Bit26a, jie26a]. Following BCIKS, we also handle repeated factors [BCI⁺23]; Section 5.7 explains how to preserve these degree savings in every characteristic. That analysis supplies the derivative-free part of the first-order count as well.

2.3 Higher derivatives approach capacity

BCPZZ use higher hidden derivatives to reach capacity [BCP⁺26a]. We keep their local interpolation conditions and choose a different weighted support to obtain explicit quantitative bounds. The first-order argument already shows the two ingredients we need: a positive coefficient surplus and degree bounds for the resulting solution families. At higher order, we track how both depend on the number of derivatives.

2.3.1 More local vanishing and a weighted coefficient count

We use the normalized derivatives defined by

$$P(c + T) = \sum_{j \geq 0} P^{[j]}(c) T^j.$$

These are the *Hasse derivatives*: $P^{[0]} = P$, $P^{[1]} = P'$, and $P^{[j]} = P^{(j)}/j!$ whenever $j!$ is invertible. The tuple $P(c), P^{[1]}(c), \dots, P^{[d]}(c)$ is the order- d jet. Here d counts derivatives; B_{jet} instead bounds the polynomial degree of the explainer in its $d + 1$ message variables. At an agreeing pair (α, b) , the higher-order backward Taylor identity is

$$P(\alpha + T) - b - \sum_{j=1}^d (-1)^{j+1} T^j P^{[j]}(\alpha + T) \equiv 0 \pmod{T^{d+1}}.$$

As at first order, we substitute the resulting expression for Y_0 into the local interpolation conditions while leaving $Y_1, \dots, Y_d$ formal. The remainder now carries T^{d+1} rather than T^2 . Its higher powers disappear sooner modulo T^m , reducing the number of local coefficients we need to set to zero. No derivative evaluations are supplied.

To exploit this saving, choose an interpolation dimension $K \geq \max\{k, d + 1\}$ and a weight bound W . We allow

$$X^x Y_0^u \prod_{j=1}^d Y_j^{b_j}, \quad \sum_{j=1}^d j b_j \leq W, \quad x + (K - 1) \left(u + \sum_j b_j \right) < mA.$$

The first restriction charges each hidden derivative according to its order. The second ensures specialization degree less than mA for every polynomial of degree less than K , hence for every actual message.

These restrictions give finite counts, just as the two degree bounds did at first order. Write $|b| = \sum_j b_j$, and let $p_d(B)$ be the number of nonnegative integer tuples b with $\sum_j j b_j \leq B$. Then

$$N_{\text{coef}} = \sum_{\sum_j j b_j \leq W} \sum_{u \geq 0} (mA - (K - 1)(u + |b|))_+,$$

$$R_{\text{loc}} \leq \sum_{s=0}^{m-1} \left\lceil \frac{m-s}{d+1} \right\rceil p_d(W + s).$$

For the first sum, fixing the derivative tuple and the Y_0 exponent leaves the displayed number of choices for x . For the second, group local output terms after writing the remainder as $T^{d+1}V$. A term containing V^v has T -degree $s + (d + 1)v$ for some $s \geq 0$, and derivative weight at most $W + s$. For fixed s , truncation modulo T^m leaves at most $\lceil (m - s)/(d + 1) \rceil$ possible values of v . We count these possible output coefficients without assuming they are independent. The test $N_{\text{coef}} > n R_{\text{loc}}$ certifies an explainer.

2.3.2 Choosing the derivative order uniformly over rates

To turn these finite counts into a rate condition, let ρ_{amb} bound $(K - 1)/n$, with $0 < \rho_{\text{amb}} < a$. Our analysis sums the remaining choices of X and Y_0 over the weighted derivative tuples, rather than replacing all of them

by a worst-case degree. Increasing W allows more source monomials but also more local output coefficients. Estimating the source sum through a simplex integral balances these effects at the choice

$$W = \left\lfloor \frac{mad}{\rho_{\text{amb}} \log(6d)} \right\rfloor,$$

and gives the following limiting lower bound on the coefficient-to-constraint ratio as m increases:

$$\Gamma(\rho_{\text{amb}}, a, d) = \frac{27}{20} \frac{\rho_{\text{amb}}(d+1)}{(6d)^{\rho_{\text{amb}}/a}} \quad (d \geq 500).$$

A strict inequality $\Gamma > 1$ therefore yields $N_{\text{coef}} > nR_{\text{loc}}$ for sufficiently large finite m , after accounting for rounding. This is the same interpolation test as before, expressed as a ratio rather than a difference. The large-order estimate does not apply at $d = 1$; we use the sharper first-order analysis there.

The formula also explains how higher derivatives close the gap to capacity. At fixed $\rho_{\text{amb}} < a$, it grows like $d^{1-\rho_{\text{amb}}/a}$. The exponent is positive, so increasing d eventually makes $\Gamma > 1$. The rate has remained in the estimate, allowing us to solve explicitly for a sufficient derivative order.

For a bound uniform over all rates, we must also control the prefactor ρ_{amb} when k/n is very small. Enlarging the interpolation message space is independently used in Jeronimo's capacity argument [Jer26]. Our choice enlarges it on the low-rate branch: given $A/n \geq k/n + \delta$, choose K so that ρ_{amb} is bounded below in terms of δ while remaining separated from the agreement fraction a . This changes neither the evaluation set nor the requested messages; it asks the interpolant to explain a larger set of polynomials. Inserting the chosen rate and gap into the finite estimates gives the uniform sufficient order

$$d = \lceil e^{1.5/\delta} \rceil$$

for $0 < \delta < 0.24$ and sufficiently large n . [Theorem 6.8](#) gives the two parameter choices, proves a positive margin uniform over the rate, and accounts for the finite rounding and length requirements. The first-order analysis already handles larger gaps.

2.3.3 Counting the messages in the higher-order cover

Interpolation dimension and reconstruction length have different jobs. The dimension K was chosen to make the coefficient count work. For the counting cover, we use Kopparty's recurrence [Kop15] to reconstruct only $K_{\text{rec}} = \max\{k, d+1\}$ Taylor coefficients: enough to retain the initial jet and every coefficient of an actual message. We then set the coefficients of degrees $k, \dots, K_{\text{rec}} - 1$ to zero, intersecting with the degree- $< k$ subspace before counting. This linear section does not increase the cover's degree. The reconstruction assumes characteristic zero or $p > \max\{D, d, B_{\text{jet}}\}$.

To include solutions whose separant vanishes identically, repeatedly differentiate the equation with respect to its highest derivative variable still present. Each such derivative lowers total jet degree under the characteristic assumption. A solution is assigned to the first equation it satisfies whose derivative it does not satisfy identically; there the Taylor recurrence is regular. The chain has at most B_{jet} stages, and its derivative-free stages use ordinary polynomial reconstruction.

At a regular stage, one equation in $d+1$ initial coefficients leaves at most d free parameters. The later coefficients are rational functions of these parameters with degree $O(n)$ when the interpolation parameters are fixed. The same image-degree argument as at first order now uses at most d linear sections, giving degree $O(n^d)$ for the fixed-word cover. Summing the bounded number of stages, including those of lower order, preserves this scale. Agreement incidence converts this to an $O(n^d)$ list bound at fixed positive gap. Keeping the challenge adds one dimension and gives joint degree $O(n^{d+1})$. The persistent-witness argument then gives $O(n^{d+1})$ exceptional challenges: both the off-witness count and the $O(n^d)$ witnesses with at most n

accidental challenges have this scale. We apply the symbolic-family framework of Zheng and Jeronimo with these explicit degree estimates [rs-26, Jer26].

For the uniform capacity statement, we need only $p > k - 1$. If the remaining jet-degree condition fails, then $k \leq B_{\text{jet}}(\delta)$; the stated minimum length gives

$$\sqrt{\frac{k-1}{n}} \leq \sqrt{\frac{B_{\text{jet}}(\delta)}{n}} \leq \frac{\delta}{2} < \frac{A}{n}.$$

Thus the target agreement is already above Johnson, and the characteristic-free bound handles this case.

2.4 Decoding without enumerating initial values

We now turn from bounding the list to computing it. The counting argument allowed us to work over an algebraic closure and reason about every point of a candidate family. A decoder cannot simply enumerate those points: the field may be cryptographically large, and a first-order family may contain a whole curve of differential solutions. We need to use the received agreements before choosing initial values, not merely to check candidates after an expensive enumeration.

For decoding, let $F = \mathbb{F}_q$, with $q \geq n$ and the supplied fast arithmetic of Definition 9.1. The idea of using received information to bypass general factorization goes back to Augot–Pecquet and underlies CHK’s deterministic Johnson decoder [AP00, CHK26]. Our organization separates two tasks: first represent a sufficiently small finite family without finding its parameter roots, then recover the close messages directly from their agreements. The second task is shared by all three decoders, so we explain it first.

2.4.1 Shared recovery: use agreements to extract messages

We first illustrate recovery from a simple finite representation. Its input consists of a squarefree polynomial $G(u)$ and polynomials $c_0(u), \dots, c_{k-1}(u)$ over an auxiliary finite extension E of F . Each root θ of G specifies one candidate message polynomial

$$C_\theta(X) = c_0(\theta) + c_1(\theta)X + \dots + c_{k-1}(\theta)X^{k-1}.$$

Thus G specifies the allowed parameter values, while $c_j(\theta)$ is the coefficient of X^j in the candidate indexed by θ . We take these polynomials as given for the recovery step; the decoder constructions below explain how to obtain them without finding the roots.

To process the candidates together, retain u as a variable and work with

$$C(X) = \sum_{j < k} c_j(u)X^j \in (E[u]/G)[X]_{<k}.$$

Reducing a coefficient modulo G preserves its value at every root θ . Arithmetic modulo G therefore performs the same operation on all the candidates at once, without computing any θ .

Assume $k \leq A \leq n$. We maintain pairs (H, J) , where H is a nonconstant factor of G and J is a set of processed positions. The roots of H represent candidates with exactly the same agreement positions so far, namely J . Initially there is just one pair $(G, \emptyset)$, since no position has been processed.

At position i , the polynomial

$$r_i(u) = \sum_{j < k} c_j(u)\alpha_i^j - y_i$$

records the error at that position: $r_i(\theta) = 0$ precisely when $C_\theta(\alpha_i) = y_i$. For each current pair (H, J) , compute

$$H_{\text{yes}} = \gcd(H, r_i), \quad H_{\text{no}} = H/H_{\text{yes}}, \quad (H, J) \mapsto \begin{cases} (H_{\text{yes}}, J \cup \{i\}), & \text{agreement at position } i, \\ (H_{\text{no}}, J), & \text{disagreement at position } i. \end{cases}$$

Discard constant factors, which have no roots. Since G is squarefree, these two factors have disjoint root sets and their degrees sum to $\deg H$. Thus every parameter root follows exactly one branch; the sum of the degrees of all current factors stays at most $\deg G$.

Stop processing a pair as soon as $|J| = k$. Interpolate the unique $P_J \in F[X]_{<k}$ satisfying $P_J(\alpha_i) = y_i$ for $i \in J$. For every root θ of its factor H , we then have

$$(C_\theta - P_J)(\alpha_i) = 0 \quad (i \in J), \quad \deg(C_\theta - P_J) < k \implies C_\theta = P_J.$$

The implication holds because a nonzero polynomial of degree less than k cannot have k distinct roots. This is why we can recover a whole branch without finding any of its parameter roots: all its candidates are the single polynomial P_J , whose coefficients we obtain directly in F from the received pairs.

Check P_J against all n received positions, output it if at least A agree, and remove the pair (H, J) from further processing whether or not the check passes. No other polynomial is represented on that branch. Remove duplicate outputs by comparing coefficient vectors. Conversely, any represented candidate with at least $A \geq k$ agreements reaches this stopping rule by its k th agreeing position, so the procedure recovers every wanted message.

Here $E[u]/G$ has vector-space dimension $\Delta = \deg G$ over E : each coefficient of $C(X)$ is stored using Δ field elements. Shared polynomial arithmetic makes this recovery pass cost $\tilde{O}((n+k)\Delta)$ field operations. Keeping the representation small is therefore the computational counterpart of keeping the candidate list small, and it is the task on which our three decoders differ. The first-order constructions also use finite algebras with repeated factors; the same agreement histories apply once we adapt the splitting step below.

2.4.2 Zeroth order: a finite family directly from interpolation

At order zero, the Guruswami–Sudan equation is $Q(X, P(X)) = 0$. At a suitable center c , the initial value $P(c)$ must be a root of the univariate polynomial $Q(c, Y)$. Unlike the first-order initial pair, this is already a finite set. At fixed rate and positive gap above Johnson, fast interpolation constructs Q with bounded Y -degree and X -degree $O(n)$, using the polynomial-module algorithms of Jeannerod, Neiger, Schost, and Villard [JNSV17]. The remaining question is how to lift all its useful initial values without extracting these roots.

We first divide out the gcd of Q 's coefficient polynomials in X , removing factors that depend only on X . We then compute the bivariate radical, removing repeated factors, and discard factors inseparable in Y . This must be done carefully in small characteristic, but it cannot remove a wanted factor $Y - P(X)$: its derivative in Y is 1. The fixed Y -degree bounds the characteristic whenever coefficient p th roots are needed; the arithmetic implementation preserves the near-linear cost.

Call the resulting separable polynomial $T(X, Y)$. We use the resultant again, now to choose a center where no root is repeated. The polynomial

$$\text{lc}_Y(T) \text{Res}_Y(T, T_Y) \in E[X]$$

is nonzero. Choose c where it does not vanish. The leading-coefficient factor preserves the Y -degree, and the resultant factor ensures that $T(c, Y)$ and $T_Y(c, Y)$ have no common root. Thus every initial root is simple and can be lifted by the recurrence. These exclusions have total degree $O(n)$, so batched evaluation finds a suitable center in near-linear time; a small extension supplies enough possible centers for all sufficiently large lengths. At fixed rate and gap, the remaining bounded lengths use direct interpolation from subsets of the received positions.

Now take $G(u)$ to be the monic version of $T(c, u)$ and compute in $E[u]/G$. The element $T_Y(c, u)$ is invertible there, so the usual implicit power-series recurrence lifts the initial value u to all k message coefficients simultaneously. At order zero this recurrence divides by $T_Y(c, u)$, not by Taylor indices or factorials, which is

why it also works in small characteristic. Fast series arithmetic keeps this lifting near-linear [NRS17], and the shared agreement routine then recovers exactly the close messages.

Here $\Delta = \deg G$ is bounded independently of n , so interpolation, preparation, lifting, and recovery together take $\tilde{O}_{\rho,a}(n \log q)$ bit operations. CHK already give deterministic polynomial time above Johnson in every characteristic; our improvement combines agreement-based recovery with the fast arithmetic pipeline, avoiding a separate finite-field root-finding step. The stated near-linear bound fixes the rate and a positive Johnson gap (Theorem 9.5).

2.4.3 First order: cut a curve down to a finite family

We return to the first-order family, where one equation constrains the two initial values $P(c), P'(c)$. There are two problems to solve. We must cut this curve down to a small finite family, and we must cover all wanted messages without repeating the curve computation at $O(n)$ different centers. We begin with the first problem, assuming a suitable regular family is available, and then explain how the common-center construction supplies the missing coverage.

Write $B = B_{\text{jet}}$ for the fixed total degree bound, and for this main construction take the characteristic larger than $k - 1$ and a bound depending on B . After a linear change of the initial coordinates, a regular family has a bounded-degree equation $h(u, v) = 0$, monic in v , and a polynomial $s(u, v)$ specifying the valid part by $s \neq 0$. Here s is the original separant transported through the coordinate change; it need not equal h_v . Its nonvanishing makes the Taylor recurrence solvable. We discard any component on which s vanishes identically. The recurrence represents its candidates as

$$C_{u,v}(X) = \frac{1}{B_0(u, v)} \sum_{j < k} N_j(u, v)(X - c)^j.$$

The denominator B_0 does not vanish on the valid part, and it and the numerators have degree $O_B(k)$. Clearing this denominator turns agreement at position i into the polynomial equation

$$g_i(u, v) := \sum_{j < k} N_j(u, v)(\alpha_i - c)^j - y_i B_0(u, v) = 0.$$

We do not yet solve these equations. Our first task is to identify a small set of possible u -coordinates containing every wanted point.

The same uniqueness principle used in shared recovery bounds how many agreement tests can fail to cut the curve. Some g_i may vanish on an entire component, giving no finite cut. But at most $k - 1$ positions can agree identically on that component: otherwise interpolation would fix every coefficient of $C_{u,v}$, including its initial value and derivative, which recover (u, v) through the inverse coordinate change. Thus at every wanted point there are at least

$$t = A - k + 1$$

nonuniversal agreements, each of which cuts its component in finitely many points. We need to exploit these cuts without factoring the curve or finding which agreements hold identically on each component.

Removing universally agreeing components. A direct resultant $\text{Res}_v(h, g_i)$ can be identically zero as soon as g_i vanishes on one whole component. Instead, introduce a formal variable W and compute

$$\chi_i(u, W) = \text{Res}_v(h(u, v), W - g_i(u, v)).$$

To understand this polynomial, temporarily imagine the roots of h in v : the resultant multiplies the factors $W - g_i(u, v)$ over those roots. On a universally agreeing component, g_i is identically zero, so its factors

contribute only a power of W . The lowest nonzero coefficient in W , which we call $v_i(u)$, retains the resultant contribution from the other components. Consequently, every nonuniversal agreement point projects to a root of v_i , including at intersections of components or where roots in a projection fiber coincide. Each v_i has degree $O_B(k)$, and every wanted u -coordinate is a root of at least t distinct polynomials among $v_1, \dots, v_n$.

Selecting frequently occurring roots. Gcds and lcms let us select these frequently occurring roots without finding them. A gcd keeps roots appearing in both inputs, while an lcm keeps roots appearing in either; a sorting network built from these operations produces a polynomial H_t for roots appearing in at least t inputs. Its degree satisfies

$$\deg H_t \leq \frac{\sum_i \deg v_i}{t} = O_B(nk/t).$$

We count distinct received positions here. A root repeated many times in one v_i still certifies only one position, so high multiplicity in the product $\prod_i v_i$ would not be a valid substitute for this test.

After removing repeated factors from H_t , let G be the resulting squarefree polynomial. Keeping all points above its roots gives the finite algebra

$$\mathcal{A} = (E[u]/G)[v]/h, \quad \dim_E \mathcal{A} = \deg G \deg_v h = O_B(nk/t).$$

We remove the points where $s = 0$ and invert B_0 on what remains. This is a candidate filter, not yet a decoding decision: different v -values above the same u can contribute agreements at different positions. The shared recovery routine tests the actual candidates and therefore resolves these projection collisions.

Preserving repeated factors. Recovery also has to preserve repeated roots in the retained fibers. Even when G is squarefree, h may specialize to a polynomial such as v^2 . The residual v then vanishes at the only point, although it is not zero as an element of $E[v]/v^2$. We therefore split by whether a residual vanishes at the represented points, rather than requiring it to be zero in the algebra. For a residual e and fiber degree $b = \deg_v h$, we take

$$h_{\text{agree}} = \gcd(h, e^b \bmod h).$$

If e vanishes at a root, its b th power contains every copy of that root in h ; if not, it contains none. This keeps the full repeated factor on the agreeing side and its coprime complement on the other. Agreement histories and interpolation then work as before.

The total curve-filtering and recovery cost is

$$\tilde{O}_B\left(k^2 + nk + \frac{n^2 k}{t}\right)$$

field operations. At fixed agreement gap, $t = \Omega(n)$ and $k \leq n$, so the finite family has dimension $O_B(n)$ and the computation is quadratic up to logarithmic factors. The threshold t , not an enumeration of possible initial derivatives, is what makes this bound possible.

2.4.4 First order: one center for all close messages

We now supply the common-center coverage needed by the curve computation. The difficulty is that a message can be regular as a differential solution yet have zero separant at our chosen center. For each proposed center c , the equation $Q = Y_1^2 - Y_0$ has the solution $P_c(X) = (X - c)^2/4$, whose separant $2P'_c(X) = X - c$ vanishes at that center. Yet writing $Y_0 = z^2$, $Y_1 = z$ changes the nonconstant branch to $z' = 1/2$, which is regular even at $z = 0$. The failure here is in the coordinates, not in the message.

We first separate the messages whose separant is identically zero, together with the derivative-free factors, into an ordinary equation obtained by a resultant. This is the decoding analogue of the earlier counting

reduction, and the zeroth-order recovery procedure handles it. The remaining messages have nonzero separant as a polynomial in X , although it may vanish at the center we select.

For these messages, normalization enlarges the coordinate algebra by rational functions satisfying monic polynomial equations. One computable polynomial $H(X)$ excludes the poles of the coefficients in these monic equations and of the recorded algebraic identities. Away from its zeros, a new coordinate cannot have a pole along a wanted message: its highest power would then have a pole of strictly greater order than every other term in its monic equation, preventing cancellation. Differentiating a rational function introduces no new pole locations, so the derivatives of these coordinates are also defined there. The construction bounds $\deg H = O_B(n)$. Batched evaluation therefore finds a center c with $H(c) \neq 0$ in a working field E of more than $\deg H$ elements; a bounded-degree extension suffices if needed.

At this center, every wanted message has well-defined normalized initial coordinates. To reconstruct its later coefficients, we still need differential formulas that can be evaluated there: a rational formula may display $0/0$ even when the derivative along the message has a finite value. Write z_j for the normalized coordinates. The common-center construction in [Section F](#) computes alternative rules $z'_j = b_{\ell j}/d_\ell$, together with a certificate

$$1 = \sum_{\ell} d_{\ell} R_{\ell}.$$

The functions R_{ℓ} have no poles along any wanted message at the chosen center. Evaluating this identity there shows that some d_{ℓ} is nonzero. For that rule, the denominator has an invertible constant Taylor coefficient; expanding its reciprocal lets us determine successive coefficients of the normalized coordinates, and hence of the message.

We must still organize the initial states without running the curve filter for every alternative rule. The construction records a polynomial $g(u)$ that marks where the original coordinates or Taylor rule may fail. Working over $\mathbb{F}_q(X)$, write the original curve as $h(u, v) = 0$, monic in v , and form g as the product of its discriminant in v and the determinant of multiplication by the separant. Away from its zeros, the original coordinate algebra equals its normalization and the separant is invertible, so the original Taylor curve suffices. After specializing at $X = c$, write this polynomial as $g_c(u)$. Unlike $H(X)$, which excludes bad centers, g_c marks potentially troublesome states at our chosen center.

We apply the curve filter once to the original regular Taylor curve, covering all states with $g_c \neq 0$. Any missed wanted state lies above $g_c = 0$, a finite boundary: the specialized normalized algebra is free of rank $b = \deg_v h$ over $E[u]$, so imposing $g_c = 0$ leaves dimension $b \deg g_c = O_B(1)$. On this boundary we try the denominators in order, reconstructing on the part where the current d_{ℓ} is nonzero and passing the rest to the next rule. The certificate guarantees that every wanted boundary state is picked up. The open curve, this finite boundary, and the ordinary equation therefore cover every wanted message at the same center.

The boundary algebras may have repeated structure and are supplied by multiplication tables rather than two-variable equations. To split them, we view multiplication by the element being tested, a denominator or agreement residual, as a linear map. Once its powers stabilize, the kernel gives the part where that element is nilpotent, and the image gives the part where it is invertible. This implements the denominator selection above and the subsequent agreement tests by linear algebra instead of polynomial gcds. Since the boundary dimension is bounded, its reconstruction and recovery do not add an extra factor of n to the quadratic curve computation.

This preparation initially requires characteristic larger than a bound depending on B . If that extra condition fails while $p > k - 1$, then k itself is bounded in terms of B ; the simpler multiple-center or ordinary-decoding methods handle that case within the quadratic budget. Thus the rate-and-gap decoder requires only $p > k - 1$ for nonconstant messages. [Section 9.6](#) proves the curve filter, common-center coverage, and these remaining cases.

2.4.5 Higher order: isolate the wanted points by agreement

The shared recovery routine still applies, but the family of initial data now has more than one dimension. We obtain a finite representation by choosing agreement equations that isolate each wanted point, then applying Rojas's polynomial-system algorithm [Roj99]. There is no need for the first-order common-center construction here: trying $O(n)$ centers fits the target $n^{O(d+1)}$ running time. We describe the main case $k > d$ with $p > \max\{k - 1, B\}$, where B again bounds total degree in the derivative variables. The remaining bounded message dimensions have elementary reductions.

We first need families of initial data on which Taylor reconstruction is valid and which together cover every wanted message. Start with the differential equation supplied by the explainer. At any step, let $r \leq d$ be the highest derivative still present, and write the current equation as $V(X, Y_0, \dots, Y_r) = 0$. Its separant is $S = \partial V / \partial Y_r$. For a solution P , there are two possibilities. If $S(X, P, \dots, P^{[r]})$ is nonzero as a polynomial in X , some center c makes its value nonzero, and Taylor reconstruction applies there. Otherwise P also solves $S = 0$, so we repeat with this equation, whose degree in Y_r is smaller. Once Y_r disappears, the active order drops. The characteristic bound ensures that differentiating a nonconstant polynomial in Y_r really lowers its degree. This process therefore reaches a regular equation for each wanted message, or an ordinary equation handled by the zeroth-order decoder. There are only a parameter-bounded number of such stages. At each regular stage the separant along a message has degree $O(n)$, so testing $O(n)$ distinct centers guarantees one where it is nonzero.

Now fix a positive-order regular equation and a center where its separant is nonzero. Write $P(c + T) = \sum_j U_j T^j$, so $U_j = P^{[j]}(c)$ in our Hasse-derivative notation. Evaluating the differential equation at c gives one constraint on the initial coefficients,

$$h(U_0, \dots, U_r) := V(c, U_0, \dots, U_r) = 0.$$

Where the separant is nonzero, $\partial h / \partial U_r \neq 0$. Thus this constraint leaves r local degrees of freedom, and the Taylor recurrence determines the later coefficients through degree $k - 1$ as rational functions of $U = (U_0, \dots, U_r)$. Denote the resulting candidate polynomial by $C_U(X)$. We have reduced recovery to finding the wanted points in these r -dimensional families; agreement with the received word will provide the additional constraints.

As at first order, we clear the common denominator in $C_U(\alpha_i) - y_i$ to obtain an agreement polynomial $g_i(U)$. We would like to solve the square system

$$h(U) = 0, \quad g_i(U) = 0 \quad (i \in J), \quad |J| = r.$$

Counting equations and variables does not by itself justify this step: dependent equations can leave a curve of solutions. What matters is that some subset of a wanted message's agreeing positions gives independent constraints at its initial data.

To prove this, consider an infinitesimal change of the initial coefficients along $h = 0$. Because the coefficient map retains those initial coefficients, a nonzero change produces a nonzero polynomial change in the reconstructed message, still of degree less than k . It cannot vanish at k distinct evaluation points. Hence the linearized agreement tests at any k agreeing positions detect every nonzero direction in the family, and some r of them are independent. Together with h , these equations have full-rank Jacobian at the wanted point, making it a simple isolated root. This is the reason an isolated-root solver can recover it.

We do not know the agreeing positions in advance, so we try all r -subsets of the received positions. Some resulting systems may still have positive-dimensional solution sets. Rojas's algorithm is useful precisely because it allows those components while representing every isolated affine solution. For $s = r + 1$ variables and equations of degree at most $L = O_{d,B}(n)$, the required dense degree bound gives cost and representation size $(L + 1)^{O(s)}$. Its output consists of univariate polynomials $H, H_0, \dots, H_r$: every wanted isolated point

occurs as $(H_0(\theta), \dots, H_r(\theta))$ for some root θ of H . We need this symbolic representation, not the roots themselves.

We remove repeated factors from H , discard parameter values where the chart denominator vanishes, and substitute the coordinate polynomials H_j into the Taylor formulas. This gives a finite candidate family of the form already handled by agreement recovery. The solver may return extra points, and several subsets may find the same message; the full agreement check and coefficient comparison remove both kinds of excess. The solver also requires a sufficiently large working field. Since $q \geq n$, an extension of degree bounded in terms of the fixed parameters suffices; its construction and arithmetic fit the same complexity bound.

There are at most n^r subsets per chart, and each system representation and recovery costs $n^{O(r+1)}$. Including the $O(n)$ centers gives $O(n) n^r n^{O(r+1)} = n^{O(r+1)}$ field operations, hence $n^{O(d+1)} \tilde{O}(\log q)$ bit operations. Thus agreement has two roles: a suitable subset isolates every wanted point, and the shared recovery step obtains its message without enumerating field-valued initial data. [Section 9.7](#) proves the bound, including the field-size and arithmetic details.

3 Interpolation and agreement

We first define agreement and common witnesses, then describe the local constraints that turn an agreement into a root of high multiplicity. Comparing the number of these constraints with the available coefficients will give the interpolation principle used at every derivative order.

Throughout the paper, $\mathbb{F}$ denotes an arbitrary field and $\mathbb{F}_q$ a finite field, $\mathbb{N} = \{0, 1, 2, \dots\}$, and $[n] = \{1, \dots, n\}$. The local interpolation argument in [Section 3.5](#) works in arbitrary characteristic; root-finding statements specify their characteristic hypotheses. We write $F[X]_{\leq D}$ for the vector space of univariate polynomials of degree at most D , including the zero polynomial. We take $\deg 0 = -\infty$, so the degree inequalities below also include it.

We write $x_+ = \max\{x, 0\}$, $\mathbf{1}_{\{\mathcal{P}\}}$ for the indicator of a condition, and $|S|$ for cardinality. For a nonnegative vector, $|x|$ denotes the sum of its coordinates. The notation $\Pr[\mathcal{A}]$ and $\mathbb{E}[U]$ denotes probability and expectation; vol denotes Euclidean volume. Logarithms are natural unless a base is shown. A subscript in $O_\delta(\cdot)$ permits dependence on that parameter.

3.1 Reed–Solomon codes and agreement

Fix a field F and pairwise distinct evaluation points $\alpha = (\alpha_1, \dots, \alpha_n) \in F^n$. The Reed–Solomon code of message dimension k on this evaluation set is

$$\text{RS}_\alpha[n, k] := \{(P(\alpha_1), \dots, P(\alpha_n)) : P \in F[X], \deg P < k\}. \quad (1)$$

Thus the code has rate $\rho = k/n$; no multiplicative or additive structure is assumed of the evaluation set.

For $u, v \in F^n$, their agreement is the number of coordinates on which they coincide:

$$\text{Agr}(u, v) := \{i \in [n] : u_i = v_i\}, \quad \text{agr}(u, v) := |\text{Agr}(u, v)|. \quad (2)$$

Given a received word $y \in F^n$ and an integer A , the agreement list at ambient degree D is

$$\mathcal{L}_D(y, A) := \{P \in F[X]_{\leq D} : \text{agr}((P(\alpha_i))_{i=1}^n, y) \geq A\}. \quad (3)$$

We distinguish this ambient degree bound from the actual message dimension k , which requires degree less than k . The finite-field case is $F = \mathbb{F}_q$. When k is fixed, write $\mathcal{L}(y, A) = \mathcal{L}_{k-1}(y, A)$.

3.2 Polynomial received curves and common witnesses

Fix an integer $\ell \geq 1$ and coefficient words $w_0, \dots, w_\ell \in F^n$. Their *polynomial received curve* is the word $w(z)$ whose i -th coordinate is

$$w_i(z) = \sum_{t=0}^{\ell} z^t w_{t,i}. \quad (4)$$

For a polynomial P and a tuple $P_0, \dots, P_\ell$, write

$$S_z(P) = \{i : P(\alpha_i) = w_i(z)\}, \quad S(P_0, \dots, P_\ell) = \{i : P_t(\alpha_i) = w_{t,i} \text{ for every } 0 \leq t \leq \ell\}.$$

Definition 3.1 (Full common-witness agreement). A pair (z, P) has the *full common-witness property* for the received curve if there are degree- $< k$ polynomials $P_0, \dots, P_\ell$ such that

$$P = \sum_{t=0}^{\ell} z^t P_t, \quad S_z(P) = S(P_0, \dots, P_\ell). \quad (5)$$

For a collection C of pairs (z, P) , the property holds outside $\mathcal{E} \subseteq F$ if it holds for every pair in C with $z \notin \mathcal{E}$.

For a received line $f + zg$, this is the case $\ell = 1$ with $(w_0, w_1) = (f, g)$. We retain the shorthand $S(F_0, G_0) = S(P_0, P_1)$ in this case. A full agreement-set line-MCA bound E means that every received line has this property outside one set $\mathcal{E}$ of size at most E for the collection of all pairs with $\deg P < k$ and $|S_z(P)| \geq A$. The exception set is chosen before the candidate P . The equality of sets is stronger than a bound on their cardinalities: it rules out extra agreements created only by cancellation at the chosen challenge. Over $\mathbb{F}_q$, E is a count and E/q is the corresponding probability.

The constant-code endpoint is independent of interpolation and geometry. Its collision argument also handles every polynomial received curve.

Lemma 3.2 (Constant-code polynomial-curve MCA). *Let F be any field, let $1 \leq A \leq n$, and retain a received curve of degree $\ell \geq 1$. For every challenge, the number of constants $c \in F$ with $|S_z(c)| \geq A$ is at most $\lfloor n/A \rfloor$. Moreover, there is an exceptional set $\mathcal{E} \subseteq F$ such that*

$$|\mathcal{E}| \leq \begin{cases} \ell \binom{n}{2}, & A = 1, \\ \lfloor \ell \binom{n}{2} / (A - 1) \rfloor, & A \geq 2, \end{cases} \quad (6)$$

and every pair (z, c) with $z \notin \mathcal{E}$ and $|S_z(c)| \geq A$ has the full common-witness property of [Definition 3.1](#) for $k = 1$.

Proof. Put $v_i = (w_{0,i}, \dots, w_{\ell,i})$. For a fixed challenge, the agreement sets of distinct constants are disjoint, which proves the list bound. Call a pair $\{i, j\}$ a collision at z if $v_i \neq v_j$ but

$$\sum_{t=0}^{\ell} z^t w_{t,i} = \sum_{t=0}^{\ell} z^t w_{t,j}.$$

Each such pair collides at at most ℓ challenges. If the desired conclusion fails for a constant c , its agreement set contains two distinct tuples and hence a collision. When $A \geq 2$, a partition of at least A agreeing positions into at least two tuple classes has at least $A - 1$ cross-class pairs. These pairs are disjoint for distinct constants at the same challenge. Double counting collision pairs therefore gives (6); for $A = 1$, charge just one collision pair.

Outside the resulting exceptional set, all tuples on $S_z(c)$ are equal to some $(c_0, \dots, c_\ell)$. No position with a different tuple can evaluate to c , since that would be a collision. Thus $c = \sum_t z^t c_t$ and $S_z(c) = S(c_0, \dots, c_\ell)$. $\square$

3.3 Hasse derivatives and multiplicity

We use the standard Hasse-derivative convention [DKSS13].

Definition 3.3 (Hasse derivative). For $P \in \mathbb{F}[X]$, its j -th Hasse derivative $P^{[j]}$ is defined by the polynomial identity

$$P(X + T) = \sum_{j \geq 0} P^{[j]}(X) T^j. \quad (7)$$

Equivalently, if $P(X) = \sum_i p_i X^i$, then

$$P^{[j]}(X) = \sum_{i \geq j} \binom{i}{j} p_i X^{i-j}. \quad (8)$$

The definition is valid in every characteristic and gives $\deg P^{[j]} \leq D - j$ whenever $\deg P \leq D$, with a zero derivative when $j > D$.

Hidden derivatives and decoding parameters. An ordinary RS symbol is one value $P(\alpha_i)$. A multiplicity-code symbol also transmits derivatives at that point [GW13, Kop15]. Our received word supplies only values; the derivatives used in an interpolation equation remain unknown. The maximum index d in $Q(X, Y_0, \dots, Y_d)$ is its derivative order, while the exponent of a variable Y_j measures its algebraic degree. For example, Y_1^2 uses the first derivative quadratically, not the second derivative. A family containing every close message can contain additional polynomials; decoding must recover exactly those with sufficient agreement. Section 9 gives the bit-complexity model, separately from the characteristic assumptions used to bound lists and MCA errors.

Ordinary Taylor expansion uses derivatives at X to obtain the value at $X + T$. The backward identity below uses derivatives at $X + T$ to recover the value at X . The hidden-derivative constraints use this inversion of Equation (7). We include the short proof because every sign and power of T enters the local multiplicity calculation.

Lemma 3.4 (Backward Taylor identity). *Every $P \in \mathbb{F}[X]$ satisfies*

$$P(X) = \sum_{j \geq 0} (-T)^j P^{[j]}(X + T) \quad \text{in } \mathbb{F}[X, T]. \quad (9)$$

Consequently, for every $d \geq 0$,

$$P(X + T) \equiv P(X) + \sum_{j=1}^d (-1)^{j+1} T^j P^{[j]}(X + T) \pmod{T^{d+1}}. \quad (10)$$

Proof. Apply Equation (7) with X replaced by $X + T$ and the increment replaced by $-T$. This gives Equation (9). Isolating its $j = 0$ term and discarding the terms with $j > d$ gives Equation (10). $\square$

Definition 3.5 (Multiplicity). For $F \in \mathbb{F}[X]$ and $\alpha \in \mathbb{F}$, the multiplicity of F at α is

$$\text{mult}_\alpha(F) := \max\{s \in \mathbb{N} : T^s \text{ divides } F(\alpha + T)\}. \quad (11)$$

We set $\text{mult}_\alpha(0) = \infty$.

We use the standard fact that a nonzero univariate polynomial cannot have more roots, counted with multiplicity, than its degree. In the form needed below, if $\alpha_1, \dots, \alpha_t$ are distinct and $\text{mult}_{\alpha_i}(F) \geq m$ for every i , then either $F = 0$ or $\deg F \geq mt$.

For a bivariate polynomial $Q(X, Y)$, multiplicity at least s at (α, y) means that every coefficient of $U^a V^b$ with $a + b < s$ vanishes in $Q(\alpha + U, y + V)$. This is the notion used by the order-zero interpolation argument. If $P(\alpha) = y$, both T and $P(\alpha + T) - y$ are divisible by T . Substituting them for U, V therefore shows that $Q(X, P(X))$ has a root of multiplicity at least s at $X = \alpha$.

3.4 Weighted degree

Definition 3.6 (Weighted degree). For nonnegative integer weights $w_0, \dots, w_s$, the weighted degree of a monomial $Z_0^{e_0} \cdots Z_s^{e_s}$ is $\sum_{i=0}^s w_i e_i$. The weighted degree $\text{wdeg}_{w_0, \dots, w_s} Q$ of a nonzero polynomial Q is the maximum weighted degree of one of its monomials with nonzero coefficient. We set $\text{wdeg}_{w_0, \dots, w_s}(0) = -\infty$.

The weights relevant to polynomial differential equations record the actual degree lost under Hasse differentiation. Namely, if $d \leq D$, $Q \in \mathbb{F}[X, Y_0, \dots, Y_d]$, and $P \in \mathbb{F}[X]_{\leq D}$, then

$$\deg_X Q(X, P(X), P^{[1]}(X), \dots, P^{[d]}(X)) \leq \text{wdeg}_{1, D, D-1, \dots, D-d} Q. \quad (12)$$

Indeed, the variable X contributes degree one and the substitution for Y_j contributes degree at most $D - j$. This elementary accounting is the reason to distinguish the actual derivative weights from the common upper bound D used in the weighted-support certificate.

3.5 Local interpolation constraints

BCPZZ's backward-Taylor substitution turns agreement into multiplicity without requiring derivative evaluations [BCP⁺26a, Section 3.1]. Fix an order $d \geq 0$ and a target multiplicity $m \geq 1$. For formal variables $Y = (Y_1, \dots, Y_d)$, write $B_d(T, Y) = \sum_{j=1}^d (-1)^{j+1} T^j Y_j$; an empty sum is zero. The identity in Equation (10) says that, at an agreement, the remainder after these d terms is divisible by T^{d+1} . We encode that divisibility directly in the local substitution.

Definition 3.7 (Local hidden-derivative constraints). For $(\alpha, y) \in \mathbb{F}^2$ and $Q \in \mathbb{F}[X, Y_0, \dots, Y_d]$, put

$$\Phi_{\alpha, y}^{d, m}(Q) := Q(\alpha + T, y + B_d(T, Y) + T^{d+1}V, Y_1, \dots, Y_d) \pmod{T^m} \quad (13)$$

in $\mathbb{F}[T, V, Y_1, \dots, Y_d]/(T^m)$. The multiplicity- m , order- d constraints at (α, y) are

$$\Phi_{\alpha, y}^{d, m}(Q) = 0. \quad (14)$$

These are homogeneous linear conditions on any fixed interpolation space. They require every coefficient in $V, Y_1, \dots, Y_d$ to be divisible by T^m before reduction.

Lemma 3.8 (Local multiplicity [BCP⁺26a, Lemma 3.1]). *If $P(\alpha) = y$ and Q satisfies (14), then*

$$Q(\alpha + T, P(\alpha + T), P^{[1]}(\alpha + T), \dots, P^{[d]}(\alpha + T)) \equiv 0 \pmod{T^m}. \quad (15)$$

Thus each agreement gives a root of multiplicity at least m after substituting the candidate and its derivatives into Q .

Proof. The backward Taylor identity makes

$$V_P(T) = \frac{P(\alpha + T) - y - \sum_{j=1}^d (-1)^{j+1} T^j P^{[j]}(\alpha + T)}{T^{d+1}} \quad (16)$$

a polynomial. Substitute $Y_j = P^{[j]}(\alpha + T)$ and $V = V_P(T)$ into (14). The second input to Q becomes $P(\alpha + T)$, proving the claim in every characteristic. $\square$

For the rank calculations we retain the equivalent remainder variable $E = T^d V$. Substitute $Y_0 = y + B_d(T, Y) + TE$ and retain the rows

$$T^i E^u Y^j \quad \text{with } i + du < m. \quad (17)$$

Rows with $i < u$ are structurally zero, since each E comes with a factor of T . On the remaining rows, the substitution relabels

$$(i, u, \mathbf{j}) \mapsto (i + du, u, \mathbf{j}). \quad (18)$$

Its image is exactly the possible nonzero rows of (13), whose T -degrees are at least $(d + 1)u$ and below m . Thus it preserves every matrix entry, kernel, rank, and jet-degree grouping. If y depends on a challenge, its challenge degrees are preserved as well. We can therefore compute ranks in the E coordinates while using Lemma 3.8 for every order.

3.6 The common interpolation principle

An interpolation polynomial is useful because many agreements force one polynomial identity. We call it an explainer, following the terminology introduced above.

Definition 3.9 (Explainer). Fix a field F , distinct evaluation points $\alpha_1, \dots, \alpha_n \in F$, a received word $w \in F^n$, integers $K \geq 1$ and $1 \leq A \leq n$, and an order bound $d \geq 0$. A nonzero polynomial $Q \in F[X, Y_0, \dots, Y_d]$ is an *explainer* for w at agreement threshold A and degree bound K if every $P \in F[X]$ with $\deg P < K$ satisfies

$$|\{i : P(\alpha_i) = w_i\}| \geq A \implies Q(X, P, P^{[1]}, \dots, P^{[d]}) \equiv 0.$$

The implication is one-way: an explainer can have polynomial solutions that are far from the received word. Agreement must still be imposed when recovering or counting the requested messages. The ambient degree bound K is part of the definition and may exceed the actual message dimension k ; explaining all degree- $< K$ candidates also explains the degree- $< k$ candidates.

There are two budgets to compare. The dimension of a polynomial space counts the coefficients we may choose; the ranks of local linear maps count the conditions those coefficients must satisfy. The following form of the interpolation principle separates this comparison from our choices of space and local constraints [GS99, BCP⁺26a].

Proposition 3.10 (Global interpolation principle). *Let $V \subseteq F[X, Y_0, \dots, Y_d]$ be finite dimensional, and let Φ_i be linear maps on V , one for each received pair (α_i, y_i) . Suppose that $Q \in \ker \Phi_i$ and $P(\alpha_i) = y_i$ imply*

$$(X - \alpha_i)^m \mid Q(X, P, P^{[1]}, \dots, P^{[d]}).$$

Assume every $Q \in V$ specializes at every degree- $\leq D$ polynomial to degree less than mA . If

$$\dim V > \sum_{i=1}^n \text{rank } \Phi_i, \quad (19)$$

then V contains an explainer for y at degree bound $D + 1$ and agreement threshold A .

Proof. The stacked map has rank at most the sum of the local ranks, so its kernel contains a nonzero Q . At A distinct agreement positions, the specialization has at least mA roots counted with multiplicity. Its degree is smaller than mA , so it vanishes identically. $\square$

The space need not be spanned by monomials, and the local conditions need not be independent. Monomial spaces make the degree and dimension counts explicit. Translation invariance is useful for proving a rank bound uniform over received pairs; it is not an additional hypothesis of the principle itself.

We next keep the received word's challenge symbolic, so the same interpolation principle can support MCA. We then begin the solution counts with ordinary bivariate equations in Section 5.7, sharpening the bounds at the existing Johnson threshold. These ordinary bounds also handle the resultant equation in the first-order proof and the terminal equation in its successive-stage alternative.

3.7 One equation for every challenge

Following the earlier function-field construction [rs-26], we keep the challenge symbolic. Choosing an unrelated explainer for each word $f + zg$ would lose the relationship between its candidates and those at other challenges. We therefore seek one equation that retains Z as a variable and explains every close candidate at every challenge. Its degree in Z must be small enough to give a useful exceptional-count bound.

Definition 3.11 (Symbolic explainer for a received line). Fix $f, g \in F^n$, an agreement threshold A , a degree bound $K \geq 1$, and an order bound $d \geq 0$. A nonzero polynomial $Q \in F[X, Z, Y_0, \dots, Y_d]$ is a *symbolic explainer* for $f + Zg$ at these bounds if, for every $z \in F$ and every $P \in F[X]$ with $\deg P < K$,

$$|S_z(P)| \geq A \implies Q(X, z, P, P^{[1]}, \dots, P^{[d]}) \equiv 0.$$

Here nonzero means that Q is nonzero as a polynomial in all its variables. The construction below gives more: $Q(X, z, Y)$ remains nonzero at every challenge, and the explaining identity holds after any extension of the field. Formal nonvanishing of $Q(X, z, Y)$ and vanishing after substituting a close message are the two different requirements already present for a single received word.

We track three bounds: the derivative order d , the total jet-degree bound B_{jet} in $Y_0, \dots, Y_d$, and the challenge-degree bound B_Z . The derivative order controls how many initial Taylor coefficients are needed to describe a solution. The total jet degree controls the algebraic degrees and the number of differentiation stages in the reconstruction argument. The challenge degree measures dependence on Z . To bound it, we interpolate over $F(Z)$ and look for a polynomial kernel vector after clearing denominators. The next lemma shows how the coefficient-constraint margin controls the degree of that vector.

A graded polynomial kernel vector. The interpolation columns and constraint rows need not consume the same amount of challenge degree. The following lemma assigns a cost to each column and row, then counts only the polynomial coefficients that remain within a common height H . This one count will give both the uniform height bound below and the sharper finite tests in [Proposition 5.10](#).

Lemma 3.12 (Graded degree bound for a polynomial kernel vector). *Let F be a field, let I, J be finite sets, and let $\mathbf{A}(Z) \in F[Z]^{I \times J}$. Fix integer column costs c_j , row costs r_i , and a height H . Suppose*

$$\deg \mathbf{A}_{ij} \leq c_j - r_i, \quad i \in I, j \in J,$$

where a negative upper bound means that the entry is zero. If

$$\sum_{j \in J} (H - c_j + 1)_+ > \sum_{i \in I} (H - r_i + 1)_+, \quad (20)$$

then there is a nonzero vector $v \in F[Z]^J$ satisfying $\mathbf{A}(Z)v = 0$, with

$$\deg v_j \leq H - c_j \quad \text{when } c_j \leq H, \quad v_j = 0 \quad \text{when } c_j > H.$$

Its entries can be chosen to have greatest common divisor one in $F[Z]$. For that choice, $v(z) \neq 0$ for every z in every field extension of F .

Proof. For each active column j , give v_j degree at most $H - c_j$; set the other entries to zero. The left side of [Equation \(20\)](#) counts the scalar coefficients of these unknown polynomials. In row i , every product satisfies

$$\deg(\mathbf{A}_{ij}v_j) \leq (c_j - r_i) + (H - c_j) = H - r_i.$$

This row therefore imposes at most $(H - r_i + 1)_+$ scalar equations. The strict surplus in [Equation \(20\)](#) gives a nonzero kernel vector.

Divide its entries by their greatest common divisor. Since $F[Z]$ is an integral domain, the polynomial identities remain valid, and division does not increase any degree. Bézout's identity for the primitive entries shows that they have no common zero over any field extension. $\square$

Taking equal costs recovers the uniform entry-degree bound needed below. If $\mathbf{A}$ has N_{col} columns, rank $R < N_{\text{col}}$ over $F(Z)$, and entry degrees at most B_{entry} , choose R original rows spanning its row space and put

$$B_{\text{ker}} := \left\lfloor \frac{RB_{\text{entry}}}{N_{\text{col}} - R} \right\rfloor. \quad (21)$$

Taking $c_j = B_{\text{entry}}$, $r_i = 0$, and $H = B_{\text{ker}} + B_{\text{entry}}$ reduces the strict surplus to

$$(N_{\text{col}} - R)(B_{\text{ker}} + 1) > RB_{\text{entry}}.$$

Thus the lemma gives a primitive kernel vector whose entries have degree at most B_{ker} . Since the selected rows span the full row space over $F(Z)$, this vector annihilates the original matrix as well.

Proposition 3.13 (Constructing a symbolic explainer). *Let $Q \subseteq F[X, Y_0, \dots, Y_d]$ be a finite monomial space invariant under translations in X and Y_0 . Suppose its total jet degree is at most B_{jet} , its specialization degree at degree- $< K$ polynomials is less than mA , and its dimension N_{coef} exceeds nR_{loc} , where R_{loc} is the rank of its local map at $(0, 0)$. For any distinct $\alpha_i \in F$ and $f, g \in F^n$, there is $Q \in F[Z] \otimes_F Q$ such that*

$$\deg_Z Q \leq \left\lfloor \frac{nR_{\text{loc}}B_{\text{jet}}}{N_{\text{coef}} - nR_{\text{loc}}} \right\rfloor, \quad Q(X, z, Y) \neq 0 \text{ for every } z \in \overline{F},$$

and every degree- $< K$ polynomial P agreeing with $f + zg$ in at least A positions satisfies

$$Q(X, z, P, P^{[1]}, \dots, P^{[d]}) = 0. \quad (22)$$

These assertions hold after any extension of the base field.

Thus the same support explains the whole line, including the actual degree- $< k$ messages when $k \leq K$. Its specialization-degree bound and local multiplicity remain unchanged. What is new is the degree in Z : a larger coefficient-constraint margin $N_{\text{coef}} - nR_{\text{loc}}$ permits a smaller bound on it. The support need not be retuned to obtain a symbolic explainer, but retuning can improve its exceptional-count bound.

Proof. Use the local constraints at $(\alpha_i, f_i + Zg_i)$. In their matrices, the challenge enters only through powers of $f_i + Zg_i$, whose exponents are bounded by the source Y_0 -degree, hence by B_{jet} . Translation gives an automorphism of the source space over $F(Z)$, and the local map factors as translation followed by the map at $(0, 0)$. Its rank is therefore R_{loc} , also after scalar extension. Let $R \leq nR_{\text{loc}}$ be the rank of the stacked matrix. Its entries have degree at most B_{jet} . Apply the equal-cost specialization of [Lemma 3.12](#). Since $R/(N_{\text{coef}} - R)$ increases with R , the resulting primitive kernel vector has degree at most the bound in the statement. It has no common root among its entries and gives Q , nonzero at every specialization. The specialized matrix identities give all local constraints. The multiplicity and specialization-degree argument of [Proposition 3.10](#) proves (22). $\square$

4 First-order interpolation and the agreement curve

We use the common interpolation principle with one hidden derivative. Our goal is a rate-dependent agreement threshold below Johnson, together with explicit bounds on both the decoding list and the exceptional challenges in line MCA. We choose the allowed monomials, count their coefficients and local constraints, and optimize the resulting agreement threshold. These are interpolation arguments, valid in every characteristic. The next section turns the equation into list and MCA bounds.

4.1 From agreement to a differential equation

At first order, the local remainder map of [Section 3.5](#) substitutes

$$Q(\alpha + T, y + TY_1 + T^2V, Y_1) \pmod{T^m}.$$

Following the hidden-derivative construction [[Nas26](#), [BCP⁺26a](#)], we require this polynomial to vanish. For the rank calculation it is convenient to use $E = TV$, equivalently $Y_0 = y + T(Y_1 + E)$, and retain the rows $T^i E^j Y_1^b$ with $i + j < m$. The relabeling in (18) preserves their rank. By [Lemma 3.8](#), each agreement gives multiplicity at least m in $Q(X, P, P^{[1]})$. If its degree is less than mA , then A agreements force this specialization to vanish identically. We now choose the monomials of Q so that these local conditions leave a nonzero polynomial.

4.2 Choosing the allowed monomials

We need more coefficients to choose than independent local conditions. The allowed monomials determine both counts. We begin with a simple space that yields a clean rate curve, then retain sharper degree information when evaluating finite parameters. We reserve m for interpolation multiplicity. The two jet-degree bounds have separate notation: B_∂ bounds the exponent of the first-derivative variable Y_1 , and B_{jet} bounds the total degree in Y_0, Y_1 . Thus

$$\deg_{Y_1} Q \leq B_\partial, \quad \deg_{Y_0, Y_1} Q \leq B_{\text{jet}}.$$

These bounds play different roles. The multiplicity m makes each agreement point a root of multiplicity at least m after specialization. The first-derivative degree bound B_∂ controls the regular factor families and the ordinary resultant. In the successive-stage alternative, it also limits how many times reconstruction can differentiate in Y_1 . The total jet-degree bound B_{jet} controls the algebraic degree throughout reconstruction. Smaller bounds reduce the cost of counting solutions, but also remove interpolation coefficients. The dimension comparison checks whether enough coefficients remain.

4.2.1 A support with an explicit local rank bound

Fix $\delta > 0$ and integers $m \geq 1$, $B_\partial \geq 0$, and $B_{\text{jet}} \geq 1$. For $n \geq 2$, $1 \leq k \leq n$, and $k + \delta n \leq A \leq n$, put $K = \max\{k, 2\}$ and $D = K - 1$. Define the first-order support by

$$\mathcal{V}_{m, B_\partial, B_{\text{jet}}} = \text{span}_F \{X^x Y_0^a Y_1^b : b \leq B_\partial, a + b \leq B_{\text{jet}}, x + D(a + b) < mA\}. \quad (23)$$

The bound on the exponent of Y_1 limits the local constraints while leaving the Y_0 exponent available to enlarge the space. The weighted cutoff then enforces the specialization-degree requirement.

Write $u_+ = \max\{u, 0\}$ and define

$$R_{\text{loc}}(m, B_\partial) = \sum_{s=0}^{m-1} [(s+1)(B_\partial+1) - (2s-m+1)_+(s+B_\partial-m+1)_+] \\ \stackrel{B_\partial \leq m/2}{=} (B_\partial+1) \left[\frac{m(m+1-B_\partial)}{2} + \frac{B_\partial(2B_\partial+1)}{6} \right], \quad (24)$$

$$N_{\text{jet}}(t) = \min\{t, B_\partial\} + 1. \quad (25)$$

Here $N_{\text{jet}}(t)$ counts the allowed jet monomials of total degree t . The closed form for R_{loc} uses $B_\partial \leq m/2$.

To read the rank sum, we apply the local divisibility kernel from Naskręcki's construction [[Nas26](#)] one Taylor degree at a time. At degree s , the containing rectangle has $(s+1)(B_\partial+1)$ monomials. The divisibility relation contributes a kernel only after both $2s \geq m$ and $s + B_\partial \geq m$; its dimension is the product of the

two positive-part factors in (24). For example, if $m = 12$, $B_\partial = 5$, and $s = 9$, the rectangle has $10 \cdot 6 = 60$ monomials, while the kernel has $(18 - 12 + 1)(9 + 5 - 12 + 1) = 7 \cdot 3 = 21$. This degree therefore contributes $60 - 21 = 39$ to the local rank. The saving occurs in the high Taylor degrees, where the two substituted variables are forced to agree to the remaining multiplicity.

Lemma 4.1 (First-order dimension and rank). *Over any field, the space in (23) is invariant under translations in X and Y_0 , has total jet degree at most B_{jet} , and specializes to degree less than mA at every polynomial of degree less than K . Its order-one, multiplicity- m local constraint map has rank at most $R_{\text{loc}}(m, B_\partial)$, uniformly over the received point. Its coefficient count $N_{\text{coef}} = \dim \mathcal{V}$ satisfies*

$$N_{\text{coef}} = \sum_{t=0}^{B_{\text{jet}}} N_{\text{jet}}(t)(mA - Dt)_+ \geq n \sum_{t=0}^{B_{\text{jet}}} N_{\text{jet}}(t)(m\delta + (k/n)(m - t))_+.$$

Proof. Translations lower x or a and preserve every support inequality. For $\deg P \leq D$, specialization of a monomial has degree at most $x + Da + (D - 1)b \leq x + D(a + b) < mA$. There are $N_{\text{jet}}(t)$ jet monomials of total degree t . Each admits $(mA - Dt)_+$ nonnegative integer choices of x . Since $D \leq k$ and $A \geq k + \delta n$, the dimension formula gives the stated lower bound term by term.

Translation reduces the rank calculation to the received point $(0, 0)$. First substitute $X = T$ and $Y_0 = TU$, and reduce modulo T^m . At T -degree $s < m$, the image is contained in the rectangle spanned by $U^a Y_1^b$ with $0 \leq a \leq s$ and $0 \leq b \leq B_\partial$. The second substitution $U = Y_1 + E$, followed by reduction modulo E^{m-s} , imposes exactly the order-one local constraints at this T -degree. Its kernel on the whole rectangle consists of multiples of $(U - Y_1)^{m-s}$. For a nonzero polynomial H , multiplication by this factor increases both its separate U -degree and its separate Y_1 -degree by $m - s$. Thus the kernel dimension on the rectangle is $(s - (m - s) + 1)_+(B_\partial - (m - s) + 1)_+$. Subtracting it from $(s + 1)(B_\partial + 1)$ and summing over s proves the rank bound. The actual support can omit rectangle monomials, so this is an upper bound for its rank. The argument uses polynomial divisibility and inverts no binomial coefficients, hence holds in every characteristic. Evaluating the finite sum gives (24). $\square$

4.3 The rate curve and finite interpolation parameters

Optimizing this support with the rectangle rank bound gives the explicit two-regime sufficient threshold $a_1(\rho)$. Fix $0 < \rho < a < 1$ and normalize the two degree bounds by m :

$$\lambda_\partial := \frac{B_\partial}{m}, \quad \frac{B_{\text{jet}}}{m} \longrightarrow \frac{a}{\rho}.$$

The first quantity is the normalized first-derivative degree bound. For the limiting calculation, replace A by an and D by ρn . This only decreases the coefficient count when $A \geq an$ and $D = k - 1 \leq \rho n - 1$; we recover that finite saving below. To pass from the discrete coefficient count to an integral, consider one total jet degree $t \approx um$. The exponent of Y_1 ranges from zero to $\min\{t, B_\partial\}$, giving approximately $m \min\{u, \lambda_\partial\}$ choices. Once those jet exponents are fixed, the weighted cutoff leaves approximately $nm(a - \rho u)_+$ choices for the exponent of X . The positive part ends the support at $u = a/\rho$; writing the integral to infinity only records that the integrand is zero after this endpoint. Finally, consecutive values of t are spaced by $1/m$ in the u coordinate, so summing the layers contributes the third factor of m . Dividing the resulting global count by nm^3 gives the limiting coefficient density

$$\text{Coef}(\rho, a, \lambda_\partial) = \int_0^\infty \min\{u, \lambda_\partial\}(a - \rho u)_+ du = \frac{\lambda_\partial a^2}{2\rho} - \frac{a\lambda_\partial^2}{2} + \frac{\rho\lambda_\partial^3}{6}, \quad 0 < \lambda_\partial < a/\rho. \quad (26)$$

The $+1$ in $N_{\text{jet}}(t)$ includes the endpoint exponent and contributes only a lower-order term. Thus this lower estimate for the global coefficient count is $nm^3 \text{Coef} + o(nm^3)$.

The local constraints use the same cubic scale before multiplication by the n received positions. At normalized Taylor degree $u = s/m$, the containing rectangle contributes $m^2 u \lambda_\partial$ coordinates. The kernel is nonzero only when both $2s \geq m$ and $s + B_\partial \geq m$. Its dimension is a product whose normalized factors are $(2u - 1)_+$ and $(u + \lambda_\partial - 1)_+$. Summing the m Taylor degrees $0 \leq s < m$ gives the limiting local rank

$$\text{Rank}(\lambda_\partial) = \int_0^1 [u \lambda_\partial - (2u - 1)_+ (u + \lambda_\partial - 1)_+] du.$$

For every $\lambda_\partial \geq 0$, direct integration gives

$$\text{Rank}(\lambda_\partial) = \begin{cases} \lambda_\partial/2 - \lambda_\partial^2/2 + \lambda_\partial^3/3, & \lambda_\partial \leq 1/2, \\ \lambda_\partial/4 + 1/24, & \lambda_\partial \geq 1/2. \end{cases}$$

We retain this rectangle rank bound in both regimes. Since the local rank bound is $m^3 \text{Rank} + o(m^3)$ and there are n positions, both sides of the interpolation inequality have global scale nm^3 . The normalized coefficient-constraint margin is

$$\text{Margin}_{\text{int}}(\rho, a, \lambda_\partial) := \text{Coef}(\rho, a, \lambda_\partial) - \text{Rank}(\lambda_\partial).$$

We need this margin to be positive. The change in the rank formula at $\lambda_\partial = 1/2$ produces one rate cutoff:

$$\rho_c := 11 - 3\sqrt{13} \approx 0.183346. \quad (27)$$

At the threshold above this rate, an optimizing derivative degree satisfies $\lambda_\partial \leq 1/2$. In this range the margin factors as

$$\text{Margin}_{\text{int}}(\rho, a, \lambda_\partial) = \frac{\lambda_\partial}{2} \left[\frac{a^2}{\rho} - 1 + (1 - a)\lambda_\partial - \frac{2 - \rho}{3} \lambda_\partial^2 \right].$$

Because $\lambda_\partial > 0$, positivity of the margin is equivalent to positivity of the bracket. Maximizing the bracket therefore locates the smallest agreement threshold certified by this support and rectangle rank bound, even though it need not maximize the full margin, whose prefactor is $\lambda_\partial/2$. The bracket is maximized at $\lambda_\partial = 3(1 - a)/(2(2 - \rho))$ and is positive exactly when

$$\frac{a^2}{\rho} + \frac{3(1 - a)^2}{4(2 - \rho)} > 1. \quad (28)$$

As λ_∂ tends to zero, the divided margin recovers the Johnson condition $a^2 > \rho$. Optimizing the derivative degree adds the positive second term in (28), permitting smaller agreement. Solving the boundary equation gives the second branch of the curve.

As a numerical reading of the limiting count, take $\rho = 0.25$, $a = 0.48$, and $\lambda_\partial = 0.45$. Then $\text{Coef} = 0.162556875$ and $\text{Rank} = 0.154125$, leaving $\text{Margin}_{\text{int}} = 0.008431875 > 0$. This example illustrates the asymptotic surplus; it is not a finite interpolation certificate, which also has rounding and lower-order terms.

Below ρ_c , the optimum lies in the other rank regime, $\lambda_\partial > 1/2$. We find the threshold by requiring both zero margin and stationarity in the derivative-degree choice:

$$\text{Margin}_{\text{int}} = 0, \quad \frac{\partial \text{Margin}_{\text{int}}}{\partial \lambda_\partial} = \frac{(a - \rho \lambda_\partial)^2}{2\rho} - \frac{1}{4} = 0.$$

Since $\lambda_\partial < a/\rho$, the second equation gives $a - \rho \lambda_\partial = \sqrt{\rho/2}$. Substitution into the first yields

$$a = \sqrt{\rho/2} + \rho \lambda_\partial, \quad 4\rho \lambda_\partial^3 + 6\sqrt{2\rho} \lambda_\partial^2 = 1. \quad (29)$$

We can express this cubic more simply by measuring the threshold relative to $\sqrt{\rho/2}$. Write $a = \sqrt{\rho/2}(1+u)$, so $u = \sqrt{2\rho} \lambda_\partial$. The two equations reduce to

$$u^2(u+3) = \sqrt{\rho/2}. \quad (30)$$

The left side increases strictly from zero to infinity for $u > 0$, so there is a unique positive solution u_ρ , readily computed by bisection.² The resulting curve is therefore

$$a_1(\rho) = \begin{cases} \sqrt{\rho/2}(1+u_\rho), & 0 < \rho < \rho_c, \\ \frac{3\rho + 2\sqrt{\rho(5-\rho)(2-\rho)}}{8-\rho}, & \rho_c \leq \rho < 1. \end{cases} \quad (31)$$

At the branch boundary $\lambda_\partial = 1/2$, we have $u = \sqrt{\rho/2}$. Equation (30) becomes $\sqrt{\rho/2}(\sqrt{\rho/2}+3) = 1$, giving $\rho_c = 11 - 3\sqrt{13}$ and $a_1(\rho_c) = 4 - \sqrt{13}$.

The cubic also explains the low-rate behavior without solving it. Since $u_\rho + 3 > 3$, the cubic gives $0 < u_\rho \leq (\rho/2)^{1/4}/\sqrt{3}$, so

$$\sqrt{\rho/2} < a_1(\rho) \leq \sqrt{\rho/2} + \frac{(\rho/2)^{3/4}}{\sqrt{3}}, \quad \frac{a_1(\rho)}{\sqrt{\rho}} \rightarrow \frac{1}{\sqrt{2}} \text{ as } \rho \rightarrow 0.$$

Thus the leading agreement threshold is smaller than Johnson's by a factor $1/\sqrt{2}$ at low rates. This is a square-root threshold, still well above the capacity threshold ρ . For every $0 < \rho < 1$, we have $\rho < a_1(\rho) < \sqrt{\rho}$. On the upper branch, evaluate the left side of (28) at $a = \rho$ and $a = \sqrt{\rho}$; on the lower branch, use (29). The improvement over the exact finite Johnson threshold $\sqrt{(k-1)/n}$ holds for sufficiently large n at each fixed rate.

An explicit finite choice above the curve. Write $\eta_1 = a - a_1(\rho) > 0$. We now choose the interpolation parameters and quantify their growth as η_1 decreases. For a block length $n \geq 2/\rho$, put $\rho_n = \rho - 1/n$. The exact message-degree bound is $D = k - 1 \leq \rho_n n$. Retaining this one-degree saving will also control the MCA bound when the gap is smaller than $1/n$. Our choices depend on (ρ, a, n) , and precede the field, evaluation set, and received word. To keep the parameters bounded when a approaches one, set $a_{\text{cert}} = \min\{a, (1 + a_1(\rho))/2\}$. Use the local-rank sum $R_{\text{loc}}(m, B_\partial)$ from (24), and set

$$\lambda_\partial = \begin{cases} \frac{a_1(\rho) - \sqrt{\rho/2}}{\rho}, & 0 < \rho < \rho_c, \\ \frac{3(1 - a_{\text{cert}})}{2(2 - \rho)}, & \rho_c \leq \rho < 1, \end{cases}$$

For this choice, the exact limiting coefficient-constraint margin is

$$\text{Margin}_{\text{int}} = \begin{cases} \frac{\lambda_\partial}{2} (a_{\text{cert}} - a_1(\rho)) \left[\frac{a_{\text{cert}} + a_1(\rho)}{\rho} - \lambda_\partial \right], & 0 < \rho < \rho_c, \\ \frac{\lambda_\partial}{2} \left[\frac{a_{\text{cert}}^2}{\rho} - 1 + \frac{3(1 - a_{\text{cert}})^2}{4(2 - \rho)} \right], & \rho_c \leq \rho < 1. \end{cases} \quad (32)$$

We keep this derivative-degree ratio and use the larger finite-length margin

$$\text{Margin}_{\text{int}}^{(n)} = \text{Coef}(\rho_n, a_{\text{cert}}, \lambda_\partial) - \text{Rank}(\lambda_\partial). \quad (33)$$

²Solving the cubic explicitly gives the equivalent $a_1(\rho) = \sqrt{2\rho} \cos(\frac{1}{3} \arccos(\sqrt{\rho/8} - 1))$ for $0 < \rho < \rho_c$.

The remaining parameters are

$$\begin{aligned}
C_{\text{rank}} &= 2\lambda_{\partial} + 3, & m &= \left\lceil \frac{4C_{\text{rank}}}{\text{Margin}_{\text{int}}^{(n)}} \right\rceil, & B_{\partial} &= \lfloor \lambda_{\partial} m \rfloor, & B_{\text{jet}} &= \lceil ma_{\text{cert}}/\rho_n \rceil, \\
v_{\text{coef}} &= \sum_{t=0}^{B_{\text{jet}}} (\min\{t, B_{\partial}\} + 1)(ma_{\text{cert}} - \rho_n t)_+, \\
B_Z &= \max \left\{ 1, \left\lfloor \frac{R_{\text{loc}}(m, B_{\partial}) B_{\text{jet}}}{v_{\text{coef}} - R_{\text{loc}}(m, B_{\partial})} \right\rfloor \right\}.
\end{aligned} \tag{34}$$

Both branches in (32) are positive because $a_{\text{cert}} > a_1(\rho)$. Here v_{coef} is a per-position lower bound on the number of interpolation coefficients, while R_{loc} bounds the independent conditions imposed at one position. Their difference is the finite coefficient-constraint margin. The denominator is positive and $0 \leq B_{\partial} \leq B_{\text{jet}}$; Lemma 4.2 proves both facts.

Lemma 4.2 (Finite coefficient-constraint margin above the first-order curve). *The parameters in (34) satisfy $0 \leq B_{\partial} \leq B_{\text{jet}}$ and*

$$v_{\text{coef}} - R_{\text{loc}}(m, B_{\partial}) \geq m^3 \text{Margin}_{\text{int}}^{(n)} - C_{\text{rank}} m^2 \geq 3m^3 \text{Margin}_{\text{int}}^{(n)} / 4 > 0.$$

For fixed ρ , writing $\bar{\eta}_1 = \eta_1 + 1/n$, they have $m, B_{\partial}, B_{\text{jet}} = O_{\rho}(\bar{\eta}_1^{-1})$ and $B_Z = O_{\rho}(\bar{\eta}_1^{-2})$.

Proof. Integrating the decreasing positive-part function over unit boxes under the sum gives $v_{\text{coef}} \geq m^3 \text{Coef}(\rho_n, a_{\text{cert}}, \lambda_{\partial})$. The boxes with $0 \leq b \leq \lfloor \lambda_{\partial} m \rfloor$ cover the strip $0 \leq b \leq \lambda_{\partial} m$, and the total-degree cutoff includes every positive summand. For the rank, put $r_{\text{dens}}(u) = u\lambda_{\partial} - (2u - 1)_+(u + \lambda_{\partial} - 1)_+$. The summand at s is at most

$$m^2 r_{\text{dens}}(s/m) + m(s/m + \lambda_{\partial}) + 1.$$

In general, r_{dens} has Lipschitz constant at most $1 + \lambda_{\partial}$. Comparing the left Riemann sum with its integral and summing the remaining terms gives $R_{\text{loc}}(m, B_{\partial}) \leq m^3 \text{Rank}(\lambda_{\partial}) + (2\lambda_{\partial} + 3)m^2$. The definitions of C_{rank} and m give $m \text{Margin}_{\text{int}}^{(n)} \geq 4C_{\text{rank}}$ and prove the claim. Also $\lambda_{\partial} < a_{\text{cert}}/\rho < a_{\text{cert}}/\rho_n$ gives $B_{\partial} \leq B_{\text{jet}}$. If $B_{\partial} = 0$, the same counts give an ordinary interpolant, which is covered by the same bounds.

In either rate regime, (32) has a simple zero at $a = a_1(\rho)$. On the clipped interval used above, $\text{Margin}_{\text{int}} = \Theta_{\rho}(\eta_1)$, and λ_{∂} stays between positive ρ -dependent constants. For $\rho_n \leq r \leq \rho$, we have $r \geq \rho/2$ and $\lambda_{\partial} < a_{\text{cert}}/r$, so

$$-\partial_r \text{Coef}(r, a_{\text{cert}}, \lambda_{\partial}) = \frac{\lambda_{\partial} a_{\text{cert}}^2}{2r^2} - \frac{\lambda_{\partial}^3}{6} > \frac{\lambda_{\partial} a_{\text{cert}}^2}{3r^2}.$$

This derivative is bounded above and below by positive constants depending only on ρ . Integrating over an interval of length $1/n$ proves

$$\text{Margin}_{\text{int}}^{(n)} = \Theta_{\rho}(\eta_1 + 1/n).$$

Consequently $m, B_{\partial}, B_{\text{jet}} = O_{\rho}(\bar{\eta}_1^{-1})$, while $R_{\text{loc}} = O_{\rho}(m^3)$ and

$$B_Z \leq 1 + \frac{4R_{\text{loc}}(m, B_{\partial}) B_{\text{jet}}}{3m^3 \text{Margin}_{\text{int}}^{(n)}} = O_{\rho}(\bar{\eta}_1^{-2}).$$

In particular, replacing $\text{Margin}_{\text{int}}^{(n)}$ by $\text{Margin}_{\text{int}}$ would only increase m and B_{∂} ; the retuning does not strengthen the characteristic requirement. $\square$

5 From equations to list and MCA bounds

We have constructed a first-order equation at agreement below Johnson. Its solution set can still contain whole curves, so we must use the received agreements again to bound the list. We develop the shared incidence argument, handle ordinary equations, and then count the first-order families. The Johnson specialization uses the same transfer without hidden derivatives.

The principal parameters in these counting arguments have the following roles. The symbol D is defined locally because some arguments use the actual message degree $k - 1$, while padded constructions use the ambient degree $K - 1$. Reconstruction lengths such as K_{rec} are defined separately where they are used.

Parameter	Meaning
k	actual message dimension; candidates satisfy $\deg P < k$
K	ambient interpolation dimension, sometimes larger than k
D	local polynomial-degree bound, normally $k - 1$ or $K - 1$ as specified
d	highest Hasse derivative order used by the explainer
B_{jet}	total degree bound in the jet variables $Y_0, \dots, Y_d$
B_{∂}	first-order bound on the exponent of Y_1
B_Z	degree bound in the symbolic challenge Z

5.1 Agreement geometry

The interpolation step places every close polynomial in an algebraic candidate family. We now explain the geometric argument that turns such a family into a list bound or a common-witness statement. Dimension counts the free parameters in a family, while degree bounds how many components and isolated intersections remain after imposing agreement equations. These are the only geometric quantities used by the later transfer arguments.

An *affine algebraic set* is a common zero set of polynomials in finitely many coordinates. We work over an algebraically closed field, meaning that every nonconstant univariate polynomial splits into linear factors. A *locus* is the set satisfying the stated conditions, which may include inequalities such as $S \neq 0$. A *hypersurface* is the zero set of one nonconstant polynomial; a *hyperplane* is the degree-one case. The *Zariski closure* of a set is the smallest algebraic set containing it, and an *irreducible component* is a maximal piece that cannot be written as a union of two proper algebraic subsets.

The degree of an irreducible s -dimensional algebraic set is the number of points, counted with multiplicity, obtained by intersecting its projective closure with a general linear subspace of codimension s . For a set with several components, total degree sums their degrees, including components of different dimensions. A rational map has coordinates that are ratios of polynomials and is used where its denominators are nonzero. Its image closure is the Zariski closure of the values it takes. The appendix proves the rational-image estimates that produce the candidate families; here we state and prove the incidence and transfer lemmas that use these bounds.

5.2 Incidence and common-witness transfer

We prove a dimension-sensitive agreement count, then use it to identify common witnesses and bound exceptional challenges. The same incidence argument applies to a fixed received word and to the joint family with the challenge retained.

All algebraic sets below are over an algebraically closed field E . Their *total degree* is the sum of the degrees of their distinct irreducible components, including components of smaller dimension. We use the following forms of the refined Bézout inequality [Hei83, Ful98]: a hypersurface defined by a nonzero polynomial of degree b has total degree at most b ; a proper hyperplane section of an irreducible set has

smaller dimension and total degree at most that of the set; and intersecting with a linear subspace does not increase total degree. More generally, successive cuts of degrees at most $b \geq 1$ multiply total degree by at most b at each cut, retaining a component when the cutting polynomial vanishes identically on it.

For Reed–Solomon agreement equations, each independent evaluation imposes one linear constraint on the coefficient vector. We retain the resulting dimension-dependent saving in the incidence count.

Each successive agreement cut reduces the number of free parameters. The resulting incidence bound contains one factor per remaining dimension: for integers $r \geq 0$ and $k \leq T \leq n$, define

$$P_r(T) := \prod_{i=1}^r \frac{n-k+i}{T-k+i}, \quad P_0(T) := 1. \quad (35)$$

Each factor is at least one; the lemma below shows how their product converts a degree bound into a count of points with T agreements.

Lemma 5.1 (Dimension-sensitive agreement incidence). *In the coefficient space of degree- $< k$ polynomials, an algebraic set of dimension at most r and total degree at most B has at most $BP_r(T)$ points agreeing with a fixed word in at least T positions.*

In the joint space of (z, P) , exclude the graph lines with at least L common agreements, where $k \leq L \leq A \leq n$. For a set of dimension at most $r+1$ and degree at most B , the number of remaining points with at least A agreements is at most

$$B \frac{n-L+1}{A-L+1} P_r(A).$$

The same bound holds on the lifted polynomial-curve locus $(z, z^2, \dots, z^\ell, P)$, excluding the corresponding persistent graphs.

Proof. We induct on the dimension of an irreducible component C ; a zero-dimensional component is a point. If $\dim C = t \geq 1$ and c agreement hyperplanes contain C , Vandermonde inversion gives $c \leq k - t$: any $c \leq k$ evaluation equations leave dimension $k - c$. Each point with T agreements lies on at least $T - c$ of the remaining hyperplanes. Their proper sections have smaller dimension and total degree at most $\deg(C)$, so induction first proves finiteness and then bounds their point incidences by

$$(T - c)N(C) \leq (n - c) \deg(C) P_{t-1}(T).$$

Since $(n - c)/(T - c) \leq (n - k + t)/(T - k + t)$, this gives $N(C) \leq \deg(C) P_t(T)$. Larger dimension drops obey the same bound because all incidence factors are at least one. Summing over components proves the fixed-word assertion.

In the joint space, imposing $c \leq k$ agreement equations instead leaves dimension $k + 1 - c$. A component of dimension $t \geq 2$ lies in at most $k + 1 - t$ hyperplanes and incurs the factor $(n - k + t - 1)/(A - k + t - 1)$. At dimension one, a component contained in L agreement hyperplanes is one of the excluded graphs: any k of the equations determine its coefficient polynomials by Vandermonde inversion. Every other curve incurs at most $(n - L + 1)/(A - L + 1)$. The same induction, with zero-dimensional components as its base, proves the joint bound and finiteness. On the lifted curve locus, solving c evaluation equations still leaves the challenge and $k - c$ free coefficients, so its dimension is $k + 1 - c$. This proves the final assertion by the identical induction. $\square$

These two incidence statements are the counting bounds for candidate covers: one bounds a fixed-word list, and the other bounds points away from common-witness graphs. To obtain MCA, we must also count those graphs and their accidental agreements. The joint degree controls the points away from the graphs; a reduced transcendental fiber controls the number of graphs. Components confined to one challenge miss that

transcendental fiber; the joint incidence count still includes their high-agreement points among the off-graph exceptions. The next lemma needs only coverage of the wanted pairs, not a characterization of all differential solutions.

Lemma 5.2 (Geometric transfer on a polynomial received curve). *Let F be a field, let E be an algebraically closed extension, and let $1 \leq k \leq L \leq A \leq n$ and $\ell \geq 1$. Fix distinct $\alpha_1, \dots, \alpha_n \in F$ and words $w_0, \dots, w_\ell \in F^n$. Work in the lifted locus*

$$\Lambda_\ell = \{(z, z^2, \dots, z^\ell, P) : \deg P < k\}.$$

For nonnegative integers r_v , let $V_v \subseteq \Lambda_\ell$ be finitely many closed algebraic sets satisfying

$$\dim V_v \leq r_v + 1, \quad \deg V_v \leq J_v.$$

Let z_ be transcendental over E , and form the fibers after base change to an algebraic closure of $E(z_*)$. Suppose that the reduced algebraic set underlying $(V_v)_{z_*}$ has dimension at most r_v and total degree at most F_v .*

Let C be a collection of pairs (z, P) with $z \in F$ and $P \in F[X]$ of degree less than k . Suppose that, outside a finite preliminary set $\mathcal{E}_{\text{pre}} \subseteq F$, every $(z, P) \in C$ gives the point $(z, z^2, \dots, z^\ell, P) \in \bigcup_v V_v$. Then there is $\mathcal{E} \subseteq F$ of size at most

$$|\mathcal{E}_{\text{pre}}| + \frac{n-L+1}{A-L+1} \sum_v P_{r_v}(A) J_v + \ell(n-L) \sum_v P_{r_v}(L) F_v \quad (36)$$

such that the following holds. For every $(z, P) \in C$ with $z \notin \mathcal{E}$ and

$$|\{i : P(\alpha_i) = \sum_{t=0}^{\ell} z^t w_{t,i}\}| \geq A,$$

there are $P_0, \dots, P_\ell \in F[X]$ of degree less than k such that

$$P = \sum_{t=0}^{\ell} z^t P_t, \quad \{i : P(\alpha_i) = \sum_{t=0}^{\ell} z^t w_{t,i}\} = \{i : P_t(\alpha_i) = w_{t,i} \text{ for every } t\}.$$

Proof. For each position use the affine agreement hyperplane

$$H_i = \{P(\alpha_i) = w_{0,i} + z w_{1,i} + \sum_{t=2}^{\ell} T_t w_{t,i}\}.$$

Within Λ_ℓ , any k such equations determine, by Vandermonde interpolation, a graph

$$\Gamma_{P_0, \dots, P_\ell} = \{(z, z^2, \dots, z^\ell, \sum_{t=0}^{\ell} z^t P_t) : z \in E\}.$$

It is isomorphic to the affine line through its z coordinate, so every positive-dimensional irreducible closed subset is the whole graph.

In each V_v , retain the graphs contained in at least L of the H_i . Distinct graphs give distinct points over z_* : their coefficient polynomials in z cannot coincide at a transcendental element. The fixed-word incidence bound on the reduced fiber gives at most $P_{r_v}(L) F_v$ retained graphs. The joint incidence bound gives at most

$$\frac{n-L+1}{A-L+1} P_{r_v}(A) J_v$$

remaining high-agreement points. This count includes vertical components: they miss the transcendental fiber and cannot contain a graph projecting onto the challenge line.

Let $\mathcal{E} \subseteq F$ contain $\mathcal{E}_{\text{pre}}$ and all base-field challenges from the remaining points or accidental agreements on retained graphs. Outside a graph's common set, an agreement equation is a nonzero polynomial in z of degree at most ℓ . There are at most $n - L$ such positions, costing at most $\ell(n - L)$ challenges per graph. Summation proves (36).

Outside $\mathcal{E}$, each candidate lies on a retained graph with no agreement outside its common set. Any k common positions recover $P_0, \dots, P_\ell$ by interpolation over F , proving the decomposition, descent, and equality of agreement sets. $\square$

5.3 Ordinary equations in every characteristic

First-order Taylor reconstruction can leave solutions where the derivative of the equation vanishes. Eliminating the derivative variable places these solutions in an ordinary polynomial equation. We need a common-witness bound for that equation, including in small characteristic.

We prove such a transfer for any nonzero ordinary equation. Following the function-field approach of BCIKS and the persistent-witness argument of Zheng [BCI⁺23, rs-26], we track its solutions over the challenge. A factorwise Frobenius pullback handles inseparability; bounding message and challenge degrees separately then controls the joint family. The same transfer will yield our quantitative Johnson theorem.

Messages have degree at most D and dimension $k = D + 1$. We write

$$\rho = \frac{D+1}{n}, \quad \rho_- := \frac{D}{n} = \rho - \frac{1}{n}.$$

Thus ρ is the code rate used elsewhere in the paper, while the exact degree ratio ρ_- is the reduced-rate parameter in the BCHKS cutoffs. Here B bounds the interpolation polynomial's degree in the message variable Y , and H bounds its degree in the challenge Z . For integers $D \geq 1$, $n \geq A > D$, $B \geq 1$, $H \geq 0$, a received-curve degree $\ell \geq 1$, and $D + 1 \leq L \leq A$, define

$$\Psi_D(B) = 1 + (2D - 1)(2B - 1) + 2(B - 2D - 1)_+, \quad (37)$$

$$\mathbf{E}_{\text{ord}}^{(\ell)}(D, A, n; B, H; L) = (2B - 1)H + \frac{n - L + 1}{A - L + 1} (\ell B + H \Psi_D(B)) + \ell(n - L)B, \quad (38)$$

$$\mathbf{E}_{\text{ord}}^{(\ell)}(D, A, n; B, H) := \mathbf{E}_{\text{ord}}^{(\ell)}(D, A, n; B, H; D + 1). \quad (39)$$

This applies the candidate-cover MCA bound of Lemma 5.2: the middle line combines the preliminary, joint-image, and accidental agreement costs. The free threshold L balances points off persistent witness graphs against accidental agreements on them. We abbreviate the choice $L = D + 1$ by omitting L from the notation. More explicitly, $(2B - 1)H$ excludes challenges at which the content or resultant degenerates. Away from those challenges, a solution outside a persistent witness graph has at least $A - L + 1$ proper agreement cuts; the ratio in the middle term counts these points from the joint image degree. On a retained graph, each of the $n - L$ remaining coordinates contributes at most ℓ accidental challenge values, giving the final term.

Lemma 5.3 (Ordinary MCA transfer in every characteristic). *Let F be a field, let $1 \leq D \leq n - 2$, and let $D + 1 \leq L \leq A \leq n$. Fix distinct $\alpha_1, \dots, \alpha_n \in F$ and a received curve $w(Z) = \sum_{t=0}^{\ell} Z^t w_t$, where $\ell \geq 1$ and $w_t \in F^n$. Suppose the nonzero polynomial $Q \in F[X, Z, Y]$ satisfies*

$$\deg_Y Q \leq B_{\text{jet}}, \quad \deg_Z Q \leq B_Z, \quad B_{\text{jet}} \geq 1.$$

There is a set $\mathcal{E}_Q \subseteq F$ of size at most

$$\mathbf{E}_{\text{ord}}^{(\ell)}(D, A, n; B_{\text{jet}}, B_Z; L) \quad (40)$$

such that every $z \in F \setminus \mathcal{E}_Q$ and every $P \in F[X]_{\leq D}$ satisfying

$$Q(X, z, P(X)) = 0, \quad |S_z(P)| \geq A,$$

has the full common-witness property of [Definition 3.1](#), with message dimension $D + 1$. No restriction on the characteristic or on ℓ is needed. The line case is $\ell = 1$.

To prove the lemma, we first pull each inseparable factor back through Frobenius. Its order-zero Taylor recurrence has leading coefficient one in every characteristic. A mixed-degree image bound then gives joint degree $\ell B_{\text{jet}} + B_Z \Psi_D(B_{\text{jet}})$ and reduced generic-fiber degree B_{jet} , which are exactly the two degree budgets used above.

This transfer bounds the candidates that both solve an ordinary equation and have many agreements. The equation need not explain every close polynomial; that additional implication is used only when we derive the Johnson theorem. We retain the powers of the challenge as extra coordinates for a received polynomial curve, making agreement linear in the enlarged space. The line case used by Johnson follows by setting the received degree to one.

Proof of Lemma 5.3. We may work over an algebraic closure $\overline{F}$ when constructing an envelope; the final candidates and received words remain over F . Remove repeated factors from Q and separate its content in Y . Let $C(X, Z)$ be that content, and let

$$F_1(X, Z, Y), \dots, F_r(X, Z, Y)$$

be the distinct irreducible factors of positive Y -degree. Write

$$j_i = \deg_Y F_i, \quad h_i = \deg_Z F_i, \quad h_C = \deg_Z C.$$

Thus, over $\overline{F}$, the polynomial has the form $Q = C \prod_i F_i^{e_i}$ with $e_i \geq 1$. The squarefree product divides Q , so

$$\sum_i j_i \leq B_{\text{jet}}, \quad h_C + \sum_i h_i \leq B_Z, \quad \sum_i h_i j_i \leq B_Z B_{\text{jet}}. \quad (41)$$

Choose one nonzero coefficient of C as a polynomial in X . There are at most h_C challenges for which that coefficient, and hence possibly $C(X, z)$, vanishes identically. We include them in the preliminary set $\mathcal{E}_{\text{pre}}$. Away from them, every root identity for Q is a root identity for at least one F_i .

For each factor F_i , construct its separable Frobenius pullback as follows. In characteristic zero put $s_i = 1$ and $G_i = F_i$. In characteristic $p > 0$, take the largest $s_i = p^{v_i}$ such that $F_i(X, Z, Y) = G_i(X, Z, Y^{s_i})$, and put $b_i = \deg_U G_i = j_i/s_i$. Then $\partial_U G_i \neq 0$. Applying inverse Frobenius to the coefficients of G_i gives an irreducible polynomial $\tilde{G}_i(T, W, U)$, separable in U , of bidegree at most (h_i, b_i) in (W, U) , and

$$\tilde{G}_i(T, W, U)^{s_i} = F_i(T^{s_i}, W^{s_i}, U). \quad (42)$$

Hence $F_i(X, z, P(X)) = 0$ and $w^{s_i} = z$ imply

$$\tilde{G}_i(T, w, P(T^{s_i})) = 0. \quad (43)$$

Frobenius is bijective on $\overline{F}$, and on every finite base field.

Choose t transcendental over $\overline{F}$ and work over an algebraic closure of $\overline{F}(t)$. By Gauss's lemma, $\tilde{G}_i(t, W, U)$ is coprime to its U -derivative, so

$$R_i(W) = \text{Res}_U(\tilde{G}_i(t, W, U), \partial_U \tilde{G}_i(t, W, U)) \neq 0, \quad (44)$$

$$\deg_W R_i \leq (2b_i - 1)h_i. \quad (45)$$

The roots yield at most this many original challenges $z = w^{s_i}$; add their base-field images to $\mathcal{E}_{\text{pre}}$.

Away from these roots, the source curve

$$C_i : \quad \tilde{G}_i(t, W, u) = 0 \quad (46)$$

has nonzero separant $S = \partial_U \tilde{G}_i(t, W, u)$. Put $K'_i = s_i D + 1$. The order-zero Taylor recurrence reconstructs $R(t + \xi) = \sum_{r < K'_i} c_r \xi^r$ from $c_0 = u$ and gives, for $1 \leq r < K'_i$,

$$c_r = \frac{A_r(W, u)}{S(W, u)^{2r-1}}, \quad \deg_u A_r \leq (2r-1)(b_i-1) + 1, \quad \deg_W A_r \leq (2r-1)h_i.$$

This is the denominator induction of [Lemma A.4](#), retaining W . Its leading binomial coefficient is always $\binom{r}{0} = 1$, so no characteristic restriction occurs. With denominator $S^{2K'_i-3}$, the map to $(z, z^2, \dots, z^\ell, P)$ has degree bounds

$$\beta_i = 1 + (2K'_i - 3)(b_i - 1), \quad \alpha_i = s_i \ell + (2K'_i - 3)h_i \quad (47)$$

in u and W , respectively. The $s_i \ell$ term accounts for the output powers of $z = W^{s_i}$. The map outputs the original P : for $x_0 = t^{s_i}$,

$$P((t + \xi)^{s_i}) = \sum_{r=0}^D P^{[r]}(x_0) \xi^{s_i r}, \quad (48)$$

so the coefficients with indices $s_i r$ determine P by an invertible linear change of coordinates. Only the auxiliary reconstruction degree has increased; the output degree remains at most D . Hence every root pair of F_i outside the resultant set lies in this rational image.

Applying [Lemma A.3](#) with $d = 0$ gives an image of dimension at most one and degree

$$\begin{aligned} J_i &\leq h_i \beta_i + b_i \alpha_i \\ &= \ell j_i + h_i [1 + (2Ds_i - 1)(2b_i - 1)] \\ &\leq \ell j_i + h_i \Psi_D(j_i). \end{aligned} \quad (49)$$

The identity $s_i b_i = j_i$ cancels the inseparability factor from the mixed degree bound.

To prove the final inequality, rearrange that bracket as

$$1 + (2Ds_i - 1)(2b_i - 1) = 1 + (2D - 1)(2j_i - 1) + 2(s_i - 1)(b_i - D). \quad (50)$$

If $s_i = 1$ or $b_i \leq D$, its last term is nonpositive. Otherwise $s_i \geq 2$ and $b_i \geq D + 1$, and

$$(s_i - 1)(b_i - D) = j_i - (b_i + (s_i - 1)D) \leq j_i - 2D - 1.$$

This proves the bound by $\Psi_D(j_i)$ in both cases. The function Ψ_D is increasing on the positive integers, so $\sum_i h_i \Psi_D(j_i) \leq B_Z \Psi_D(B_{\text{jet}})$.

For a transcendental output challenge z , there is set-theoretically one w with $w^{s_i} = z$, and at most b_i possible values of u . The rational image is constructible and dense in each curve component of its closure. Its finite omitted subset and any vertical components miss this fiber, whose reduced degree is therefore at most b_i . Purely inseparable scheme multiplicity is not charged.

Let V be the union of the image closures, intersected with the degree- $\leq D$ coefficient space. By (41),

$$\deg V \leq D_{\text{joint}} := \sum_i J_i \leq \ell B_{\text{jet}} + B_Z \Psi_D(B_{\text{jet}}), \quad (51)$$

and the reduced transcendental fiber has degree at most $\sum_i b_i \leq B_{\text{jet}}$. The saved powers in the comparison with BCHKS come from reconstructing through the actual message degree D and using the whole-factor bidegree image bound, as detailed in [Section 5.7](#). Apply [Lemma 5.2](#) with

$$k = D + 1, \quad r_1 = 0, \quad J_1 = D_{\text{joint}}, \quad F_1 = B_{\text{jet}}.$$

Since $P_0(A) = P_0(L) = 1$, it contributes $\frac{n-L+1}{A-L+1}D_{\text{joint}} + \ell(n-L)B_{\text{jet}}$ and gives witnesses over F with equality of agreement sets. Finally,

$$|\mathcal{E}_{\text{pre}}| \leq h_C + \sum_i (2b_i - 1)h_i \leq (2B_{\text{jet}} - 1)B_Z. \quad (52)$$

These bounds prove (40). □

5.4 First-order solution counting

The interpolation construction supplies an equation for every close message. We now bound the solutions whose separant is nonzero after substituting the candidate, place the solutions whose separant vanishes identically on an ordinary equation, and then apply the agreement counts above.

5.4.1 Recovering a polynomial from initial Taylor data

The interpolation identity is not yet a list bound. For example, the equation $Q = Y_1$ has every constant polynomial as a solution when the characteristic is zero or greater than the message degree. Thus even a very simple first-order equation can have a one-parameter family of polynomial solutions. We must parameterize that family and then use the received-word agreements to leave only finitely many points.

At a Taylor center x_0 , write $P(x_0 + T) = \sum_j c_j T^j$. The initial pair (c_0, c_1) gives the value and first derivative. For a differential equation $Q(X, P, P^{[1]}) = 0$, its *separant* is $\partial Q / \partial Y_1$. When its value at (x_0, c_0, c_1) is nonzero, the equation successively determines the later coefficients. We call such data regular. The coefficient multiplying the next unknown c_j is j times the separant, so $p > D$ suffices through degree D . The resulting rational formulas are the Taylor parameterizations used in the count below; their degrees, not merely the number of initial values, control the list bound.

Counting factors instead of differentiation stages. We use the regular–singular separation already present in Naskręcki’s construction [[Nas26](#)], retaining the factor degrees explicitly in the Taylor-image and resultant bounds. For $D = k - 1$ and agreement threshold A , put $\lambda_{\text{agr}} = (n - D)/(A - D)$. We first remove repeated factors, which does not change the solution set but prevents multiplicity from making a separant vanish for a purely formal reason. Write the squarefree part as UV , where U contains the factors independent of Y_1 and every factor of V involves Y_1 . On the locus where $\partial V / \partial Y_1$ is nonzero, one rational Taylor map covers its regular curve components, with image degree $O(DB_{\text{jet}}B_\partial)$. Each received coordinate cuts this image by the linear equation $P(\alpha_i) = y_i$; the incidence count uses the excess $A - D$ to bound how many points satisfy at least A such equations.

The remaining solutions either satisfy $U = 0$, or satisfy both $V = 0$ and $\partial V / \partial Y_1 = 0$. Eliminating Y_1 puts them on the ordinary equation

$$U \text{Res}_{Y_1}(V, \partial V / \partial Y_1) = 0,$$

whose degree is $O(B_{\text{jet}}B_\partial)$. The ordinary transfer therefore handles the solutions whose separant vanishes identically, without another derivative recursion. Together these two pieces charge the factor degrees once, rather than once for every successive derivative of the original equation. This is the source of the quadratic dependence on the inverse gap in the list bound. [Lemma 5.6](#) retains the exact degree calculation; [Theorem 5.7](#) uses this smaller candidate cover.

5.4.2 From the equation to list and MCA bounds

We now state the counting bound used by both first-order theorems. The ordinary transfer from [Lemma 5.3](#) handles the solutions where regular first-order reconstruction fails. The remaining solutions form one regular family. Its degree controls the list; retaining the challenge gives the joint degree needed for MCA. The proof below factors the equation, constructs this family, and inserts its degrees into the shared geometric transfer. We first record the specialized Taylor and image-degree estimates that retain the separate degree in the hidden derivative.

Taylor and image-degree bounds. The separate bound $\deg_{Y_1} Q \leq B_\partial$ also bounds the exponent of Y_1 in Taylor reconstruction. We record the two degree estimates needed below before applying them factor by factor. The same estimates also support the successive-stage alternative used for individual finite certificates. In both arguments, ordinary equations are handled by [Lemma 5.3](#).

To see how a separate degree bound helps, consider polynomials in Y_0, Y_1 of total degree at most B and Y_1 -degree at most $C \leq B$. Their monomials fill a triangle with one corner removed. The exact count is

$$N_\Delta(B, C) = \sum_{v=0}^C (B - v + 1) = (C + 1)(B + 1) - \frac{C(C + 1)}{2}.$$

The missing corner also reduces the degree of a family parameterized by these polynomials. The following refinement of [Lemma A.3](#) makes this consequence precise.

Lemma 5.4 (Rational images with a separate derivative-degree bound). *Let $G \in E[Z, Y_0, Y_1]$ be nonzero, of challenge degree at most B_Z , total jet degree at most j , and Y_1 -degree at most $r \leq j$. Consider a rational map on a union of irreducible components of $V(G)$. Suppose that its numerators and common denominator have challenge degree at most D_Z , total jet degree at most b , and Y_1 -degree at most c , where $D_Z \geq 1$ and $1 \leq c \leq b$. Assume the denominator is nonzero on each selected component. Put*

$$D_{\text{fiber}}(j, r; b, c) = jc + r(b - c).$$

The image closure has degree at most

$$B_Z(2bc - c^2) + 2D_Z D_{\text{fiber}}(j, r; b, c).$$

After fixing a generic challenge, the image of the remaining curve components has degree at most $D_{\text{fiber}}(j, r; b, c)$.

Proof. Use the monomial embedding and Hilbert-function comparison from [Lemma A.3](#). For an irreducible factor G_0 with actual degrees $(d_Z, d_{\text{jet}}, d_\partial)$, the only changed input is the dimension of the truncated source space. With bounds $(D_Z N, bN, cN)$ it is $(D_Z N + 1)N_\Delta(bN, cN)$, so multiplication by G_0 leaves quotient dimension at most

$$(D_Z N + 1)N_\Delta(bN, cN) - (D_Z N - d_Z + 1)N_\Delta(bN - d_{\text{jet}}, cN - d_\partial).$$

Its quadratic coefficient is

$$\frac{d_Z(2bc - c^2)}{2} + D_Z(d_{\text{jet}}c + d_\partial(b - c)).$$

The argument of [Lemma A.3](#) turns twice this coefficient into the image-degree bound, including with base points or positive-dimensional fibers. When $c = b$, the full triangle gives the same coefficient. At a fixed generic challenge, N_Δ alone has linear coefficient $d_{\text{jet}}c + d_\partial(b - c)$, giving the curve bound. Finally sum over the selected factors: their three degrees add to at most (B_Z, j, r) , and both bounds are linear in these source degrees. $\square$

The improvement over the total-degree bounds is explicit:

$$jb - D_{\text{fiber}}(j, r; b, c) = (j - r)(b - c),$$

$$B_Z b^2 + 2D_Z jb - (B_Z(2bc - c^2) + 2D_Z D_{\text{fiber}}(j, r; b, c)) = B_Z(b - c)^2 + 2D_Z(j - r)(b - c).$$

Both differences are nonnegative. Thus retaining the separate degree can only improve the bounds, and recovers the old bounds when $c = b$.

Lemma 5.5 (Degrees of a regular first-order Taylor map). *Let $D \geq 1$, $\ell \geq 1$, and $\tau = \max\{0, 2D - 3\}$. Suppose a first-order equation $T \in F[X, Z, Y_0, Y_1]$ has total jet degree at most j , Y_1 -degree at most $1 \leq r \leq j$, and challenge degree at most H . Fix a Taylor center $a \in F$, and put $T_a = T(a, Z, Y_0, Y_1)$ and $q = (\partial T / \partial Y_1)(a, Z, Y_0, Y_1)$. Assume characteristic zero or $p > D$. On every component of $V(T_a)$ where q is nonzero, Taylor reconstruction through message degree D , with the coordinates $(Z, Z^2, \dots, Z^\ell, P)$ retained, has numerator and common-denominator degrees at most*

$$b = 1 + \tau(j - 1), \quad c = \min\{b, \tau(r - 1) + D\}, \quad D_Z = \ell + \tau H \quad (53)$$

in total jet degree, Y_1 -degree, and challenge degree, respectively.

Proof. Before specializing $X = a$, differentiation along a solution is

$$\partial_X + Y_1 \partial_{Y_0} - \frac{T_X + Y_1 T_{Y_0}}{T_{Y_1}} \partial_{Y_1}.$$

For Taylor order $s \geq 2$, a common denominator is q^{2s-3} after specialization. The same induction bounds the numerator's total jet degree by $1 + (2s - 3)(j - 1)$. The second numerator has Y_1 -degree at most $r + 1$, and each further differentiation adds at most $2r - 1$. Its numerator degree is therefore at most

$$(2s - 3)(r - 1) + s = (2s - 3)r - s + 3.$$

If $D = 1$, no coefficient beyond the initial pair is reconstructed: the Taylor map is the identity on (Y_0, Y_1) , so $(b, c, D_Z) = (1, 1, \ell)$. For $D \geq 2$, the largest denominator exponent through degree D is $2D - 3 = \tau$. Passing to the common denominator q^τ through degree D gives separate degree at most $\tau(r - 1) + D$; the total-degree induction gives $1 + \tau(j - 1)$. Clearing the common denominator adds at most τH in the challenge, while the retained powers of Z add at most ℓ . Division by $s!$ uses only $p > D$. $\square$

In the lemma below, write $B = B_{\text{jet}}$, $M = B_\partial$, and $H = B_Z$ for the three degree bounds. With these bounds and the received-curve degree fixed, an agreement threshold exceeding the message degree by a fixed fraction of n gives lists of size $O(n)$ and exceptional sets of size $O(n^2)$. The formulas retain the constants needed for finite applications.

Let $1 \leq D \leq n - 2$, $D + 1 \leq A \leq n$, $B \geq 1$, $1 \leq M \leq B$, and $H \geq 0$. Fix distinct evaluation points $\alpha_1, \dots, \alpha_n \in F$ and a received curve $w(Z) = \sum_{t=0}^\ell Z^t w_t$, with $\ell \geq 1$ and $w_t \in F^n$. Put $\lambda_{\text{agr}} = (n - D)/(A - D)$. Define the regular image and ordinary-equation degree bounds by

$$\begin{aligned} \tau &= \max\{0, 2D - 3\}, & u &= 1 + \tau(B - 1), & v &= \min\{u, \tau(M - 1) + D\}, \\ F_{\text{reg}} &= Bv + M(u - v), & S &= \max\{B, (2M - 1)B - M^2\}, & H_s &= (2M - 1)H. \end{aligned} \quad (54)$$

For this received curve, put

$$J_\ell = H(2uv - v^2) + 2(\ell + \tau H)F_{\text{reg}},$$

$$E_{0,\ell}(s, h) = \min_{D+1 \leq L_0 \leq A} E_{\text{ord}}^{(\ell)}(D, A, n; s, h; L_0).$$

The minimum is over integer thresholds, independently of the regular-family threshold L below.

Lemma 5.6 (Squarefree first-order counting). *With these parameters, let $Q \in F[X, Z, Y_0, Y_1]$ be nonzero, with total jet degree at most B , Y_1 -degree at most M , and challenge degree at most H . Assume characteristic zero or $p > \max\{D, M\}$. For each integer $D + 1 \leq L \leq A$, outside a set of at most*

$$E_{0,\ell}(S, H_s) + \frac{n-L+1}{A-L+1} \lambda_{\text{agr}} J_\ell + \ell(n-L) \frac{n-D}{L-D} F_{\text{reg}} \quad (55)$$

challenges, every degree- $\leq D$ solution of $Q(X, z, P, P^{[1]}) = 0$ with at least A agreements has the full common-witness property of [Definition 3.1](#), with message dimension $D + 1$. For a nonzero fixed-word equation with the same jet-degree bounds, the number of close solutions is at most

$$\lambda_{\text{agr}} F_{\text{reg}} + S \leq 4\lambda_{\text{agr}} DBM + 2BM + B. \quad (56)$$

Proof of Lemma 5.6. Work over $\overline{F}$. Write the squarefree part of Q as UV , where U contains every factor independent of Y_1 and V is the product of the distinct factors of positive Y_1 -degree. Put $j = \deg_{Y_0, Y_1} V$, $r = \deg_{Y_1} V$, and $b_U = \deg_{Y_0} U$. Then $b_U + j \leq B$, $r \leq M$, and $\deg_Z U + \deg_Z V \leq H$. If $r = 0$, every solution solves the ordinary equation U ; its degree bounds are at most (S, H_s) . Assume $r > 0$. Because $p > M$ in positive characteristic, V is separable over $\overline{F}(X, Z, Y_0)$: each irreducible factor has positive degree less than p , and the factors are distinct.

First cover the solutions P for which $\partial_{Y_1} V(X, z, P(X), P^{[1]}(X))$ is nonzero as a polynomial in X . Extend the coefficient field to the algebraic closure of $\overline{F}(c)$, where c is transcendental over $\overline{F}$, and use c as the Taylor center. Transcendence preserves each such nonzero identity, so its value at $X = c$ is nonzero. Apply [Lemma 5.5](#) over this field to obtain one rational map on the regular components of $V = 0$. Its numerator and denominator bounds are u in total jet degree, v in Y_1 , and $\ell + \tau H$ in the challenge, retaining the coordinates $(Z, Z^2, \dots, Z^\ell, P)$. Applying [Lemma 5.4](#) to these components gives reduced generic-fiber degree at most F_{reg} and joint degree at most J_ℓ . The joint image has dimension at most two, and its generic challenge fiber has dimension at most one. After the linear intersection with degree- $\leq D$ coefficient space, [Lemma 5.2](#) applies to this one regular family with $k = D + 1$, $r_1 = 1$, $J_1 = J_\ell$, and $F_1 = F_{\text{reg}}$. Since $P_1(A) = \lambda_{\text{agr}}$ and $P_1(L) = (n - D)/(L - D)$, it gives the last two terms of (55). For a fixed received word, apply the Taylor construction separately to its nonzero fixed-word equation. Its image degree is at most F_{reg} , so [Lemma 5.1](#) bounds its regular candidates by $\lambda_{\text{agr}} F_{\text{reg}}$. This does not require a degree bound on every special fiber of the symbolic family.

To cover the other solutions, form the nonzero polynomial

$$R = \text{Res}_{Y_1}(V, \partial_{Y_1} V).$$

The total jet-degree bound sharpens its ordinary degree. Writing $V = \sum_{i=0}^r a_i Y_1^i$, we have $\deg_{Y_0} a_i \leq j - i$. Every monomial in this resultant has coefficient degree $2r - 1$ and coefficient-index weight r^2 ; these identities follow from $\text{Res}(V, V') = a_r \text{Disc}(V)$. Consequently

$$\deg_{Y_0} R \leq (2r - 1)j - r^2, \quad \deg_Z R \leq (2r - 1) \deg_Z V.$$

If both specialized equations vanish, so does their resultant, including a specialization that lowers the Y_1 degree: the Sylvester matrix is specialized at its original degrees. Every solution not covered by the regular map thus solves UR . The degree budgets give

$$\begin{aligned} \deg_{Y_0}(UR) &\leq b_U + (2r - 1)j - r^2 \leq (2r - 1)B - r^2 \leq S, \\ \deg_Z(UR) &\leq \deg_Z U + (2r - 1) \deg_Z V \leq H_s. \end{aligned}$$

Here $(2r - 1)B - r^2$ increases for $1 \leq r \leq M \leq B$. The ordinary transfer, [Lemma 5.3](#), contributes $E_{0,\ell}(S, H_s)$ by choosing its integer threshold independently. Its content term includes challenges at which UR vanishes

identically, so no specialization hypothesis is missing. For a fixed word, the same construction over $\overline{F}(X)$ gives at most S ordinary roots. This proves the first list inequality; the second uses $u \leq 2DB$, $v \leq 2DM$, and $S \leq B + 2BM$.

The common transfer supplies witnesses over F , also for infinite fields. Factorization is used only for counting; it imposes no factorization requirement on the decoder. $\square$

For $M = 0$, apply [Lemma 5.3](#) directly with degrees (B, H) ; the fixed-word list is at most B by ordinary factor counting. This case needs no positive derivative-degree assumption.

Theorem 5.7 (First-order list decoding above the rate curve). *Fix $0 < \rho < a < 1$ with $a > a_1(\rho)$. Let $n \geq 2$, $2 \leq k \leq pn$, and $an \leq A \leq n$ be integers, and choose $(m, B_\partial, B_{\text{jet}}, B_Z)$ by (34). Over every field of characteristic zero or $p > \max\{k - 1, B_\partial\}$, and every set of n distinct evaluation points, each received word has at most*

$$\Lambda \leq 4D\lambda_{\text{agr}}B_{\text{jet}}B_\partial + 2B_{\text{jet}}B_\partial + B_{\text{jet}} = O_\rho\left(\frac{n}{(\eta_1 + 1/n)^2}\right) = O_\rho(n/\eta_1^2), \quad (57)$$

close degree- $< k$ polynomials at agreement threshold A .

Proof. The coefficient count uses $D \leq \rho_n n$ and $A \geq an$, so the positive finite margin supplies a nonzero first-order equation. Apply [Lemma 5.6](#) with $B = B_{\text{jet}}$ and $M = B_\partial$. Since $D \leq n$, $\lambda_{\text{agr}} \leq 1/(a - \rho) = O_\rho(1)$, and $B_{\text{jet}}, B_\partial = O_\rho((\eta_1 + 1/n)^{-1})$, its list bound has the stated dependence. The argument also applies over infinite fields by bounding every finite subset. $\square$

A small refinement at high rates. We retain a_1 as our main curve, although the rectangle rank bound leaves a small improvement at high rates. Keeping the total-degree restriction in (62) lowers the limiting threshold when $\rho > 8 - 3\sqrt{6} \approx 0.651531$; further weighted restrictions on the monomials can improve it slightly more. These particular refinements do not affect rates $1/2$ and below, which are our main focus. The finite tests in (62) already retain the total-degree and specialization cutoffs when choosing parameters for a concrete code.

5.5 Following candidates along a received line

For one received word we counted the solutions of $Q(X, P, P^{[1]}) = 0$. For a received line, we keep the challenge symbolic and ask which solutions arise from common witnesses. We use the full agreement sets $S_z(P)$ and $S(F_0, G_0)$ defined in [Section 3](#). Following the earlier function-field constructions [[BCI⁺23](#), [rs-26](#)], the same coefficient-constraint margin also bounds the equation's challenge degree. We use the symbolic construction of [Proposition 3.13](#).

The geometric transfer in [Lemma 5.2](#) now identifies common witnesses and bounds the remaining challenges. For the first-order regular family, the two inputs are the joint degree J_1 and reduced generic-fiber degree F_{reg} from [Lemma 5.6](#). The ordinary resultant equation contributes its separate exceptional count. We insert these bounds below, using $L = D + \lceil (A - D)/2 \rceil$ for the regular family and $L_0 = k$ for the ordinary tail. The finite certificates later optimize the two thresholds independently.

Theorem 5.8 (First-order line MCA above the rate curve). *Under the numerical and characteristic hypotheses of [Theorem 5.7](#), put*

$$b = B_{\text{jet}}(2B_\partial + 1), \quad h = B_Z(2B_\partial + 1), \\ E_{\text{ord}} = (2b - 1)h + \lambda_{\text{agr}}(h + b + 4Dbh) + (n - D - 1)b.$$

For every received line there is one exceptional set $\mathcal{E}$ with

$$\begin{aligned} |\mathcal{E}| &\leq E_{\text{ord}} + (48D^2B_Z + 16D)\lambda_{\text{agr}}^2 B_{\text{jet}} B_{\partial} + 8D(n - D - 1)\lambda_{\text{agr}} B_{\text{jet}} B_{\partial} \\ &= O_{\rho}\left(\frac{n^2}{(\eta_1 + 1/n)^4}\right) = O_{\rho}(n^2/\eta_1^4) \end{aligned} \quad (58)$$

such that every $z \notin \mathcal{E}$ and every degree- $< k$ candidate P with $|S_z(P)| \geq A$ admit degree- $< k$ witnesses F_0, G_0 with $P = F_0 + zG_0$ and $S_z(P) = S(F_0, G_0)$. Over $\mathbb{F}_q$, divide the bound by q for the exceptional probability.

Proof. Apply [Lemma 5.6](#) to the symbolic interpolant with $B = B_{\text{jet}}$, $M = B_{\partial}$, and $H = B_Z$. Choose $L = D + \lceil (A - D)/2 \rceil$. The regular joint degree is at most $8DB_{\text{jet}}B_{\partial} + 24D^2B_ZB_{\text{jet}}B_{\partial}$, and both threshold incidence factors are at most $2\lambda_{\text{agr}}$. Bounding the ordinary bidegrees by (b, h) and choosing $L_0 = k$ gives the displayed finite bound, including solutions whose separant vanishes identically and exceptional specializations. For $B_{\partial} = 0$, the ordinary transfer gives it directly.

To obtain the gap exponent, write $\bar{\eta}_1 = \eta_1 + 1/n$. The regular factors contribute $O_{\rho}(n^2\bar{\eta}_1^{-4})$. The ordinary equation has $b = O_{\rho}(\bar{\eta}_1^{-2})$ and $h = O_{\rho}(\bar{\eta}_1^{-3})$, so its contribution is $O_{\rho}(n\bar{\eta}_1^{-5})$. The finite-length adjustment is needed precisely here: $n\bar{\eta}_1 \geq 1$ gives

$$n\bar{\eta}_1^{-5} \leq n^2\bar{\eta}_1^{-4}.$$

Thus the fourth-power bound holds at every permitted block length, including $n\eta_1 < 1$. $\square$

The characteristic requirement has two separate purposes. Recovering coefficients through degree D divides by $1, \dots, D$, so needs $p > D$. Separability of each positive- Y_1 factor needs $p > B_{\partial}$. There is no restriction on B_{jet} : the ordinary tail uses the characteristic-free theorem. The squarefree and successive-stage polynomial-curve bounds in [Lemmas 5.6](#) and [5.9](#) have the same requirement. If $p \leq D$, even the equation $P' = 0$ has the family $P = H(X^p)$, whose later coefficients are not determined by its initial value. This explains the limit of this reconstruction argument, without asserting an impossibility theorem for every small-characteristic code.

5.5.1 A second counting bound for finite parameters

Factoring before reconstruction gives the clean gap exponents above. At a particular finite support, the successive-stage argument can give a smaller bound. Each derivative stage gives a regular family. We stop when Y_1 disappears and use the ordinary transfer for the terminal equation; this avoids differentiating in Y_0 , which would unnecessarily require the total jet degree to be below the characteristic, and counts the ordinary roots only once.

Let $2 \leq k \leq A \leq n$, $D = k - 1$, $\ell \geq 1$, $B_{\text{jet}} \geq 1$, $0 \leq B_{\partial} \leq B_{\text{jet}}$, and $B_Z \geq 0$. Fix a received curve $w(Z) = \sum_{t=0}^{\ell} Z^t w_t$ with $w_t \in F^n$. The following budgets separate the regular stages from the ordinary tail. Put $\lambda_{\text{agr}} = (n - D)/(A - D)$ and $\tau = \max\{0, 2D - 3\}$. For every possible actual first-derivative degree $0 \leq d \leq B_{\partial}$, and $0 \leq i < d$, set

$$j_i = B_{\text{jet}} - i, \quad r_i = d - i, \quad b_i = 1 + \tau(j_i - 1), \quad c_i = \min\{b_i, \tau(r_i - 1) + D\}, \quad D_{\text{fiber}, i} = j_i c_i + r_i(b_i - c_i).$$

Here j_i and r_i are the remaining total and first-derivative degrees. The quantities b_i and c_i bound the corresponding degrees of the reconstruction map, and $D_{\text{fiber}, i}$ is the contribution of this stage to the reduced generic-fiber degree. Summing the regular stages gives the joint and reduced generic-fiber budgets

$$D_{\text{joint}}(d) = \sum_{i < d} [B_Z(2b_i c_i - c_i^2) + 2(\ell + \tau B_Z)D_{\text{fiber}, i}], \quad D_{\text{fiber}}(d) = \sum_{i < d} D_{\text{fiber}, i}.$$

For $D + 1 \leq L \leq A$, the ordinary-tail contribution is

$$E_{\text{tail}}^{(\ell)}(b, L) = \begin{cases} E_{\text{ord}}^{(\ell)}(D, A, n; b, B_Z; L), & b \geq 1, \\ B_Z, & b = 0. \end{cases}$$

We minimize its threshold independently of the regular family's threshold:

$$\overline{E}_{\text{tail}}^{(\ell)}(b) = \min_{D+1 \leq L_0 \leq A} E_{\text{tail}}^{(\ell)}(b, L_0).$$

Combining this tail with the regular-family incidence bound gives

$$E_d^{(\ell)}(L) = \overline{E}_{\text{tail}}^{(\ell)}(B_{\text{jet}} - d) + \frac{n - L + 1}{A - L + 1} \lambda_{\text{agr}} D_{\text{joint}}(d) + \ell(n - L) \frac{n - D}{L - D} D_{\text{fiber}}(d). \quad (59)$$

Lemma 5.9 (First-order counting with an ordinary tail). *With these parameters, let $Q \in F[X, Z, Y_0, Y_1]$ be nonzero, of total jet degree at most B_{jet} , degree in Y_1 at most B_{∂} , and challenge degree at most B_Z . Put $d_Q = \deg_{Y_1} Q$. Assume characteristic zero or $p > \max\{D, d_Q\}$. For a received-word-independent support certificate, the sufficient characteristic assumption is $p > \max\{D, B_{\partial}\}$. For each z outside one set of size at most $\max_{0 \leq d \leq B_{\partial}} \min_{k \leq L \leq A} E_d^{(\ell)}(L)$, every degree- $< k$ solution of $Q(X, z, P, P^{[1]}) = 0$ with at least A agreements with $w(z)$ has the full common-witness conclusion of [Definition 3.1](#). For a nonzero fixed-word equation, its close polynomial solutions number at most $\lambda_{\text{agr}} D_{\text{fiber}}(B_{\partial}) + B_{\text{jet}} - B_{\partial}$.*

Proof of Lemma 5.9. If $D = n - 1$, then the hypotheses force $k = A = n$. Interpolate the coefficient words separately as $P_0, \dots, P_{\ell} \in F[X]_{\leq D}$. A candidate with A agreements agrees everywhere, so it equals $\sum_t z^t P_t$ and its full agreement set is the common agreement set of the P_t . Thus the empty exceptional set suffices. A fixed word likewise has at most one candidate, which is within the displayed list bound. We may therefore assume $D \leq n - 2$ below.

Take $d = d_Q$. Its successive Y_1 derivatives end at the nonzero leading coefficient $U = (1/d!) \partial_{Y_1}^d Q \in F[X, Z, Y_0]$. For a solution of $Q = 0$, either there is an index $0 \leq i < d$ such that $\partial_{Y_1}^i Q$ vanishes identically after substitution while $\partial_{Y_1}^{i+1} Q$ is nonzero as a polynomial in X , or the solution satisfies $U(X, z, P) = 0$. The first case supplies a regular order-one equation at stage i , and the second is an ordinary equation of degree at most $B_{\text{jet}} - d$ in Y_0 .

At a regular stage, [Lemma 5.5](#), with $(j, r, H) = (j_i, r_i, B_Z)$, gives the degree bounds $(b_i, c_i, \ell + \tau B_Z)$. [Lemma 5.4](#) now gives exactly $D_{\text{joint}}(d)$ and $D_{\text{fiber}}(d)$. No bound $p > B_{\text{jet}}$ enters this calculation.

The ordinary equation may have inseparable factors. Apply the ordinary transfer of [Section 5.7](#) to its roots that also meet the agreement threshold; it need not explain every close candidate. Its content and resultant exceptions cost at most $(2b - 1)B_Z$ for $b = B_{\text{jet}} - d \geq 1$, and its joint and generic-fiber degrees are bounded by $\ell b + B_Z \Psi_D(b)$ and b . For $b = 0$, only the at most B_Z identically vanishing challenge specializations remain. These give $\overline{E}_{\text{tail}}^{(\ell)}(b)$ by choosing its threshold independently in the free-threshold form of [Lemma 5.3](#). For the order-one families, apply [Lemma 5.2](#) with $k = D + 1$, $r_1 = 1$, and the summed joint and reduced generic-fiber degrees $D_{\text{joint}}(d)$ and $D_{\text{fiber}}(d)$. Its two incidence factors are exactly the displayed multipliers of those degrees. This proves $E_d^{(\ell)}(L)$ for the actual $d = d_Q$; maximizing over the possible actual degree gives the received-word-independent bound.

For a fixed word, the ordinary equation has at most $B_{\text{jet}} - d$ polynomial roots over $F(X)$, and the order-one families contribute at most $\lambda_{\text{agr}} D_{\text{fiber}}(d)$. As d increases, all common stage bounds increase and a new stage contributes at least one; since $\lambda_{\text{agr}} \geq 1$, $\lambda_{\text{agr}} D_{\text{fiber}}(d) + B_{\text{jet}} - d$ is nondecreasing. This proves the list bound. $\square$

The actual degree d_{∂} can be smaller than the allowed degree B_{∂} . Thus a uniform MCA bound cannot simply use tail degree $B_{\text{jet}} - B_{\partial}$. The finite maximum above handles this exactly.

5.6 Sharper finite parameters

The rate curve describes when a suitable first-order space exists. To determine the constants for a particular code, we use its integers (n, k, A) directly, preserving the actual rate and the coefficient-constraint margin. We give a finite certificate whose validity can be checked with integer arithmetic, without inspecting the evaluation points or received words.

For each support choice $(m, B_\partial, B_{\text{jet}})$, we compute the coefficient count N_{coef} , the local-rank bound R_{loc} , and the challenge-degree bound B_Z . The checks below make the support an explainer; the preceding transfer then supplies Λ and E . No new decoding or MCA theorem is needed for each parameter choice.

Fix $n \geq 2$, $1 \leq k < A \leq n$, $m \geq 1$, $B_\partial \geq 0$, and $B_{\text{jet}} \geq 1$, and put $K = \max\{k, 2\}$ and $D = K - 1$. Replace B_∂ by $\min\{B_\partial, B_{\text{jet}}\}$ throughout: no monomial of total jet degree at most B_{jet} can have a larger derivative exponent. Thus we may assume $0 \leq B_\partial \leq B_{\text{jet}}$. One can use either the equal-weight support of (23) or the larger derivative-weighted support

$$\mathcal{W} = \text{span}\{X^x Y_0^a Y_1^b : b \leq B_\partial, a + b \leq B_{\text{jet}}, x + Da + (D - 1)b < mA\}. \quad (60)$$

The latter uses the actual degree drop under differentiation. For each total jet degree $t = a + b$ and derivative exponent b , the allowed X exponents contribute $(mA - Dt + b)_+$ coefficients. For the local rank, group the substituted monomials by their T exponent s and total jet degree t . The rank of a group is at most the smaller of its numbers of source and target coefficients. These two counts give

$$N_{\text{coef}} = \sum_{t=0}^{B_{\text{jet}}} \sum_{b=0}^{\min(t, B_\partial)} (mA - Dt + b)_+, \quad (61)$$

$$R_{\text{loc}} = \sum_{s=0}^{m-1} \sum_{t=0}^{B_{\text{jet}}} \mathbf{1}_{\{s+(D-1)t < mA\}} \min\{m - s, (\min(t, s) - \max(0, t - B_\partial) + 1)_+\}. \quad (62)$$

For the equal-weight support, replace the dimension summand by $(mA - Dt)_+$ and the rank summand by

$$\min\{m - s, (\min(t, s) - \max(0, t - B_\partial, s + Dt - mA + 1) + 1)_+\}.$$

Unlike (24), these bounds retain the total-degree bound and the specialization cutoff. No restriction $B_\partial \leq m/2$ is needed.

Counting coefficients in the challenge as well. For a received curve of degree ℓ , an entry from source jet degree t to output degree u has challenge degree at most $\ell(t - u)$. A column budget $B_Z - \ell t$ therefore leaves row budget $B_Z - \ell u$. The graded kernel lemma counts these remaining coefficients directly; negative budgets omit a column or row.

Proposition 5.10 (A finite first-order certificate). *Use either support and its corresponding counts above, with $N_{\text{coef}} > nR_{\text{loc}}$. Let $N_{\text{coef}}(t)$ count source monomials of total jet degree t , and let $R_{\text{loc}}(t)$ be the sum over s of the rank summands at that degree. Thus $N_{\text{coef}} = \sum_t N_{\text{coef}}(t)$ and $R_{\text{loc}} = \sum_t R_{\text{loc}}(t)$. For a degree- ℓ received curve, any integer $B_Z \geq 0$ satisfying*

$$\sum_{t=0}^{B_{\text{jet}}} N_{\text{coef}}(t)(B_Z - \ell t + 1)_+ > n \sum_{t=0}^{B_{\text{jet}}} R_{\text{loc}}(t)(B_Z - \ell t + 1)_+ \quad (63)$$

certifies a sound first-order interpolant of challenge degree at most B_Z , total jet degree at most B_{jet} , and Y_1 -degree at most $\min(B_\partial, B_{\text{jet}})$, nonzero at every specialization. The construction holds over every field,

uniformly over evaluation sets and received curves. Characteristic restrictions enter only when reconstructing and counting polynomial solutions.

There is also a column-only test. If $N_{Y_0}(a)$ counts monomials with Y_0 exponent a , it suffices that

$$\sum_{a=0}^{B_{\text{jet}}} N_{Y_0}(a)(B_Z - \ell a + 1)_+ > nR_{\text{loc}}(B_Z + 1). \quad (64)$$

One may use whichever test gives the smaller certified challenge-degree bound.

The row test uses column and row costs ℓt and ℓu in Lemma 3.12; the column test uses cost ℓa for a Y_0 -exponent a and zero row cost.

Proof of Proposition 5.10. Both supports are translation invariant and specialize to degree less than mA . At the origin, a source monomial has local image $T^{x+a}(Y_1 + E)^a Y_1^b$ modulo terms of total (T, E) -degree at least m . Group by $s = x + a$ and total jet degree $t = a + b$, assigning E degree one. The possible a lie between $\max(0, t - B_\partial)$ and $\min(t, s)$. The target block has at most $m - s$ coordinates. For the derivative-weighted support the degree cutoff becomes $s + (D - 1)t < mA$; for the equal-weight support it also requires $a \geq s + Dt - mA + 1$. This proves the rank bounds degree by degree.

In each homogeneous image at the origin, choose at most $R_{\text{loc}}(t)$ output coordinates whose projection is injective on that image over F . Translation is an automorphism of the support, so every translated local image over $F(Z)$ is the scalar extension of this same fixed image. The selected coordinates therefore detect all local constraints, without challenge-dependent row operations. An entry from source degree t to output degree u is zero when $u > t$ and otherwise has challenge degree at most $\ell(t - u)$. Apply Lemma 3.12 with column cost ℓt , row cost ℓu , and height B_Z . Its two coefficient counts are exactly the sides of (63).

For (64), entries in a column of Y_0 -exponent a have challenge degree at most ℓa . Choose at most nR_{loc} original rows spanning the stacked matrix over $F(Z)$ and apply the same lemma with column cost ℓa and row cost zero. The resulting primitive kernel vector annihilates the whole matrix and is nonzero at every challenge. The multiplicity argument of Proposition 3.13 proves soundness in both cases. $\square$

To find a challenge-degree bound, write $W_{\text{coef}} = \sum_t t N_{\text{coef}}(t)$ and $W_{\text{rank}} = \sum_t t R_{\text{loc}}(t)$. For $B_Z \geq \ell B_{\text{jet}}$, the row test reduces to

$$(N_{\text{coef}} - nR_{\text{loc}})(B_Z + 1) > \ell(W_{\text{coef}} - nW_{\text{rank}}).$$

Hence the following value is sufficient:

$$\max \left\{ \ell B_{\text{jet}}, \left\lceil \frac{\ell(W_{\text{coef}} - nW_{\text{rank}})}{N_{\text{coef}} - nR_{\text{loc}}} \right\rceil \right\}.$$

Smaller values can be checked on the linear pieces of (63), whose breakpoints are $\ell t - 1$; the truncated test need not be monotone. The column test has the analogous all-active formula with $W_{Y_0} = \sum_a a N_{Y_0}(a)$ and no row-degree subtraction.

Using a finite certificate. The calculation follows one sequence.

1. Choose either the equal-weight support or the derivative-weighted support, and compute its corresponding N_{coef} and R_{loc} .
2. Check the strict surplus $N_{\text{coef}} > nR_{\text{loc}}$.
3. Certify a challenge-degree bound B_Z using either (63) or (64).

4. Evaluate every applicable counting transfer. For $k \geq 2$ and characteristic zero or $p > \max\{k-1, B_\partial\}$, the squarefree formula of [Lemma 5.6](#) uses $(B, M, H) = (B_{\text{jet}}, B_\partial, B_Z)$ in (54) and then minimizes (55) over $k \leq L \leq A$. When $B_\partial = 0$, use the ordinary transfer directly. In either use of an ordinary tail, optimize its threshold L_0 independently of the regular family's threshold L . The successive-stage formula of [Lemma 5.9](#) is another option for polynomial received curves.
5. For the successive-stage formula, minimize $E_{d_\partial}^{(\ell)}(L)$ separately over L for each possible actual derivative degree $0 \leq d_\partial \leq B_\partial$, then take the maximum over d_∂ . Finally, retain the smaller valid result among the applicable transfers.

Neither counting argument assumes that the interpolant uses every allowed derivative power.

Retaining the degree of a received curve. Both formulas account for the received degree ℓ and use the Frobenius transfer for their ordinary part. Thus they require only $p > \max\{k-1, B_\partial\}$ for a support certificate, with no further condition on ℓ . We optimize the successive-stage objective $E_d^{(\ell)}(L)$ of (59), using the actual message degree $D = k-1$. For fixed d , both transfer objectives are convex on $[k, A]$: their nonlinear terms are positive multiples of $1 + (n-A)/(A-L+1)$ and $(n-k+1)/(L-k+1) - 1$, and any remaining term is linear. Thus monotone forward differences find an integer minimizer in $O(\log n)$ exact comparisons. When $A = n$, choose $L = n$.

The balancing threshold also has a continuous expression. For a fixed regular family, write $J = D_{\text{joint}}(d)$, $F = D_{\text{fiber}}(d)$, $N = n - D$, $t = A - D$, and $x = L - D$. Up to terms independent of x , its cost is

$$\frac{U}{t+1-x} + \frac{V}{x}, \quad U = \frac{N}{t}J(n-A), \quad V = \ell N^2 F.$$

For $U, V > 0$, the minimum occurs at $x_* = (t+1)\sqrt{V}/(\sqrt{U} + \sqrt{V})$, clipped to $[1, t]$; an integer minimizer is one of its two neighboring integers. If $U = 0 < V$, use $x = t$; if $V = 0 < U$, use $x = 1$; if both vanish, any threshold works. This explains the balance without changing the exact search.

A line term of s bits is obtained by taking $q \geq 2^s \left\lceil \max_{0 \leq d \leq B_\partial} \min_{k \leq L \leq A} E_d^{(1)}(L) \right\rceil$ in addition to the characteristic hypothesis; affine sampling uses denominator $q-1$.

The list bound from the same certificate. The support also bounds the candidates for one received word. We record its finite value here; [Corollary 7.5](#) later transfers it to arrays without a loss depending on the number of rows.

Lemma 5.11 (A finite first-order list-size bound). *Suppose the support hypotheses of [Proposition 5.10](#) hold, $k \geq 2$, and the characteristic is zero or greater than $\max\{k-1, B_\partial\}$. Let $D_{\text{fiber}}(B_\partial)$ be the successive-stage bound in [Lemma 5.9](#), let F_{reg}, S be as in (54) with $(B, M) = (B_{\text{jet}}, B_\partial)$, and put $\lambda_{\text{agr}} = (n-k+1)/(A-k+1)$. For $B_\partial \geq 1$, every received word has list size at most*

$$\mathcal{L} = \left\lfloor \min \{ \lambda_{\text{agr}} F_{\text{reg}} + S, \lambda_{\text{agr}} D_{\text{fiber}}(B_\partial) + B_{\text{jet}} - B_\partial \} \right\rfloor$$

at agreement threshold A . If $B_\partial = 0$, the list size is at most B_{jet} in every characteristic.

Proof. For $B_\partial \geq 1$, apply the fixed-word conclusions of [Lemmas 5.6](#) and [5.9](#) and take the smaller integer bound. For $B_\partial = 0$, the nonzero interpolant is bivariate with Y_0 -degree at most B_{jet} . Every close polynomial satisfies $Q(X, P(X)) = 0$, and a nonzero polynomial over $F(X)$ has at most its degree many roots. Both arguments hold over every field extension. $\square$

The ordinary bound is eligible when the certified interpolant is independent of Y_1 . It is not an alternative list bound for an arbitrary derivative-dependent certificate. This distinction matters for the list-dependent application checks in [Section 8.2](#).

Two fixed supports give the uniform gap-6/25 list and MCA guarantees stated in [Theorem 5.13](#); its certificate is in [Section C.1](#). The applications in [Section 8](#) instead tune these finite inequalities at their actual parameters.

5.7 The Johnson specialization

The interpolation polynomial is the ordinary Guruswami–Sudan polynomial, with the challenge kept symbolic as in Ben-Sasson, Carmon, Haböck, Kopparty, and Saraf (BCHKS) [[GS99](#), [BCH⁺26](#)]. BCHKS already aggregate all useful factors, improving the earlier function-field accounting of BCIKS [[BCI⁺23](#)]. We refine what is charged to each factor: a factorwise Frobenius pullback makes each algebraic branch separable, Taylor reconstruction gives a bidegree-sensitive image bound, and projection back to the original challenge coordinate preserves the ordinary witness lines. We then apply the solution-family and persistent-witness incidence mechanism from the earlier low-rate MCA argument [[rs-26](#)], now with separate joint and generic-fiber degrees. Summing the image degrees before this incidence bound yields the quantitative improvement. The shorter reconstruction horizon and shared numerator accounting have finite-instance antecedents in the Proximity Prize submissions of BitWonka and jieyilong [[Bit26a](#), [jie26a](#)]. The ordinary transfer above includes inseparable branches, and the following choice gives uniform parameters; [Section 10.5](#) compares the constructions.

Theorem 5.12 (Quantitative Johnson MCA in every characteristic). *Let F be any field, let $\alpha_1, \dots, \alpha_n \in F$ be distinct, and suppose $1 \leq D \leq n - 2$. Put $\rho = (D + 1)/n$ and $\rho_- = D/n$. Choose $\eta_0 > 0$ with $a := \sqrt{\rho_-} + \eta_0 \leq 1$, and define*

$$A = \lceil an \rceil, \quad m = \max \left\{ \left\lceil \frac{\sqrt{\rho_-}}{2\eta_0} \right\rceil, 3 \right\}, \quad t = m + \frac{1}{2}, \quad (65)$$

and

$$B_{\text{jet}} = \left\lceil \frac{t}{\sqrt{\rho_-}} \right\rceil - 1, \quad B_Z = \left\lceil \frac{t^2}{3\rho_-} \right\rceil - 1, \quad \lambda_{\text{agr}} = \frac{n - D}{A - D}. \quad (66)$$

Then every received word $y \in F^n$ satisfies the classical list bound

$$|\mathcal{L}_D(y, A)| \leq \min \left\{ B_{\text{jet}}, \left\lceil \frac{n(A - D)}{A^2 - nD} \right\rceil \right\}. \quad (67)$$

Moreover, for every $f, g \in F^n$ there is a set $\mathcal{E} \subseteq F$ such that

$$|\mathcal{E}| \leq B_{\text{cf}} := E_{\text{ord}}^{(1)}(D, A, n; B_{\text{jet}}, B_Z) < \frac{8nt^3}{3\rho_-}. \quad (68)$$

For every $z \notin \mathcal{E}$ and every $P \in F[X]_{\leq D}$ whose full agreement set

$$S_z(P) = \{i : P(\alpha_i) = f_i + zg_i\}$$

has size at least A , there are $F_0, G_0 \in F[X]_{\leq D}$ with

$$P = F_0 + zG_0, \quad S_z(P) = \{i : F_0(\alpha_i) = f_i \text{ and } G_0(\alpha_i) = g_i\}.$$

All assertions are uniform over the prescribed evaluation set. If F is finite and the challenge is uniform in F , the exceptional probability is at most $B_{\text{cf}}/|F|$. In particular, at fixed rate the exceptional count is $O_\rho(n/\eta_0^3)$, giving probability $O_\rho(n/(q\eta_0^3))$ over $\mathbb{F}_q$.

The saving is easiest to see before rounding. Define the BCHKS weighted specialization cutoff $D_X := t\sqrt{Dn}$. Its degree cutoffs have scales

$$D_X = \Theta(nt), \quad B_{\text{jet}} = \Theta(t), \quad B_Z = \Theta(t^2),$$

where $t = \Theta(1/\eta_0)$ at fixed rate. Here D_X is the weighted specialization and lifting cutoff of the interpolant, whereas D is the actual message degree. For a positive- Y factor R_i , let $j_i = \deg_Y R_i$ and let h_i^{tot} be its content-free total (Y, Z) -degree in BCHKS. If $R_i(x_0, Y, Z)$ has a local factor of Y -degree b_{ij} , their local sufficient Hensel threshold is

$$2D_X j_i b_{ij} h_i^{\text{tot}}.$$

BCHKS already sum the local degrees $\sum_j b_{ij} = j_i$, so their charge for R_i is $2D_X j_i^2 h_i^{\text{tot}}$. Summing all useful factors gives the leading global scale $D_X B_{\text{jet}}^2 B_Z = \Theta(nt^5)$.

Our factor degree called h_i is instead $\deg_Z F_i$; separating it from the Y -degree is part of the bidegree calculation. [Section 5.3](#) proves the image bound

$$J_i \leq \ell j_i + h_i \Psi_D(j_i) \leq \ell j_i + h_i(1 + 4D j_i).$$

After summing the factors, the joint degree is at most $\ell B_{\text{jet}} + B_Z \Psi_D(B_{\text{jet}})$. The incidence multiplier and the preliminary and accidental-agreement terms are exactly those in (38); at fixed rate they do not change the leading $DB_{\text{jet}}B_Z = \Theta(nt^3)$ scale.

These formulas explain the two saved powers of t . Reconstructing only the actual degree- D message removes the ratio $D_X/D = \Theta(t)$, while the whole-factor bidegree image removes the remaining factor $j_i = O(B_{\text{jet}}) = O(t)$. We use the BCHKS interpolant and factor aggregation; the savings come from reconstructing only the message coefficients and bounding message and challenge degrees separately. The agreement threshold remains Johnson. The smaller multiplicity in (65) improves the finite constant but does not cause the change from η_0^{-5} to η_0^{-3} .

The two list estimates in (67), from Guruswami–Sudan factor counting and the pairwise-agreement Johnson argument, are classical. In positive characteristic, we pull each inseparable factor back through Frobenius before Taylor reconstruction and project the resulting curve to the original (z, P) coordinates before testing agreement. Writing s_i for the Frobenius degree and b_i for the reduced fiber degree, the identity $s_i b_i = j_i$ cancels the inseparability factor from the image bound. The order-zero recurrence has leading binomial coefficient $\binom{r}{0} = 1$, so it divides by no characteristic-sensitive integer. The transfer proof is above; the integral rounding check is in [Section B](#).

Here “order zero” refers to interpolation without derivative variables. The formulas above deliberately exclude the degree-zero code $D = 0$, where ρ_- makes them undefined. At the other endpoint $D = n - 1$, every word is a Reed–Solomon codeword and line MCA holds with $\mathcal{E} = \emptyset$ by interpolating f and g separately. The factorwise argument also extends to higher-degree received curves in every characteristic; [Lemma 5.3](#) and [Section 7.2](#) give the transfer and its dependence on the curve degree.

5.8 A uniform first-order guarantee

The two fixed supports certified in [Section C.1](#) give a simple bound at every rate once the gap to capacity is at least 0.24. They also supply the large-gap branch of our capacity theorems.

Theorem 5.13 (Uniform first-order bounds at gap 0.24 from capacity). *Let $\delta \geq 6/25$. Let $n \geq 2$, $1 \leq k \leq n$, and let F be a field. If $k \geq 2$, assume that its characteristic is zero or greater than $k - 1$; no restriction is needed for $k = 1$. Fix distinct $\alpha_1, \dots, \alpha_n \in F$ and an integer threshold $k + \delta n \leq A \leq n$. Then the following hold.*

(i) Every $y \in F^n$ has at most

$$307n \quad (69)$$

degree- $< k$ polynomials agreeing with it in at least A positions.

(ii) For every $f, g \in F^n$, there is a set $\mathcal{E} \subseteq F$ of size at most $1325775n^2$ such that every $z \notin \mathcal{E}$ and every degree- $< k$ polynomial P with $|S_z(P)| \geq A$ admit degree- $< k$ polynomials F_0, G_0 satisfying

$$P = F_0 + zG_0, \quad S_z(P) = S(F_0, G_0).$$

For $F = \mathbb{F}_q$, the exceptional probability is at most $1325775n^2/q$ for lines.

At gap 0.24, the agreement fraction $\rho + 0.24$ lies below the asymptotic Johnson threshold $\sqrt{\rho}$ precisely when $0.16 < \rho < 0.36$. The theorem supplies uniform finite bounds at all rates; the comparison with the finite Johnson threshold also retains the correction from k to $k - 1$.

6 Higher derivatives and decoding up to capacity

One derivative already improves Johnson, but its curve remains above capacity. To reach smaller gaps, we allow derivatives through order d . The route is the same: construct an explainer and a candidate cover, then use the counting bounds of Lemmas 5.1 and 5.2 for lists and common witnesses. What changes is the coefficient count. We choose the derivative monomials so that this count can be expressed through an elementary simplex integral, and keep the rate in the resulting inequality.

6.1 A weighted support and its finite counts

The local remainder constraints of Section 3.5 give the same multiplicity guarantee at every derivative order. We therefore need a space with more coefficients than conditions. For integers $D \geq d, m, W \geq 1$, and an integer threshold $A_0 \leq A$, use

$$V = \text{span} \left\{ X^x Y_0^u \prod_{j=1}^d Y_j^{b_j} : \sum_{j=1}^d j b_j \leq W, \quad x + D \left(u + \sum_{j=1}^d b_j \right) < m A_0 \right\}. \quad (70)$$

All exponents are nonnegative integers. The second inequality ensures specialization degree less than $m A_0$; the first controls the number of derivative monomials reachable by the local map. Unlike the first-order support, it allows the derivative degree to be distributed among several orders. The space is invariant under translations in X, Y_0 .

Let $p_d(B) = \#\{b \in \mathbb{N}^d : \sum_j j b_j \leq B\}$, and write $|b| = \sum_j b_j$. The exact coefficient count and a local-rank bound are

$$N_{\text{coef}} = \sum_{\sum_j j b_j \leq W} \sum_{u \geq 0} (m A_0 - D(u + |b|))_+, \quad (71)$$

$$R_{\text{loc}} \leq \sum_{s=0}^{m-1} \left\lceil \frac{m-s}{d+1} \right\rceil p_d(W+s). \quad (72)$$

To see the rank bound, a reachable term $T^i E^u Y^z$ satisfies

$$i \geq u, \quad \sum_j j z_j \leq W + i - u, \quad i + du < m.$$

Set $s = i - u$. There are at most $\lceil (m - s)/(d + 1) \rceil$ possible u , and at most $p_d(W + s)$ possible z . This counts potentially nonzero coefficients; it does not assume that they are independent.

The finite test $N_{\text{coef}} > nR_{\text{loc}}$ already gives interpolation at any order. To understand which orders suffice at a given rate, we estimate these sums while retaining the degree left after choosing b .

6.2 From the finite counts to a derivative-order condition

Fix $0 < \rho < a < 1$, with $D \leq \rho n$ and $A_0 \geq an$. Put $w = W/m$, and define

$$S_B = \{u \in \mathbb{R}_{\geq 0}^d : \sum_j j u_j \leq B\}.$$

Let U be uniform on S_w , the image of S_W under scaling by $1/m$. Flooring coordinates and integrating the remaining Y_0 -degree give

$$\frac{N_{\text{coef}}}{nm^2 \text{vol}(S_W)} \geq \frac{1}{2\rho} \mathbb{E}[(a - \rho|U|)_+^2], \quad \text{vol}(S_W) = \frac{W^d}{(d!)^2}. \quad (73)$$

The square records the degree available to X and Y_0 . Keeping this positive-part square is stronger than replacing $|U|$ by its mean.

To evaluate this expectation, set $v_i = \sum_{j=i}^d U_j$. Then $v_1 \geq \dots \geq v_d \geq 0$, $\sum_i v_i \leq w$, and $|U| = v_1$. This change of variables has determinant one. Sorting a uniform point of the ordinary simplex therefore shows that

$$|U| \stackrel{\text{law}}{=} wM_d,$$

where M_d is the largest of its first d coordinates, with a final slack coordinate making their sum one. The needed coefficient count is a moment of this maximum, rather than a new probabilistic assumption on the evaluation set.

Lemma 6.1 (A simplex moment bound). *For every integer $d \geq 500$, put $Z_d = dM_d - \log d$. Then*

$$\mathbb{E}[(\log 6 - Z_d)_+^2] > \frac{27}{10}. \quad (74)$$

Proof. Let $(T_1, \dots, T_d)$ be uniform on the simplex $T_i \geq 0$, $\sum_i T_i \leq 1$, and put $M_d = \max_i T_i$. Write $H_d = \sum_{j=1}^d 1/j$ and $H_d^{(2)} = \sum_{j=1}^d 1/j^2$ for the first- and second-order harmonic sums. For $t \geq 0$, inclusion-exclusion and a translation of the simplex give

$$\Pr[M_d > t] = \sum_{j=1}^d (-1)^{j+1} \binom{d}{j} (1 - jt)_+^d.$$

Integrating this identity for the first two moments yields

$$\mathbb{E}[M_d] = \frac{H_d}{d+1}, \quad \mathbb{E}[M_d^2] = \frac{H_d^2 + H_d^{(2)}}{(d+1)(d+2)}.$$

For example, the required alternating sums follow by expanding $1 - (1 - x)^d$ and integrating after division by x ; a second integration supplies the sum with denominator j^2 . These formulas give the mean and variance of $Z_d = dM_d - \log d$ used in Lemma 6.1. For $t \geq 0$, the union bound gives

$$\Pr[Z_d > t] \leq d \left(1 - \frac{\log d + t}{d}\right)_+^d \leq e^{-t}.$$

To use the variance, split the second moment at a threshold θ :

$$\mathbb{E}[(\theta - Z_d)_+^2] = (\theta - \mathbb{E}[Z_d])^2 + \text{Var}(Z_d) - \mathbb{E}[(Z_d - \theta)_+^2].$$

At $\theta = \log 6$, the preceding tail bound gives

$$\mathbb{E}[(Z_d - \log 6)_+^2] = \int_0^\infty 2u \Pr[Z_d > \log 6 + u] du \leq \frac{1}{3}.$$

Thus we keep the variance contribution while losing at most $1/3$ to the upper tail. The finite estimates in [Lemma D.1](#) give

$$(\log 6 - \mathbb{E}[Z_d])^2 + \text{Var}(Z_d) - \frac{1}{3} > \frac{27}{10},$$

which proves the claim. Replacing Z_d by its mean would discard precisely the variance used here. $\square$

Choose w close to $ad/(\rho \log(6d))$. In the limit, this makes $\lambda = d/w = \rho \log(6d)/a$. On the common normalization of (73), the moment and cube-covering bounds give

$$\liminf_{m \rightarrow \infty} \frac{N_{\text{coef}}}{nm^2 \text{vol}(S_W)} \geq \frac{27\rho}{20\lambda^2}, \quad \limsup_{m \rightarrow \infty} \frac{R_{\text{loc}}}{m^2 \text{vol}(S_W)} \leq \frac{e^\lambda}{(d+1)\lambda^2}.$$

When $R_{\text{loc}} > 0$, dividing these limiting estimates yields

$$\liminf_{m \rightarrow \infty} \frac{N_{\text{coef}}}{nR_{\text{loc}}} \geq \frac{27}{20} \rho(d+1)e^{-\lambda} = \frac{27}{20} \frac{\rho(d+1)}{(6d)^{\rho/a}} = \Gamma(\rho, a, d).$$

These are limiting bounds; [Lemma 6.2](#) retains the cube-covering and rounding corrections at finite m . Thus the simple sufficient condition is $\Gamma(\rho, a, d) > 1$, where

$$\Gamma(\rho, a, d) = \frac{27}{20} \frac{\rho(d+1)}{(6d)^{\rho/a}}. \quad (75)$$

A strict inequality $\Gamma > 1$ leaves a positive coefficient-constraint margin. For fixed $0 < \rho < a < 1$, this expression grows like $d^{1-\rho/a}$. Its positive exponent ensures that sufficiently many derivatives make the condition hold, explaining why the construction reaches every positive agreement gap from capacity. Equivalently, when the denominator is positive, it asks for

$$a > a_d(\rho) := \frac{\rho \log(6d)}{\log((27/20)\rho(d+1))}.$$

This solved expression is useful only when $a_d(\rho) < 1$. The exact finite test (71)–(72) remains available at smaller orders and when this sufficient curve is too coarse.

Lemma 6.2 (Finite partition ratio). *Let $d \geq 500$, $0 < \rho < a \leq 1$, and $m \geq 1$ be an integer for which $W = \lfloor mad/(\rho \log(6d)) \rfloor \geq 1$. Set*

$$w = W/m, \quad \lambda = d/w, \quad S_d = d(d+1)/2, \quad \gamma_m = \frac{27}{20} \rho(d+1) \frac{e^{-\lambda(1+S_d/m)}}{1 + (d+1)\lambda/m}.$$

For the space (70), with $D \geq d$, $D \leq \rho n$, and $A_0 \geq an$, its dimension and local rank obey $N_{\text{coef}} > \gamma_m n R_{\text{loc}}$ whenever $R_{\text{loc}} > 0$; its dimension is positive also when $R_{\text{loc}} = 0$. Moreover $\gamma_m \rightarrow \Gamma(\rho, a, d)$ as $m \rightarrow \infty$. If $d \geq 519$ and $m = \lceil 300d^2 \log(6d) \rceil$, then

$$\gamma_m > \Gamma(\rho, a, d) e^{-1677/10^6}. \quad (76)$$

The strict inequality $\Gamma > 1$ therefore gives a terminating search for a finite m with positive margin. When $\Gamma > e^{1677/10^6} (151/150)$, the displayed closed choice certifies $N_{\text{coef}} > (151/150)nR_{\text{loc}}$ and challenge degree less than $150B_{\text{jet}}$. The proof retaining every rounding term is in [Section D.2](#).

6.3 Differential equations, lists, and correlated agreement

We now turn the explainer into list and common-witness bounds. For a fixed word, we parameterize each regular solution family by its initial Taylor values, impose the actual message-degree restriction, and count its agreements. Keeping the challenge symbolic gives a joint family; its degree and the degree of a generic fiber control the exceptional challenges.

6.3.1 Counting fixed-word solutions

A regular solution is determined by a short initial Taylor jet. Its remaining coefficients are rational functions of that jet, so their image closure contains the solutions assigned to that stage. We retain each stage's actual degree instead of charging every stage the largest bound. Starting from Q , repeatedly differentiate with respect to the highest derivative variable still present. This is the *separant chain*. Under the characteristic assumption below, each derivative is nonzero and lowers the total degree, until no Y variables remain. A solution belongs to the first stage whose equation vanishes identically after substitution while its separant is nonzero as a polynomial in X . We record that stage by (j, r) , its total degree and highest derivative index.

For reference, the incidence factor used below is

$$P_r(T) = \prod_{i=1}^r \frac{n-k+i}{T-k+i}, \quad P_0(T) = 1.$$

It records one agreement ratio for each free initial derivative value.

Theorem 6.3 (Direct agreement bound for differential roots). *Let $1 \leq k \leq K \leq n$, $0 \leq d < K$, and let $Q \in F[X, Y_0, \dots, Y_d]$ be nonzero of total jet degree $B_{\text{jet}} \geq 1$. Assume the characteristic is zero or exceeds $\max\{K-1, B_{\text{jet}}\}$. For distinct $\alpha_1, \dots, \alpha_n \in F$, $y \in F^n$, and $k \leq A \leq n$, write $\mathcal{L}$ for the set of degree- $< k$ solutions of $Q(X, P, P^{[1]}, \dots, P^{[d]}) = 0$ with at least A agreements. Then $\mathcal{L}$ is finite. If (j, r) runs over the actual regular stages of its separant chain and $b_j = 1 + 2K(j-1)$, its size is at most*

$$\sum_{\text{actual stages } (j,r)} P_r(A) j b_j^r \leq P_d(A) \sum_{j=1}^{B_{\text{jet}}} j b_j^d \leq B_{\text{jet}}^2 \left(\frac{n[1 + 2K(B_{\text{jet}} - 1)]}{A - k + 1} \right)^d.$$

Proof. Starting at Q , differentiate each positive-jet-degree polynomial in its highest jet variable. The characteristic hypothesis ensures that this derivative is nonzero and strictly decreases total jet degree. The chain ends at a nonzero polynomial in X , which has no solutions. Assign each candidate to the first stage T for which $T(X, P, P^{[1]}, \dots) = 0$ identically in X while $S(X, P, P^{[1]}, \dots)$ is nonzero as a polynomial in X .

Choose a transcendental over $\overline{F}$ and work over $E = \overline{F}(a)$. Transcendence of a preserves this nonvanishing, so the value of the substituted separant at $X = a$ is nonzero. Thus this one center is regular for every candidate at its assigned stage.

Fix a stage of jet degree j and highest jet index $r \leq d$, and put $b_j = 1 + 2K(j-1)$. By Lemma A.4, its regular initial-jet locus has dimension at most r and total degree at most j , and its map to the K Taylor coefficients has numerator and common-denominator degrees at most b_j . The denominator vanishes identically on none of the retained components. Hence Lemma A.2 gives an image closure of dimension at most r and total degree at most $j b_j^r$.

Intersect this image with the linear subspace of coefficient vectors representing polynomials of degree less than k . Translation from ordinary to Taylor coefficients preserves this degree filtration, and a linear intersection does not increase dimension or total degree. In this message space, Lemma 5.1 bounds the points agreeing with y in at least A positions by

$$P_r(A) j b_j^r.$$

Every candidate assigned to this stage is one of these points.

The positive stage degrees are a strictly decreasing subset of $\{1, \dots, B_{\text{jet}}\}$. Since $r \leq d$, $b_j \geq 1$, and every incidence factor is at least one, summing gives the actual-stage and common-order bounds in the statement, and proves finiteness. Finally,

$$\sum_{j=1}^{B_{\text{jet}}} j b_j^d \leq \frac{B_{\text{jet}}(B_{\text{jet}} + 1)}{2} b_{B_{\text{jet}}}^d \leq B_{\text{jet}}^2 b_{B_{\text{jet}}}^d, \quad P_d(A) \leq \left(\frac{n}{A - k + 1} \right)^d,$$

which gives the stated coarse bound. $\square$

Interpolation may use a larger dimension than reconstruction needs. For an equation explaining degree- $< k$ messages, choose $K_{\text{rec}} = \max\{k, d + 1\}$. The cover proof needs soundness only for those messages, so this choice gives the characteristic assumption $p > \max\{k - 1, d, B_{\text{jet}}\}$ without changing the interpolation support.

Corollary 6.4 (Field-independent output bound). *If an order- d explainer has total jet degree at most B_{jet} , ambient bound $K \leq n$, and $A - k \geq \delta n > 0$, then, under the characteristic hypotheses of [Theorem 6.3](#),*

$$|\mathcal{L}(y, A)| \leq B_{\text{jet}}^2 (2B_{\text{jet}}/\delta)^d n^d. \quad (77)$$

This also bounds the output of any exact decoder for the same list.

Proof. For positive jet degree use $1 + 2K(B_{\text{jet}} - 1) \leq 2B_{\text{jet}}n$ and $A - k + 1 \geq \delta n$ in [Theorem 6.3](#). An explainer of jet degree zero has no polynomial solutions. $\square$

6.3.2 Following the challenge through the cover

For a received curve, the Taylor formulas depend on both the challenge Z and the initial derivative values u . Keeping their degrees separate lets us charge the challenge degree only once in the joint bound. The reduced generic fiber measures a different quantity: the candidate family above a transcendental challenge, not at every specialization.

Lemma 6.5 (Joint envelope and generic fiber). *Let $1 \leq K \leq n$, $0 \leq d < K$, $\ell \geq 1$, and let $Q \in F[X, Z, Y_0, \dots, Y_d] \setminus \{0\}$ have total jet degree at most $B_{\text{jet}} \geq 1$ and challenge degree at most $B_Z \geq 0$. Assume characteristic zero or greater than $\max\{K - 1, B_{\text{jet}}\}$. For each actual separant-chain stage of jet degree j and highest jet index r , put $b_j = 1 + 2K(j - 1)$ and*

$$J_{j,r}^{(\ell)} = B_Z b_j^{r+1} + (r + 1)j(\ell + 2KB_Z) b_j^r, \quad F_{j,r} = j b_j^r. \quad (78)$$

There are an algebraically closed extension E/F , a set $Z_0 \subseteq \overline{F}$ with $|Z_0| \leq B_Z$, and stage images $V_{j,r} \subseteq E^{\ell+K}$ of dimension at most $r + 1$ and total degree at most $J_{j,r}^{(\ell)}$. Their union contains every lifted root pair $(z, z^2, \dots, z^\ell, P)$ over $\overline{F}$ with $\deg P < K$ and $z \notin Z_0$. Over a challenge transcendental over E , each reduced geometric fiber has dimension at most r and degree at most $F_{j,r}$. In particular, writing $B_{\text{lift}} = 1 + 2K(B_{\text{jet}} - 1)$, the union has joint degree at most

$$D_{\text{joint}}^{(\ell)} = B_{\text{jet}} B_{\text{lift}}^d [B_Z B_{\text{lift}} + (d + 1)B_{\text{jet}}(\ell + 2KB_Z)]$$

and reduced generic-fiber degree at most $B_{\text{jet}}^2 B_{\text{lift}}^d$.

For counting, one transcendental Taylor center works for the entire family: every nonzero separant identity over $\overline{F}$ remains nonzero there. This auxiliary extension does not enter the degree bound. The decoder constructs finite centers instead ([Section 9](#)).

Proof. Choose a transcendental over $\overline{F}$ and put $E = \overline{F(a)}$. Successive highest-jet derivatives of Q are nonzero and strictly decrease total jet degree, ending at $T_* \in F[X, Z] \setminus \{0\}$. Each has challenge degree at most B_Z , so $T_*(X, z)$ vanishes identically on a set Z_0 of size at most B_Z . If Q has jet degree zero, the empty union suffices.

Outside Z_0 , every root pair is regular at some stage T , with separant S . For its degrees (j, r) , the locus

$$U^\circ = \{(z, u) : T(a, z, u) = 0, S(a, z, u) \neq 0\}$$

has closure of bidegree at most (B_Z, j) . Write $h_* = K - 1 - r$. Retaining Z in (123) gives

$$c_{r+h} = \frac{A_h(Z, u)}{S(a, Z, u)^{2h-1}}, \quad \deg_u A_h \leq (2h-1)(j-1) + 1, \quad \deg_Z A_h \leq (2h-1)B_Z.$$

The last bound follows by clearing S^{2h-2} in the numerator induction and charging one further B_Z for a coefficient of T . With common denominator S^{2h_*-1} , the map to $(Z, Z^2, \dots, Z^\ell, c_0, \dots, c_{K-1})$ has numerator and denominator degrees at most $\ell + 2KB_Z$ in Z and b_j in u . If $h_* = 0$, denominator one satisfies the same bounds. Transcendence of a puts every regular root pair in this image. Lemma A.3 with $d = r$ gives the degree $J_{j,r}^{(\ell)}$ and dimension at most $r + 1$.

Over a transcendental challenge, the source is an open part of a nonzero hypersurface of u -degree at most j , and the specialized map has degree at most b_j . Its image closure therefore has dimension at most r and degree at most jb_j^r by Lemma A.2. This is the reduced generic fiber of the joint image closure: the constructible image contains a dense open subset of each component of its closure [Mil09, Chapter 10]. Vertical components miss the generic fiber; on each remaining component, the omitted subset has smaller fiber dimension, so the image remains dense. Adding powers of Z does not change this fiber. No claim is made about its degree at arbitrary special challenges. Finally, there are at most B_{jet} stages, their jet degrees sum to at most B_{jet}^2 , and $r \leq d$. Summing gives the two coarse bounds. $\square$

6.3.3 From the cover to common witnesses

We can now apply the common geometric transfer. Its three costs are the preliminary exclusions, candidates off persistent witness graphs, and accidental agreements on those graphs. Setting $\ell = 1$ below gives the line bound.

Theorem 6.6 (Symbolic transfer for polynomial curves). *Let F be a field, $1 \leq k \leq K \leq n$, $0 \leq d < K$, and $k \leq L \leq A \leq n$, and integers $B_{\text{jet}} \geq 1$ and $B_Z \geq 0$. Fix $\ell \geq 1$, distinct evaluation points $\alpha_1, \dots, \alpha_n \in F$, and coefficient words $w_0, \dots, w_\ell \in F^n$, giving the received curve $w(z) = \sum_{t=0}^\ell z^t w_t$ and agreement sets $S_z(P)$. Suppose $\text{char } F = 0$ or $\text{char } F > \max\{K-1, B_{\text{jet}}\}$. Let $Q \in F[X, Z, Y_0, \dots, Y_d]$ be nonzero, of total jet degree at most B_{jet} and challenge degree at most B_Z . Suppose every degree- $< k$ polynomial P with $|S_z(P)| \geq A$ satisfies the symbolic soundness condition (22). Use B_{lift} and $D_{\text{joint}}^{(\ell)}$ from Lemma 6.5, and put $\lambda_{\text{off}} = (n - L + 1)/(A - L + 1)$. There is an exceptional set $\mathcal{E} \subseteq F$ of size at most*

$$B_Z + \lambda_{\text{off}} P_d(A) D_{\text{joint}}^{(\ell)} + \ell(n - L) P_d(L) B_{\text{jet}}^2 B_{\text{lift}}^d \quad (79)$$

such that every pair (z, P) outside this set with $\deg P < k$ and $|S_z(P)| \geq A$ has the full common-witness property of Definition 3.1. There is no additional characteristic restriction involving ℓ .

Proof. Apply Lemma 6.5 at received degree ℓ and intersect its stage images with the linear subspace $\deg P < k$. These sections do not increase the joint or reduced generic-fiber degrees. The lemma supplies a preliminary set Z_0 of at most B_Z challenges, joint degree at most $D_{\text{joint}}^{(\ell)}$, and reduced generic-fiber degree at most $B_{\text{jet}}^2 B_{\text{lift}}^d$. Use Lemma 5.2 with

$$r_1 = d, \quad J_1 = D_{\text{joint}}^{(\ell)}, \quad F_1 = B_{\text{jet}}^2 B_{\text{lift}}^d, \quad \mathcal{E}_{\text{pre}} = Z_0 \cap F.$$

Symbolic soundness gives candidate coverage; the transfer gives (79), witnesses over F , and equality of full agreement sets. Its accidental-agreement polynomials have degree at most ℓ , so no characteristic restriction involving ℓ is needed. The actual-stage bounds (78) can be used before summing when sharper constants are wanted. $\square$

6.4 List and line-MCA bounds from one parameter choice

We choose the interpolation parameters once, before the field, evaluation points, and received word. A positive coefficient surplus supplies the fixed-word equation. Retaining the size of that surplus also bounds the challenge degree, so the same choice gives line MCA.

Theorem 6.7 (List decoding and MCA from the derivative-order condition). *Fix $0 < \rho < a < 1$ and an integer $d \geq 500$ with $\Gamma(\rho, a, d) > 1$. There are explicit integers $m, B_{\text{jet}}, N_{\rho, a, d}$ and a certified source-to-rank ratio $\gamma > 1$ with the following property. Let n, k, A be integers with $n \geq N_{\rho, a, d}$, $1 \leq k \leq \rho n$, and $an \leq A \leq n$. Over any field of characteristic zero or $p > \max\{k - 1, d, B_{\text{jet}}\}$, fix n distinct evaluation points and put $\delta = a - \rho$. Then every received word has list size*

$$|\mathcal{L}(y, A)| \leq C_L n^d, \quad C_L = B_{\text{jet}}^2 (2B_{\text{jet}}/\delta)^d. \quad (80)$$

For the symbolic equation choose the challenge-degree bound

$$B_Z = \max\{1, \lceil B_{\text{jet}}/(\gamma - 1) \rceil\}. \quad (81)$$

Every received line has an exceptional set of size at most $C_E n^{d+1}$ outside which all close candidates have the full common-witness property, where

$$C_E = B_Z + 2^d B_{\text{jet}}^{d+2} \delta^{-d} \left[\frac{B_Z(d+1)(3d+5)}{\delta} + 3 \right]. \quad (82)$$

Over $\mathbb{F}_q$, the exceptional probability is at most $C_E n^{d+1}/q$.

Proof. Increase m until the lower ratio in Lemma 6.2 is greater than one; this terminates by (75). Set $W = \lfloor mad/(\rho \log(6d)) \rfloor$, and let $B_{\text{jet}} = \lceil 2m/\rho \rceil$. For example, take

$$N_{\rho, a, d} = \left\lceil \max \left\{ \frac{d+1}{\rho}, B_{\text{jet}} + 1, \frac{1}{1-\rho} \right\} \right\rceil.$$

Then, for $n \geq N_{\rho, a, d}$,

$$D = \lfloor \rho n \rfloor \geq \max\{d+1, \rho n/2\}, \quad K = D+1 \leq n, \quad n > B_{\text{jet}}, \quad n \geq 2m.$$

Use $A_0 = \lceil an \rceil \leq A$. The space (70) has $N_{\text{coef}} > nR_{\text{loc}}$, jet degree less than $mA_0/D \leq 2m/\rho$, and specialization degree less than mA_0 . Its explainer is satisfied by every degree- $< k$ message because $k \leq D < K$. For reconstruction use $K_{\text{rec}} = \max\{k, d+1\} \leq K$ as explained after Theorem 6.3. Applying Corollary 6.4 with this dimension and $\delta = a - \rho$ gives (80).

Retain the certified ratio $\gamma > 1$. By Proposition 3.13, the bound (81) suffices for the symbolic explainer. Thus a smaller surplus can increase the exceptional-count constant even at the same derivative order. The same support gives a symbolic explainer of jet degree at most B_{jet} and challenge degree at most B_Z . Use reconstruction dimension $K_{\text{rec}} = \max\{k, d+1\} \leq K$ and apply Theorem 6.6 with $\ell = 1$ and $L = k + \lfloor d(A-k)/(d+1) \rfloor$. Its incidence factors obey

$$\lambda_{\text{off}} \leq \frac{d+1}{\delta}, \quad P_d(A) \leq \delta^{-d}, \quad P_d(L) < 3\delta^{-d}.$$

The last estimate uses $(1 + 1/d)^d < 3$; integer rounding is absorbed by the $+j$ in each denominator of the product P_d . Since $K_{\text{rec}} \leq K \leq n$ and $B_Z \geq 1$, the joint and fiber degrees satisfy

$$B_{\text{lift}} \leq 2B_{\text{jet}}n, \quad D_{\text{joint}}^{(1)} \leq B_Z(3d+5)2^d B_{\text{jet}}^{d+2} n^{d+1}, \quad B_{\text{jet}}^2 B_{\text{lift}}^d \leq 2^d B_{\text{jet}}^{d+2} n^d.$$

Insert these bounds into (79) with $\ell = 1$ and use $n - L \leq n$. The recognition argument is unchanged: agreement on k positions determines common witnesses, and the exceptional set removes off-line candidates and accidental agreements. $\square$

The finite stage bounds retain more information than these simple constants. Our applications use the applicable first-order estimates at their actual parameters. Computing the lists is a separate task: [Section 9](#) uses the received agreements during recovery to avoid polynomial dependence on the field size.

6.5 Fixed-rate asymptotics and uniform gap bounds

Rewriting Γ gives the exact identity

$$\Gamma(\rho, a, d) = \frac{9\rho}{40} (6d)^{(a-\rho)/a} \left(1 + \frac{1}{d}\right). \quad (83)$$

Consequently the single sufficient condition

$$d \geq \max \left\{ 500, \frac{1}{6} \left(\frac{40}{9\rho} \right)^{a/(a-\rho)} \right\} \implies \Gamma(\rho, a, d) \geq 1 + \frac{1}{d} > 1. \quad (84)$$

This gives an explicit sufficient derivative order. When Γ only slightly exceeds one, the rate theorems use the terminating multiplicity search from [Section D.2](#); the closed multiplicity used for the stronger uniform certificate requires additional margin.

We can read (84) at either an additive or a multiplicative gap. For fixed $0 < \rho < 1$ and $0 < \delta < 1 - \rho$, substitute $a = \rho + \delta$ and put $c(\rho) = \rho \log(40/(9\rho))$. The sufficient order is

$$d = \left\lceil \max \left\{ 500, \frac{20}{27\rho} \exp \left(\frac{c(\rho)}{\delta} \right) \right\} \right\rceil. \quad (85)$$

For fixed multiplicative slack $0 < \vartheta < 1$, instead set $\rho = (1 - \vartheta)a$ with $0 < a < 1$. Since $(a - \rho)/a = \vartheta$, every integer

$$d \geq \max \left\{ 500, \frac{1}{6} \left(\frac{40}{9\rho} \right)^{1/\vartheta} \right\} \quad (86)$$

satisfies the same condition. Both choices inherit the list and line-MCA bounds of [Theorem 6.7](#), with its length and characteristic hypotheses. At fixed ϑ , this bound on d grows polynomially as $a \rightarrow 0$, whereas the uniform order $\lceil e^{1.5/\delta} \rceil$ with $\delta = \vartheta a$ grows exponentially.

6.6 Uniform finite parameters over all rates

The fixed-rate order bound does not by itself give one choice uniform as the rate tends to zero: it contains a prefactor proportional to $1/\rho$. We remove that dependence by enlarging the ambient interpolation dimension only in the low-rate branch. The resulting finite theorem records the support margin separately from the later reconstruction algorithm.

For the small-gap range $0 < \delta < 6/25$, set

$$\begin{aligned} d(\delta) &= \lceil e^{1.5/\delta} \rceil, & m(\delta) &= \lceil 300d(\delta)^2 \log(6d(\delta)) \rceil, \\ B_{\text{jet}}(\delta) &= \left\lceil \frac{m(\delta)}{\delta^2} \right\rceil - 1, & N(\delta) &= \max \left\{ \left\lceil \frac{m(\delta)}{\delta^2} \right\rceil, \left\lceil \frac{4B_{\text{jet}}(\delta)}{\delta^2} \right\rceil \right\}. \end{aligned} \quad (87)$$

Theorem 6.8 (Uniform finite choices up to capacity). *For $0 < \delta < 6/25$, use the parameters in (87). Let F be a field, fix $n \geq N(\delta)$ distinct evaluation points in F , and let $1 \leq k < A \leq n$ satisfy $A - k \geq \delta n$.*

Interpolation. For every received word $y \in F^n$ there is an order- $d(\delta)$ explainer of jet degree at most $B_{\text{jet}}(\delta)$, with ambient dimension $k \leq K \leq n$, $d(\delta) < K$, constructed from a support with source-to-rank ratio greater than $151/150$. This interpolation assertion is characteristic independent. For every received line, the same support gives a symbolic explainer of challenge degree less than $150B_{\text{jet}}(\delta)$.

Counting. In characteristic zero or $p > k - 1$ when $k \geq 2$, and in every characteristic when $k = 1$, the list and line-MCA bounds are

$$|\mathcal{L}(y, A)| \leq \max \left\{ B_{\text{jet}}(\delta)^2 (2B_{\text{jet}}(\delta)/\delta)^{d(\delta)}, \frac{4}{3\delta} \right\} n^{d(\delta)}, \quad |\mathcal{E}| \leq C_\delta n^{d(\delta)+1},$$

where $C_\delta = \max\{C_E, 343/3\}$ and C_E is (82) with $d = d(\delta)$, $B_{\text{jet}} = B_{\text{jet}}(\delta)$, and $B_Z = 150B_{\text{jet}}(\delta)$.

For $6/25 \leq \delta < 1$, let $n \geq 2$, $1 \leq k \leq n$, and $k + \delta n \leq A \leq n$, and fix n distinct evaluation points in a field F . If $k \geq 2$, assume its characteristic is zero or greater than $k - 1$; no restriction is needed when $k = 1$. Then Theorem 5.13 gives list bound $307n$ and line exceptional bound $1325775n^2$.

Proof. Fix $0 < \delta < 6/25$ and its displayed $d(\delta), m(\delta), N(\delta)$, abbreviated here by d, m, N . Then $d \geq 519 > 500$. Write $\rho_* = k/n$ and use one of the following ambient choices:

$$(\rho, D, a) = \begin{cases} (\rho_*, k, \rho_* + \delta), & \rho_* \geq \delta^2, \\ (2\delta^2, \lfloor 2\delta^2 n \rfloor, \delta), & \rho_* < \delta^2. \end{cases}$$

In both cases set $K = D + 1$, $A_0 = \lceil an \rceil$ and $W = \lfloor mad/(\rho \log(6d)) \rfloor$. The low-rate choice deliberately explains a larger message space, keeping the ambient degree bound D at least $\delta^2 n$ even when k is small. The length hypothesis gives

$$k \leq D, \quad \delta^2 n \leq D \leq \rho n, \quad D \geq m > d, \quad K \leq n, \quad A_0 \leq A \leq n.$$

Here $N \geq B_{\text{jet}} + 1$ and $\delta^2 n \geq m \geq 1$, so $\lfloor 2\delta^2 n \rfloor \geq \delta^2 n$; for $K \leq n$, use the nonvacuous gap in the high-rate branch and $2\delta^2 < 1/8$ in the low-rate branch. This padding makes the jet-degree bound uniform over the rate:

$$D \geq \delta^2 n \implies \deg_{\text{jet}} Q < mA_0/D \leq m/\delta^2,$$

so its integer degree is at most $B_{\text{jet}}(\delta) < n$.

It remains to give a rate-uniform coefficient-constraint margin. Put $B = 27/20$ and $C = 40/9$. In the high-rate branch,

$$(\rho_* + \delta) \log \Gamma \geq \delta \log d - G(\rho_*, \delta), \quad G(\rho, \delta) = \rho \log(C/\rho) + \delta \log(1/(B\rho)).$$

For $\delta^2 \leq \rho \leq 1 - \delta$, we claim $G(\rho, \delta) \leq \log C$. On $[\delta, 1 - \delta]$, the function is concave and its derivative at the right endpoint is at least $\log C - 4/3 > 0$, so its maximum is there. At that endpoint,

$$G(1 - \delta, \delta) = \log C - \delta \log 6 - \log(1 - \delta) \leq \log C,$$

using $-\log(1 - \delta) \leq 4\delta/3 < \delta \log 6$. On $[\delta^2, \delta]$, convexity reduces to the endpoints; the remaining one satisfies

$$G(\delta^2, \delta) < \frac{\log C}{16} + \frac{\log 4}{8} + \frac{\log 4}{2} < 1 < \log C.$$

Here we used $\delta < 1/4$ and monotonicity of $-\delta \log \delta$ and $-\delta^2 \log \delta$ on this range. Since $\delta \log d \geq 3/2$ and $\rho_* + \delta \leq 1$,

$$\log \Gamma \geq \frac{3}{2} - \log(40/9).$$

In the low-rate branch the same lower bound follows from the stronger estimate

$$\log \Gamma \geq F(\delta) := \frac{3}{2\delta} - 3 + \log(27/10) + 2 \log \delta - 2\delta \log 6.$$

Its derivative is negative for $0 < \delta \leq 1/4$, and $F(1/4) > 3/10 > 3/2 - \log(40/9)$. Applying [Lemma 6.2](#) in either branch gives

$$\log \frac{N_{\text{coef}}}{nR_{\text{loc}}} > \frac{3}{2} - \log(40/9) - \frac{1677}{10^6} > \log(151/150),$$

when $R_{\text{loc}} > 0$. The case $R_{\text{loc}} = 0$ is immediate. The last strict comparison is the rational inequality in [Lemma D.2](#). The symbolic height follows from [Lemma 3.12](#).

For $k = 1$, [Lemma 3.2](#) gives list bound n and line exceptional bound $(n - 1)/(2\delta)$ in every characteristic, both smaller than the displayed bounds. For $k \geq 2$ in characteristic zero or $p > B_{\text{jet}}$, apply the list and line-counting bounds with $K_{\text{rec}} = \max\{k, d + 1\}$, the actual gap $A - k \geq \delta n$, and degrees $B_{\text{jet}}(\delta)$ and $B_Z = 150B_{\text{jet}}(\delta)$. Since $B_{\text{jet}} > d$, their characteristic assumption is $p > \max\{k - 1, B_{\text{jet}}\}$; using $K_{\text{rec}} \leq K \leq n$ preserves the constants. It remains to prove the counting claims when $k - 1 < p \leq B_{\text{jet}}$. Then $k \leq B_{\text{jet}}$, so the length choice gives

$$\frac{k - 1}{n} \leq \frac{\delta^2}{4}, \quad \frac{A}{n} - \sqrt{\frac{k - 1}{n}} \geq \frac{\delta}{2}.$$

Apply [Theorem 5.12](#) with $D = k - 1$ and $a = A/n$. Its multiplicity is $m = 3$, and its bounds give

$$|\mathcal{L}(y, A)| \leq \frac{4}{3\delta}, \quad |\mathcal{E}| < \frac{343}{3} \frac{n^2}{k - 1} \leq \frac{343}{3} n^2.$$

Since $d(\delta) \geq 1$, these fit the stated bounds. The two characteristic cases explain the maxima in the constants: large characteristic uses the differential equation, while bounded message dimension is already above Johnson at the chosen length. The larger-gap branch is [Theorem 5.13](#). $\square$

The corresponding exact decoding bounds, including the supplied-field model, characteristic assumptions, and bounded-dimension reconstruction, are stated and proved in [Section 9.8](#).

At gaps at least $1/4$, ordinary list decoding already reaches the target because $\max_{0 \leq \rho \leq 1} (\sqrt{\rho} - \rho) = 1/4$; the first-order line bound remains useful there.

6.7 The resulting capacity guarantees

The uniform parameters give one exceptional-count bound at every rate. The next statement records the common witnesses and separates the characteristic assumption from the field size needed for a small error.

Theorem 6.9 (Mutual correlated agreement up to capacity). *For every fixed $\delta \in (0, 1)$, there are an integer $N_\delta^{\text{CA}} \geq 4$ and a function $E_\delta(n) = n^{O_\delta(1)}$ with the following guarantee. Let $n \geq N_\delta^{\text{CA}}$, $1 \leq k \leq n$, and $k + \delta n \leq A \leq n$ be an integer threshold. Let F have characteristic zero or $p > k - 1$, with no restriction if $k = 1$. Fix distinct $\alpha_1, \dots, \alpha_n \in F$. For every $f, g \in F^n$, there is $\mathcal{E} \subseteq F$ with $|\mathcal{E}| \leq E_\delta(n)$ such that, for every $z \notin \mathcal{E}$ and every $P \in F[X]$ of degree less than k with at least A agreements with $f + zg$, there are $F_0, G_0 \in F[X]$ of degree less than k satisfying*

$$P = F_0 + zG_0, \\ \{i : P(\alpha_i) = f_i + zg_i\} = \{i : F_0(\alpha_i) = f_i \text{ and } G_0(\alpha_i) = g_i\}.$$

For $F = \mathbb{F}_q$, the exceptional probability is at most $E_\delta(n)/q$.

The explicit choices in [Theorem 6.8](#) give $E_\delta(n) = C_\delta n^{d(\delta)+1}$ at small gaps and a quadratic bound already at gap $6/25$. Under the theorem's characteristic hypothesis, a target MCA error $\epsilon_{\text{MCA}} > 0$ follows from $q \geq E_\delta(n)/\epsilon_{\text{MCA}}$; in particular, a σ -bit allocation at small gaps follows from $q \geq 2^\sigma C_\delta n^{d(\delta)+1}$. The characteristic supports the counting argument; the field size makes its probability bound small, and the bound is nontrivial when $q > E_\delta(n)$. The gap is fixed before the block length grows; the constants and eventual length threshold may depend on it. [Section 10.3](#) compares the constructions and their parameter dependence in detail.

7 Agreement for larger families of received words

The first-order line theorem relates candidates for a combination of two words. We now extend this guarantee to combinations of several words and to simultaneous agreement across several rows. The same reductions also apply to our ordinary and higher-order line bounds, retaining the hypotheses of the chosen bound. Polynomial received curves additionally require control of the symbolic interpolation degree; their statements specify that extra input.

We use one observation to recover full agreement sets from subset MCA. At a good parameter, let S be a candidate's full agreement set, with $|S| \geq A \geq k$. Subset MCA supplies degree- $< k$ witnesses for every coefficient word on S . Their prescribed combination equals the candidate by interpolation on k positions. Their common agreement set contains S , and the identity gives the reverse inclusion. This argument also applies row by row to interleaved words.

7.1 Affine families

We next allow the received word to depend on several independently chosen coefficients, rather than a single challenge. The question is whether one can retain the line theorem's error bound independently of the number of coefficients. A coordinate-by-coordinate extension of the line theorem would lose a factor equal to the number of parameters. We instead use the line-to-affine reduction of Bordage, Chiesa, Guan, and Manzur [[BCGM25](#), Lemma 7.1]. The following form fixes the agreement threshold; interpolation then recovers the full agreement set.

Lemma 7.1 (Line-to-affine MCA transfer [[BCGM25](#), Lemma 7.1]). *Let $C \subseteq \mathbb{F}_q^n$ be a linear code and $1 \leq A \leq n$. Suppose that for every $f, g \in \mathbb{F}_q^n$, at most an ε fraction of $z \in \mathbb{F}_q$ admit a set $S \subseteq [n]$ with $|S| \geq A$ such that*

$$(f + zg)|_S \in C|_S, \quad f|_S \notin C|_S \text{ or } g|_S \notin C|_S.$$

For every $s \geq 1$ and $a, u_1, \dots, u_s \in \mathbb{F}_q^n$, the fraction of $t \in \mathbb{F}_q^s$ admitting such an S with

$$\left(a + \sum_{j=1}^s t_j u_j \right)|_S \in C|_S, \quad a|_S \notin C|_S \text{ or some } u_j|_S \notin C|_S$$

is at most $\varepsilon/(1 - 1/q)$.

Proof. Let B be the bad parameters and choose a witnessing set S_t for each. In $\mathbb{F}_q^{S_t}/C|_{S_t}$, some $[u_j]$ is nonzero: otherwise $[a] + \sum_j t_j [u_j] = 0$ would force $[a] = 0$, contradicting badness. Thus the map $\lambda \mapsto \sum_j \lambda_j [u_j]$ is nonzero, and its kernel has at most q^{s-1} elements. Averaging gives a direction λ for which the set B' of bad parameters with $(\sum_j \lambda_j u_j)|_{S_t} \notin C|_{S_t}$ has size at least $(1 - 1/q)|B|$. On every line $v + z\lambda$, each point of B' is bad for the scalar received line with direction $\sum_j \lambda_j u_j$. The line hypothesis gives $\Pr_z[v + z\lambda \in B'] \leq \varepsilon$. Averaging over uniform v yields $|B|/q^s \leq |B'|/(q^s(1 - 1/q)) \leq \varepsilon/(1 - 1/q)$. $\square$

Corollary 7.2 (Mutual agreement for affine families). *Suppose the given degree- $< k$ Reed–Solomon code has full agreement-set line MCA at threshold $A \geq k$ with exceptional count E . Let $F = \mathbb{F}_q$, $s \geq 1$, and let $a, u_1, \dots, u_s \in \mathbb{F}_q^n$. For $w(t) = a + \sum_{j=1}^s t_j u_j$, there is $\mathcal{E}_s \subseteq \mathbb{F}_q^s$ of size at most $Eq^s/(q-1)$ such that the following holds. For every $t \notin \mathcal{E}_s$ and every degree- $< k$ polynomial P with full agreement set*

$$S = \{i : P(\alpha_i) = w(t)_i\}, \quad |S| \geq A,$$

there are degree- $< k$ polynomials $F_0, \dots, F_s \in \mathbb{F}_q[X]$ with

$$P = F_0 + \sum_{j=1}^s t_j F_j, \quad S = \{i : F_0(\alpha_i) = a_i \text{ and } F_j(\alpha_i) = u_{j,i} \text{ for every } j \geq 1\}.$$

In particular, unless common witnesses on at least A positions exist, the probability that $w(t)$ has A agreements with a codeword is at most $E/(q-1)$ for uniform t , independently of s . For $s = 1$, the sharper bound E/q remains valid.

Proof. Take C to be the degree- $< k$ Reed–Solomon code on the given points. The line theorem implies the hypothesis of [Lemma 7.1](#) with $\varepsilon = E/q$: any set S witnessing an agreement extends to the full agreement set of a polynomial, and the line theorem supplies constituent witnesses on that full set, hence on S . The lemma therefore bounds the bad affine parameters by $Eq^s/(q-1)$.

Outside them, the interpolation observation at the start of this section gives the polynomial identity and equality of the full sets. $\square$

Applying [Corollary 7.2](#) gives affine error $E/(q-1)$ with the threshold and all numerical hypotheses of the corresponding scalar theorem. The choices are

ordinary	$E = B_{\text{cf}}$	every characteristic,
first order	$E = E_1$	$p > \max\{k-1, B_\partial\}$,
uniform gap $\delta \geq 0.24$	$E = 1325775n^2$	$p > k-1$,
capacity	$E = E_\delta(n)$	$p > k-1$,

where B_{cf} and E_1 are given by (68) and (58). The four rows use [Theorems 5.8, 5.12, 5.13](#) and [6.9](#), respectively. Constant codes need no characteristic condition in the uniform and capacity rows.

7.2 Polynomial curves

Proof systems often combine several committed words using powers of one random challenge, producing $w_0 + zw_1 + \dots + z^\ell w_\ell$. As z varies, these combinations trace a polynomial curve in the space of received words. We ask whether a nearby polynomial at a typical point of this curve comes from a tuple of polynomials agreeing with all its coefficient words on a common set of coordinates.

An affine-family bound does not settle this question: an exceptional set small in the ambient affine space can contain the whole curve. Instead, we construct a symbolic explainer for the curve, retaining the powers of Z in the interpolation argument. Its defining identity is the same as for a line, with $f + zg$ replaced by the received curve at z . Agreement is linear in these lifted coordinates, while the Taylor parameterization controls the degree of the resulting family. This gives an exceptional-count bound linear in the curve degree, including for positive derivative order. The choice of coefficients $1, Z, \dots, Z^\ell$ is also called powers batching.

We use the received-curve and agreement-set notation of [Section 3.2](#). The constant-code endpoint is already covered by [Lemma 3.2](#); here we treat positive message degree.

The general symbolic transfer is [Theorem 6.6](#). Its three terms count preliminary exclusions, candidates outside persistent common-witness graphs, and accidental agreements on those graphs.

Controlling challenge degree. A degree- ℓ received curve multiplies every translation degree by ℓ . The polynomial-kernel argument therefore has the same form, with these scaled degrees. [Proposition 5.10](#) gives the finite row and column tests explicitly. A certified ratio $\gamma > 1$ gives $B_Z \leq \lfloor \ell B_{\text{jet}} / (\gamma - 1) \rfloor$ by the same polynomial-kernel argument. In particular, the uniform support of [Theorem 6.8](#) gives $B_Z < 150\ell B_{\text{jet}}$. Both the joint degree and the accidental-agreement count then grow linearly with ℓ , without a characteristic restriction on ℓ .

Corollary 7.3 (Powers batching from a certified support). *Fix the parameters of [Theorem 6.6](#), with characteristic zero or greater than $\max\{K - 1, B_{\text{jet}}\}$. Suppose a fixed support constructs a symbolic explainer satisfying that theorem's hypotheses for every scalar received curve of degree at most ℓ , and certifies $N_{\text{coef}} \geq \gamma n R_{\text{loc}}$ for some $\gamma > 1$. The polynomial-kernel construction gives*

$$B_Z \leq \left\lfloor \frac{\ell B_{\text{jet}}}{\gamma - 1} \right\rfloor.$$

Put $c = B_{\text{jet}} / (\gamma - 1)$, and let C_{pow} be the coefficient of ℓ in (79) after substituting $B_Z = c\ell$. Then at most ℓC_{pow} challenges are exceptional: outside them, every degree- $< k$ candidate with at least A agreements has the full common-witness property of [Definition 3.1](#). No further characteristic restriction depends on ℓ . Over $\mathbb{F}_q$, the error is at most $\ell C_{\text{pow}} / q$, with the same guarantee at every interleaving width.

Proof. Each term of (79) is nondecreasing in B_Z and becomes linear in ℓ after this substitution. Apply [Theorem 6.6](#); its guarantee for every scalar received curve then gives the interleaved conclusion by [Corollary 7.7](#). $\square$

For first-order supports, [Lemma 5.9](#) retains separate derivative degrees and gives sharper stage budgets for the whole received curve; (59) is the finite objective minimized by the tuner.

For first order, [Lemma 5.6](#) supplies the alternative (55) under the hypothesis $p > \max\{k - 1, B_{\partial}\}$ (or characteristic zero). It separates regular solutions from one ordinary resultant equation, then uses the Frobenius transfer for the latter. The finite tuner takes the minimum of the applicable bounds.

At order zero, [Lemma 5.3](#) gives the sharper bound

$$E_{0,\ell} = (2B_{\text{jet}} - 1)B_Z + \frac{n - k + 1}{A - k + 1} (\ell B_{\text{jet}} + B_Z \Psi_{k-1}(B_{\text{jet}})) + \ell(n - k)B_{\text{jet}}, \quad (88)$$

for $2 \leq k < A \leq n$, in every characteristic, with Ψ_D as in [Lemma 5.3](#); this display chooses $L = k$. The Frobenius pullback accounts for inseparable factors without increasing the count of challenges. Thus the ordinary case has neither the positive-order characteristic condition nor a restriction on the received curve degree.

7.3 Interleaved codes

For a code $C \subseteq F^n$, its width- s interleaving is

$$C^{\equiv s} := \{M \in F^{s \times n} : M_{j,*} \in C \text{ for every } j \in [s]\}.$$

Thus interleaving replaces a scalar-valued word by an s -tuple of words on the same coordinate set. Agreement is simultaneous: a column agrees only if every row agrees there. There are two questions. How many polynomial tuples can agree with one received array? And do the candidates along a family of arrays come from common witnesses? The first is a list bound; the second is MCA. We transfer both without a factor depending on the number of rows.

Lists through extension fields. A field-independent scalar list bound already controls interleaved Reed–Solomon lists. The reason is that a column can be encoded as one element of an extension field, preserving equality exactly.

Lemma 7.4 (Extension-field encoding of interleaved lists). *Let F be a field with distinct evaluation points $\alpha_1, \dots, \alpha_n$, and fix a dimension k , an integer threshold A , and a width $s \geq 1$. Suppose an extension E/F contains s elements linearly independent over F , and every scalar received word over E has at most B degree- $< k$ candidates at agreement threshold A . Then the interleaved list over F has at most B candidates at column agreement threshold A . In particular, a scalar bound valid over every extension of F transfers at every width.*

Proof. Choose linearly independent elements $\beta_1, \dots, \beta_s \in E$. The map

$$(P_1, \dots, P_s) \mapsto \sum_{j=1}^s \beta_j P_j$$

is injective on degree- $< k$ row tuples. Package the received columns using the same elements. Linear independence makes equality in E equivalent to equality in every row, so this maps the interleaved list injectively into the scalar list. Apply its bound B . For arbitrary F , the extension $F(\xi)$ with a transcendental ξ supplies the independent elements $1, \xi, \dots, \xi^{s-1}$. For finite $F = \mathbb{F}_q$, a basis of $E = \mathbb{F}_{q^s}$ gives the usual bijective encoding [ABF26, Lemma 2.21]. Only injectivity is needed for the bound. $\square$

Corollary 7.5 (A first-order list-size bound at every interleaving width). *Under the hypotheses of Lemma 5.11, its scalar list bound also holds at every row-wise interleaving width, with agreement measured in columns. For $B_\partial = 0$, this gives the bound B_{jet} in every characteristic.*

Proof. The scalar bound holds after every field extension, which preserves characteristic. Apply Lemma 7.4. $\square$

Consistency along lines and affine families. Applying the scalar theorem separately to each row would multiply the exceptional-count bound by s . Linearity permits a sharper argument: a suitable scalar projection of the rows preserves every bad challenge of one received polynomial curve. There are at most q challenges, so the projection need avoid at most q proper subspaces. Jo’s interleaving theorem therefore transfers the original bound without a width loss [Jo26a, Theorem 4.4 and Corollaries 4.5–4.6]. The same projection proves the polynomial-family case below; lines have $\ell = 1$.

Lemma 7.6 (Interleaving preserves polynomial-family MCA). *Let $C \subseteq \mathbb{F}_q^n$ be a linear code, $1 \leq A \leq n$, and $\ell \geq 1$. Suppose that, for every $w_0, \dots, w_\ell \in \mathbb{F}_q^n$, at most E challenges z admit a set S of size at least A such that $(\sum_{t=0}^\ell z^t w_t)|_S \in C|_S$ but some $w_t|_S \notin C|_S$. Then the same bound holds for $C^{\equiv s}$ at every width $s \geq 1$, with agreement measured in columns.*

Proof. At each bad challenge z , choose a witnessing set S_z and let $u_{t,i}$ be the class of coefficient row i in $\mathbb{F}_q^{S_z}/C|_{S_z}$. For each row, $\sum_t z^t u_{t,i} = 0$, while some $u_{t,i}$ is nonzero. Thus the projections $\lambda \in \mathbb{F}_q^s$ satisfying $\sum_i \lambda_i u_{t,i} = 0$ for every t form a proper subspace K_z . At most q proper subspaces have union of size at most $1 + q(q^{s-1} - 1) < q^s$. One projection therefore avoids every K_z . It commutes with the polynomial combination and preserves every bad challenge in a single scalar received curve. Its bound E proves the claim. $\square$

Corollary 7.7 (MCA for interleaved Reed–Solomon codes). *Suppose every scalar received polynomial curve of degree at most $\ell \geq 1$ has full agreement-set MCA at threshold $A \geq k$ with exceptional-count bound E_ℓ .*

Then the same bound holds for interleaved received curves of every width, with degree- $< k$ polynomial witnesses for all $\ell + 1$ coefficient arrays and equality of full common column-agreement sets. The exceptional probability is at most E_ℓ/q , independently of width. In particular, a scalar line bound E gives interleaved line error E/q and affine-family error $E/(q - 1)$, independently of both width and the number of affine parameters.

Proof. Apply Lemma 7.6 to the subset-MCA consequence of the scalar guarantee. The interpolation observation at the start of Section 7, applied row by row, gives the polynomial identity and equality of full column-agreement sets. Lines use $\ell = 1$. For affine families, apply Lemma 7.1 to the interleaved code: its quotient-space proof works over $(\mathbb{F}_q^S)^S/C|_S$ unchanged. $\square$

7.4 Multilinear folds

A multilinear fold starts with 2^h received words and repeatedly replaces two children u, v by $(1 - r)u + rv$. One challenge is shared by every node at a level. We exploit this sharing: the nodes at that level form the rows of one interleaved word, so the interleaving bound charges the level once, regardless of its number of nodes. Full common-agreement-set equality lets us repeat this argument through all levels.

Write $P|_\alpha = (P(\alpha_i))_{i=1}^n$ for a polynomial's evaluation word. Use Agr for the agreement set defined in Section 3.

Corollary 7.8 (MCA for multilinear folds). *Let the degree- $< k$ Reed–Solomon code over $\mathbb{F}_q$ satisfy the full agreement-set line-MCA conclusion at threshold $A \geq k$ with exceptional-count bound E , as in Corollary 7.7. Fix $h \geq 1$ and words $(u_\sigma)_{\sigma \in \{0,1\}^h}$. Sample independent uniform challenges $r_1, \dots, r_h \in \mathbb{F}_q$, and define the folded words from the leaves upward by $W_\sigma = u_\sigma$ for $|\sigma| = h$ and*

$$W_\tau = (1 - r_{j+1})W_{\tau 0} + r_{j+1}W_{\tau 1} \quad (|\tau| = j < h). \quad (89)$$

Outside at most hEq^{h-1} challenge tuples, every degree- $< k$ polynomial P agreeing with $W_\emptyset$ on at least A positions admits degree- $< k$ polynomials $(P_\sigma)_\sigma$ such that, writing $\ell_0(r) = 1 - r$ and $\ell_1(r) = r$,

$$P = \sum_{\sigma \in \{0,1\}^h} \left(\prod_{j=1}^h \ell_{\sigma_j}(r_j) \right) P_\sigma, \quad (90)$$

$$\text{Agr}(P|_\alpha, W_\emptyset) = \bigcap_{\sigma \in \{0,1\}^h} \text{Agr}(P_\sigma|_\alpha, u_\sigma). \quad (91)$$

For $h = 0$, there is one word and no challenge; the conclusion holds with $P_\emptyset = P$ and no exceptions. The exceptional probability is therefore at most hE/q for every $h \geq 0$. In particular, a height-three fold costs $3E/q$. The same bound holds when the input words are themselves interleaved, independently of their width.

Proof. At depth j , form the arrays

$$U_j = (W_{\tau 0})_{|\tau|=j}, \quad V_j = (W_{\tau 1} - W_{\tau 0})_{|\tau|=j}.$$

They depend only on $r_{j+2}, \dots, r_h$. Conditional on these challenges, r_{j+1} is uniform, so Corollary 7.7 gives one exceptional set of size at most E for $U_j + r_{j+1}V_j$. This set works simultaneously for every tuple of polynomial candidates. A union bound over the h levels gives

$$\Pr[\text{some level is exceptional}] \leq hE/q. \quad (92)$$

Fix a challenge tuple outside these events and a root candidate P . Let S be its full agreement set. Inductively maintain a tuple $(P_\tau)_{|\tau|=j}$ whose full common column-agreement set is S . Since $|S| \geq A$,

interleaved MCA splits this entire tuple into its children and preserves that common set. Rewriting its witnesses for U_j, V_j as witnesses for the two children gives $P_\tau = (1 - r_{j+1})P_{\tau 0} + r_{j+1}P_{\tau 1}$. At the leaves these identities compose to (90), and the invariant gives (91). Individual rows may agree at additional positions; only their common set is preserved.

Each uniform challenge tuple has probability q^{-h} , giving the stated cardinality bound. Interleaved input words are handled by including their original row index in the arrays at each level. $\square$

8 First-order bounds and smaller proofs

The first-order list and exceptional-count bounds in [Proposition 5.10](#) and [Lemma 5.6](#) reduce query counts below those justified by Johnson-based soundness analyses in ProveKit, ZisK, and LambdaVM. The sharp line and powers-family bounds require characteristic zero or $p > \max\{k - 1, B_\partial\}$ when $k \geq 2$; constant codes need no restriction. These are characteristic conditions, distinct from extension-field cardinality.

Why the applications use first order. Higher derivatives lower the agreement threshold but enlarge the candidate and exceptional-count bounds. Already at second order these are $O(n^2)$ and $O(n^3)$, respectively ([Theorems 6.3](#) and [6.6](#)). At the field sizes below, the resulting Λ/q and E/q terms can erase the query saving. First order beats Johnson while retaining linear-size lists and quadratic exceptional counts.

The rate curve suggests an agreement threshold, but we need finite coefficient, rank, and challenge-degree counts to check the protocol’s error budget at its actual field size. At $a = A/n$, a certificate supplies a list bound Λ and exceptional counts E . The list and MCA terms may use different supports certifying the same (n, k, A) , provided each meets its own field hypotheses.

8.1 From certificates to proof bytes

The accompanying implementation [[DKT26](#)] searches supports, checks the strict coefficient-constraint margin, evaluates both challenge-degree tests, and minimizes the applicable transfer bounds using integer and rational arithmetic. Every successful output is a certificate. Failure within a bounded search is not an impossibility result. Arnon, Boneh, and Fenzi isolate the resulting tradeoff in a protocol whose knowledge-soundness error is $\epsilon_{\text{mca}} + \Lambda/q + a^t$ [[ABF26](#), Construction 6.2 and Lemmas 6.6, 6.8]. Our applications use their own protocol bounds. A line combination contributes E_{line}/q , a powers combination its separately certified E_ℓ/q , an affine reduction $E_{\text{line}}/(q - 1)$, and an h -level binary fold hE_{line}/q ([Corollary 7.8](#)); the remaining collision and identity terms depend on Λ .

For query-grinding acceptance probability p_{grind} , the query term is $p_{\text{grind}}a^t$. When it shares an allocation ϵ with the algebraic terms, we require

$$\epsilon_{\text{alg}} + p_{\text{grind}}a^t \leq \epsilon.$$

We choose a and t only after adding every algebraic term and enforce separate error allocations when the protocol’s phases have them. The zkVM applications retain powers batching derived from a few transcript challenges, including LambdaVM’s optimized sampling [[Lam26b](#)]; we charge its curve MCA, and any grinding credit, only where the protocol checks the corresponding query challenge.

Each comparison preserves the affected component’s existing error allocation, composition rule, and grinding convention. [Table 2](#) identifies the measured objects and baselines. Separate 128-bit allocations do not assert a 128-bit bound on their sum, and the LambdaVM measurement and local certificate do not establish a new end-to-end security theorem for the VM.

We apply this accounting to hash-based systems whose Johnson analyses leave room to reduce queries [[DKT26](#)]. Net bytes subtract added evaluations and nonces from removed query values and authentication data. Shared Merkle paths use expected sizes; separately transmitted paths use direct counts.

Measured object	Baseline	Size samples	Percentage denominator
ProveKit passport proof	Original outer and internal schedules	Ten proofs per configuration	Original compressed mean
ProveKit lookup proof	Original witness and blinding schedules	Thirty proofs per configuration	Original compressed mean
ZisK compressed final proof	Proof from the released 54-query prover	One proof and its verified 51-query conversion	Original native proof bytes
LambdaVM CPU subproof	Original 219-query CPU profile	One proof per profile on the same benchmark	Original serialized CPU bytes

Table 2: Interpretation of the proof-size measurements. ZisK sizes precede any PLONK wrapping. LambdaVM changes only the CPU subproof inside the complete VM proof. Its repeated timing measurements are a separate experiment and do not increase the size-comparison sample count.

System	Shipped	Johnson floor	Ours	Security check
ZisK	54	53	51	Each phase: 128 bits
LambdaVM CPU	219	216	208	Local sum: 128.063 bits

Table 3: Query counts with unchanged grinding. The finite-Johnson column is a lower bound from the query term alone, not a certified configuration. Our counts also satisfy the algebraic checks below. ZisK’s successive-stage and squarefree transfers both certify 51 queries; the sharper transfer adds margin. LambdaVM uses the squarefree transfer and two early evaluations. ProveKit’s multiround measurements are in Table 4.

ZisK and LambdaVM use cubic Goldilocks, $q = (2^{64} - 2^{32} + 1)^3$, while BN254 has size about 2^{254} [Wor26a, Pro26b, Lam26c].

8.2 ProveKit: passport and lookup proofs

ProveKit turns a computation into constraints on a witness, then uses Spartan and WHIR to prove that those constraints hold. World Foundation reports that the toolkit is integrated into World ID and describes its passport benchmark as checking document signatures, expiry, and a minimum age without revealing the passport [Wor26d, Wor26c]. We retune the zkWHIR witness commitments in this public v1 benchmark, and separately test lookup proofs over the cubic extension of the Goldilocks field [Wor26e, Wor26b]. The first saves 79.4 KB in a complete compressed BN254 proof; the second saves 16.7 KB in a compressed cubic-Goldilocks proof. Table 4 gives the exact measurements and sample counts.

For each queried code, we choose an agreement threshold, bound its close candidates and exceptional combinations, and reduce the query count only when every changed term fits its original 128-bit allocation. The Goldilocks example adds one out-of-domain evaluation to each witness commitment, which the measured sizes include.

Which error terms must be rechecked? Consider a WHIR code with n rows and degree bound $k - 1$. Its rows contain eight field elements, representing a vector of length $8k$. At agreement $a = A/n$, our finite certificate supplies a list bound Λ and a line exceptional count E . The interleaved list has the same bound as the scalar list (Corollary 7.5). The folding distribution needs a further step: WHIR combines the eight columns with the equality-polynomial weights of three random challenges. These are the weights of three successive binary folds. Corollary 7.8 bounds their exceptional probability by $3E/q$. Thus the folding and

identity checks, and the test distinguishing candidate vectors by u out-of-domain evaluations, satisfy

$$\epsilon_{\text{fold}} \leq \frac{3E + 2\Lambda}{q}, \quad \epsilon_{\text{OOD}} \leq \binom{\Lambda}{2} \left(\frac{8k - 1}{q} \right)^u. \quad (93)$$

The second bound uses the degree of the full vector polynomial: any two distinct candidates differ by a nonzero polynomial of degree at most $8k - 1$. Increasing u suppresses this collision term when the candidate list is too large for one evaluation.

The transition to the next code uses both the source query count and the target list bound. If the source has t queries and the target uses u' out-of-domain evaluations with list bound Λ' , its constraint-combination term is at most

$$\frac{2(t + u')\Lambda'}{q}. \quad (94)$$

We recompute this term whenever either code changes. In particular, retuning several rounds requires the new target lists, rather than the Johnson lists that the original parameters used. The remaining vector and linear-form combination terms retain their original field-size bounds [Wor26e, Wor26f].

For each queried code, we keep its existing grinding threshold T . The verifier accepts a 64-bit hash at most T , so its exact acceptance probability is $(T + 1)/2^{64}$. The query check is therefore

$$\frac{T + 1}{2^{64}} \left(\frac{A}{n} \right)^t \leq 2^{-128}. \quad (95)$$

The first code has $T = 18786624067678312$; subsequent codes use their respective original thresholds. All probabilities here use the usual uniform-field challenge model. The implementation’s field-sampling error is negligible relative to these allocations; the accompanying analysis includes its bound [DKT26].

The BN254 passport circuit. The reconstructed launch circuit has 294402 constraints and 522648 witness elements. Lookup challenges separate the witness into parts of lengths 249345 and 273303, each committed with a padded vector length 2^{19} . The first code in each commitment has $n = 2^{18}$ and $k = 2^{16}$. The first finite certificate uses agreement $123208/2^{18}$ and support $(m, B_\partial, B_{\text{jet}}) = (384, 168, 688)$, permitting 109 queries in place of 127. We retune the later codes as well, keeping one out-of-domain evaluation in each code (Section H.1).

How much of this improvement remains after optimizing the Johnson baseline? At the same grinding probability and error allocation, even the limiting agreement $\sqrt{65535/262144}$ requires 119 queries in (95): 118 do not suffice. The count 119 is also achievable: at agreement $1/2$, the finite Johnson bounds of Theorem 5.12 satisfy the initial round’s list, MCA, and out-of-domain checks.³ Our certified choice of 109 therefore saves ten initial queries, or 8.4%, over this optimized Johnson count. The complete-proof measurements below use the implemented configuration with 127 initial queries and include the later rounds and internal proofs as well.

Each witness opening also contains an internal zkWHIR proof about its masking polynomials. We retune these internal proofs as well. Their initial commitment batches two vectors, requiring a separate line-MCA bound E/q , followed by the multilinear-fold bound $3E/q$. Cancellation among claimed evaluations and linear-form claims contributes separate identity-test terms. All changed checks meet their existing allocations, including the grouped initial batching error $(E + 1)/q$. The masking randomness, privacy parameters, and out-of-domain counts are unchanged. Section H.1 records both schedules.

We generated and verified ten complete proofs per passport configuration on the same input. Random query positions share different portions of their Merkle paths, and compression adds further variation, so we compare the sample means in Table 4.

³The rounded Johnson parameters are $m = 65536$, $B_{\text{jet}} = 131074$, and $B_Z = 5726797827$, with list bound 65537. The query, folding, and out-of-domain terms have respectively 128.94, 183.01, and 203.60 bits of security.

Cubic Goldilocks. To test the improvement over a smaller field, we use ProveKit’s lookup benchmark over the cubic extension of Goldilocks [Wor26b]. It proves that 4000 values belong to a table of 2000 entries, using the LogUp lookup argument. It has 8001 constraints and two witness parts of length 8001, each padded to 2^{13} . Some witness values depend on lookup challenges in the cubic extension, so this circuit commits extension-field elements, each occupying 24 bytes. Its first code has $n = 4096$ and $k = 1024$. The certificate at $A = 1933$, with support $(96, 42, 174)$, permits 109 initial queries and two out-of-domain evaluations. We also retune the later codes and the separate blinding commitment (Section H.1). The additional evaluations cost 48 bytes in total. The separate derivative-degree bound matters in the second code, where $(n, k, A) = (2048, 128, 453)$ with support $(38, 23, 133)$. The squarefree bound gives

$$\Lambda \leq \left\lfloor \frac{2659384185}{326} \right\rfloor = 8157620,$$

and 137.08 bits of separation with one out-of-domain evaluation. For the same retuned configuration, the successive-stage transfer gives 93954343, while using only total jet degree with the same support gives a list bound of 507709435 and only 125.16 bits. Retaining the separate derivative degree already gives 130.03 bits with the successive-stage bound, enough for this 55-query configuration without another evaluation or more grinding. The squarefree bound increases that margin to 137.08 bits; the accompanying artifact records all three bounds [DKT26].

Proof	Parameters	Raw bytes	Compressed bytes	Saved KB	Saved %
Passport, BN254	Original	722305.0	716276.5	–	–
	Outer and internal codes retuned	641748.2	636865.6	79.4	11.1
Lookup, Goldilocks ³	Original	207314.7	161184.8	–	–
	All witness codes retuned	183419.2	144508.4	16.7	10.3

Table 4: Verified ProveKit proof-size measurements. Passport entries are means over ten proofs per configuration; lookup entries are means over thirty and include retuning the initial blinding queries. Raw passport sizes count the uncompressed proof encoding; raw lookup sizes count the transcript and authentication data. Compressed sizes count the complete proof files. Saved KB uses 1000 bytes per KB and compares compressed means. Sample standard deviations (raw, compressed), in row order, are (1649.6, 2056.2), (1616.6, 1818.9), (776.6, 967.4), and (730.7, 895.3) bytes.

The lookup experiment uses ProveKit’s non-hiding Goldilocks backend; it does not implement a private passport proof over that field [Wor26b, Wor26a]. Section H.1 gives the original and revised query counts and the exact error margins for the measured configurations.

8.3 ZisK: a smaller compressed final proof

ZisK is an open-source zkVM.⁴ Its compressed final proof certifies a recursively aggregated execution [Zis26b]. We use 51 queries in place of 54, saving a measured 13.2 KB (4.63%) in the compressed final proof.

We use the released compressed-final parameters: an LDE of size 2^{19} , code dimension 2^{15} , and cubic Goldilocks challenges [Zis26c]. The change is applied after the execution has been aggregated, so it does not depend on which computations were proved before aggregation.⁵

⁴Its repository describes ongoing security and correctness audits toward production [Zis26a].

⁵If this proof is subsequently wrapped using PLONK, our saving concerns the inner proof.

Keeping the existing batching and transcript. ZisK combines opening words using two nested powers challenges. Within each opening-point group it combines words by powers of one challenge; it then combines the groups by powers of a second challenge. Fixed opening-point weights are included in the words being combined. To account for the shared inner challenge only once, we apply the interleaved curve bound to all inner groups together. Conditional on that challenge, we apply the scalar curve bound to the outer combination. The inner bound is uniform over every tuple of candidates, including those reconstructed using the outer challenge.

For the compressed final proof, the nontrivial inner degrees are 1, 2, 103, 24, and the outer degree is 4. At agreement $A/n = 124136/2^{19}$, the support $(m, B_\partial, B_{\text{jet}}) = (10, 6, 37)$ supplies certificates for the two batching stages and the three FRI folds. Pad the inner coefficient arrays with zeros to degree 103 and apply [Corollary 7.7](#). The padded coefficients recover degree- $< k$ polynomials vanishing on the common agreement set; since $A \geq k$, they are zero, leaving witnesses for exactly the original words. A union bound over the shared inner challenge and the outer challenge gives batching probability at most $(E_{103} + E_4)/q$, namely

$$\frac{4073087021134687044}{(2^{64} - 2^{32} + 1)^3} < 2^{-130.17}.$$

Every fold has more than 140.12 bits. These checks use no grinding credit. The query term satisfies

$$2^{-22}(124136/2^{19})^{51} < 2^{-128}.$$

This query calculation uses the protocol’s computational grinding convention and models hash and transcript outputs as independent uniform field elements. The implemented sampler allows repeated queries: it slices query indices from the low 63 bits of successive Goldilocks outputs [[Pro26a](#)]. For these parameters it uses 16 outputs. Accounting also for the implemented grinding predicate, a conservative correction replaces the query term by

$$(124136/2^{19})^{51} \frac{2^{42}}{p} \left(\frac{2^{64}}{p} \right)^{16} < 2^{-128}, \quad p = 2^{64} - 2^{32} + 1.$$

The factor $2^{64}/p$ bounds the sampling bias per field output; the grinding predicate accepts 2^{42} of the p possible outputs. These checks concern the stated transcript model, not a proof that the concrete hash produces independent uniform outputs. Thus each batching, folding, and query phase retains its separate 128-bit allocation. This 51-query support also satisfies the earlier successive-stage curve bound: its batching error is below $2^{-128.46}$. The query reduction here comes from finite parameter tuning; the squarefree transfer supplies additional algebraic margin.

Below the finite Johnson threshold. ZisK’s implemented Johnson analysis uses agreement $\sqrt{\rho} + \eta_0$, with $\rho = 1/16$ and positive gap η_0 , and selects 54 queries [[Zis26c](#)]. Our replacement also improves on what the sharper finite-length Johnson threshold can permit. At $a_J = \sqrt{(k-1)/n}$, even the query term alone at 52 queries is

$$2^{-22}a_J^{52} = 2^{-22} \left(\frac{2^{15} - 1}{2^{19}} \right)^{26} > 2^{-128}.$$

Consequently this threshold requires at least 53 queries with the same grinding. Our certificate permits smaller agreement and reaches 51, while preserving the algebraic error allocations as well.

From queries to proof size. We generate and verify a compressed final proof for a 679-step execution using the released prover. Keeping its first 51 query openings gives a proof accepted by the matching 51-query verifier: the transcript generates these queries as a prefix of the original sequence. We preserve all other proof contents and save both proofs in ZisK’s native format. Their measured sizes are 286013 and 272783 bytes, a

reduction of 13230 bytes (4.63%). This measures the complete compressed final proof, including its public values, rather than just counting field values and authentication hashes. The public artifact supplies both verified proofs, the pinned configurations, and the reproduction procedure.

8.4 LambdaVM: smaller CPU subproofs

LambdaVM is an experimental Rust RISC-V zkVM developed with LambdaClass and 3MI Labs [Lam26a]. Its execution proof contains CPU, arithmetic, memory, and coprocessor table subproofs linked by LogUp lookups [Lam26d]. We modify only the CPU subproof, whose trace constrains control flow and links instruction decoding, arithmetic, and memory requests to the corresponding tables. On a benchmark with a 32768-row CPU trace, removing eleven queries and adding two early vector evaluations reduces the CPU subproof by 59 832 bytes (59.8 KB, or 4.85%) [Dao26b].

The default profile uses cubic Goldilocks, half-rate codes, 20 grinding bits, and 219 queries for a 128-bit target [Lam26c, Lam26d]. We keep the field, rate, powers batching, grinding, and terminal codeword length 256 fixed. The CPU certificate changes the query count to 208 and the agreement threshold to $A/n = 45690/65536$; its exact-arithmetic check is included in the artifacts [DKT26]. The support $(m, B_\partial, B_{\text{jet}}) = (32, 9, 44)$ uses the squarefree curve bound (55). The received-word batching curve has degree $\ell = 50$, while the resulting symbolic explainer has the separately certified challenge-degree bound $B_Z = 36670$. This MCA certificate uses dimension $T = 32768$. The list used to select a main-column tuple after DEEP reconstruction has dimension $T + 3 = 32771$, and the same support gives $\Lambda = 113769033$ as a valid integer upper bound. We use this larger-dimension list in every list-dependent term of (96).

Fixing the CPU trace before lookup. A list of close polynomials leaves several possible traces behind one commitment. The shared LogUp challenges must not help the prover select among them, so we order the transcript as

CPU main commitment $\longrightarrow$ two early points and claimed values $\longrightarrow$ LogUp challenges.

Lemma 8.1 proves that this order fixes a unique candidate trace, except for the collision event bounded below. The verifier samples distinct points s_1, s_2 outside the CPU LDE domain and receives the values of all $w = 38$ main columns there. The two vectors contain $2w$ extension-field elements and occupy 1856 serialized bytes.

The later out-of-domain point z retains the usual polynomial-identity check. For each main column we replace its DEEP quotient by

$$\frac{f(X) - I_f(X)}{(X - z)(X - s_1)(X - s_2)},$$

where I_f interpolates the three claimed values. Successful low-degree reconstruction produces a polynomial of degree at most $T + 2$ that agrees with the committed word and takes both early values. The following lemma proves that these values select at most one candidate, uniformly over the prover’s choices of claimed values. Fix the main commitment, and let C be its degree-at-most- $T + 2$ 38-column tuples with at least A simultaneous agreements. **Corollary 7.5** at dimension $T + 3$ gives $|C| \leq \Lambda$. This family is fixed before the early evaluations and LogUp challenges.

Lemma 8.1 (Selecting the CPU trace before lookup). *Sample two distinct points uniformly without replacement outside the LDE domain, assuming $q - n \geq 2$. Except with probability at most*

$$\epsilon_{\text{anchor}} = \binom{\Lambda}{2} \left(\frac{T + 2}{q - n - 1} \right)^2,$$

the following holds simultaneously for all claimed vectors, including those chosen after the early points s_1, s_2 are revealed:

$$\text{for every } (v_1, v_2) \in (\mathbb{F}_q^{38})^2, \quad \#\{f \in C : f(s_1) = v_1, f(s_2) = v_2\} \leq 1.$$

If the later OOD point is distinct from the early points and DEEP extraction succeeds with the degree-three main-column divisors, its reconstructed main-column tuple is this member.

Proof. Two members differ in a coordinate of degree at most $T + 2$. At each sample its conditional vanishing probability is at most $(T + 2)/(q - n - 1)$. Union-bounding over $\binom{\Lambda}{2}$ pairs gives the collision estimate for all claimed vectors simultaneously.

Later extraction yields a quotient polynomial h of degree less than T . Undoing its divisor reconstructs

$$f(X) = (X - z)(X - s_1)(X - s_2)h(X) + I_f(X).$$

Its degree is at most $T + 2$; it has the common agreements and both early values. The tuple therefore belongs to C and, outside the collision event, is uniquely determined. Without such a member, extraction cannot succeed. $\square$

Reduction modulo $X^T - 1$ preserves the trace-domain values and bus contributions, giving one degree- $< T$ trace fixed before LogUp. The local error bound charges both the collision event and rejection when the OOD point equals an early point.

Retaining powers batching. The CPU table has 38 main columns, 10 auxiliary columns, one next-row accumulator opening, and two composition parts, for $b = 51$ DEEP terms. Combining them with $1, \gamma, \dots, \gamma^{b-1}$ therefore uses the curve theorem at degree 50. The parameter search includes this degree dependence; neither the number of DEEP terms nor the FRI degree bound changes.

Checking the complete CPU error bound. Write $a = A/n$ and use $A_j = \lceil An_j/n \rceil$ on a folded domain of size n_j . The inequalities $2A_{j+1} \geq A_j$ reconstruct through every binary fold, including the last uncommitted one. If E_{init} and E_j are the initial and fold exceptional counts, their contribution is

$$\epsilon_{\text{MCA}} = \frac{E_{\text{init}}}{q} + \sum_j \frac{E_j}{q},$$

where E_{init} is the degree-50 curve bound and $q = (2^{64} - 2^{32} + 1)^3$. We use the independent uniform transcript-output model and the existing computational 20-bit grinding convention [Lam26b]. Rejection sampling makes the field challenges uniform. The $t = 208$ query indices are sampled uniformly with replacement from the $n/2$ row pairs; at most $A/2$ pairs can have both rows agreeing, so each query contributes at most A/n . The early points are instead sampled without replacement outside the LDE domain, as required by Lemma 8.1. The complete local bound is

$$\epsilon_{\text{CPU}} \leq \epsilon_{\text{MCA}} + 2^{-20}a^t + \frac{49\Lambda}{q} + \frac{(4T + 6)\Lambda + 2n + 2}{q - n - T} + \epsilon_{\text{anchor}}. \quad (96)$$

The five terms cover exceptional challenges, acceptance by all query tests, cancellation among the CPU constraints, failure of the OOD identity test, and collision at the two early points. Only the query term receives the existing grinding credit.

The constraint and OOD terms. We now justify the two algebraic terms in (96). The CPU table has maximum constraint degree 3, no transition end exemptions, and one LogUp boundary constraint. With two early points, every reconstructed trace coordinate has degree at most $T + 2$. A constraint product therefore has degree at most $3(T + 2)$, and clearing its denominator with $X^T - 1$ raises this to at most $4T + 6$. The two recovered composition parts have degree at most T ; recombining them as $C_0(X^2) + XC_1(X^2)$ and multiplying by $X^T - 1$ gives the smaller bound $3T + 1$. Thus $D_{\text{res}} = 4T + 6$ bounds the cleared CPU residual.

Fix the CPU trace and auxiliary commitments before sampling the uniform constraint-combination challenge β . An invalid trace candidate violates a constraint at a trace-domain point, so its rational constraint has a pole there. Combining the CPU constraints with powers of β retains a pole except at at most 49 challenge values. A union bound over the fixed trace-candidate list gives $49\Lambda/q$; the composition polynomial cannot cancel the pole.

All committed words, including the composition words, are fixed before the OOD challenge. Apply the width-independent list bound to this full tuple at dimension $T + 3$ and common agreement A , giving at most Λ candidates. For each invalid tuple, the cleared OOD residual is a nonzero polynomial of degree at most $4T + 6$. The OOD sampler has at least $q - n - T$ choices after excluding the LDE and trace domains. At most $(4T + 6)\Lambda$ choices are roots of the residuals, at most $2n$ have their square in the LDE domain, and at most two coincide with the early points. Hence its failure probability is at most

$$\frac{(4T + 6)\Lambda + 2n + 2}{q - n - T},$$

which is the OOD term in (96).

The bound in (96) has 128.063 bits of security (the negative base-two logarithm, rounded to three decimals), preserving the 128-bit local target.

The reduction to 208 queries requires agreement below finite Johnson: even without charging any algebraic error, agreement at or above that threshold needs at least 216 queries with the same grinding.

Measured CPU-subproof reduction. We implement the two-evaluation profile and compare it with the original profile on the same assembly benchmark and empty input. The benchmark executes 32768 instructions and produces one 32768-row CPU chunk; both complete proofs contain 26 table subproofs. Every non-CPU table retains 219 queries. The fork’s release CLI generates, deserializes, and verifies both proofs under their respective profiles [Dao26b]. Serializing each complete CPU subproof with `rkyv` gives

$$1\,233\,024 \text{ bytes} \longrightarrow 1\,173\,192 \text{ bytes},$$

a 4.85% reduction, including the added evaluations. These sizes include all CPU commitments, claims, and openings; only the shared VM-level metadata is excluded. The complete VM proofs decrease by the same 59 832 bytes, from 36 868 888 to 36 809 056 bytes (approximately 0.162% of the original complete-VM proof). This is a measured A/B instance, not an average over workloads. The experimental profile checks the supported CPU trace shape rather than applying this fixed certificate to arbitrary executions.

In this proof, each removed query saves 5608 serialized bytes, and the two early claim vectors add 1856 bytes, giving

$$11 \cdot 5608 - 1856 = 59\,832 \text{ bytes}.$$

These sizes include the query records, vector metadata, and alignment used by the serializer.

Cost of the added evaluations. We measure the same benchmark on an Apple M4 Max with 16 threads, using release builds and the baseline grinding factor 20. Over 20 paired rounds, the change in proving time is -15.2 ms, with a 95% interval of $[-49.4, +18.9]$ ms. Whole verification, measured over 50 paired rounds,

changes by +0.02 ms with interval $[-0.62, +0.52]$ ms. We therefore detect no end-to-end latency regression on this benchmark. These are paired, round-level bootstrap intervals; they do not establish equal performance on other workloads [Dao26a].

Section H.2 records the instruction counts, component timings, and memory diagnostics.

9 Efficient decoding over large fields

The preceding sections bound the agreement lists. We now compute them from ordinary received values. The hidden-derivative interpolant supplies a differential equation satisfied by every wanted message, but it can have many other solutions. We keep its initial values symbolic and use received-word agreement to select the messages before extracting any parameter roots. At order zero, fast GS interpolation and a single regular fiber give near-linear bit complexity at a fixed gap above Johnson. Given a first-order explainer of bounded total jet degree and X -degree $O(n)$, a fixed linear gap $A - k + 1 = \Omega(n)$ permits deterministic quadratic-time reconstruction. Our first-order interpolation bound supplies such an explainer above $a_1(\rho)$. At each fixed higher order, the same Taylor construction reduces decoding to polynomial systems in a bounded number of variables.

9.1 The computational problem

Kopparty’s reconstruction enumerates initial Taylor data and then filters the resulting solutions [Kop15, Section 4.2]. His full-field multiplicity code has length q , so a power of q is polynomial in that length. For ordinary RS words of length n over an independently large prime field, it can be exponential in $\log q$. The stated BCPZZ and Jeronimo decoders have running times $q^{O_{\text{params}}(1)}$ [BCP⁺26a, Corollary 5.1][Jer26, Theorem 4.4]; a field-independent list bound alone does not remove this cost [Jer26, Remark 6.4].

As explained in Section 2, the differential equation can have many solutions that do not agree with the received word. Our algorithm represents finite families algebraically and splits them by agreement. This agreement-to-interpolation principle has precedents in ordinary RS reconstruction [AP00, CHK26]; the families here can be nonlinear.

Arithmetic model. Write $q = p^e$ and $b_q = \lceil \log_2 q \rceil$. The algorithms do not need a particular basis, but they do need a representation that exposes the operations used in their bit-complexity analysis.

Definition 9.1 (Decoder-admissible field presentation). A *decoder-admissible presentation* of $K = \mathbb{F}_q$ supplies p, e and presentation data of size $\tilde{O}(b_q)$ with the following deterministic operations.

- (i) Every element has a canonical $O(b_q)$ -bit encoding. Equality, addition, subtraction, multiplication, inversion of a nonzero element, and embedding an element of the prime field each cost $\tilde{O}(b_q)$ bit operations.

- (ii) There is a bijection

$$\text{elt}_K : \{0, \dots, q - 1\} \longrightarrow K$$

whose value at any index is computed in $\tilde{O}(b_q)$ bit operations.

- (iii) After one reusable preprocessing step of cost $\tilde{O}(pb_q)$, a batch of $N \geq 1$ elements can be mapped to their unique p -th roots in $\tilde{O}(pNb_q)$ bit operations. Equivalently, preprocessing and one batch cost $\tilde{O}(p(N + 1)b_q)$. The preprocessing is charged only if inverse Frobenius is invoked.

Primality, the cardinality and validity of the presentation, and distinctness of the evaluation points are input promises.

The factor p in the last operation does not introduce an independent dependence on the field size in the algorithms below. In the bounded- Y -degree separable preprocessing, a nontrivial p -th-root step can occur only when p is at most that fixed degree. In the general univariate radical routine, such a step instead implies $p \leq D$, where D is the current polynomial degree, and batches inverse Frobenius over $O(D/p)$ coefficients. Thus each use is absorbed by a polynomial-degree cost already charged in [Lemmas E.4](#) and [E.9](#).

The definition specifies the decoder's operational needs rather than the algebraic form of the field. Polynomial bases, fast absolute normal bases, and supplied quotient extensions instantiate it under the hypotheses proved in [Corollary E.2](#) and [Lemmas E.1](#) and [E.3](#). In particular, it places no restriction on the extension degree e . We do not charge for constructing or converting the supplied presentation from the cardinality q alone.

Remark 9.2 (Slower explicit presentations). Correctness and the field-operation counts do not depend on [Definition 9.1](#). The same algorithms work with any effective finite-field representation after charging its actual costs for arithmetic, enumeration, and inverse Frobenius. For example, a basis with structure constants gives deterministic time polynomial in n and in the length of that presentation [[Len91](#)]. Such a description can be much larger than $\log q$, so it does not imply the displayed quasilinear dependence on $\log q$.

The positive-order theorems state their additional lower bound on p explicitly. A field element uses $O(b_q)$ bits under [Definition 9.1](#). We return each degree- $< k$ polynomial as its k coefficients, including the zero vector. Throughout this section, $\tilde{O}$ hides polylogarithmic factors in n and b_q , not another power of $\log q$. The decoder does not enumerate MCA's exceptional challenges.

The intermediate routines below construct a candidate cover: every degree- $< k$ polynomial with at least A agreements is represented by at least one returned family, which may also contain duplicate or extension-field points and Taylor truncations that are not differential roots. *Exactness* is supplied only by [Algorithm 1](#), which extracts a polynomial from k base-field agreements and checks it against the complete received word.

The decoder may receive either a certified monomial support, from which it constructs an explainer Q , or a supplied sparse nonzero $Q \in \mathbb{F}_q[X, Y_0, \dots, Y_d]$ with $\deg_X Q \leq Cn$ and $\deg_Y Q \leq B_{\text{jet}}$. The supplied equation must satisfy $Q(X, P, \dots, P^{[d]}) = 0$ for every degree- $< k$ message with at least A agreements. [Algorithm 6](#) in [Section E.1](#) records the complete positive-order procedure and its treatment of the finite exceptional cases.

9.2 Fast interpolation for a fixed received word

The coefficient-count arguments above prove that the local constraints have a nonzero solution. Solving their displayed scalar matrix by Gaussian elimination would nevertheless cost cubic time. For a fixed received word, the same constraints form a polynomial-module interpolation problem: the unknown coefficients of each jet monomial are one univariate polynomial in X , while multiplication by X on every local truncated Taylor expansion is a Jordan block. The minimal-basis algorithm of Jeannerod, Neiger, Schost, and Villard [[JNSV17](#), Theorem 1.5] then finds the required relation in time quasilinear in the total local output size.

Lemma 9.3 (Fast fixed-word interpolation). *Fix $d, B \geq 0, m, s \geq 1$, and distinct jet monomials $M_1, \dots, M_s \in K[Y_0, \dots, Y_d]$ of total degree at most B . Let $w_1, \dots, w_s, W$ be nonnegative integers and put*

$$\mathcal{V} = \left\{ \sum_{j=1}^s f_j(X) M_j : \deg f_j + w_j < W \text{ whenever } f_j \neq 0 \right\}.$$

Assume $W, w_j = O(n)$. Given $n \geq 1$ received pairs (α_i, y_i) over K , whose abscissae need not be distinct, one can deterministically find a nonzero $Q \in \mathcal{V}$ with $\Phi_{\alpha_i, y_i}^{d, m}(Q) = 0$ for every i , or certify that none exists, in $\tilde{O}_{\text{params}}(n)$ field operations. The assertion holds in every characteristic, and costs $\tilde{O}_{\text{params}}(n \log q)$ bits over a finite field of size q supplied in a decoder-admissible presentation.

Algorithm 4 in Section E.1 records the corresponding minimal-basis procedure.

Proof. For each received position, use the full finite-dimensional target

$$\mathcal{R}_i = (K[T]/T^m) \otimes_K \text{span}_K \{V^u Y_1^{b_1} \cdots Y_d^{b_d} : u + b_1 + \cdots + b_d \leq B\}. \quad (97)$$

The substitution in Equation (13) has total degree at most B in $V, Y_1, \dots, Y_d$, so every local image lies in $\mathcal{R}_i$. More specifically, it separates the unknown univariate coefficient from its jet monomial:

$$\Phi_{\alpha_i, y_i}^{d, m}(f_j(X)M_j) = f_j(\alpha_i + T)\Phi_{\alpha_i, y_i}^{d, m}(M_j). \quad (98)$$

Thus multiplication by X on each copy of $K[T]/T^m$ is represented by the Jordan block with eigenvalue α_i . On $\mathcal{R} = \bigoplus_i \mathcal{R}_i$, the local constraints are exactly the single module relation

$$\sum_{j=1}^s f_j \cdot e_j = 0. \quad (99)$$

When several pairs have the same abscissa, we retain one summand $\mathcal{R}_i$ and one Jordan block for each pair. Equal eigenvalues do not merge their local constraints, and the block-diagonal Jordan representation used by the minimal-basis algorithm allows these repetitions. The module has dimension $\sigma = nm \binom{B+d+1}{d+1}$, with its multiplication matrix supplied in Jordan form. Forming all e_j costs $O_{d, m, B}(ns)$ field operations. Apply JNSV with shift $(w_1, \dots, w_s)$, normalized by subtracting its minimum. Their bound is

$$\tilde{O}(s^{\omega-1}(\sigma + \xi)), \quad \xi = \sum_j (w_j - \min_h w_h) = O_{\text{params}}(n). \quad (100)$$

Here $s \leq \binom{B+d+1}{d+1}$ is fixed, giving the claimed cost. Choosing a row of minimum shifted degree in the returned basis therefore finds a relation of minimum shifted degree among all nonzero relations. Its shifted degree lies below W exactly when $\mathcal{V}$ contains a nonzero polynomial satisfying all the local constraints, and then the returned Q is such a polynomial. The construction uses no division by the characteristic. $\square$

Every monomial support used by our decoder theorems has this form: after a jet monomial is fixed, its allowed X -powers are an initial interval. The coefficient/rank certificate guarantees that the minimum shifted degree is below W . This lemma concerns one fixed received word. The symbolic interpolation used for MCA instead controls degrees in the challenge and continues to use the polynomial-kernel argument of Section 3.7.

9.3 Agreement recovery in finite algebras

Suppose we have represented the candidate messages by a polynomial $C(X) \in (E[u]/G)[X]_{<k}$, where G is squarefree. Each root of G gives a candidate when substituted for u . To recover the close messages, we could try to find these roots and check the candidates individually. Received agreement gives us a more direct route: at position i , we split G by the residual $C(\alpha_i) - y_i$,

$$G_0 = \gcd(G, C(\alpha_i) - y_i), \quad G_1 = G/G_0$$

The roots of G_0 give precisely the candidates that agree there. We repeat this split while recording agreements on each piece. Once a piece has k agreements, we interpolate from those received values: they determine the same degree- $< k$ polynomial at every root of the piece. We have recovered its coefficients without finding any parameter root, and one check against all n positions decides whether to return it.

To use the same idea at positive order, we must also allow a second parameter whose equation can have repeated roots. Our families then have the form

$$\mathcal{A} = (E[u]/G(u))[v]/h(u, v), \quad C(X) \in \mathcal{A}[X]_{<k}, \quad (101)$$

where G is monic and squarefree and h is monic of bounded positive v -degree; h need not be fiberwise squarefree. A geometric point is a simultaneous choice of u, v in an algebraic closure satisfying $G = h = 0$, or equivalently an algebra map $\mathcal{A} \rightarrow \overline{E}$. Substituting these values in C gives the candidate at that point. The tower has dimension $\deg G \deg_v h$ over E , and the ordinary univariate representation $E[u]/G$ is the case $h = v$. The parameter-bounded boundary families in Section F instead come with multiplication tables. They may retain nilpotents.

The agreement test still asks whether a residual vanishes at a point, but we must take care with repeated roots. For example, v vanishes at the only point of $E[v]/v^2$, even though it is nonzero in the algebra. We want that point on the agreeing side of the split. Raising a residual e to $b = \deg_v h$ before taking the gcd achieves exactly this:

$$h_{\text{nil}} = \gcd(h, e^b \bmod h).$$

If e vanishes at a root of h , this power contains every copy of that root in h ; if it does not vanish, it contains none. We keep the full repeated factor on the agreeing side, where e is nilpotent, and its coprime complement on the other side, where e is a unit. Lemma E.8 implements this gcd over the squarefree base without factoring G . The recovery argument below needs only the resulting partition of points.

Lemma 9.4 (Agreement recovery from a finite family). *Let $1 \leq k \leq A \leq n$ and let the received pairs lie in $\mathbb{F}_q$. Let a collection of towers (101) have fiber degrees at most a fixed b and total dimension Δ . Once its coefficient representations are available, exactly the base-field polynomials represented by its geometric points that have at least A agreements can be recovered deterministically in $\tilde{O}_b((n+k)\Delta)$ operations in E , and $\tilde{O}(k\Delta)$ base-field coefficient comparisons. For arbitrary finite commutative E -algebras supplied by multiplication tables and of total dimension δ , the corresponding bound is $\tilde{O}(n\delta^{O(1)} + (n+k)\delta)$ operations in E . If the collection covers every wanted message, this is the full list. No parameter-root extraction or irreducible factorization is required.*

Proof. We first justify the split used by the algorithm. For towers, apply Algorithm 9. For a multiplication-table algebra $\mathcal{R}$ of dimension $\delta_{\mathcal{R}}$, let m_e be multiplication by the residual. The Fitting decomposition for this linear map stabilizes by exponent $\delta_{\mathcal{R}}$ and gives

$$\mathcal{R} = \ker(m_e^{\delta_{\mathcal{R}}}) \times \text{im}(m_e^{\delta_{\mathcal{R}}}).$$

Both summands are ideals. On the first, $e^{\delta_{\mathcal{R}}} = 0$, so the residual is nilpotent. On the second, multiplication by e is bijective, so the residual is a unit. Linear algebra computes the two ideals, their identities, and their coordinate projections. Thus both representations supply the same partition into agreeing and disagreeing geometric points, preserving total dimension.

The invariant is that every geometric point of a live component agrees at every position in its history. On a nilpotent branch, the residual maps to zero under every map to an algebraic closure; on a unit branch, it maps to a nonzero value. Once a component records k agreements, evaluate C at any of its geometric points. Vandermonde invertibility identifies the resulting degree- $< k$ polynomial with the unique interpolant $P \in \mathbb{F}_q[X]_{<k}$. This conclusion is geometric: $C - P$ can still have nilpotent coefficients in the component algebra. The algorithm therefore checks P against the full received word before returning it.

Agreement histories are persistent linked prefixes. When a component splits, both children share the old record, and only a nilpotent child allocates one new node. The k indices are materialized only when a

Algorithm 1 Exact recovery from finite algebraic candidate families

Require: finite algebras $\mathcal{A}$ carrying $C(X) \in \mathcal{A}[X]_{<k}$, represented as bounded-fiber towers or multiplication-table algebras, and α, y, k, A

Ensure: exactly the represented base-field polynomials with at least A agreements

```
1: procedure RECOVERAGREEMENT(FAMILIES,  $\alpha, y, k, A$ )
2:   materialize all  $e_i = C(\alpha_i) - y_i$  by shared multipoint evaluation
3:   LIVE  $\leftarrow \{(\mathcal{A}, \perp) : \mathcal{A} \in \text{FAMILIES}\}$ ; OUTPUT  $\leftarrow \emptyset$ 
4:   for  $i = 1, \dots, n$  do
5:     NEXT  $\leftarrow \emptyset$ 
6:     transfer the coordinate representation of  $e_i$  to all live factors
7:     for  $(\mathcal{R}, J) \in \text{LIVE}$  do
8:        $(N, \mathcal{U}) \leftarrow \text{SPLITNILUNIT}(\mathcal{R}, e_i \bmod \mathcal{R})$  ▷ Lemma 9.4
9:       for  $\mathcal{R}_0 \in N$  do
10:         $J_0 \leftarrow \text{EXTENDHISTORY}(J, i)$  ▷ persistent linked prefix
11:        if  $|J_0| = k$  then
12:          interpolate  $P \in \mathbb{F}_q[X]_{<k}$  through  $\{(\alpha_j, y_j) : j \in J_0\}$ 
13:          if  $P$  has at least  $A$  agreements then
14:            insert its  $k$  coefficients into OUTPUT
15:          end if
16:        else
17:          append  $(\mathcal{R}_0, J_0)$  to NEXT
18:        end if
19:      end for
20:      for  $\mathcal{R}_1 \in \mathcal{U}$  do
21:        append  $(\mathcal{R}_1, J)$  to NEXT
22:      end for
23:    end for
24:    LIVE  $\leftarrow \text{NEXT}$ 
25:  end for
26:  return DEDUPLICATE(OUTPUT)
27: end procedure
```

component is finalized, so splitting never copies an $O(k)$ list. The full check in [Algorithm 1](#) gives soundness, while every wanted point follows a nilpotent or unit branch until it records k of its at least A agreements. The batching and dimension accounting in [Lemma E.8](#) give the tower bound. For multiplication-table algebras, dense linear algebra and coordinate transfers cost $n\delta^{O(1)}$ because the live dimensions sum to at most δ . Coordinatewise multipoint evaluation, interpolation, full agreement checks, and deduplication contribute $\tilde{O}((n+k)\delta)$, giving the multiplication-table bound. Growing tower families continue to use the structured gcd decomposition of [Lemma E.8](#); the dense multiplication-table computation is needed only for algebras supplied in that representation. $\square$

9.4 Near-linear zeroth-order decoding

At order zero, fast Guruswami–Sudan interpolation and our shared agreement recovery give a deterministic near-linear decoder. The main point is to recover messages without extracting the roots of the initial univariate equation. We retain those roots together in a finite algebra and let received agreements determine the output coefficients. This use of received information builds on Augot–Pecquet and CHK [[AP00](#), [CHK26](#)]; fast

interpolation and lifting supply the arithmetic ingredients [JNSV17, NRS17].

Why order zero has its own route. The shared fast interpolation routine constructs the bounded- Y -degree GS equation. We then remove repeated factors globally, discard inseparable factors that contain no polynomial message, and use the resulting discriminant to choose one center regular for every retained branch. There is then no separant chain to traverse. The initial equation already defines a finite family, which we lift using the order-zero coefficient recurrence and pass to RECOVERAGREEMENT as a single bounded-dimensional algebra. We therefore need neither the first-order curve filter nor the higher-order polynomial-system solver; the decoding proof also does not use the separate list-size or MCA bounds.

Theorem 9.5 (Deterministic near-linear decoding above Johnson). *Fix $0 < \rho < 1$ and $\sqrt{\rho} < a \leq 1$. Over every finite field $\mathbb{F}_q$ supplied in a decoder-admissible presentation, with $q \geq n$ and unrestricted characteristic, the complete degree- $< k$ agreement list at threshold an can be computed deterministically in $\tilde{O}_{\rho,a}(n \log q)$ bit operations, for every prescribed set of n distinct evaluation points and $1 \leq k < n$ with $(k-1)/n \leq \rho$. The dependence on the positive Johnson gap is contained in the constant.*

The parameters used by the algorithm depend only on ρ, a . Choose positive integers m, ℓ satisfying

$$\sum_{j=0}^{\ell} (ma - j\rho)_+ > m(m+1)/2, \quad N_* := 2\ell m + 1. \quad (102)$$

Here $(x)_+ = \max\{x, 0\}$. The proof below establishes that such m, ℓ exist. The threshold N_* ensures enough centers in an extension of degree at most two; it imposes no characteristic restriction. For shorter words, interpolating from subsets of received positions gives an exact method.

Proof. We first construct the interpolation equation, then obtain one regular fiber containing the initial value of every wanted message. Lifting in its quotient algebra gives coverage; the shared recovery lemma supplies exactness.

Fast interpolation. For $k = 1$, sorting the received values and counting their frequencies returns exactly the sufficiently frequent constants in $\tilde{O}(n \log q)$ bits. Henceforth assume $k \geq 2$. With $\ell = \lfloor ma/\rho \rfloor$, the sum in (102) is $m^2 a^2 / (2\rho) + O_{\rho,a}(m)$. Since $a^2 > \rho$, taking m sufficiently large gives the required strict inequality. Fix these parameters for the rest of the proof. Put $A = \lceil an \rceil$ and $W = mA - 1$. The number of monomials with Y -degree at most ℓ and $(1, k-1)$ -weighted degree at most W is

$$\sum_{j=0}^{\ell} \max\{W - j(k-1) + 1, 0\} \geq n \sum_{j=0}^{\ell} (ma - j\rho)_+ > nm(m+1)/2.$$

Thus there are more coefficients than the $nm(m+1)/2$ multiplicity constraints. Apply Lemma 9.3 with jet monomials $1, Y, \dots, Y^\ell$, shift $(0, k-1, \dots, \ell(k-1))$, and strict cutoff $mA = W + 1$. It finds a nonzero row of weighted degree at most W in $\tilde{O}_{\rho,a}(n)$ field operations. For a wanted message P , every agreeing position is a zero of $Q(X, P(X))$ of multiplicity at least m . Its degree is at most $W < mA$, so $Q(X, P(X)) = 0$, the usual GS root identity [GS99].

If $n < N_*$, subset interpolation already returns exactly the list. Indeed, $k-1 \leq \rho n < an \leq A \leq n$, so every wanted polynomial has at least k agreeing positions. Enumerating their subsets and filtering the interpolants costs $\tilde{O}_{\rho,a}(\log q)$ bits for these bounded lengths. Henceforth $n \geq N_*$; this is a cardinality condition, not a bound on p .

Algorithm 2 Deterministic zeroth-order specialization at fixed Johnson gap

Require: $\mathbb{F}_q, \alpha, y, k, A$ as in [Theorem 9.5](#), $A = \lceil an \rceil$; fixed m, ℓ, N_* from (102)

Ensure: the exact degree- $< k$ agreement list at threshold A

```
1: procedure ZEROORDERDECODE( $\alpha, y, k, A$ )
2:   if  $k = 1$  then
3:     return the distinct values occurring at least  $A$  times in sorted  $y$ 
4:   end if
5:   if  $n < N_*$  then
6:     return EXACTBYSUBSETS( $\alpha, y, k, A$ )
7:   end if
8:    $W \leftarrow mA - 1$ 
9:    $Q \leftarrow \text{FASTFIXEDWORDEXPLAINER}(\alpha, y, 0, m, (1, Y, \dots, Y^\ell), (0, k - 1, \dots, \ell(k - 1)), mA)$ 
10:  Divide  $Q$  by the gcd of its coefficients in  $\mathbb{F}_q[X]$ 
11:   $T \leftarrow \text{SEPARABLEPART}(Q, \ell)$  ▷ Algorithm 7; preserves every polynomial message
12:  if  $\deg_Y T = 0$  then
13:    return  $\emptyset$ 
14:  end if
15:   $D \leftarrow \text{lc}_Y(T) \text{Res}_Y(T, \partial_Y T)$ ;  $M \leftarrow \deg D$ 
16:   $E \leftarrow \mathbb{F}_q$  if  $q > M$ ; otherwise construct the quadratic extension of Section E.7
17:  Evaluate  $D$  at  $M + 1$  distinct elements of  $E$  and choose  $c$  with  $D(c) \neq 0$ 
18:   $h(z) \leftarrow T(c, z)/\text{lc}_Y(T)(c)$ ;  $\mathcal{A} \leftarrow E[z]/h(z)$ 
19:   $F(Z) \leftarrow$  the order-zero lift with  $F(0) = z$  from Lemma E.5, truncated modulo  $Z^k$ 
20:  Take the degree- $< k$  representative of  $F$  and set  $C(X) \leftarrow F(X - c)$ 
21:  return RECOVERAGREEMENT( $\{(\mathcal{A}, C)\}, \alpha, y, k, A$ )
22: end procedure
```

One regular fiber for every message. Divide Q by its content as a polynomial in Y , namely the gcd of its X -polynomial coefficients. This removes only X -dependent factors. [Lemma E.4](#) first computes the bivariate radical, then removes factors inseparable in Y . Such a factor cannot be $Y - P(X)$, whose Y -derivative is 1, so every wanted message remains a solution of the resulting equation. The resulting primitive polynomial T divides the primitive Q and is separable over $\mathbb{F}_q(X)$. Thus $\deg_Y T \leq \ell$ and $\deg_X T \leq W < mn$ in every characteristic. If T has Y -degree zero, no wanted message exists.

Otherwise $D = \text{lc}_Y(T) \text{Res}_Y(T, T_Y)$ is nonzero, with degree $M \leq 2\ell mn$. The gcd, primitive part, and resultant computations cost $\tilde{O}_{\rho,a}(n)$ field operations: their polynomial matrices have bounded size and their entries have X -degree $O_{\rho,a}(n)$. We evaluate D simultaneously at $M + 1$ distinct centers and select the first nonzero value. If $q > M$, these centers lie in $\mathbb{F}_q$. Otherwise $q \leq M \leq 2\ell mn$, so the deterministic quadratic construction of [Section E.7](#) costs $\tilde{O}_{\rho,a}(n)$ field operations. The inequality $q^2 \geq n^2 > 2\ell mn \geq M$ supplies enough centers in the extension. Batched evaluation also costs $\tilde{O}_{\rho,a}(n)$ operations. This search finds a nonvanishing evaluation of D ; it does not find its roots.

At the chosen c , the polynomial $h(Y) = T(c, Y)/\text{lc}_Y(T)(c)$ has the same degree as T in Y and is squarefree. Thus $\mathcal{A} = E[z]/h$ has dimension at most ℓ . After extending scalars to an algebraic closure $\bar{E}$, this algebra is a product of $\deg h$ copies of $\bar{E}$: sending z to a root of h gives the corresponding projection. We perform arithmetic modulo h without constructing these projections or finding the roots.

Simultaneous lifting and exact recovery. Coprimality of h and $T_Y(c, Y)$ gives a Bézout identity, so $T_Y(c, z)$ is invertible in $\mathcal{A}$. We can therefore apply [Lemma E.5](#) at order zero: the equation for each new coefficient has

this same unit multiplier. The recurrence determines a unique F with $F(0) = z$ and $T(c + Z, F) = 0 \bmod Z^k$. It divides by no Taylor index or factorial, so it works in every characteristic after our separable preprocessing.

For every wanted P , the value $P(c)$ is a root of h . Projecting the lifted F along $z \mapsto P(c)$ gives the unique regular lift $P(c + Z) \bmod Z^k$. Taking the degree- $< k$ representative before shifting back therefore represents $P(X)$ exactly. Other projections may give truncations that do not solve T as polynomial identities; we only require their degree- $< k$ representatives and check actual agreement. Apply [Lemma 9.4](#) to this one univariate algebra, using its ordinary-univariate case. Since $\dim_E \mathcal{A} \leq \ell$ and $k \leq A \leq n$, recovery costs $\tilde{O}_{\rho,a}(n)$ operations and returns exactly the base-field messages, also when the center field E is quadratic.

Bit complexity. All polynomial arithmetic above has degree $O_{\rho,a}(n)$ over a field or an algebra of bounded dimension. Computing the shifted coefficients of $T(c + Z, Y)$, evaluating the recurrence through degree $k - 1$, and shifting the resulting representative back each take $\tilde{O}_{\rho,a}(n)$ field operations. Division-free Taylor shifts apply even if the X -degrees exceed the characteristic ([Section E.3](#)). Each operation in $\mathcal{A}$ uses $O_{\rho,a}(1)$ operations in E . Arithmetic in the supplied presentation and its possible quadratic quotient extension costs $\tilde{O}(\log q)$ bits per operation. The inverse-Frobenius calls in preprocessing cost $\tilde{O}_{\rho,a}(n \log q)$ bits in total by [Definition 9.1](#) and [Lemma E.4](#). Input processing, agreement recovery, and output comparisons have the same total bound, proving $\tilde{O}_{\rho,a}(n \log q)$ bit operations. $\square$

For the table's exact threshold $a = \sqrt{(k - 1)/n} + \eta_0$, write $\rho = k/n$. Since $\sqrt{\rho} - \sqrt{\rho - 1/n} \leq 1/(n\sqrt{\rho})$, lengths $n \geq 2/(\eta_0\sqrt{\rho})$ have gap at least $\eta_0/2$ above $\sqrt{\rho}$. Apply the theorem at that smaller threshold and filter by A ; shorter lengths use bounded subset interpolation. This gives $\tilde{O}_{\rho,\eta_0}(n \log q)$ as stated in the table.

Lemma 9.6 (Bounded message dimension at fixed agreement). *Fix $K_0 \geq 1$ and $0 < a_0 \leq 1$. For $k \leq K_0$ and $k \leq A \leq n$ with $A \geq a_0 n$, the exact agreement list can be recovered over every decoder-admissibly presented field with $q \geq n$, in every characteristic, in $\tilde{O}_{K_0,a_0}(n \log q)$ bit operations.*

Proof. We use the ordinary decoder once the length is large enough that the bounded message dimension lies below a fixed Johnson threshold. If $n \geq 4K_0/a_0^2$, then

$$\frac{k - 1}{n} \leq \frac{a_0^2}{4} < a_0^2.$$

Apply [Theorem 9.5](#) at surrogate rate $a_0^2/4$ and agreement a_0 , then retain only outputs with at least A agreements. The remaining lengths are bounded in terms of K_0, a_0 and use subset interpolation. Constant codes can also be recovered by frequency counting. $\square$

9.5 Positive-order Taylor charts and tower recovery

With derivatives, we must retain several initial values instead of the single value used by the ordinary decoder. We construct the later Taylor coefficients as rational functions of these initial values, so one family represents every message whose separant is nonzero at the chosen center. Agreement will then reduce this family to the finite algebras accepted by [Algorithm 1](#): a curve filter handles first order, and isolated polynomial systems handle higher order. We construct the family directly from the explainer, using no factorization from the separate counting proof.

Fix an expansion center c and a separant-chain equation $T(X, Y_0, \dots, Y_r)$ of total jet degree at most B_{jet} . The initial jet $U = (U_0, \dots, U_r)$ of a prospective message must satisfy

$$H(U) := T(c, U) = 0, \quad S(U) := T_{Y_r}(c, U) \neq 0.$$

The nonzero partial derivative S , called the separant, permits the equation to determine later Taylor coefficients. For example, at first order the coefficient of Z in $T(c + Z, U_0 + U_1Z + wZ^2 + \dots, U_1 + 2wZ + \dots)$ is

$$T_X(c, U) + T_{Y_0}(c, U)U_1 + 2S(U)w.$$

Setting it to zero determines w . Kopparty's recurrence continues this calculation [Kop15]. We retain the initial values as variables, as in earlier symbolic solution families [rs-26, Jer26].

Assume $r < k \leq n$ and $p > \max\{k - 1, B_{\text{jet}}\}$. We truncate at the actual message bound k , even if interpolation used a larger ambient bound K . The resulting family has the form

$$C_U(c + Z) = \sum_{j < k} c_j(U)Z^j, \quad c_j(U) = \frac{N_j(U)}{B_0(U)}, \quad c_j(U) = U_j \ (j \leq r). \quad (103)$$

Every wanted polynomial whose separant is nonzero at this initial jet equals its own truncation. Other points need not give polynomial solutions of T : if a truncation has the required agreements, it is already a valid degree- $< k$ decoding output. Thus no equations setting the coefficients of degrees $k, \dots, K - 1$ to zero are needed.

To represent this family computationally, retain the reduced union H_* of the components of $H = 0$ on which S is not identically zero. An invertible linear change of the initial coordinates,

$$U = M(x, z), \quad x = (x_1, \dots, x_r),$$

makes the equation of H_* monic in the last coordinate z ; call that equation $h(x, z)$. The valid open set is still determined by the original separant, now represented by

$$s(x, z) = S(M(x, z)) \bmod h.$$

In general s is not h_z : the coordinate change makes the equation monic, while s records the partial derivative that makes the Taylor recurrence solvable. The coefficient formulas and agreement equations below are transported through the same map M .

Algorithm 8 constructs this entire family in $\widetilde{O}_{r, B_{\text{jet}}}(k^{r+1} + \deg_X T)$ field operations. It returns $h(x, z)$, $s(x, z)$, and the numerators in (103); one may take

$$B_0 = s^{2k} \bmod h, \quad \deg N_j, \deg B_0 \leq L := 1 + 2k(B_{\text{jet}} - 1).$$

It inverts only the separant, so its formulas cover every point with $s \neq 0$, including ramified projection fibers. To make Kopparty's recurrence fast, we cache products of completed coefficient blocks using relaxed series arithmetic [vdH02]. The appendix proves the resulting cost over the truncated parameter algebra, where the coefficients remain symbolic.

Agreement at position i is now the polynomial equation

$$g_i(x, z) := \sum_{j < k} N_j(x, z)(\alpha_i - c)^j - y_i B_0(x, z) = 0. \quad (104)$$

All these polynomials have total degree at most $L = O_{B_{\text{jet}}}(n)$. Multipoint evaluation of the scalar coefficient polynomials constructs them together in $\widetilde{O}_{r, B_{\text{jet}}}(n^{r+1})$ operations. We always retain the condition $s \neq 0$: clearing a common denominator can otherwise create spurious agreement along $s = 0$.

Why the chart loop covers all solutions. Assign a wanted polynomial to the first stage T_j for which $T_j(X, P, \dots) = 0$ but $S_j(X, P, \dots)$ is not identically zero. Such a stage exists: each differentiation lowers total jet degree, and the chain ends in a nonzero polynomial in X , which has no polynomial solutions. The nonzero separant specialization has degree at most H_j , so it is nonzero at one of the $H_j + 1$ prescribed centers. Algorithm 8 retains its component. This is the constructive content of (122) and Lemmas A.1 and E.6. The precise center fields and bounded-instance dispatch are given in Sections E.7 and E.8.

9.6 First-order recovery from a curve

At first order, the two initial values $P(c), P'(c)$ lie on a curve. We use the received word to cut this curve down to a finite family: every wanted point must satisfy many agreement equations that do not hold identically on its component. We then arrange to use one center for all wanted messages, recovering the states where the separant vanishes at that center from bounded finite boundary families.

Theorem 9.7 (First-order reconstruction with an explicit agreement gap). *Fix $B_{\text{jet}}, C \geq 1$. Let $\mathbb{F}_q$ be supplied in a decoder-admissible presentation with $q \geq n$, $1 \leq k < n$, $k \leq A \leq n$, and $t = A - k + 1$. If $k \geq 2$, assume $p > \max\{k - 1, B_{\text{jet}}\}$. Given distinct evaluation points $\alpha_i \in \mathbb{F}_q$ and a word $y \in \mathbb{F}_q^n$, suppose a supplied nonzero $Q \in \mathbb{F}_q[X, Y_0, Y_1]$ has $\deg_X Q \leq Cn$, $\deg_{Y_0, Y_1} Q \leq B_{\text{jet}}$, and satisfies $Q(X, P, P') = 0$ for every degree- $< k$ polynomial with at least A agreements. Then the exact list, as duplicate-free coefficient vectors, is computable deterministically in*

$$\tilde{O}_{B_{\text{jet}}, C} \left(\left(n^2 + \frac{n^3}{t} \right) \log q \right)$$

bit operations. In particular, for $t \geq \varepsilon n$ the bound is $\tilde{O}_{B_{\text{jet}}, C}((1 + \varepsilon^{-1})n^2 \log q)$. Constructing Q from a fixed certified interpolation support also fits this bound. At $A = k$, the assertion remains conditional on the supplied bounded-degree explainer; its degree bounds can depend on the interpolation gap. Over prime fields $q \geq n$, the characteristic hypothesis can be weakened to $p > k - 1$.

9.6.1 Recovering a finite candidate family from a curve

The first step assumes a center at which the transformed separant is nonzero and the Taylor recurrence gives a curve of candidates. We use the received agreements to cut that curve down to a finite family before choosing a center that works for all wanted messages.

Lemma 9.8 (Agreement recovery on a bounded-degree curve). *Let $h(u, v) \in E[u, v]$ be reduced, monic of v -degree b , and of total degree at most B_{jet} . Suppose a degree- $< k$ message family on the open set $h = 0$, $s \neq 0$ has coefficients $N_j(u, v)/B_0(u, v)$, where B_0 is invertible on this open set, $\deg_v N_j < b$, and the numerator and denominator degrees are $O_{B_{\text{jet}}}(k)$. The degrees of h, s are bounded in terms of B_{jet} . Assume $2 \leq k \leq A \leq n$ and that the coefficient map*

$$(u, v) \mapsto (N_0/B_0, \dots, N_{k-1}/B_0)$$

is nonconstant on every geometric irreducible component of the open curve. Put $t = A - k + 1$. Its base-field polynomial images with at least A received agreements can be recovered exactly in

$$\tilde{O}_{B_{\text{jet}}}(k^2 + nk + n^2 k/t)$$

operations over E , provided the coefficient polynomials are supplied. Here E is the base field or the affordable quadratic extension from [Section E.7](#).

Both first-order routes use the original Taylor family, where $c_0 = U_0$ and $c_1 = U_1$. These coefficients recover (u, v) through the invertible coordinate change, so they cannot both be constant on a curve component. This verifies the hypothesis on the polynomial image and explains why we handle constant codes before entering the first-order routine.

We will retain a parameter only when at least t different agreement polynomials detect it. This distinction prevents ramification or repeated roots in one residual from being counted as several received positions. The resulting parameter algebra has dimension $O_{B_{\text{jet}}}(nk/t)$; the final agreement check still removes parameters whose different branches contributed different positions.

The curve filter therefore needs a polynomial that records how many distinct received positions detect a parameter. A product of the residual polynomials does not do this: one residual can contribute a large multiplicity by itself. The following threshold construction instead sorts all factor valuations at once.

Lemma 9.9 (Polynomial threshold). *Let K be a field, let $f_1, \dots, f_n \in K[U]$ be nonzero monic polynomials, and put $L = \sum_i \deg f_i$. For every $1 \leq t \leq n$, one can construct a nonzero monic polynomial H_t in $\tilde{O}(L + n)$ operations in K such that, for every $\beta \in \bar{K}$,*

$$H_t(\beta) = 0 \iff \#\{i : f_i(\beta) = 0\} \geq t.$$

Moreover, $t \deg H_t \leq L$.

Proof. Pad the input with copies of 1 to the next power of two and run a fixed bitonic sorting network [Bat68]. We orient every comparator from its smaller-output wire to its larger-output wire and replace it by

$$(f, g) \mapsto (\gcd(f, g), \text{lcm}(f, g)), \quad \text{lcm}(f, g) = fg / \gcd(f, g),$$

with both outputs monic. For each irreducible $\phi \in K[U]$, this comparator acts on valuations by

$$(v_\phi(f), v_\phi(g)) \mapsto (\min\{v_\phi(f), v_\phi(g)\}, \max\{v_\phi(f), v_\phi(g)\}).$$

The network therefore sorts the ϕ -valuations for every irreducible ϕ at once. Let H_t be the output on the t -th largest wire. Its ϕ -valuation is positive exactly when at least t different inputs contain ϕ , and it satisfies

$$tv_\phi(H_t) \leq \sum_i v_\phi(f_i).$$

The first assertion follows after passing to $\bar{K}$ and taking $\phi = U - \beta$; summing the displayed inequality over the irreducible factors gives $t \deg H_t \leq L$. Notice that repeated factors within one input never increase the number of positions being counted.

A comparator preserves total degree because

$$\deg \gcd(f, g) + \deg \text{lcm}(f, g) = \deg f + \deg g.$$

Thus each network layer has total degree L . Fast gcds, exact divisions, and products cost $\tilde{O}(L + n)$ per layer, and the $O(\log^2 n)$ layers are absorbed by the soft notation. No factorization is computed. $\square$

Proof of Lemma 9.8. Write the family as $C_{u,v}(X) = \sum_{j < k} N_j(u, v)(X - c)^j / B_0(u, v)$. Multipoint evaluation in X constructs the agreement polynomials

$$g_i(u, v) = \sum_{j < k} N_j(u, v)(\alpha_i - c)^j - y_i B_0(u, v)$$

in $\tilde{O}_{B_{\text{jet}}}(k^2 + nk)$ operations, including monic reduction modulo h . The received points are distinct and lie in the base field. The transformed denominator s and h have no common curve component; remove such components first if necessary, since their open part is empty. We keep all projection fibers, including ramified fibers. The finite-algebra localization later removes the points where $s = 0$.

Why universal agreements need no separate cases. Fix a geometric curve component D . If k agreement polynomials vanished identically on D , interpolation at those k distinct evaluation points would make every coefficient of $C_{u,v}$ constant on D . This contradicts the hypothesis on the coefficient map. Hence at most $k - 1$ received positions agree identically on D . We use this bound without finding those positions or partitioning the curve by their agreement histories.

Algorithm 3 Filter first-order curve candidates by distinct agreements

Require: first-order chart $(h, s, B_0, N_0, \dots, N_{k-1}, g_1, \dots, g_n)$ satisfying [Lemma 9.8](#), and A, k

Ensure: finite families covering every polynomial image with at least A agreements

```
1: procedure FIRSTORDERCURVECANDIDATES( $\mathcal{C}, A, k$ )
2:    $t \leftarrow A - k + 1$ 
3:    $h \leftarrow h / \gcd_{E(u)[v]}(h, s)$ ; return  $\emptyset$  if  $h = 1$ 
4:   for  $i = 1, \dots, n$  do
5:      $\chi_i(u, W) \leftarrow \text{Res}_v(h(u, v), W - g_i(u, v))$ 
6:      $v_i(u) \leftarrow$  the monic normalization of the lowest nonzero  $W$ -coefficient of  $\chi_i$ 
7:   end for
8:    $H_t \leftarrow \text{POLYNOMIALTHRESHOLD}(v_1, \dots, v_n; t)$  ▷ Lemma 9.9
9:   if  $H_t = 1$  then
10:    return  $\emptyset$ 
11:  end if
12:   $G \leftarrow \text{RADICAL}(H_t)$  ▷ Lemma E.9
13:   $\mathcal{T} \leftarrow \text{LOCALIZEFIBER}((E[u]/G)[v]/h, s)$  ▷ Algorithm 9
14:  invert  $B_0$  on every retained tower and materialize  $C(X)$ 
15:  return  $\mathcal{T}$ 
16: end procedure
```

How one coefficient detects a nonuniversal agreement. For each position, form the characteristic polynomial

$$\chi_i(u, W) = \text{Res}_v(h(u, v), W - g_i(u, v)).$$

This is the characteristic polynomial of multiplication by g_i in the free $E[u]$ -module $E[u, v]/h$. Let $v_i(u)$ be its lowest nonzero coefficient in W , normalized to be monic. In particular, $v_i = 1$ if g_i vanishes on the whole curve and $\chi_i = W^b$.

To see why this coefficient is the right one, factor h only conceptually over an algebraic closure. Let $h_i^{(0)}$ be the product of the geometric components on which g_i vanishes identically, and let $h_i^{(1)}$ contain the others. These products are Galois-stable and descend, after monic normalization, to $E[u, v]$. Multiplicativity of the resultant gives

$$\chi_i(u, W) = W^{\deg_v h_i^{(0)}} \text{Res}_v(h_i^{(1)}, W - g_i).$$

The second factor has a nonzero constant coefficient as a polynomial in $E[u][W]$. Up to a nonzero scalar, that coefficient is $\text{Res}_v(h_i^{(1)}, g_i)$, and hence it is v_i . If (β, γ) lies on a component of $h_i^{(1)}$ and $g_i(\beta, \gamma) = 0$, specialization makes this resultant vanish. Monicity of $h_i^{(1)}$ makes specialization legitimate. The argument therefore also covers ramified fibers and intersections of curve components. We use the factorization only for this proof; the algorithm extracts v_i directly from χ_i .

The Sylvester matrices have bounded size and entries of u -degree $O_{B_{\text{jet}}}(k)$. Thus $\deg v_i = O_{B_{\text{jet}}}(k)$, and all n characteristic polynomials cost $\tilde{O}_{B_{\text{jet}}}(nk)$ operations.

Threshold and recovery. Let (β, γ) represent a wanted polynomial, and let D be a curve component through it. Of its at least A agreements, at most $k - 1$ are universal on D . The preceding argument therefore makes β a root of at least $t = A - k + 1$ different polynomials v_i . Apply [Lemma 9.9](#) to obtain H_t , then take the single characteristic-safe radical $G = \text{rad}(H_t)$. If $L = \sum_i \deg v_i$, the retained base satisfies

$$\deg G \leq \deg H_t \leq L/t = O_{B_{\text{jet}}}(nk/t).$$

Over the reduced base $E[u]/G$, Algorithm 9 removes the full primary factors above the geometric zeros of s and retains the nilpotents everywhere else. The resulting collection of algebras has total dimension $\Delta = O_{B_{\text{jet}}}(nk/t)$, and s is a unit on each piece. Reduce the coefficient polynomials into these algebras and invert B_0 to materialize the k message coefficients.

Different branches above the same u can contribute different agreements. We therefore keep every retained branch and let Algorithm 1 perform the exact test. Every wanted point is retained by the threshold test and localization, which proves completeness; the final base-field agreement check proves soundness. The threshold and radical cost $\tilde{O}_{B_{\text{jet}}}(nk)$, while materialization and recovery cost

$$\tilde{O}_{B_{\text{jet}}}(k^2 + (n+k)\Delta + nk) = \tilde{O}_{B_{\text{jet}}}(k^2 + nk + n^2k/t).$$

This proves the claimed bound and exactness. $\square$

The preceding calculation works once the chosen center makes the Taylor recurrence available for the message under consideration. We now choose one center that covers all close messages, without repeating the curve calculation at $O(n)$ centers.

9.6.2 Choosing one center for all close messages

Why normalized coordinates matter. One center need not work in the original coordinates. For example, $Q = Y_1^2 - Y_0$ explains $P_c(X) = (X - c)^2/4$, but its separant $2P'_c = X - c$ vanishes at c . On the nonconstant branch, writing $Y_0 = z^2$ and $Y_1 = z$ gives $z' = 1/2$, which has no pole even at $z = 0$. Here the characteristic is greater than two; the constant solution belongs to the ordinary tail. The failure of the original recurrence was a failure of coordinates, not of the solution.

We first use an ordinary equation to recover the solutions whose separant vanishes identically after substitution into the differential equation. For the remaining curve we enlarge its coordinate algebra by the rational functions satisfying monic polynomial equations over a chosen parameter ring. This finite enlargement is its *normalization*; the monic equations will prevent the new coordinates from acquiring poles along a wanted message at a good center. To determine their later coefficients, we need denominators for their derivatives that are nonzero at the initial state. An ideal identity proves that the available denominators cannot all vanish along a wanted message. On the finite boundary, we split successively according to whether each denominator is nilpotent or a unit, and use that denominator's existing rational rule on its unit factor. This preparation produces one polynomial $H(X)$ of degree $O_{B_{\text{jet}},C}(n)$: every center with $H(c) \neq 0$ works simultaneously for all remaining wanted messages. Testing centers requires only polynomial evaluation; Taylor reconstruction is performed once on the open curve and on a parameter-bounded collection of finite boundary factors. Section F proves the preparation using finite linear algebra and univariate module arithmetic, without factoring the curve or computing a Gröbner basis.

To prove coverage, take a reduced branch V whose separant V_Z is nonzero as a polynomial and satisfies $\gcd(V, V_Z) = 1$, put $K = \mathbb{F}_q(X)$ and $O = K[Y, Z]/(V)$, and let N be its componentwise normalization. On the total quotient algebra, the differential equation defines

$$DY = Z, \quad DZ = -\frac{V_X + ZV_Y}{V_Z}. \quad (105)$$

Choose the finite separable coordinate u and a free $K[u]$ -basis $e_1, \dots, e_b$ of N supplied by the appendix, and set

$$a_0 = u, \quad a_i = e_i \quad (1 \leq i \leq b), \quad f_i = Da_i \quad (0 \leq i \leq b). \quad (106)$$

The same computation supplies a polynomial $g \in K[u]$ such that

$$N_g = O_g, \quad (V_Z)^{-1} \in O_g, \quad f_i \in g^{-2}N \quad (0 \leq i \leq b), \quad (107)$$

Define the fractional N -ideal and its inverse by

$$J = N + \sum_{i=0}^b N f_i, \quad I = (N : J) = \{c \in N : c f_i \in N \text{ for every } 0 \leq i \leq b\}. \quad (108)$$

Componentwise, N is a Dedekind domain. The nonzero fractional ideal J is therefore invertible and $I = J^{-1}$, whence

$$IJ = N. \quad (109)$$

Equivalently, localize at every height-one prime, where a fractional ideal is principal, and intersect the local equalities inside the total quotient algebra [Sut21, Corollary 3.3].

The appendix computes generators $c_1, \dots, c_m$ of I and records

$$b_{\ell i} := c_{\ell} f_i \in N \quad (0 \leq i \leq b). \quad (110)$$

It also supplies $d_{\ell,*}, d_{\ell i} \in N$ satisfying

$$1 = \sum_{\ell=1}^m c_{\ell} \left(d_{\ell,*} + \sum_{i=0}^b d_{\ell i} f_i \right) = \sum_{\ell} c_{\ell} d_{\ell,*} + \sum_{\ell,i} d_{\ell i} b_{\ell i}. \quad (111)$$

The identity is the semantic reason that one of the derivative denominators must be nonzero at a wanted state; the appendix proves that all of this data is computable within the claimed bounds.

Lemma 9.10 (Common-center preparation). *Let $V \in \mathbb{F}_q[X, Y, Z]$ have positive Z -degree, X -degree less than N_X , and total (Y, Z) -degree at most B , and satisfy*

$$\gcd(V, V_Z) = 1 \quad \text{in } K[Y, Z], \quad K = \mathbb{F}_q(X).$$

Fix $k \geq 2$ and assume $p > \max\{k-1, \kappa(B)\}$, where $\kappa(B) = 2(B+1)^3$. In $\widetilde{O}_B(N_X)$ operations in $\mathbb{F}_q$, one can compute the normalized module and multiplication table, original open chart, denominators c_{ℓ} , products $b_{\ell i}$, and coverage certificate constructed above, together with a nonzero $H \in \mathbb{F}_q[X]$ of degree $O_B(N_X)$. Consider any extension $E/\mathbb{F}_q$, center $a \in E$ with $H(a) \neq 0$, and polynomial $P \in \mathbb{F}_q[X]$ of degree less than k satisfying

$$V(X, P, P') = 0, \quad V_Z(X, P, P') \neq 0. \quad (112)$$

Here the second condition means that the resulting polynomial in X is not identically zero. The normalized state coordinates $a_i(P)$ and their derivatives $f_i(P)$ have no poles at a . At that center, either $g(u(P))(a) \neq 0$ and the original open chart contains the initial state, or some $c_{\ell}(P)(a) \neq 0$ and the ℓ -th rational rule $Da_i = b_{\ell i}/c_{\ell}$ is defined there. The same a works for every such P .

For such a center, write g_a and $c_{\ell}(a)$ for the specializations of the recorded g and c_{ℓ} . One can construct possibly nonreduced multiplication-table algebras $\mathcal{U}_{\ell}$ of total dimension at most $b \deg g = O_B(1)$, with $c_{\ell}(a)$ a unit on each nonzero $\mathcal{U}_{\ell}$, and polynomials $C_{\ell}(X) \in \mathcal{U}_{\ell}[X]_{<k}$ obtained from the corresponding rational rules $Da_i = b_{\ell i}/c_{\ell}$. These families represent every wanted message whose initial state lies on $g_a = 0$. Their construction costs $\widetilde{O}_B(N_X + k)$ operations in E , and agreement recovery from them costs $\widetilde{O}_B(n + k)$ further operations. In the decoder-admissible field presentations used below, the corresponding total bit cost is $O_B((N_X + n + k) \log q)$.

Proof. The finite normalization and boundary-factor computations are given in Section F.2. That construction clears the coefficient denominators and records the module, multiplication, inverse-chart, and coverage identities before specializing them at a . It remains to prove that these specialized certificates cover every wanted message.

Let P satisfy (112). The evaluation $Y \mapsto P, Z \mapsto P'$ gives a K -point of the coordinate algebra O . At its prime, V_Z is nonzero, so the plane curve is smooth there and the local ring equals its normalization. Evaluation therefore extends to the relevant component of N and sends every normalized basis element e_i to an element of K . After applying this map, the cleared monic equation for e_i supplied by the appendix construction has coefficients in $\mathbb{F}_q[X, H^{-1}]$. This localization of the PID $\mathbb{F}_q[X]$ is integrally closed, hence

$$e_i(P) \in \mathbb{F}_q[X, H^{-1}]. \quad (113)$$

Differentiating $V(X, P, P') = 0$ and using $V_Z(X, P, P') \neq 0$ identifies the value of DZ in (105) with P'' . Thus evaluation commutes with D on $O_{V_Z} = N_{V_Z}$, which contains the normalized coordinates and their derivatives. Differentiation preserves $\mathbb{F}_q[X, H^{-1}]$, and consequently

$$f_i(P) = \frac{d}{dX} a_i(P) \in \mathbb{F}_q[X, H^{-1}]. \quad (114)$$

The coordinate $a_0(P) = u(P, P')$ and its derivative are already polynomials. The recorded coefficients $c_\ell, b_{\ell i}, d_{\ell,*}, d_{\ell i}$ are $\mathbb{F}_q[X, H^{-1}]$ -linear combinations of the normalized basis elements; the construction includes their coefficient denominators in H . They therefore have no poles wherever H is nonzero. Thus every term of the ideal identity (111) has a value at a . Specializing that identity forces some $c_\ell(P)(a) \neq 0$; otherwise its right side would vanish. This proves denominator coverage, including states where fractional coordinate formulas display $0/0$. When $g(u(P))(a) \neq 0$, the identity $N_g = O_g$ already gives the original chart.

Suppose instead that the initial state lies on $g_a = 0$. In the residue field of its geometric point, each $c_\ell(a)$ is either zero or nonzero. Following the successive Fitting decompositions, the point stays in the nilpotent factor until the first index for which $c_\ell(P)(a) \neq 0$, and then enters $\mathcal{U}_\ell$. On this factor the corresponding rational rule has a unit denominator. The recurrence is therefore defined and coefficientwise unique through degree $k - 1$, so its Y -coordinate recovers P . The final multiplication-table agreement recovery removes any other initial states that do not produce close base-field polynomials. $\square$

9.6.3 Assembling the decoder

The appendix records the resulting one-center algorithm: it separates the solutions whose separant vanishes identically into an ordinary equation, applies the Taylor curve on the open set where the transformed separant is nonzero, and adds the finite boundary families from Lemma 9.10. The next proposition combines these pieces and proves their cost.

Proposition 9.11 (Normalized quadratic first-order reconstruction). *Fix $B_{\text{jet}}, C \geq 1$, let $2 \leq k \leq A \leq n$, and put $t = A - k + 1$. Let a supplied nonzero $Q \in \mathbb{F}_q[X, Y_0, Y_1]$ explain every wanted message, with $\deg_X Q \leq Cn$ and total jet degree at most B_{jet} . Assume a decoder-admissibly presented field with $q \geq n$ and $p > \max\{k - 1, \kappa(B_{\text{jet}})\}$, where $\kappa(B) = 2(B + 1)^3$. Put $N_X = 1 + \deg_X Q$. On the decoder's large branch, where the base field or the quadratic extension from Section E.7 contains more than $\deg H$ elements, Algorithm 10 returns the exact wanted list in*

$$\tilde{O}_{B_{\text{jet}}, C} \left(N_X + nk + \frac{n^2 k}{t} \right) \quad (115)$$

operations over that field, and hence the same expression times $\tilde{O}(\log q)$ in the supplied-presentation bit model. The treatment of bounded lengths makes the field condition effective with constants depending only on (B_{jet}, C) .

Proof. We first separate the messages covered by an ordinary equation. Write $Y = Y_0$, $Z = Y_1$, and work over $K = \mathbb{F}_q(X)$. After removing the X -content of Q , let V be its reduced part in $K[Y, Z]$, and put

$$W = \gcd(V, V_Z), \quad V_{\text{reg}} = V/W.$$

Let $R \in \mathbb{F}_q[X, Y]$ be the primitive denominator-cleared form of

$$\begin{cases} W, & \deg_Z V_{\text{reg}} = 0, \\ W \text{Res}_Z(V_{\text{reg}}, (V_{\text{reg}})_Z), & \deg_Z V_{\text{reg}} > 0. \end{cases}$$

Since the state degree is less than the characteristic, the gcd with the two state partial derivatives removes each multiplicity exactly. Every irreducible factor of W has zero Z -derivative, and its degree is less than p , so W is independent of Z . The resultant in Algorithm 10 is nonzero. If a solution of $V = 0$ does not lie on $W = 0$ but has identically zero separant, it satisfies this resultant. Thus the ordinary branch and the branch with nonzero separant polynomial cover every solution.

The gcds, primitive reductions, and resultant cost $\tilde{O}_{B_{\text{jet}}}(N_X)$. The ordinary equation R has bounded message degree and X -degree $O_{B_{\text{jet}}}(N_X)$, but can have repeated factors. Remove its X -content, apply Lemma E.4, then use the regular-fiber and recurrence steps of Algorithm 2 and Lemma 9.4. These cost $\tilde{O}_{B_{\text{jet}}}(N_X + nk)$; no Johnson hypothesis is needed once R is supplied.

For the branch with nonzero separant polynomial, Lemma 9.10 and batched evaluation of the nonvanishing polynomial H cost $\tilde{O}_{B_{\text{jet}}}(N_X)$. Apply Lemma E.6 with $r = 1$ to V_{reg} at a . Its first two coefficients are Y and Z , so its polynomial image is nonconstant on every curve component. The filter of Lemma 9.8 therefore applies and gives total construction and recovery cost $\tilde{O}_{B_{\text{jet}}}(N_X + k^2 + nk + n^2k/t)$.

Every remaining wanted state has $(V_{\text{reg}})_Z(a) = 0$, hence $g_a = 0$ by (107). We recover these states from the possibly nonreduced boundary factors of Lemma 9.10 in $\tilde{O}_{B_{\text{jet}}}(N_X + n + k)$ operations. That lemma proves coverage, and the shared agreement routine checks and deduplicates the outputs of all three parts. Since $k \leq n$, summing these costs gives (115).

The field construction and bit conversion are those of Sections E.7 and E.8. Because $N_X = O_C(n)$, affordable quadratic extensions suffice outside parameter-bounded lengths; the latter use subset interpolation. This includes the charged inverse-Frobenius operations over the supplied base presentation and the constructed bounded-degree quotient extensions. $\square$

Proof of Theorem 9.7. Constant codes use frequency counting. Put $\kappa(B_{\text{jet}}) = 2(B_{\text{jet}} + 1)^3$. Handle the parameter-bounded field and length exceptions by Section E.8. For $p > \kappa(B_{\text{jet}})$, apply Proposition 9.11, whose cost is

$$\tilde{O}_{B_{\text{jet}}, C}\left((N_X + nk + n^2k/t) \log q\right), \quad N_X = 1 + \deg_X Q.$$

Since $N_X = O_C(n)$ and $k \leq n$, this gives the claimed bound.

If instead $B_{\text{jet}} < p \leq \kappa(B_{\text{jet}})$, then $p > k - 1$ implies $k \leq \kappa(B_{\text{jet}})$. Use the separant chain and center cover of Section 9.5. Retain the message dimension in the Taylor bound: Lemma E.6 costs $\tilde{O}_{B_{\text{jet}}}(N_X + k^2)$ per center, and its coefficient numerators have degree $O_{B_{\text{jet}}}(k)$. The curve filter in Lemma 9.8 therefore costs $\tilde{O}_{B_{\text{jet}}}(nk + k^2 + n^2k/t)$ per center. Ordinary stages use a bounded-dimensional family and fit the same bound. There are $O_{B_{\text{jet}}, C}(n)$ center-stage pairs. Since k is now bounded in terms of B_{jet} , their total is $\tilde{O}_{B_{\text{jet}}, C}(n^2 + n^3/t)$ field operations. The field constructions and treatment of bounded lengths are those of Sections E.7 and E.8; they do not infer bounded block length from bounded characteristic. Finally, Lemma 9.3 constructs a fixed-support explainer in soft-linear time, and Section E.7 supplies the bit conversion, including inverse Frobenius over extension fields. For the final prime-field assertion, failure of $p > B_{\text{jet}}$ implies $n \leq q = p \leq B_{\text{jet}}$, so subset interpolation has parameter-bounded cost. $\square$

Corollary 9.12 (Deterministic quadratic decoding above the first-order curve). *Fix $0 < \rho < a < 1$ with $a > a_1(\rho)$. Let $n \geq 2$, $1 \leq k \leq \rho n$, and $an \leq A \leq n$ be integers. Over a decoder-admissibly presented field $\mathbb{F}_q$ with $q \geq n$ and $p > k - 1$ when $k \geq 2$, the complete degree- $< k$ agreement list at threshold A is computable deterministically in*

$$\tilde{O}_{\rho,a}(n^2 \log q)$$

bit operations. Constant codes need no characteristic restriction.

Proof. Choose the interpolation parameters of [Theorem 5.7](#); their degree bounds are bounded uniformly in n, k in terms of (ρ, a) . When $p > B_{\text{jet}}$, construct the explainer by [Lemma 9.3](#) and apply [Theorem 9.7](#); its degree bounds depend only on (ρ, a) and $A - k + 1 \geq (a - \rho)n$. For the remaining $p \leq B_{\text{jet}}$, the hypothesis gives $k \leq B_{\text{jet}}$. Apply [Lemma 9.6](#) with $K_0 = B_{\text{jet}}$ and $a_0 = a$. $\square$

Corollary 9.13 (Uniform quadratic decoding at gap $6/25$). *Over a decoder-admissibly presented field $\mathbb{F}_q$ with $q \geq n$ and $p > k - 1$ when $k \geq 2$, the exact agreement list is computable deterministically in $\tilde{O}(n^2 \log q)$ bit operations whenever $1 \leq k \leq n$ and $k + 6n/25 \leq A \leq n$, with an absolute hidden constant.*

Proof. The support $(m, B_\partial, B_{\text{jet}}) = (12, 4, 22)$ in [Theorem 5.13](#) gives $\deg_X Q < 12A \leq 12n$. For $p > 22$, apply [Theorem 9.7](#) with $C = 12$ and $t \geq 6n/25$. If $p \leq 22$, then $k \leq 22$; [Lemma 9.6](#) applies with $K_0 = 22$ and $a_0 = 6/25$. Bounded lengths and constant codes use [Section E.8](#). $\square$

9.7 Higher-order recovery by isolated systems

For higher order we retain the same Taylor families and agreement recovery, but use isolated polynomial-system roots to obtain finite representations. We prove completeness by enumerating all coordinate subsets of the received word having the required size.

Lemma 9.14 (Agreement cuts isolate regular candidates). *At a regular order- r chart point representing a degree- $< k$ polynomial, with $r < k$, any k agreeing evaluations have differentials spanning its r -dimensional cotangent space. Thus some r agreement equations, together with $h = 0$, isolate the point simply.*

Proof. A nonzero tangent direction gives a nonzero degree- $< k$ polynomial, since (103) retains the initial jet. It cannot vanish at k distinct evaluations. Therefore no proper cotangent subspace contains differentials from k agreeing positions. Choose r independent differentials and adjoin h , whose gradient is nonzero at a regular point, to obtain a nonsingular ambient Jacobian. $\square$

The all-subsets decoder. Algorithm 5 in [Section E.1](#) records the isolated-system procedure. It enumerates all r -subsets of agreement equations at each order- r chart. When an extension is needed, its degree is parameter-bounded and its cardinality is $n^{O(r+1)}$; otherwise $E' = E$. The bounded irreducibility search is analyzed in [Section E.7](#). The procedure neither extracts parameter roots nor imposes tail equations beyond degree $k - 1$.

Theorem 9.15 (Deterministic reconstruction in arbitrary fixed order). *Fix integers $d \geq 0$, $B_{\text{jet}} \geq 1$, and $C \geq 1$. Let $\mathbb{F}_q$ be supplied in a decoder-admissible presentation with $q \geq n$, $1 \leq k \leq n$, and $k \leq A \leq n$. If $k \geq 2$, assume $p > \max\{k - 1, B_{\text{jet}}\}$. Given distinct evaluation points $\alpha_1, \dots, \alpha_n \in \mathbb{F}_q$, a received word $y \in \mathbb{F}_q^n$, and a nonzero $Q \in \mathbb{F}_q[X, Y_0, \dots, Y_d]$ with*

$$\deg_X Q \leq Cn, \quad \deg_{Y_0, \dots, Y_d} Q \leq B_{\text{jet}},$$

where the second bound is the total jet degree, suppose every degree- $< k$ polynomial with at least A agreements solves its differential equation. Represent Q sparsely, with one coefficient and exponent tuple per nonzero monomial. Then the complete list can be found deterministically in

$$C_{d,B_{\text{jet}},C} n^{O(d+1)} \tilde{O}(\log q)$$

bit operations. No positive linear gap between A and k is needed.

Proof. Apply the finite-case procedures of [Section E.8](#); on the large branch every active order satisfies $r < k$. Follow the separant chain of [Section 9.5](#). The degree bounds allow at most $(\lfloor Cn \rfloor + 1) \binom{B_{\text{jet}}+d+1}{d+1}$ monomials, with $O_{d,B_{\text{jet}},C}(\log(n+1))$ -bit exponents. Thus reading the supplied sparse input fits the bound; any additional encoding padding is charged separately to input-reading time. At every separant stage, the specialization on a degree- $< k$ message has degree at most $Cn + (B_{\text{jet}} - 1)(k - 1)$. One more than this number of centers covers every regular candidate. The fast Taylor lemma and this cover construct all regular families in $n^{O(d+1)}$ operations.

By [Lemma 9.14](#), every wanted point is a simple isolated root of one system enumerated by [Algorithm 5](#). Rojas's algorithm supplies finite univariate representations covering these roots in $n^{O(r+1)}$ operations per system ([Section E.6](#)). The representations may contain extra points or repeated polynomials; [Algorithm 1](#) returns exactly those whose coefficient vectors lie in the base field and pass the full agreement check.

Representation construction, substitution, and recovery each have $n^{O(r+1)}$ cost; there are $O(n)$ centers and at most n^r systems per chart. This proves the bit bound using [Section E](#). In particular, multivariate substitution of Taylor coefficients into a returned representation is included in the preparation cost. $\square$

The same reconstruction applies to the rate-dependent certificate of [Theorem 6.7](#). Fix $0 < \rho < a < 1$ and an integer $d \geq 500$ with $\Gamma(\rho, a, d) > 1$, and use its finite parameters $m, B_{\text{jet}}, N_{\rho,a,d}$. Over a decoder-admissibly presented $\mathbb{F}_q$ with $q \geq n$ and $p > \max\{k - 1, d, B_{\text{jet}}\}$, for integers $n \geq N_{\rho,a,d}$, $1 \leq k \leq \rho n$, and $an \leq A \leq n$, the exact degree- $< k$ agreement list at threshold A is computable deterministically in $C_{\rho,a,d} n^{O(d+1)} \tilde{O}(\log q)$ bit operations. Indeed, the certified support has bounded jet degree and X -degree $O_{\rho,a,d}(n)$: construct its explainer by [Lemma 9.3](#), then apply [Theorem 9.15](#) when $k > d$, or [Lemma 9.6](#) with $K_0 = d$ and $a_0 = a$ otherwise. If ρ, a have no finite input representation, fix rational bounds with the same positive margin in advance.

9.8 Uniform capacity decoding

We now combine the finite interpolation parameters from the higher-order analysis with the reconstruction routines proved in this section. The small-gap branch uses isolated-system recovery; the gap-6/25 branch and all larger gaps use the first-order curve decoder. The interpolation theorem [Theorem 6.8](#) supplies the equation and the list-size bound; the recovery routines supply the algorithm and its running time. The runtime constants depend on the fixed gap, with no polynomial bound asserted on their inverse-gap dependence. The decoder also accepts $A > n$ and immediately returns the empty list; the theorem states the nontrivial range $A \leq n$.

Theorem 9.16 (All-rate Reed–Solomon list decoding). *For every fixed gap $\delta \in (0, 1)$, there are an integer N_δ and a deterministic decoder with the following guarantee. Let $n \geq N_\delta$, let $1 \leq k \leq n$, and let $\mathbb{F}_q$ be supplied in a decoder-admissible presentation in the sense of [Definition 9.1](#), with $q \geq n$. For $k \geq 2$, require $p > k - 1$. Constant codes require no characteristic assumption. Given any n distinct evaluation points $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{F}_q^n$, any received word $y \in \mathbb{F}_q^n$, and any integer agreement threshold*

$$k + \delta n \leq A \leq n,$$

the decoder, given this presentation and (n, k, A, α, y) , returns exactly

$$\mathcal{L}(y, A) = \{P \in \mathbb{F}_q[X] : \deg P < k, \#\{i : P(\alpha_i) = y_i\} \geq A\}. \quad (116)$$

The output list size is $n^{O_\delta(1)}$, independently of q , and the running time is $B_\delta n^{O(\exp(1.5/\delta))} \tilde{O}(\log q)$ bit operations. The constants and exponents, as well as N_δ , depend only on δ , uniformly over the rate k/n and the evaluation set.

Proof. Fix δ . Enlarge N_δ to dominate the finite threshold in [Theorem 6.8](#) and every parameter-dependent field, grid, and center condition used by the applicable reconstruction routine. If $k = 1$, frequency counting returns exactly the constants occurring at least A times, in every characteristic. We may therefore assume $2 \leq k < A \leq n$.

Suppose first that $0 < \delta < 6/25$. If $p \leq B_{\text{jet}}(\delta)$, then $p > k - 1$ gives $k \leq B_{\text{jet}}(\delta)$. Apply [Lemma 9.6](#) with $K_0 = B_{\text{jet}}(\delta)$ and $a_0 = \delta$ to recover the exact list in $\tilde{O}_\delta(n \log q)$ bit operations. The list bound follows from [Theorem 6.8](#).

For $p > B_{\text{jet}}(\delta)$, the finite choices in [Theorem 6.8](#) give a certified order- $d(\delta)$ explainer of jet degree at most $B_{\text{jet}}(\delta)$ and X -degree $O_\delta(n)$. The fast interpolation routine [Lemma 9.3](#) constructs it in $\tilde{O}_\delta(n \log q)$ bit operations. Under the displayed characteristic hypothesis, if $k \leq d(\delta)$, apply [Lemma 9.6](#) with $K_0 = d(\delta)$ and $a_0 = \delta$. Otherwise [Theorem 9.15](#) returns the complete agreement list. Both branches fit the finer bound

$$B_\delta n^{O(d(\delta)+1)} \tilde{O}(\log q) = B_\delta n^{O(\exp(1.5/\delta))} \tilde{O}(\log q)$$

bit operations. The list-size bound is the one already proved in [Theorem 6.8](#) and is independent of q .

For $6/25 \leq \delta < 1$, use the fixed first-order support from [Theorem 5.13](#). It is valid at the requested threshold because $A - k \geq \delta n \geq 6n/25$; after decoding at the smaller threshold, we retain only candidates meeting A . [Corollary 9.13](#) gives the complete list in $\tilde{O}(n^2 \log q)$ bit operations under the stated characteristic assumption. The uniform counting theorem gives list bound $307n$ under the same assumption.

For completeness, if one runs the decoder below N_δ , the explicit EXACTBYSUBSETS branch in [Section E.8](#) is exact and has polynomial cost in $n \log q$ for this fixed finite range. This branch is not used to validate a failed interpolation certificate. Together the cases prove the theorem. $\square$

Taking $A = k + \lceil \delta n \rceil$ and writing $\rho = k/n$, the theorem corrects every pattern of at most

$$n - k - \lceil \delta n \rceil = \lfloor (1 - \rho - \delta)n \rfloor$$

errors whenever this radius is nonnegative. This is the usual capacity radius with gap δ . The threshold A is an integer input to the decoder; the fixed real parameter δ selects its finite parameters. If $A > n$, the output is empty.

Both the list size and the decoding time are polynomial in the binary input length $n \log q$ for every fixed gap, without an upper bound on the field size. [Section 9](#) explains how agreement constraints avoid enumerating field elements. Its bounded-length fallback also gives an exact polynomial-time decoder below N_δ ; the explicit list-size estimates retain their length hypotheses.

10 Detailed comparisons with prior work

To place our results alongside earlier work, we distinguish three questions: how close to capacity the agreement bounds reach, how their parameters depend on the gap, and how efficiently we can recover the list. We first compare decoding algorithms and quantitative capacity guarantees for ordinary Reed–Solomon values. We then turn to MCA and the restrictions imposed by known counterexamples, before examining finite certificates and their use in proof systems. Throughout, we keep the choice of code explicit: a guarantee for random evaluation points or derivative-valued symbols need not apply to the ordinary evaluation domain already fixed by a protocol.

10.1 Decoding from ordinary evaluations

The list-decoding question, introduced by Elias and Wozencraft [Eli57, Woz58], has two distinct aspects: bounding the number of close codewords and finding them efficiently. For Reed–Solomon codes, Sudan and Guruswami–Sudan address both through polynomial interpolation [Sud97, GS99]. Earlier results beyond Johnson rely on suitable evaluation sets [RW14, ST23, GLS⁺24, GST23]. In particular, Brakensiek, Gopi, and Makam prove capacity list bounds for random Reed–Solomon codes over sufficiently large fields [BGM23], and subsequent work reduces the required field size [GZ23, AGL24, AGG⁺25]. These results establish small lists for the chosen code families, but do not supply efficient decoding on every prescribed evaluation set.

Other routes toward capacity change the code, as in Parvaresh–Vardy and folded Reed–Solomon codes [PV05, GR08], or by transmitting derivatives, as in multiplicity codes [KSY14, GW13, Kop15]. Modern algorithms for folded and multiplicity codes compute affine candidate spaces and select messages using received-coordinate constraints, with efficient dependence on the field’s bit length [KRZSW23, GHKS24, AHS26, GHKS26]. We use the same broad principle of selecting candidates by agreement, but ordinary RS input does not supply folded blocks or derivative tuples. The hidden-derivative equation instead leaves a nonlinear family, which our recovery step must represent and filter.

Recovering candidates without general factorization. Using agreement to avoid factorization already has precedents for ordinary RS input. Augot–Pecquet use received pairs to initialize Hensel lifting [AP00]; their analyzed reconstruction treats Sudan’s radius, without the general GS multiplicity extension, and incurs an extra factor of n . CHK obtains deterministic $\text{poly}(n, \log q)$ decoding at every agreement strictly above Johnson over every finite field [CHK26]. CHK also credits the Kalev–Ta-Shma multiplicity-code decoder, which uses the supplied derivative information to initialize differential reconstruction and recurses through singular cases [KTS22, CHK26]. Our task is to retain the efficiency of agreement-based recovery when no derivative tuple is supplied and the initial data must remain symbolic.

Arithmetic cost versus bit cost. For a near-linear bound, we must also account for the cost of obtaining the initial roots. Fast interpolation and power-series reconstruction already provide near-linear arithmetic [Ale05, JNSV17, NRS17]. GHKS also state a deterministic near-linear Johnson decoder; setting their multiplicity parameter to one gives ordinary RS codes [GHKS24, Theorem 1.5]. Their interpolation has near-linear field-operation cost, while their reconstruction invokes the Roth–Ruckenstein–Alekhnovich root routine as a black box [GHKS24, Section 10, Theorem 10.8]. Alekhnovich’s base case uses univariate factorization [Ale05, Appendix, Figure 5], so this arithmetic bound alone does not supply an unconditional deterministic near-linear bit bound for unrestricted prime q . Neiger, Rosenkilde, and Schost make this distinction explicit by charging finite-field root finding in addition to series arithmetic [NRS17, Theorem 1.2].

Theorem 9.5 gives an unconditional deterministic $\tilde{O}_{\rho, \eta_0}(n \log q)$ -bit implementation in every characteristic, for $q \geq n$ supplied in a decoder-admissible presentation and without a finite-field root-finding oracle using the agreement-based recovery of Section 2.4. It improves CHK’s stated running time when the rate and positive Johnson gap are fixed; CHK permits every strictly positive gap. The small-characteristic preprocessing, including removal of inseparability in the message variable [Lec08], is proved in Section E.2.

Decoding hidden-derivative equations. Beyond Johnson, we retain the symbolic solution families used by Zheng and Jeronimo [rs-26, Jer26], but impose received agreements before extracting individual candidates. The first-order implementation is quadratic in n (Theorem 9.7); higher-order recovery uses Rojas’s isolated-root representations [Roj99] and takes $n^{O(d+1)} \tilde{O}(\log q)$ time, including the solver cost (Theorem 9.15). Section 2.4 explains the constructions and their inherited ingredients. Table 5 summarizes the resulting comparison.

Decoder	Agreement regime	Large-field cost
Fast GS [JNSV17, NRS17]	Positive Johnson gap	Las Vegas $\tilde{O}(n \log q + (\log q)^2)$
CHK [CHK26]	Johnson	Deterministic $\text{poly}(n, \log q)$
GHKS [GHKS24, Theorem 1.5]	Positive Johnson gap	Stated deterministic $n \text{poly}(\log n, \log q)$; reconstruction invokes univariate factorization
BCPZZ [BCP ⁺ 26a, Corollary 5.1]	Capacity	Deterministic $q^{O_{\rho, \delta}(1)}$; prime $q \geq C_{\rho, \delta} n$
Jeronimo [Jer26, Theorem 4.4]	Capacity	Deterministic $q^{O_{\text{params}}(1)}$; prime $q \geq n$
This work, zeroth order	Positive Johnson gap	Deterministic $\tilde{O}_{\rho, \eta_0}(n \log q)$
This work, first order	$a > a_1(\rho)$	Deterministic $\tilde{O}_{\rho, a}(n^2 \log q)$
This work, order d	Through capacity	Deterministic $C_{\text{params}} n^{O(d+1)} \tilde{O}(\log q)$

Table 5: Decoder comparison for ordinary RS values, with rate and positive gaps fixed. Johnson means the exact threshold $\sqrt{(k-1)/n}$, asymptotic to $\sqrt{\rho}$; d is our derivative order. Our complete-list bounds use decoder-admissible presentations with $q \geq n$ and the characteristic assumptions in Section 9 (none at order zero). The fast GS bit bound assumes prime $q \geq n$; CHK allows every finite field. Folded and multiplicity codes supply different received symbols and are compared in the prose above.

10.2 Capacity and the required derivative order

Reaching capacity leaves a quantitative question: how many hidden derivatives suffice at a given gap? We compare sufficient orders under two parameterizations. An additive gap fixes agreement $\rho + \delta$; a multiplicative gap fixes the ratio between the rate and the agreement. These lead to different comparisons, especially at low rates.

Reaching all rates. BCPZZ’s revised theorem reaches every fixed rate by padding the evaluation set until their low-rate theorem applies; they credit this reduction to Alrabiah, Goyal, and Guruswami [BCP⁺26a, Section 5]. Mishra’s extension instead leaves the evaluation set unchanged and treats the derivative order as a free parameter [Mis26]. It proves the existence of a sufficient order through an existential parameter threshold. Jeronimo also leaves the derivative order free, but makes the uniformity across rates explicit by enlarging the ambient message space. At agreement count A and additive slack γ , he uses degree bound $K = \lceil (1 - \gamma/2)A \rceil$ and then filters the degree- $< K$ output back to the requested degree bound. The interpolation then depends on the ratio $A/(K-1) > 1/(1 - \gamma/2)$ rather than on k/n , so one order and multiplicity serve every rate [Jer26, Lemmas 2.2 and 4.1]. Our support instead keeps the rate in the sufficient condition (75); the uniform choice raises the ambient dimension above $k+1$ only when $k < \delta^2 n$ (Theorem 6.8).

A fixed additive gap from capacity. To compare our order $\lceil \exp(1.5/\delta) \rceil$ with BCPZZ, we first express their parameter choices in terms of δ ; they do not state the resulting bound explicitly in this form. At fixed positive rate ρ and sufficiently small gap δ , their all-rate reduction sets $\theta = \delta/(2(\rho + \delta)) = \Theta_{\rho}(\delta)$. Choose its low-rate agreement parameter η to be half the upper bound

$$\left(\frac{\theta^3(1-\theta)}{768} \right)^{(5+\theta)/(1-\theta)}$$

from their Theorem 4.1. For small enough δ , this also satisfies $\eta < \rho + \delta$, as required by their reduction. Then $\log(1/\eta) = O_{\rho}(\log(1/\delta))$, and their Algorithm 1 chooses $d = \lceil \eta^{-3/\theta} \rceil$ for this low-rate instance [BCP⁺26a,

Algorithm 1, Theorem 4.1, and Section 5]. We obtain

$$\log d = O_\rho\left(\delta^{-1} \log(1/\delta)\right).$$

Our explicit uniform certificate gives $d = \lceil \exp(1.5/\delta) \rceil$ for $0 < \delta < 6/25$, with order one sufficient at larger gaps (Theorems 5.13 and 6.8). We are comparing sufficient parameter choices here, not proving a lower bound on the derivative order needed by the earlier supports.

A fixed multiplicative gap. At small agreement, it is also useful to fix the rate as a fraction of the agreement rather than fix their difference. Jeronimo’s parameters apply to this regime and to list recovery, where each coordinate supplies several possible symbols. At fixed multiplicative slack $\vartheta > 0$, his sufficient order for agreement ε and input-list bound L is

$$O_\vartheta\left((1 + L/\varepsilon)^{1/\vartheta} \log^{5/\vartheta}(e + L/\varepsilon)\right) \quad [\text{Jer26, Corollary B.1}].$$

This is polynomial in inverse agreement at fixed ϑ . For singleton lists, our rate-dependent condition gives polynomial dependence without the logarithmic factor: at agreement a and rate at most $(1 - \vartheta)a$, every order $d \geq \max\{500, \frac{1}{6}(40/(9(1 - \vartheta)a))^{1/\vartheta}\}$ suffices for list decoding and line MCA ((86)). This is the relevant comparison at small agreement and fixed ϑ ; our uniform $\lceil e^{1.5/\delta} \rceil$ bound instead chooses one order for every rate at a fixed additive gap. Appendix G gives our bounded-input list-recovery extension in the same point–symbol formulation, with explicit dependence on the total input-list size. The main results do not require that extension.

Public chronology of the capacity results. We record the public dates to distinguish earlier and concurrent results in this closely spaced sequence. On September 4, 2026, BCPZZ’s original low-rate paper and the earlier MCA manuscript in Zheng’s repository were public. The same day, Mishra’s all-rate pull request was merged, and our ArkLib history recorded an initial all-rate list bound [BCP⁺26b, rs-26, Mis26, Ark26b]. On September 5, BCPZZ posted its all-rate revision, Jeronimo submitted his all-rate paper, and our ArkLib history recorded line and affine MCA [BCP⁺26a, Jer26, Ark26a]. ECCC published Jeronimo’s submission on September 6 [Jer26]. These dates establish public provenance, not the order in which the underlying results were discovered. The works disclose contemporaneous AI assistance; our own disclosure describes the systems’ contributions and our responsibility for the results, proofs, and exposition.

10.3 Mutual correlated agreement

For MCA, we must compare more than the agreement threshold: the exceptional-challenge probability and the agreement sets covered by the guarantee also matter. BCIKS establish correlated agreement for arbitrary Reed–Solomon codes by decoding over a function field [BCI⁺23]; Haböck and BCGM establish its mutual form [Hab25, BCGM25]. MCA controls every agreeing set at each good challenge, allowing a proof system to follow the whole list through a combination. We express that control as equality of full agreement sets, which interpolation uniqueness recovers from subset MCA (Corollary 7.2).

Agreement threshold and exceptional probability. BCHKS obtain Johnson-regime error $O_\rho(n/(q\eta_0^5))$ in every characteristic [BCH⁺26, Theorem 4.6]. Our ordinary theorem sharpens this to $O_\rho(n/(q\eta_0^3))$; first derivatives lower the required agreement, with quadratic exceptional counts and a large-characteristic hypothesis.

Relative versus vanishing improvements. Earlier beyond-Johnson theorems separate these two improvements in another way. Jo’s MCA theorem allows a fixed number of additional error positions, with error $O_{r,h}(n^6/q)$ at rate $1/r$ [Jo26b]. The relative improvement therefore tends to zero with the block length, even though the exceptional count stays polynomial. Chojecki obtains shortening bounds at fixed relative excess, but with exceptional counts exponential in n [Cho26]. Our first-order theorem instead improves the agreement threshold by a fixed positive fraction at fixed rate, while retaining error $O_{\rho,\eta_1}(n^2/q)$ (Theorem 5.8).

Padding the earlier MCA guarantee. Zheng’s earlier low-rate manuscript supplies the symbolic interpolation, Taylor reconstruction, and incidence argument behind our MCA analysis [rs-26]. Before comparing our quantitative bounds, we explain why its low-rate guarantee already implies a qualitative all-rate statement. We show that every unexplained agreement set meets the padded threshold and remains unexplained after padding.

Fix $\rho, \delta > 0$ with $\rho + \delta \leq 1$ and let $k = \lfloor \rho n \rfloor$. Choose fixed $\theta \in (0, 1)$ with $\rho/(1 - \theta) < \rho + \delta$, then choose ε small enough for the low-rate theorem and for $c = (1 - \theta)\varepsilon < \rho$. Set $N = \lceil k/c \rceil$. For sufficiently large n ,

$$N \geq n, \quad \lfloor cN \rfloor = k, \quad \lceil \varepsilon N \rceil \leq \lceil (\rho + \delta)n \rceil.$$

If $q \geq N$, append distinct evaluation points and extend both received words by zeros. An unexplained agreement set on the original coordinates remains unexplained in the padded code: its polynomial degree bound is still k , and its agreement meets the padded threshold. Restricting any padded witness equality to the original coordinates would also witness the original candidate. Thus the original bad-parameter count is at most $N^{O_{\theta,\varepsilon}(1)} = n^{O_{\rho,\delta}(1)}$. When $n \leq q < N = O_{\rho,\delta}(n)$, the corresponding polynomial-over- q probability bound can be made vacuous by increasing its constant. The witness-preservation argument is essential: passing to a subcode alone would not establish MCA. Our quantitative contribution goes beyond this fixed-rate consequence by choosing explicit gap-only parameters and sharpening the exceptional-count bounds.

Polynomial combinations and other code families. Jeronimo proves capacity MCA along polynomial curves, including a list-recovery extension [Jer26]. His symbolic interpolant has challenge degree $O(n)$ and its solutions are covered by varieties of bounded dimension and polynomial degree [Jer26, Proposition 5.2 and Theorem 5.5]. We make the challenge degree independent of n at fixed parameters by retaining the coefficient–constraint margin (Equation (81)). We then track the agreement above the witness threshold and the degree in the derivative variables separately, which improves the finite exceptional counts (Theorem 6.6 and Lemma 5.9, with $\ell = 1$ for line MCA). These are the quantities our application certificates need.

The choice of code matters in this comparison as well. Goyal–Guruswami and Goyal–Guruswami–Sun–Wootters obtain capacity MCA with linear-in- n error at fixed parameters for folded or random Reed–Solomon codes [GG25, GGSW26]. Yuan–Zhu give direct capacity correlated agreement for random codes, with distinct sampling and output-slack hypotheses [YZ26]. We instead keep the ordinary RS evaluation set arbitrary and prescribed; Table 6 displays the resulting tradeoff in exceptional counts alongside the agreement thresholds.

10.4 Dependence on the gap and the characteristic

To interpret these bounds, we need to know what changes when the gap shrinks or the characteristic is small. The counterexamples below explain why these are separate issues; they also identify which dependence on the block length is unavoidable.

Fixing the gap before the length grows. Our capacity theorem fixes $\delta > 0$ before allowing n to grow. This order of quantifiers matters for both finite-alphabet effects and the dependence of the exceptional count on δ .

Result	Code family	Error radius	MCA error
BCHKS [BCH ⁺ 26]	Arbitrary RS; every characteristic	$1 - \sqrt{\rho} - \eta_0$	$O_\rho(n/(\eta_0^5 q))$
Theorem 5.12	Arbitrary RS; every characteristic	$1 - \sqrt{\rho} - \eta_0$	$O_\rho(n/(\eta_0^3 q))$
Theorem 5.8	Arbitrary RS; characteristic zero or $p > \max\{k - 1, B_\partial\}$	$1 - a_1(\rho) - \eta_1$	$O_\rho(n^2/(\eta_1^4 q))$
Capacity MCA [GG25, GGSW26]	Folded RS or random RS; respective hypotheses	$1 - \rho - \eta$	$O_{\text{params}}(n/q)$
Earlier low-rate MCA [rs-26]	Arbitrary RS; prime fields	$1 - \varepsilon$, low-rate regime	$n^{O_{\theta, \varepsilon}(1)}/q$
Jeronimo [Jer26]	Arbitrary RS; prime $q \geq n$, or characteristic $p > \max\{k - 1, B_\gamma\}$ [Jer26, Section 8.4]	$1 - \rho - \delta$	$n^{O_\delta(1)}/q$
Theorem 6.9	Arbitrary RS; characteristic zero or $p > k - 1$, with no restriction if $k = 1$	$1 - \rho - \delta$	$n^{O_\delta(1)}/q$ over F_q

Table 6: Radius and soundness are separate comparisons. Johnson radii are written here in the fixed-rate asymptotic form $1 - \sqrt{\rho}$; the exact finite-length ordinary RS threshold uses agreement $\sqrt{(k - 1)/n}$. The random-code row is a high-probability statement over the code; the other rows are uniform over their stated code families. Probability expressions with denominator q apply over finite F_q ; the characteristic-zero cases instead retain the corresponding exceptional-set cardinality. Length and field thresholds are as in the respective theorems.

Crites–Stewart disprove the original capacity conjectures and propose entropy-corrected versions [CS25]. At error radius τ , the finite-alphabet condition is $H_q(\tau) \leq 1 - \rho$, where H_q is the q -ary entropy function. Since $H_q(\tau) \leq \tau + \log_q 2$, agreement $\rho + \delta$ leaves the required entropy slack once $\log_q 2 < \delta$. Our assumption $q \geq n$ ensures this eventually for every fixed δ .

That observation does not give a bound uniform in a shrinking gap. Prime-field constructions with gap $\Theta_\rho(1/\log n)$ have exceptional counts $n^{c-o(1)}$ for arbitrarily large fixed c [KKH26, Kam26]. In our theorem, both the polynomial exponent and the required length may depend on the fixed gap. The fixed-gap guarantee is therefore distinct from a claim of uniform polynomial dependence on its inverse.

A large field need not have large characteristic. Extending a small prime field can reduce an exceptional probability by enlarging its denominator q , but it does not change the characteristic. This distinction is essential beyond Johnson: characteristic-two and base-code constructions obstruct MCA [BCH⁺26, DG25, FS25]. There is also an obstruction to small lists themselves. Ben-Sasson, Kopparty, and Radhakrishnan use subspace polynomials over extension fields of fixed small characteristic to construct received words with superpolynomially many nearby RS codewords beyond Johnson [BKR10].

These examples rule out an extension based on field cardinality alone, but do not establish the precise characteristic threshold used in our proof. Our first-order line and polynomial-curve bounds require $p > \max\{k - 1, B_\partial\}$, while the general higher-order curve theorem requires $p > \max\{K - 1, B_{\text{jet}}\}$. Those hypotheses ensure the separability and Taylor reconstruction needed by the argument. Our uniform first-order and capacity statements need only $p > k - 1$: when the remaining degree guard fails, k is bounded in terms of the fixed parameters, and a characteristic-free argument gives the required bounds. This does not remove the degree guard from the sharper inverse-gap or general polynomial-curve estimates. The large-list obstruction, the MCA obstructions, and these reconstruction requirements are different reasons to track characteristic

explicitly.

How small can the exceptional probability be? At a fixed positive error radius, every linear code has MCA error at least $\min\{\lfloor \tau n \rfloor / q, 1\}$ in the range of [ABF26, Lemma 4.16]. Thus our ordinary theorem has the optimal linear dependence on n . This lower bound leaves two questions open: whether the Johnson-gap dependence can be improved further, and whether the quadratic dependence on n in our first-order theorem can be reduced.

10.5 Ideas from the Proximity Prize submissions

The Proximity Prize focuses on a different quantitative question: how strong a proximity guarantee can we certify for one fixed Reed–Solomon instance? Arnon, Boneh, and Fenzi formulate the challenge and its cryptographic motivation [ABF26]. The submissions first sharpened Johnson-regime estimates, then crossed Johnson using hidden derivatives, and subsequently developed stronger interpolation and candidate-counting arguments. We separate these ideas from their numerical instantiations; the latter are recorded in the submission concordance in the public artifact [DKT26].

The benchmark and its score. The challenge, hosted at <https://better.codes>, fixes

$$n = 2^{18}, \quad k = 2^{17}, \quad p = 2130706433, \quad q = p^6.$$

Each codeword consists of eight polynomials of degree less than k over $\mathbb{F}_q$, evaluated on the same multiplicative subgroup of size n in the KoalaBear base field $\mathbb{F}_p$. A coordinate is an eight-tuple, and agreement means equality of that whole tuple. Thus the alphabet rate is $1/2$ and the total message dimension is $8k = 2^{20}$ [Pro26c].

A lower-track submission supplies a formally checked safe relative error radius τ and a score b , in bits. The integer agreement threshold is $A = n - \lfloor \tau n \rfloor$. At this radius, the MCA-plus-list bound for the protocol’s combination-round extractor must be at most 2^{-128} . Independently, the 128-query term must satisfy

$$(1 - \tau)^{128} \leq 2^{-b}.$$

A larger certified radius gives a smaller probability that all spot checks miss disagreements, hence a larger score. The improvement is a stronger *lower bound on the safe proximity threshold*, not an attack or a measured proof-size reduction. Nor is b the security of the complete protocol: the reduction-error requirement and the query term are separate parts of the certificate [Pro26d]. For example, our bounds certify the reduction requirement whenever $(E + \Lambda)/q \leq 2^{-128}$, where E/q bounds MCA error and Λ bounds the list size after pairing two eight-row words, hence at interleaving width sixteen. Our scalar list bound transfers to this code without an extra factor for the interleaving width (Section 7.3). Our finite analysis allows separate interpolation supports for bounding E and Λ at the same agreement threshold. Submission [Bit26b] likewise makes this separation explicit, but a numerical comparison must include both contributions.

Which parts use the special code? The interpolation and component-counting mechanisms discussed below use distinct evaluation points, not the multiplicative-subgroup structure of this domain [Nas26, Bit26d]. Their algebraic constructions therefore apply more broadly than the challenge instance. The field enters through characteristic conditions for Taylor reconstruction, finite avoidance choices, and the probability denominator q ; the rate and length determine which numerical degree bounds fit the error budget. The sextic extension enlarges that denominator without changing the characteristic. This does not turn a fixed numerical certificate into a theorem for every field and rate. In contrast, the upper-track construction of erdkocak explicitly uses the fibers of $x \mapsto x^{512}$ on the multiplicative evaluation domain, together with choices

in the extension field [erd26b]. That obstruction is specific to the structured code, rather than a converse for arbitrary evaluation sets.

Sharper accounting at Johnson. Before the hidden-derivative crossing, BitWonka shortened the Taylor reconstruction to the coefficients needed for a message, rather than the larger interpolation-degree horizon [Bit26a]. jieyilong then used Taylor numerators shared across branches and summed resultant bounds using separate message and challenge degrees [jie26a]. Together these constructions already contain the two degree-accounting savings behind an inverse-cubic gap dependence. Their completed certificates concern the challenge’s fixed, large-characteristic instance; they do not state a uniform $O_\rho(n/(q\eta_0^3))$ theorem. In Section 5.3 we give such a theorem for every characteristic and every prescribed evaluation set, using factorwise Frobenius pullback to include inseparable branches.

Hidden derivatives and the regular–singular split. Naskręcki uses the hidden first Hasse derivative to construct a first-order differential equation satisfied by every close message [Nas26]. The submission identifies a divisibility kernel among the local interpolation constraints and separates the solutions according to whether Taylor reconstruction is regular. A resultant puts the remaining singular solutions on an ordinary polynomial equation. It also preserves the original agreement sets in the alignment argument needed for MCA. Our first-order analysis uses these mechanisms, together with the later symbolic solution-family analysis of Zheng [rs-26]. We optimize a general support family to obtain the all-rate curve $a_1(\rho)$ and prove explicit list and MCA bounds above it; the prize certificate instead establishes one finite crossing of Johnson. Our decoding theorems additionally bound the cost of recovering the list, which a candidate-counting certificate alone does not provide.

Stronger interpolation constraints. To construct an explainer, we need more available coefficients than independent local conditions. Later submissions improve both sides of this comparison: they choose coefficient-dependent monomial restrictions and exhibit additional independent vectors in the local kernel [erd26a, net26]. The linear-map formulation of Proposition 3.10 can accommodate these spaces, but our displayed support families do not establish all of their sharper rank estimates. Improving a rank estimate at leading order could change an agreement curve, whereas saving only lower-order terms improves finite parameters without changing that limit.

The construction in [net26] uses the second Hasse derivative to obtain auxiliary equations on first-order solution families, rather than count the full solution family of a second-order equation. On a regular first-order component, differentiating its equation expresses the second derivative in terms of the value and first derivative, under the submission’s large-characteristic hypotheses. Substituting this expression into the auxiliary equation and clearing denominators gives another equation on the same component. If this equation is nonzero on the component, it cuts the family; otherwise, the resulting divisibility is handled separately. The interpolation still uses a genuine second derivative, so its gains should not be identified with a lower first-derivative-only agreement curve.

Several equations and the whole interpolation kernel. One explainer can have components containing many formal solutions that are not close messages. Additional equations can cut these components, but linear independence of the equations does not imply coprimality: $Q_1 = FG_1$ and $Q_2 = FG_2$ still share the component $F = 0$. The multiple-equation constructions therefore use actual greatest common divisors and residual quotients to separate shared components from the remaining intersections [rec26, Bit26c]. Whole-kernel arguments go further. If every admissible explainer were divisible by a particular factor, the interpolation kernel would lie in the space of its multiples. A dimension bound for that space can force

an explainer that is not divisible by the factor and hence cuts its component [jac26a]. These are additional counting arguments, not consequences of selecting one nonzero kernel element as in our transfer theorems.

Using the Taylor coefficients that must vanish. A polynomial of degree less than k has zero Taylor coefficients at orders $k, k + 1, \dots$, whereas a formal solution of the differential equation need not. Later submissions turn one or more of these necessarily zero tail coefficients into additional algebraic equations [jie26b, i3426]. If such an equation is nonzero on a component, its zero set gives a smaller family to count; if it vanishes identically, that component needs another argument. Our Taylor estimates already control the degrees of the reconstructed coefficients; the additional step is to use their required vanishing to obtain smaller families, rather than retain every formal solution.

Counting components together. Even with a bound for each component, summing independent worst cases can overestimate the total: the components share the degree budget of the original equation. Later submissions keep these shared constraints in the count [sau26, jsj26, net26]. They also let different components use different interpolation equations and combine their bounds under shared degree constraints, rather than choose one equation and one worst-case bound for the whole family [jac26d, jac26b, jac26c]. The latest construction uses separate interpolation equations to bound the sum of moving-coordinate degrees over one common component family, without weighting that sum by component multiplicity [Bit26d]. Here a coordinate is moving if it is not constant on the component. These arguments exploit relations among the equations and their components that our single-explainer transfer does not use.

Finite certificates versus limiting curves. We have not integrated the multiple-equation, whole-kernel, and Taylor-tail arguments into our first-order analysis. Whether these ingredients, together with stronger interpolation, lower the threshold at rates $1/2$ and below remains open in our analysis. Better counting alone does not change the coefficient–constraint inequality that defines our curve. To lower that threshold within the present construction, we would need a leading-order improvement in the interpolation count; a different construction would require a separate analysis. At the finite level, stronger counts can already help by meeting a fixed field-size and MCA-error budget closer to a given interpolation threshold. A comparison at fixed length must therefore check those budgets, rather than compare the prize score with our limiting curve alone.

Using only the finite interpolation and counting bounds proved in this paper, a bounded parameter search certifies $A = 183210$ agreeing coordinates and hence radius $\tau = (262144 - 183210)/262144$. The resulting score is 66.15 bits, with $(E + \Lambda)/q < 2^{-128}$; our ordinary Johnson certificate attains 63.99 bits at $A = 185372$. Our calculation is an attained finite bound, not an optimality claim or a submitted prize entry. For comparison, the promoted certificate of September 15, 2026 gives 68.11 bits at $A = 181284$ agreements [Bit26d]. Its required agreement fraction is approximately 0.691544, still above our limiting threshold $a_1(1/2) \approx 0.689898$. Thus it improves on our attained finite certificate without crossing our first-order curve: it meets the prize’s fixed-length reduction-error budget closer to that curve. The limiting threshold itself is not a certificate at the prize’s fixed parameters. The public artifact records the exact certificate, search restrictions, and reproduction commands [DKT26].

10.6 From agreement bounds to proof-system parameters

Our applications combine BCGM’s affine reduction and Jo’s interleaving transfer [BCGM25, Jo26a] with the coding-to-soundness connection of DEEP-FRI and subsequent round-by-round analyses [BGKS19, GMW25]. Section 8 gives the error and byte accounting, distinguishing measured savings from comparisons against optimized Johnson parameters. Unlike the Proximity Prize, which fixes 128 queries and scores the guarantee, we choose the query count to meet each protocol’s existing error budget.

The baseline is important. Some implementations already choose queries using conjectural capacity-level bounds, while others use different code families [RIS26, Win25, Plo26, Sta26a, Sta26b]. For those systems, one must compare against the assumptions and code actually used; an improvement over a Johnson-based analysis does not by itself establish smaller proofs.

Acknowledgements

We thank Kai Zhe Zheng for close discussion and feedback, including a suggestion to name the central interpolation polynomial in BCPZZ the hidden-derivative explainer. Kominers thanks MIT Sloan’s Technological Innovation, Entrepreneurship, and Strategic Management (TIES) Group for its hospitality during this work.

Disclosure

The second and third authors are Research Partners at a16z crypto, which is an investor in various blockchain-based platforms, as well as in the crypto ecosystem more broadly. General a16z disclosures are available at <https://www.a16z.com/disclosures/>.

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A Rational-image and Taylor tools

This appendix supplies the general reconstruction and degree estimates used to build the algebraic candidate covers. We record ordinary and mixed-degree rational-image bounds, followed by the regular Taylor lift. The specialized first-order refinements appear with the first-order counting lemma in [Section 5.4.2](#).

A.1 Regular lifting and the separant chain

We count the solutions of a differential equation that agree with the received word, using Kopparty’s reconstruction from a regular initial Taylor jet [[Kop15](#), Theorem 4.3, Theorem 4.4, and Corollary 4.5]. Below the characteristic, the required binomial coefficients are nonzero, so such a jet has at most one lift. An explicit derivative chain handles solutions whose separant vanishes identically; coefficient-space incidence then gives a list bound independent of the field size.

Recall the Hasse-derivative convention of [Definition 3.3](#) and the weighted degree of [Definition 3.6](#). For nonzero $Q \in \mathbb{F}_q[X, Y_0, \dots, Y_r]$, write

$$\deg_Y Q := \deg_{Y_0, \dots, Y_r} Q, \quad \ell_D(Q) := \text{wdeg}_{1, D, D-1, \dots, D-r} Q.$$

By (12), for every $P \in \mathbb{F}_q[X]_{\leq D}$,

$$\deg_X Q(X, P, P^{[1]}, \dots, P^{[r]}) \leq \ell_D(Q). \tag{117}$$

A.2 Unique lifting below the characteristic

We first show why regular initial Taylor data determine at most one solution. Below the characteristic, each new coefficient enters the recurrence with a nonzero multiplier, so we can recover it uniquely.

Lemma A.1 (Unique lift from a regular jet). *Let E be a field of characteristic zero or $p > D$, let $0 \leq s \leq D$, and let*

$$R \in E[X, Y_0, \dots, Y_s].$$

Fix $\alpha, \beta_0, \dots, \beta_s \in E$ such that

$$R(\alpha, \beta_0, \dots, \beta_s) = 0 \quad \text{and} \quad \frac{\partial R}{\partial Y_s}(\alpha, \beta_0, \dots, \beta_s) \neq 0. \quad (118)$$

There is at most one $P \in E[X]_{\leq D}$ satisfying

$$R(X, P, P^{[1]}, \dots, P^{[s]}) \equiv 0 \quad (119)$$

and $P^{[j]}(\alpha) = \beta_j$ for every $0 \leq j \leq s$. Moreover, this polynomial can be reconstructed, or its nonexistence detected, using a number of field operations polynomial in D , $\ell_D(R)$, and the sparse support size and binary exponent lengths of R . Here coefficients in the arbitrary field E are accessed through field operations and equality tests; no bit encoding of E is assumed.

Proof. Write $P(\alpha + T) = \sum_{i=0}^D p_i T^i$, with $p_i = \beta_i$ for $i \leq s$. For $s < i \leq D$, changing p_i by γ changes the j -th Hasse derivative by

$$\gamma \binom{i}{j} T^{i-j}. \quad (120)$$

The first affected coefficient in the differential equation is at T^{i-s} , and its change is exactly

$$\gamma \binom{i}{s} \frac{\partial R}{\partial Y_s}(\alpha, \beta_0, \dots, \beta_s). \quad (121)$$

Terms with two changed factors have order at least $2(i-s) > i-s$; a lower derivative or a coefficient p_j with $j > i$ affects only higher orders. The displayed multiplier is nonzero by regularity and the characteristic hypothesis. Solving this linear equation successively therefore fixes every remaining coefficient. Truncated polynomial arithmetic and a final check of (119) give the asserted reconstruction or rejection and its polynomial cost. $\square$

A.3 Counting and enumerating solutions

The original separant can vanish identically along a solution while the separant of a later equation in the derivative chain does not. We use one deterministic chain to find that stage. Fix a nonzero $Q \in F[X, Y_0, \dots, Y_r]$ whose individual jet degrees are smaller than the positive characteristic, if any. Starting from $Q_0 = Q$, form a deterministic derivative chain. If Q_i depends on a jet variable, let

$$s_i := \max\{j : Q_i \text{ depends on } Y_j\}, \quad Q_{i+1} := \frac{\partial Q_i}{\partial Y_{s_i}}. \quad (122)$$

Every member is nonzero by the degree hypothesis, and each derivative lowers total jet degree. The chain therefore ends at a nonzero $Q_t \in F[X]$.

Every solution has a first nonzero specialization along this chain; the preceding equation vanishes identically in X and its separant does not. Under the hypotheses of Lemma A.1, a regular initial jet determines at most one solution. Enumerating all jets counts all differential roots; restricting them by received-word agreement gives the field-independent lists needed for decoding.

A.4 Degree of a rational image

Lemma A.2 (Degree of a rational image). *Let $V \subseteq E^t$ have dimension at most s and total degree at most Δ . Let $\phi = (N_1/D, \dots, N_K/D)$, where all numerators and the common denominator have degree at most $b \geq 1$, and D vanishes identically on no component of V . The closure of $\phi(V \setminus V(D))$ has dimension at most s and total degree at most Δb^s .*

Proof. For an irreducible component C of V , its image closure W is irreducible of dimension $t' \leq \dim C$. The image is constructible and contains a dense open subset of W by Chevalley's theorem [The26]. Intersect W with t' generic affine hyperplanes. Their intersection consists of $\deg W$ distinct points in that open subset. Pulling back each hyperplane gives a polynomial of degree at most b . On $C \setminus V(D)$, the resulting intersection is the disjoint union of the nonempty fibers above those points. Each fiber contains the nonempty open part of at least one irreducible component of the closed intersection. One such component cannot meet two fibers on $D \neq 0$, since its image in this finite set is irreducible. Refined Bézout therefore bounds the number of fibers by $\deg(C)b^{t'} \leq \deg(C)b^s$. Summing over C proves the degree bound, including when the fibers are positive-dimensional. $\square$

A.5 Rational images with separate degree bounds

Lemma A.3 (Rational images with two degree bounds). *Let $G \in E[Z, u_0, \dots, u_d]$ be nonzero with degree at most D_G^Z in Z and at most D_G^u in u . Let U be a union of components of $V(G)$, and let a rational map ϕ have a common denominator that vanishes identically on no component of U . If its numerators and denominator have integer degree bounds $B_\phi^Z \geq 1$ in Z and $B_\phi^u \geq 1$ in u , its image closure has dimension at most $d + 1$ and total degree at most*

$$D_G^Z (B_\phi^u)^{d+1} + (d+1) D_G^u B_\phi^Z (B_\phi^u)^d.$$

Proof. We bound degree by counting polynomial coefficients on the source hypersurface.

First take one irreducible factor G_0 of G , with degrees (D_0^Z, D_0^u) , and put $s = d + 1$. Let R_N consist of polynomials of degree at most $B_\phi^Z N$ in Z and total degree at most $B_\phi^u N$ in the s jet variables. Multiplication by G_0 injects the rectangle with degree bounds $(B_\phi^Z N - D_0^Z, B_\phi^u N - D_0^u)$ into $(G_0) \cap R_N$. For sufficiently large N , the image of R_N in $E[Z, u]/(G_0)$ therefore has dimension at most

$$(B_\phi^Z N + 1) \binom{B_\phi^u N + s}{s} - (B_\phi^Z N - D_0^Z + 1) \binom{B_\phi^u N - D_0^u + s}{s}.$$

The terms of degree $s + 1$ cancel. The remaining leading term is

$$\frac{D_0^Z (B_\phi^u)^s + s D_0^u B_\phi^Z (B_\phi^u)^{s-1}}{s!} N^s.$$

Embed the source using all monomials of bidegree at most (B_ϕ^Z, B_ϕ^u) . Since $B_\phi^Z, B_\phi^u \geq 1$, these coordinates include Z and every u_i , so this is a closed embedding. Every polynomial of ordinary degree at most N in the embedding coordinates pulls back into R_N . Comparing the resulting Hilbert-function bound with the cumulative affine Hilbert polynomial bounds the degree of the embedded hypersurface by $D_0^Z (B_\phi^u)^s + s D_0^u B_\phi^Z (B_\phi^u)^{s-1}$.

The numerators and common denominator of ϕ are linear forms in these embedding coordinates. Applying Lemma A.2 with degree one bounds the image closure by the same quantity. Sum over the selected irreducible factors of G , using $\sum D_0^Z \leq D_G^Z$ and $\sum D_0^u \leq D_G^u$. $\square$

A.6 Regular Taylor coefficients

The next lemma records the degree bounds for the regular lift used in [Lemma A.1](#). The *Taylor chart* maps the initial coefficients u to all reconstructed coefficients $c_j(u)$. Its domain excludes zeros of the separant $S = \partial T / \partial Y_r$, so the recurrence can divide by S .

Lemma A.4 (Regular Taylor chart). *Let $0 \leq r < K$ and let $T \in E[X, Y_0, \dots, Y_r]$ have total jet degree $v \geq 1$. Put $S = \partial T / \partial Y_r$. Assume the characteristic is zero or exceeds $K - 1$. Choose $a \in E$ such that $s(u) := S(a, u)$ is nonzero, and put*

$$U = \{u \in E^{r+1} : T(a, u) = 0, s(u) \neq 0\}.$$

There are polynomials $B, N_0, \dots, N_{K-1}$ of degree at most $b_v = 1 + 2K(v - 1)$, with B nonzero wherever s is nonzero, such that the rational map

$$u \longmapsto (c_0(u), \dots, c_{K-1}(u)), \quad c_j = N_j / B,$$

is injective on U , has $c_j = u_j$ for $j \leq r$, and recovers the Taylor coefficients at a of every degree- $< K$ solution of $T = 0$ whose separant specialization is nonzero at a . The closure of U has dimension at most r and sum of component degrees at most v .

Proof. Apply the triangular recurrence of [Lemma A.1](#) with degree bound $K - 1$, retaining the initial coefficients u as variables. For $1 \leq h \leq H := K - 1 - r$, it gives

$$\binom{r+h}{r} s(u) c_{r+h} + R_h(u, c_{r+1}, \dots, c_{r+h-1}). \quad (123)$$

The leading multiplier is nonzero on U . We now bound the rational functions obtained by solving this recurrence.

Induction gives a common form for each new coefficient:

$$c_{r+h} = \frac{A_h(u)}{s(u)^{2h-1}}, \quad \deg A_h \leq (2h-1)(v-1) + 1. \quad (124)$$

Indeed, a monomial $u^\beta \prod_{i < h} c_{r+i}^{e_i}$ in R_h satisfies $|\beta| + \sum_i e_i \leq v$ and $\sum_i i e_i \leq h$. After substituting the earlier rational coefficients, its denominator exponent is $\sum_i (2i-1)e_i \leq 2h-2$. If there is one later coefficient, its index is at most $h-1$; if there are at least two, subtract their number from $2 \sum_i i e_i$; if there are none, the exponent is zero. Clearing to s^{2h-2} bounds the numerator degree by

$$(2h-2)(v-1) + |\beta| + \sum_i e_i \leq (2h-1)(v-1) + 1.$$

Division by the leading scalar times s proves the induction step.

Take the uniform common denominator $B = s^{2K}$. For $j \leq r$, set $N_j = u_j s^{2K}$; for $j = r+h > r$, set $N_j = A_h s^{2K-(2h-1)}$. Since $\deg s \leq v-1$, all these numerators and the denominator have degree at most $1 + 2K(v-1) = b_v$. This convention also covers $H = 0$ and makes $B \neq 0$ equivalent to $s \neq 0$. The retained initial coordinates prove injectivity on that whole open set. The actual coefficients of a regular solution satisfy the same recurrence, so induction proves recovery. Finally, $T(a, u)$ is nonzero because its u_r derivative is $s(u)$. Its hypersurface has degree at most v and dimension at most r ; passing to the closure of its principal open subset only discards components contained in $V(s)$. $\square$

B Johnson interpolation and closed constants

This appendix records the inherited order-zero interpolant and the integral rounding behind [Theorem 5.12](#). BCHKS's count supplies one primitive polynomial $Q(X, Z, Y)$ that explains every close polynomial on the received line. The degree-sensitive transfer, including its Frobenius pullback in small characteristic, is proved with its statement in [Section 5.3](#).

B.1 The inherited order-zero interpolant

Fix the notation of [Theorem 5.12](#), and set

$$C_{\text{spec}} = t\sqrt{Dn} = tn\sqrt{\rho_-}, \quad C_{\text{jet}} = t\sqrt{n/D} = \frac{t}{\sqrt{\rho_-}}, \quad C_Z = \frac{t^2n}{3D} = \frac{t^2}{3\rho_-}.$$

Thus $B_{\text{jet}} = \lceil C_{\text{jet}} \rceil - 1$ and $B_Z = \lceil C_Z \rceil - 1$.

Proposition B.1 (BCHKS order-zero symbolic interpolant). *For every field F , every distinct $\alpha_1, \dots, \alpha_n \in F$, and every $f, g \in F^n$, there is a polynomial $Q \in F[X, Z, Y]$ such that*

$$\begin{aligned} Q(X, z, Y) &\neq 0 && \text{for every } z \text{ in every extension of } F, && (125) \\ \text{wdeg}_{1,0,D} Q &< C_{\text{spec}}, && \deg_Y Q \leq B_{\text{jet}}, && \\ \text{wdeg}_{0,1,1} Q &< C_Z, && \deg_Z Q \leq B_Z. && (126) \end{aligned}$$

For every field extension E/F , every $z \in E$, and every $P \in E[X]_{\leq D}$ with $|S_z(P)| \geq A$, one has

$$Q(X, z, P(X)) = 0. \quad (127)$$

Proof. Use the BCHKS monomial space

$$Q = \text{span}_F \{X^i Y^j Z^\ell : i + Dj < C_{\text{spec}}, j + \ell < C_Z\}.$$

At each position r , require every coefficient of $U^a V^b$, for $a + b < m$, in $Q(\alpha_r + U, Z, f_r + Zg_r + V)$ to vanish as a polynomial in Z . Translation in Y preserves total Y, Z degree, so the scalar counts are

$$\begin{aligned} N_{\text{var}} &= \sum_{j=0}^{B_{\text{jet}}} (\lceil C_{\text{spec}} \rceil - Dj)(\lceil C_Z \rceil - j), \\ N_{\text{eq}} &= n \sum_{b=0}^{m-1} (m - b)(\lceil C_Z \rceil - b). \end{aligned} \quad (128)$$

For $m \geq 3$, BCHKS's dimension estimate gives $N_{\text{var}} > N_{\text{eq}}$ for these cutoffs [[BCH⁺26](#), full version, Lemma 3.1]. Divide a nonzero solution's coefficient vector in $F[Z]$ by its greatest common divisor. The constraints and degree bounds remain valid, while Bézout's identity proves (125).

For soundness, the definition of m gives $m\eta_0 \geq \sqrt{\rho_-}/2$, and therefore

$$C_{\text{spec}} = (m + \frac{1}{2})n\sqrt{\rho_-} \leq mn(\sqrt{\rho_-} + \eta_0) = man \leq mA.$$

If $P(\alpha_r) = f_r + zg_r$, the imposed bivariate multiplicity gives a root of multiplicity at least m in $Q(X, z, P(X))$ at α_r . Its degree is less than $C_{\text{spec}} \leq mA$, so A agreements force it to vanish. This polynomial argument remains valid over every scalar extension, proving (127). $\square$

For a fixed z , every member of $\mathcal{L}_D(f + zg, A)$ gives a distinct linear factor $Y - P(X)$ of $Q(X, z, Y)$ over $F(X)$. Consequently $|\mathcal{L}_D(f + zg, A)| \leq \deg_Y Q \leq B_{\text{jet}}$. Applying the same construction with $g = 0$ proves the first term in (67) for every received word, by the Guruswami–Sudan factor argument [GS99]. For the classical pairwise bound, choose A agreements for each of L candidates. Their pairwise intersections have size at most D ; Cauchy–Schwarz gives $L^2 A^2 / n \leq LA + L(L - 1)D$. As $A^2 > nD$, this yields the other term $L \leq n(A - D)/(A^2 - nD)$ in (67).

B.2 Closed constants

Applying Lemma 5.3 with $L = D + 1$ to Proposition B.1 gives the first bound in (68). To prove the closed form, write $x = \sqrt{\rho_-}$ and $y = 1/n$. Since $1 \leq D \leq n - 2$,

$$y \leq \min\{x^2, (1 - x^2)/2\}. \quad (129)$$

The rounded parameters satisfy

$$B_{\text{jet}} < \frac{t}{x}, \quad B_Z < \frac{t^2}{3x^2}, \quad \lambda_{\text{agr}} \leq \frac{1 - x^2}{x - x^2} = \frac{1 + x}{x}.$$

Use $\Psi_D(B_{\text{jet}}) \leq 1 + 4DB_{\text{jet}}$ in (39). Retaining the negative unit in $n - D - 1$ and combining its term with $\lambda_{\text{agr}}B_{\text{jet}}$ gives

$$\frac{B_{\text{cf}}}{nt^3/x^2} < \frac{4(1 + x)}{3} + \left(\frac{2}{3x} + \frac{1 + x}{3tx} + \frac{1}{t^2} \right) y + \frac{x(1 - x^2)}{t^2}.$$

The right side decreases with $t \geq 7/2$. Substituting $t = 7/2$ and (129) reduces the claim to $U(x) < 8/3$ for

$$U(x) = \frac{4(1 + x)}{3} + \left(\frac{16}{21x} + \frac{26}{147} \right) \min\{x^2, (1 - x^2)/2\} + \frac{4x(1 - x^2)}{49}.$$

Indeed,

$$\begin{aligned} x \leq 1/\sqrt{3} : \quad & \frac{8}{3} - U(x) = \frac{196 - 320x - 26x^2 + 12x^3}{147} > 0, \\ x \geq 1/\sqrt{3} : \quad & \frac{8}{3} - U(x) = \frac{(1 - x)(-12x^3 - 25x^2 + 127x - 56)}{147x} > 0. \end{aligned}$$

In the first case the numerator decreases to $562/3 - 316/\sqrt{3} > 0$; in the second, the cubic factor has derivative at least 41 and begins at $123/\sqrt{3} - 193/3 > 0$. This proves the strict bound in (68).

C Numerical first-order certificates

The first-order counting arguments and the finite support certificate are proved with their statements in Section 5.4. Here we verify the two fixed supports that give the uniform first-order guarantee.

C.1 A uniform first-order certificate

We now show what two nearby support choices give uniformly: using only the first derivative already bounds the list by a constant times n and the exceptional challenges by a constant times n^2 , at agreement $k + 0.24n$. At rate $1/4$, this requires $\lceil 0.49n \rceil$ agreements, compared with the limiting Johnson threshold $0.50n$. This certificate supplies the larger-gap case of Theorem 6.8. Additional reference certificates and numerical refinements are provided with the computational supplement [DKT26]; the three applications instead use the finite tuner at their actual code parameters.

Proof of Theorem 5.13. It suffices to prove the claim at $\delta = 6/25$: increasing the agreement threshold only removes candidates and exceptional witnesses. For $k = 1$, Lemma 3.2 gives list size at most $\lfloor n/A \rfloor$ and exceptional count at most $\lfloor \binom{n}{2}/(A-1) \rfloor$; here $A \geq 2$. This conclusion holds in every characteristic. For $k \geq 2$, first consider characteristics $p \leq 4$. The hypothesis $p > k - 1$ implies $k \leq p \leq 3$. Counting k -subsets of agreeing positions gives

$$|\mathcal{L}(y, A)| \binom{A}{k} \leq \binom{n}{k}, \quad |\mathcal{L}(y, A)| \leq \left(\frac{n}{A-k+1} \right)^k < (25/6)^3 < 73 < 307n.$$

For MCA at $n \geq 278$, put $D = k - 1$ and $a = A/n$. Then $D/n \leq (6/25)^2/4$ and the Johnson gap is at least $3/25$. Theorem 5.12 uses $m = 3$ and gives $|\mathcal{E}| < (343/3)n^2/D \leq (343/3)n^2$. For $n < 278$, interpolate witnesses F_T, G_T on each k -subset T . At every position outside their common agreement set, the residual $(F_T(\alpha_i) - f_i) + z(G_T(\alpha_i) - g_i)$ is a nonzero affine polynomial in z . Exclude its roots for all T and i ; there are at most

$$\binom{n}{k}(n-k) \leq n^4 < 278^2 n^2 < 1325775n^2$$

such challenges. Outside this set, choose T among any close candidate's agreements. Interpolation gives $P = F_T + zG_T$, and no position outside the common agreement set can agree. Thus the full agreement sets coincide. This proves both claims in the small-characteristic case.

It remains to handle characteristic zero or $p > 4$. Use reconstruction dimension $K = k$ and interpolation weight $D_{\text{int}} = k - 1$. For the list bound, use the equal-weight support $(m, B_\partial, B_{\text{jet}}) = (12, 4, 22)$. The rank bounds grouped by total jet degree are

$$(R_{\text{loc}}(0), \dots, R_{\text{loc}}(22)) = (12, 22, 30, 36, 40, 35, 30, 25, 20, 16, 12, 9, 6, 4, 2, 1, 0, 0, 0, 0, 0, 0, 0).$$

They sum to 300, and $\sum_t t R_{\text{loc}}(t) = 1570$. With $N_{\text{jet}}(t) = \min(t, 4) + 1$ and $\rho = k/n$, put

$$N_{\text{pos}}(t) = N_{\text{jet}}(t)(72/25 + \rho(12 - t))_+.$$

The exact normalized X -coefficient length is $(12A/n - \rho t + t/n)_+$. Using the agreement lower bound, it is at least $72/25 + \rho(12 - t) + t/n$, so the number of source coefficients of jet degree t is at least $nN_{\text{pos}}(t)$. Indeed,

$$\sum_t N_{\text{jet}}(t) = 105, \quad \sum_t N_{\text{jet}}(t)(12 - t) = 5.$$

Dropping the positive part only decreases the source lower bound, so

$$N_{\text{coef}} \geq n(1512/5 + 5\rho) > 300n.$$

This proves the positive coefficient margin uniformly over the rate.

We now apply Lemma 5.6 to count the close solutions. Both interpolation and reconstruction use the actual message degree $D = D_{\text{int}} = k - 1$. In positive characteristic the requirement is therefore only $p > \max\{k - 1, 4\}$. Put $\lambda_{\text{agr}} = (n - D)/(A - D)$. Since $A - D \geq 1 + \delta n$,

$$\lambda_{\text{agr}} \leq \frac{1}{\delta}, \quad D\lambda_{\text{agr}} \leq \frac{D(n - D)}{\delta n} \leq \frac{n}{4\delta}.$$

For this support, the Taylor exponent in Lemma 5.6 gives the exact regular-fiber budget

$$F = \begin{cases} 22, & D = 1, \\ 294D - 410, & D \geq 2, \end{cases} \quad S = 138.$$

The first line is the identity map on the initial Taylor pair. Both cases give the convenient bound $F \leq 294D - 134$. Consequently

$$\Lambda \leq \lambda_{\text{agr}}(294D - 134) + 138 \leq \frac{294n^2}{4(\delta n + 1)} + 4 < \frac{1225}{4}n < 307n.$$

Here $\lambda_{\text{agr}} \geq 1$ gives the +4, while $D\lambda_{\text{agr}} \leq D(n - D)/(\delta n + 1) \leq n^2/(4(\delta n + 1))$. For $\delta = 6/25$, the penultimate inequality reduces after clearing the positive denominator to $16 < (30529/25)n$, so it holds for $n \geq 2$. This argument also holds over every extension field.

For MCA, enlarge the same equal-weight support to $(m, B_{\partial}, B_{\text{jet}}) = (12, 4, 23)$ and set $H = 276$. The added total-degree shell has local rank zero, so the rank remains 300. Since $A \geq k + (6/25)n$, the normalized source count in degree t is at least

$$N_{\text{pos},t}(\rho) = (\min\{t, 4\} + 1) \left(\frac{72}{25} + \rho(12 - t) \right)_+, \quad 0 \leq \rho = \frac{k}{n} \leq \frac{19}{25}.$$

Both the coefficient margin and the height-276 row margin are continuous and piecewise affine in ρ . Their only possible changes of slope occur at

$$\left\{ 0, \frac{19}{25} \right\} \cup \left\{ \frac{72}{25s} : 4 \leq s \leq 11 \right\}.$$

Checking these ten points gives

$$\min_{\rho} \left(\sum_{t=0}^{23} N_{\text{pos},t}(\rho) - 300 \right) = \frac{204}{55}, \quad \min_{\rho} \sum_{t=0}^{23} (N_{\text{pos},t}(\rho) - R_{\text{loc}}(t))(277 - t) = \frac{14}{5}.$$

Both minima occur at $\rho = 72/275$. Thus [Proposition 5.10](#) supplies a symbolic explainer of challenge degree at most 276 for every rate. The exact source has the additional nonnegative term t/n inside the positive part, so both margins remain valid. For the endpoint $k = 2$, feasibility gives $A \geq 3$; hence $12A \geq 36$, and the specialization cutoff is inactive through jet degree 23 in the displayed rank blocks. This verifies the coefficient and row-height tests for $D_{\text{int}} = 1$.

To pass this explainer through the successive-stage line bound, retain the actual derivative degree $r \leq 4$. Expanding the stage sums in [Lemma 5.9](#) shows that their largest values occur at $r = 4$. At $D = 1$, the identity-pair calculation gives $D_{\text{fiber}}(4) = 86$ and $D_{\text{joint}}(4) = 1276$; for $D \geq 2$, the sums are

$$D_{\text{fiber}}(4) = 724D - 962 \leq 724D, \quad D_{\text{joint}}(4) = 1149264D^2 - 3178072D + 2182340 \leq 1149264D^2.$$

The $D = 1$ values obey the same upper bounds $724D$ and $1149264D^2$. We use these bounds uniformly in what follows. Choose $L = D + \lceil (A - D)/42 \rceil$. Then

$$\frac{n - L + 1}{A - L + 1} \leq \frac{42}{41} \lambda_{\text{agr}}, \quad \frac{n - D}{L - D} \leq 42 \lambda_{\text{agr}}.$$

Apply [Lemma 5.9](#), choosing $L_0 = k$ for the optimized ordinary tail of total degree $B = 23$ and challenge degree $H = 276$, using $\Psi_D(B) \leq 1 + 4DB$. The preceding inequalities, $\lambda_{\text{agr}} \leq 1/\delta$, $D\lambda_{\text{agr}} \leq n/(4\delta)$, and $n \geq 2$ give

$$\begin{aligned} \frac{|\mathcal{E}|}{n^2} &\leq \frac{(2B - 1)H}{4} + \frac{H + B}{4\delta} + \frac{BH}{2\delta} + \frac{B}{2} + \frac{42}{41} \frac{1149264}{16\delta^2} + 42 \frac{724}{4\delta} \\ &= \frac{1304562211}{984} < 1325775. \end{aligned} \tag{130}$$

□

At rate $\rho = 1/4$, the corresponding radius is 0.51, compared with the limiting Johnson radius 0.5. More generally, the radius $1 - \rho - \delta$ exceeds $1 - \sqrt{\rho}$ exactly when $\sqrt{\rho} - \rho > \delta$. For $\delta = 0.24$ this is the open interval $0.16 < \rho < 0.36$. The distance-based Johnson radius is $1 - \sqrt{(k-1)/n}$; on a closed subinterval of this rate window, our radius exceeds it for sufficiently large n .

D Finite estimates for capacity parameters

The candidate-cover and symbolic-transfer proofs appear with their statements in [Section 6.3](#). Here we verify the numerical moment bounds and finite rounding estimates used to choose uniform capacity parameters.

D.1 Numerical bounds for the simplex moment

The proof of [Lemma 6.1](#) uses the mean and variance to retain a coefficient advantage. We verify its finite numerical input here.

Lemma D.1 (Finite moment estimates). *For $d \geq 500$ and $Z_d = dM_d - \log d$ as in [Lemma 6.1](#),*

$$(\log 6 - \mathbb{E}[Z_d])^2 + \text{Var}(Z_d) - \frac{1}{3} > \frac{27}{10}.$$

Proof. Write $H_d = \sum_{j=1}^d 1/j$ and $H_d^{(2)} = \sum_{j=1}^d 1/j^2$. For $d \geq 500$, we use

$$H_d - \log d < \frac{29}{50}, \quad H_d^{(2)} > \frac{41}{25}, \quad \log 6 > \frac{179}{100}.$$

For the first, $H_d - \log d$ decreases because $\log(1 + 1/d) > 1/(d+1)$, and the finite bounds $H_{200} < 58781/10000$ and $\log 200 > 52981/10000$ suffice. For the second, integrate x^{-2} only through the finite endpoint:

$$H_d^{(2)} \geq \sum_{j=1}^{20} \frac{1}{j^2} + \frac{1}{21} - \frac{1}{d+1} \geq \sum_{j=1}^{20} \frac{1}{j^2} + \frac{1}{21} - \frac{1}{501} > \frac{41}{25}.$$

Put $c = 29/50$, $v = 41/25$, $b = 121/100$, and $t = \log d + c$. The exact mean and variance imply

$$\mathbb{E}[Z_d] \leq c - \frac{t}{d+1}, \quad \text{Var}(Z_d) \geq v(1 - 3/d) - t^2/d.$$

Substituting these estimates gives

$$(\log 6 - \mathbb{E}[Z_d])^2 + \text{Var}(Z_d) - \frac{1}{3} > b^2 + v - \frac{1}{3} - \frac{P(t)}{d}, \quad P(t) = t^2 - 2b\frac{500}{501}t + 3v.$$

The function $P(\log d + c)/d$ decreases for $d \geq 500$: its derivative has the sign of $P'(t) - P(t)$, which is negative for $t > 6$. Since $\log 500 < 6.22$ and P increases there,

$$b^2 + v - \frac{1}{3} - \frac{P(6.8)}{500} = \frac{67667389}{25050000} > \frac{27}{10}.$$

This proves the stated finite inequality. The finite logarithm checks can be made rational: for $0 < z < 1$, truncate $2 \sum_{j \geq 0} z^{2j+1}/(2j+1)$ and bound its tail by $2z^{2s+1}/((2s+1)(1-z^2))$ after s terms. $\square$

D.2 The finite dimension ratio

We next keep the integer errors in the coefficient and rank counts. This supplies both a terminating parameter choice whenever $\Gamma > 1$ and a convenient closed choice when a larger margin is available. Cube covering gives the rank estimate

$$\frac{R_{\text{loc}}}{m^2 \text{vol}(S_W)} \leq \frac{1}{d+1} \int_0^1 \left(1 - t + \frac{d+1}{m}\right) \left(1 + \frac{t + d(d+1)/(2m)}{w}\right)^d dt. \quad (131)$$

The proof below compares this integral with the source estimate while retaining every rounding term.

Proof of Lemma 6.2. For fixed b , compare the decreasing sum over u with its integral:

$$\sum_{u \geq 0} (mA_0 - D(u + |b|))_+ \geq \frac{n}{2\rho} (ma - \rho|b|)_+^2.$$

For each point in the weighted simplex S_W , flooring coordinates gives an eligible b , decreases $|b|$, and assigns volume at most one to each lattice point. Integrating proves (73). The ordered-coordinate change of variables in the body identifies $|U| = wM_d$ in distribution. Since $w \leq ad/(\rho \log(6d))$, we have $a\lambda/\rho - \log d \geq \log 6$. Thus

$$\frac{N_{\text{cof}}}{nm^2 \text{vol}(S_W)} \geq \frac{\rho}{2\lambda^2} \mathbb{E}[(a\lambda/\rho - dM_d)_+^2] > \frac{27\rho}{20\lambda^2}.$$

For the rank, the disjoint unit cubes based at eligible integer b lie in S_{B+S_d} , proving $p_d(B) \leq (B + S_d)^d/(d!)^2$. Also $\lceil (m-s)/(d+1) \rceil \leq (m-s+d)/(d+1)$. For $t \in [s/m, (s+1)/m]$, the integrand in (131) is at least

$$\left(1 - \frac{s}{m} + \frac{d}{m}\right) \left(1 + \frac{s/m + S_d/m}{w}\right)^d.$$

Summing these intervals proves that inequality. Using $(1+x)^d \leq e^{dx}$, and then extending the nonnegative integral after the substitution $u = 1 - t$, bounds it by

$$\frac{e^{\lambda(1+S_d/m)}}{d+1} \int_0^\infty \left(u + \frac{d+1}{m}\right) e^{-\lambda u} du = \frac{e^{\lambda(1+S_d/m)}}{(d+1)\lambda^2} \left(1 + \frac{(d+1)\lambda}{m}\right).$$

Dividing the source and rank estimates proves the ratio. For $R_{\text{loc}} = 0$, the constant monomial already proves positive dimension. The limit follows from $W/m \rightarrow ad/(\rho \log(6d))$.

For the explicit choice, write $L = \log(6d)$, $w_0 = ad/(\rho L)$, $\lambda_0 = (\rho/a)L$, and $\epsilon = 1/(mw_0) \leq 1/(300d^3)$. Then

$$\lambda \leq \frac{\lambda_0}{1-\epsilon}, \quad \lambda - \lambda_0 \leq \frac{2L}{300d^3}, \quad \frac{\lambda S_d}{m} \leq \frac{1+1/d}{600(1-\epsilon)}, \quad \log\left(1 + \frac{(d+1)\lambda}{m}\right) \leq \frac{1+1/d}{300d(1-\epsilon)}.$$

For $d \geq 519$, $L < d$, and the sum of these three losses is at most

$$\frac{2}{300d^2} + \frac{1+1/d}{600(1-1/(300d^3))} + \frac{1+1/d}{300d(1-1/(300d^3))}.$$

This expression decreases with d and is less than $1677/10^6$ at $d = 519$. This proves (76). $\square$

D.3 The final scalar margin

The rate split in [Section 6.6](#) gives $\log \Gamma \geq 3/2 - \log(40/9)$. After the finite rounding loss, the following strict inequality certifies the required coefficient margin.

Lemma D.2 (A rational check of the uniform margin). *One has*

$$\frac{3}{2} - \log(40/9) - \frac{1677}{10^6} > \log(151/150) + \frac{1}{50000}.$$

Proof. Combine the logarithms as $\log(604/135)$ and put $z = 469/739$. The rational Taylor bound gives

$$\log(604/135) < 2 \sum_{j=0}^{19} \frac{z^{2j+1}}{2j+1} + \frac{2z^{41}}{41(1-z^2)} < \frac{1498303}{10^6}.$$

The second inequality is an exact rational comparison. It leaves the strict margin

$$\frac{3}{2} - \frac{1677}{10^6} - \frac{1498303}{10^6} = \frac{1}{50000}.$$

□

E Arithmetic for the decoding algorithms

We justify the quotient-ring arithmetic, polynomial-system algorithm, and auxiliary-field constructions used in [Section 9](#).

E.1 Supporting decoder pseudocode

The main text states the three procedures that carry the decoder's central ideas: exact agreement recovery, the zeroth-order specialization, and the first-order curve filter. For completeness, this subsection records the supporting interpolation, higher-order, and finite-case procedures called there.

Algorithm 4 Construct a fixed-word explainer by a minimal interpolation basis.

Require: Received pairs $(\alpha_i, y_i)_{i=1}^n$ and certified data $(d, m, B, (M_j, w_j)_{j=1}^s, W)$ as in [Lemma 9.3](#)

Ensure: A nonzero explainer in $\mathcal{V}$, or CERTIFICATE-FAILURE

```

1: procedure FASTFIXEDWORDEXPLAINER( $\alpha, y, d, m, M, w, W$ )
2:   form  $e_j = (\Phi_{\alpha_i, y_i}^{d, m}(M_j))_{i=1}^n$  in the full local modules of (97)
3:   represent multiplication by  $X$  there by its Jordan matrix  $J$ 
4:   compute a minimal interpolation basis for  $(e_1, \dots, e_s; J)$  with shift  $(w_1, \dots, w_s)$  using [JNSV17, Theorem 1.5]
5:   choose a basis row  $(f_1, \dots, f_s)$  of minimum shifted degree
6:   if  $\max_j (\deg f_j + w_j) \geq W$  then
7:     return CERTIFICATE-FAILURE
8:   end if
9:   return  $Q = \sum_{j=1}^s f_j(X) M_j$ 
10: end procedure

```

We write Opts for the supplied order, degree bounds, characteristic and field-size conditions, jet monomials and their per-monomial X -degree bounds.

Algorithm 5 Higher-order candidates from isolated agreement systems

Require: regular order- r chart $(h, s, B_0, N_0, \dots, N_{k-1}, g_1, \dots, g_n)$, k , and OPTS

Ensure: univariate finite representations covering every wanted regular chart point

```
1: procedure AGREEMENTCANDIDATES( $\mathcal{C}, k, \text{OPTS}$ )
2:   if  $r = 0$  then
3:     return the separant-filtered squarefree chart family
4:   end if
5:    $\text{SELECTIONS} \leftarrow \{J \subseteq \{1, \dots, n\} : |J| = r\}$ 
6:    $D \leftarrow$  a common degree bound for  $h, g_i$ ;  $m \leftarrow r + 1$ 
7:   choose the least bounded-degree extension  $E'/E$  with  $|E'| > (m + 1)^2 D^{2m}$ 
8:    $\text{FAMILIES} \leftarrow \emptyset$ 
9:   for  $J \in \text{SELECTIONS}$  do
10:     $\mathcal{S}_J \leftarrow \{h = 0\} \cup \{g_i = 0 : i \in J\}$ 
11:     $\mathcal{U}_J \leftarrow \text{ROJASISOLATEDROOTREPRESENTATION}(\mathcal{S}_J \text{ over } E')$ 
    The representation must contain all isolated roots; other components may have positive dimension.
12:    squarefree-reduce each parameter modulus and discard exactly the loci  $s = 0$ 
13:    substitute the parameter maps into  $N_0, \dots, N_{k-1}, B_0$  and invert  $B_0$ 
14:    append the resulting univariate families carrying  $C(X)$  to  $\text{FAMILIES}$ 
15:   end for
16:   return  $\text{FAMILIES}$ 
17: end procedure
```

For the complete procedure, N_* is the certified bounded-instance threshold. A failed interpolation certificate reports an unsupported input, not an empty list. The `EXACTBYSUBSETS` branch enumerates received-position subsets, never field elements; [Section E.8](#) proves when its cost fits the stated bound. The characteristic hypothesis is an input promise, not a reason to invoke that branch. The fixed-gap corollaries handle additional bounded-message cases before this procedure, using [Lemma 9.6](#).

Algorithm 6 Exact positive-order Reed–Solomon agreement decoding from a certified hidden-derivative support

Require: decoder-admissible presentation K of $\mathbb{F}_q$, $q \geq n$; order $d \geq 1$; distinct $\alpha_i \in \mathbb{F}_q$; $y \in \mathbb{F}_q^n$; $1 \leq k \leq n$, $k \leq A$; for $k \geq 2$, $p > \max\{k - 1, B_{\text{jet}}\}$; certified support data or Q

Ensure: under the stated input promises, exactly $\mathcal{L}(y, A)$ as duplicate-free coefficient vectors over $\mathbb{F}_q$; an interpolation failure returns CERTIFICATE-FAILURE

```

1: procedure EXACTHIDDENDERIVATIVEDECODE( $K, \alpha, y, k, A, \text{OPTS}$ )
2:   if  $A > n$  then return  $\emptyset$ 
3:   if  $k = 1$  then return values occurring at least  $A$  times in sorted  $y$ 
4:   if  $n < N_*$ , or  $k \leq d$ , or a certified field/grid/center condition implying  $n < N_*$  is violated then
5:     return EXACTBYSUBSETS( $\alpha, y, k, A$ )
6:   end if
7:   if  $Q$  is not supplied then
8:      $Q \leftarrow \text{FASTFIXEDWORDEXPLAINER}(\alpha, y, d, m, M, w, W)$  on the certified support  $\mathcal{V}$  ▷ Algorithm 4
9:     if the routine reports failure then return CERTIFICATE-FAILURE
10:  end if
11:  if  $d = 1$  and  $p > 2(B_{\text{jet}} + 1)^3$  then
12:    return COMMONCENTERFIRSTORDER( $Q, \alpha, y, k, A$ ) ▷ Proposition 9.11
13:  end if
14:   $T_0 \leftarrow Q$ ;  $j \leftarrow 0$ ; FAMILIES  $\leftarrow \emptyset$ 
15:  while  $T_j$  depends on a jet variable do
16:     $r_j \leftarrow \max\{r : T_j \text{ depends on } Y_r\}$ ;  $S_j \leftarrow \partial T_j / \partial Y_{r_j}$ ;  $T_{j+1} \leftarrow S_j$ 
17:     $H_j \leftarrow$  a proved bound on  $\deg_X S_j(X, P, \dots, P^{[r_j]})$ 
18:    choose the prescribed field  $E_j$  and  $H_j + 1$  distinct centers  $C_j \subseteq E_j$ 
19:    for  $c \in C_j$  do
20:       $\mathcal{C} \leftarrow \text{FASTREGULARTAYLORFAMILY}(E_j, T_j, c, \alpha, y, k)$  ▷ Algorithm 8
21:      if  $\mathcal{C} \neq \emptyset$  and  $r_j = 0$  then
22:        append its separant-filtered squarefree family to FAMILIES
23:      end if
24:      if  $\mathcal{C} \neq \emptyset$  and  $r_j = 1$  in first-order mode then
25:        FAMILIES  $\leftarrow$  FAMILIES  $\cup$  FIRSTORDERCURVECANDIDATES( $\mathcal{C}, A, k$ )
26:      else if  $\mathcal{C} \neq \emptyset$  and  $r_j \geq 1$  then
27:        FAMILIES  $\leftarrow$  FAMILIES  $\cup$  AGREEMENTCANDIDATES( $\mathcal{C}, k, \text{OPTS}$ )
28:      end if
29:    end for
30:     $j \leftarrow j + 1$ 
31:  end while
32:  return RECOVERAGREEMENT(FAMILIES,  $\alpha, y, k, A$ )
33: end procedure

```

E.2 A separable message equation in every characteristic

For the zeroth-order decoder, we need an equation with a regular fiber containing the initial value of every wanted message. In small characteristic, dividing Q by $\gcd(Q, Q_Y)$ alone can lose messages: for $Q = (Y - P(X))^p$, that quotient is 1. Conversely, removing repeated bivariate factors does not ensure separability in Y : the irreducible polynomial $Y^p - X$ still has zero Y -derivative. This distinction is standard in separable factorization [Lec08]. We first remove all repetitions, then discard the Y -inseparable factors, which cannot contain a polynomial message. The bounded Y -degree makes both steps inexpensive, including the required Frobenius inversions. This discard is specific to a fixed-word equation over $K[X, Y]$. It cannot replace the ordinary Frobenius pullback of Section 5.3 for a symbolic challenge Z : for example, $Y^p - Z$ acquires a constant polynomial root after every specialization $Z = z$ in a finite field, although it is Y -inseparable before specialization.

Before using inverse Frobenius in the separable preprocessing, we verify that the standard supplied field representations provide the operations required by [Definition 9.1](#). The relative quotient case contains the only nontrivial Frobenius construction needed below.

Lemma E.1 (Closure under a supplied quotient extension). *Let K_0 be decoder-admissibly presented, and let*

$$L = K_0[\theta]/(f(\theta)), \quad f(T) = \sum_{i=0}^m c_i T^i \in K_0[T]$$

for a supplied monic irreducible polynomial f . Represent elements of L in the relative power basis $1, \theta, \dots, \theta^{m-1}$. Then L is decoder-admissibly presented, without conversion to an absolute polynomial basis.

Proof. The presentation has $\tilde{O}(m \log |K_0|) = \tilde{O}(\log |L|)$ bits. Fast arithmetic and extended-gcd inversion modulo f use $\tilde{O}(m)$ operations in K_0 , hence $\tilde{O}(\log |L|)$ bit operations [\[Sch15\]](#). Concatenated base-field encodings are canonical. Mixed-radix expansion, followed by elt_{K_0} on each digit, enumerates L within the same bound.

We construct inverse Frobenius directly in the relative representation. Twist the coefficients of f by inverse Frobenius in K_0 and write

$$f^{(-1)}(Z) = \sum_{i=0}^m c_i^{1/p} Z^i, \quad \beta = \theta^{1/p} \in L.$$

Then $f^{(-1)}(\beta)^p = f(\theta) = 0$. Reducing modulo $Z^p - \theta$ gives the degree- $< p$ polynomial

$$R_f(Z) = \sum_{r=0}^{p-1} \left(\sum_j c_{pj+r}^{1/p} \theta^j \right) Z^r.$$

Since $f^{(-1)}$ is separable and $Z^p - \theta = (Z - \beta)^p$ in $L[Z]$, we recover β from

$$\gcd(R_f(Z), Z^p - \theta) = Z - \beta.$$

One batch of $m + 1$ inverse-Frobenius evaluations in K_0 , this gcd, and the powers $1, \beta, \dots, \beta^{p-1}$ cost $\tilde{O}(p \log |L|)$ bits.

For $a = \sum_{i=0}^{m-1} a_i \theta^i$, the precomputed powers give

$$a^{1/p} = \sum_{r=0}^{p-1} \beta^r \left(\sum_j a_{pj+r}^{1/p} \theta^j \right).$$

Taking p -th powers verifies the identity. For a batch of N elements, one batch of Nm coefficient roots in K_0 , together with the coordinate scans and $O(pN)$ operations in L , costs $\tilde{O}(pN \log |L|)$ bits. Thus the relative quotient presentation satisfies every clause of [Definition 9.1](#). $\square$

Corollary E.2 (Batched inverse Frobenius in a polynomial basis). *Let $K = \mathbb{F}_p[\theta] = \mathbb{F}_p[t]/f(t)$ be supplied in a polynomial basis, where f is monic irreducible of degree e and $q = p^e$. This is a decoder-admissible presentation. In particular, one can preprocess K in $\tilde{O}(p \log q)$ bit operations and then compute the unique p -th roots of any batch of N elements deterministically in $\tilde{O}(pN \log q)$ bit operations.*

Proof. Represent $\mathbb{F}_p$ by the integers in $\{0, \dots, p-1\}$. Standard modular arithmetic, integer Euclid, and the identity enumeration and embedding maps supply clauses (i) and (ii) of [Definition 9.1](#); inverse Frobenius is the identity, so clause (iii) needs no preprocessing. Apply [Lemma E.1](#) with $K_0 = \mathbb{F}_p$ and $m = e$. The preprocessing and batch estimates in its proof give the two displayed bounds. In particular, this construction does not use exponentiation by p^{e-1} . $\square$

Iterating [Lemma E.1](#) through a supplied tower of s monic quotient extensions, with s fixed, gives a decoder-admissible recursive coefficient basis, with constants depending on s ; arbitrarily deep towers require their supplied global arithmetic to satisfy [Definition 9.1](#) independently.

Lemma E.3 (Fast absolute normal-basis presentations). *Suppose $K = \mathbb{F}_{p^e}$ is supplied in an absolute normal basis*

$$\eta, \eta^p, \dots, \eta^{p^{e-1}},$$

with total presentation size $\tilde{O}(\log |K|)$, including the prime-field embedding, and multiplication in $\tilde{O}(\log |K|)$ bit operations. Then the presentation is decoder-admissible. Inverse Frobenius is a cyclic coordinate shift.

Proof. Canonical encodings, addition, equality, and enumeration come from the e prime-field coordinates. Frobenius and inverse Frobenius cyclically shift those coordinates, so a batch of N inverse-Frobenius evaluations costs $O(N \log |K|)$ bits and needs no preprocessing.

Write $1 = (\lambda, \dots, \lambda)$ in the normal basis. The supplied prime-field embedding determines the nonzero scalar $\lambda \in \mathbb{F}_p$; conversely, supplying λ determines that embedding.

It remains to account for inversion. When $e = 1$, write the coordinate of a as α . Then $a = (\alpha/\lambda)1$, so its inverse has coordinate $\lambda(\alpha/\lambda)^{-1}$, computed by integer Euclid. Assume $e \geq 2$. For $a \neq 0$, define

$$S_m(a) = a^{1+p+\dots+p^{m-1}}.$$

The recurrences

$$S_{2m}(a) = S_m(a)S_m(a)^{p^m}, \quad S_{2m+1}(a) = S_{2m}(a)a^{p^{2m}}$$

compute $S_{e-1}(a)$ with $O(\log e)$ multiplications and cyclic shifts. Thus

$$r = S_{e-1}(a)^p = a^{p+\dots+p^{e-1}}, \quad ar \in \mathbb{F}_p.$$

A Frobenius-fixed normal-basis vector has equal coordinates. If $ar = (\mu, \dots, \mu)$, then $ar = (\mu/\lambda)1$. We invert μ/λ by integer Euclid and use the supplied embedding to represent its inverse in K . Hence $a^{-1} = r(ar)^{-1}$ has $\tilde{O}(\log |K|)$ bit cost. The assumed multiplication routine now supplies the remaining clause of [Definition 9.1](#). $\square$

Lemma E.4 (Separable preprocessing with bounded Y -degree). *Let $H \in K[X, Y]$ be nonzero and primitive in Y , with $\deg_Y H \leq \ell$ and $\deg_X H \leq W$, over a decoder-admissibly presented finite field $K = \mathbb{F}_q$. One can compute a primitive divisor T of H that is separable over $K(X)$ and retains every factor $Y - P(X)$, $P \in K[X]$, in $\tilde{O}_\ell((W+1) \log q)$ deterministic bit operations. In particular, $\deg_Y T \leq \ell$ and $\deg_X T \leq W$. No irreducible factorization is required.*

Algorithm 7 Keep polynomial message factors in arbitrary characteristic

Require: Nonzero primitive $H \in K[X, Y]$, $\deg_Y H \leq \ell$; $p = \text{char } K$

Ensure: A separable divisor retaining every $Y - P(X)$ factor

```
1: procedure RADICAL( $H, \ell$ )
2:   if  $\deg_Y H = 0$  then
3:     return 1
4:   end if
5:    $G \leftarrow \gcd(H, H_X, H_Y)$ ;  $V \leftarrow H/G$ 
6:    $R \leftarrow H/\gcd(H, V^\ell)$  ▷ remove every copy of the factors visible in  $V$ 
7:   if  $\deg_Y R = 0$  then
8:     return  $V$ 
9:   end if
10:  Form  $S$  with  $S^p = R$ : divide both monomial exponents by  $p$  and apply the batched inverse Frobenius
    of Definition 9.1 to the coefficients
11:  return  $V \cdot \text{RADICAL}(S, \ell)$ 
12: end procedure

13: procedure SEPARABLEPART( $H, \ell$ )
14:    $S \leftarrow \text{RADICAL}(H, \ell)$ 
15:   return the primitive part of  $S/\gcd(S, S_Y)$ 
16: end procedure
```

Proof. All gcds and exact quotients in Algorithm 7 are in $K[X, Y]$, normalized up to a nonzero field scalar. They can be computed over $K(X)[Y]$ and made primitive by clearing denominators and removing content. To justify the algorithm, factor H conceptually as $u \prod_i h_i^{m_i}$ into distinct irreducibles. The algorithm does not compute this factorization. Primitivity gives $\deg_Y h_i \geq 1$, hence $m_i \leq \ell$. Because K is perfect, an irreducible h_i cannot have both partial derivatives zero. Thus $G = \gcd(H, H_X, H_Y)$ contains h_i to exponent $m_i - 1$ if $p \nmid m_i$, and to exponent m_i if $p \mid m_i$.

Consequently $V = H/G$ contains exactly once the factors with multiplicity not divisible by p . The gcd with V^ℓ removes all their copies from H . Every remaining multiplicity in R is divisible by p , so $R = S^p$ over K and both variable exponents are divisible by p . If R is nonconstant, then $p \leq \ell$. The factors of V and S are disjoint, and $\deg_Y S \leq \deg_Y H/p$. Induction proves that RADICAL returns the product of the distinct bivariate irreducible factors of H .

For this radical S , the quotient $S/\gcd(S, S_Y)$ retains precisely its factors with nonzero Y -derivative. By Gauss's lemma, they are separable irreducibles over $K(X)$, and their product is separable. Every wanted factor $Y - P(X)$ has derivative 1 and is therefore retained. The resulting primitive T divides H , which proves both degree bounds.

There are $O_\ell(1)$ recursion levels. All intermediate Y -degrees are bounded in terms of ℓ , including $\deg_Y V^\ell \leq \ell^2$; their X -degrees are $O_\ell(W + 1)$. Subresultant polynomial matrices have bounded size, so gcds, primitive parts, products, and exact divisions cost $\tilde{O}_\ell(W + 1)$ field operations. At every root step, $p \leq \ell$, and inverse Frobenius is applied to $O_\ell(W + 1)$ coefficients. Clause (iii) of Definition 9.1 therefore charges $\tilde{O}_\ell((W + 1) \log q)$ bits for the entire step, including its reusable preprocessing. This gives the asserted bit bound. $\square$

E.3 Fast construction of the regular Taylor family

The regular-chart lemma bounds the degrees of the rational coefficient formulas. To compute them, we expand around one parameter value, keeping enough terms in each free parameter to recover these formulas exactly.

We use Kopparty's coefficient recurrence [Kop15] and cache its products, so computing a new coefficient does not repeat all earlier work.

Lemma E.5 (Fast triangular Taylor recurrence). *Fix $r, B \geq 0$, let R be a commutative ring, and let $0 \leq r < k$. Suppose*

$$T \in R[Z, Y_0, \dots, Y_r]$$

has jet degree at most B , the initial coefficients $c_0, \dots, c_r \in R$ make the constant residual zero, and

$$s = T_{Y_r}(0, c_0, \dots, c_r) \quad \text{and} \quad \binom{m+r}{r} \quad (1 \leq m < k-r)$$

are units, with their inverses supplied. From the Z -coefficient streams of T through degree $k-r-1$, we can compute the unique $Y = \sum_{j < k} c_j Z^j$ satisfying

$$T(Z, Y, Y^{[1]}, \dots, Y^{[r]}) = 0 \pmod{Z^{k-r}}.$$

If $M_R(\ell)$ is the cost of multiplying length- ℓ streams over R , the cost is

$$O_{r,B} \left(\sum_{\substack{\ell=1,2,4,\dots \\ \ell \leq k}} \frac{k}{\ell} M_R(\ell) \right)$$

operations in the representation of R . When $r = 0$, the binomial condition holds in every characteristic.

Proof. Kopparty's coefficient lift [Kop15] gives, at stage $m \geq 1$,

$$[Z^m]T(Z, Y, Y^{[1]}, \dots, Y^{[r]}) = \binom{m+r}{r} s c_{m+r} + R_m(c_0, \dots, c_{m+r-1}). \quad (132)$$

This coefficient identity is polynomial, hence valid over R . Its leading coefficient is a unit, so it successively determines $c_{r+1}, \dots, c_{k-1}$ and proves finite-precision uniqueness. We make no claim about an infinite formal solution.

For speed, retain the coefficient stream at every multiplication gate. For inputs a, b , separate the terms involving the current coefficient:

$$[Z^m](ab) = a_0 b_m + a_m b_0 + \sum_{\substack{i,j \geq 1 \\ i+j=m}} a_i b_j. \quad (133)$$

The sum is already fixed. Following van der Hoeven's relaxed-multiplication schedule [vdH02], for each dyadic ℓ we multiply $I = [\ell, 2\ell]$ by itself and, in both orders, by each later block $J = [q\ell, (q+1)\ell]$, $q \geq 2$. These products partition all positive-index pairs. A block ending at $(q+1)\ell - 1$ first contributes at $(q+1)\ell$, so the products are ready before use. The current gate coefficients remain affine in c_{m+r} ; we substitute its solved value before finalizing them and scheduling more products. There are $O(k/\ell)$ length- $O(\ell)$ products at scale ℓ , which proves the cost bound. $\square$

Lemma E.6 (Fast regular Taylor family). *Fix r and B_{jet} . Let E be a finite field of characteristic p , let $0 \leq r < k$, and let*

$$T \in E[X, Y_0, \dots, Y_r], \quad \deg_{Y_0, \dots, Y_r} T \leq B_{\text{jet}}, \quad \deg_X T \leq D_X.$$

Fix $c \in E$, and put

$$H(u) = T(c, u_0, \dots, u_r), \quad S(u) = T_{Y_r}(c, u_0, \dots, u_r).$$

Suppose $p > \max\{k-1, B_{\text{jet}}\}$. Let H_* be the reduced union of the components of $H = 0$ on which S is not identically zero, and suppose H_* is nonconstant. There is a linear coordinate map

$$(u_0, \dots, u_r) = M(t_0, \dots, t_{r-1}, z)$$

and polynomials $B_0, N_0, \dots, N_{k-1} \in E[t_0, \dots, t_{r-1}, z]$ such that:

(i) their z -degrees are less than $b := \deg H_*$ and their total degrees are at most

$$L := 1 + 2k(B_{\text{jet}} - 1);$$

(ii) in the coordinate ring of H_* , $B_0 = S(M(t, z))^{2k}$. In its localization at S , the coefficient formulas give

$$P_{t,z}(c + Z) = \sum_{j < k} \frac{N_j(t, z)}{B_0(t, z)} Z^j;$$

(iii) this series recovers every regular degree- $< k$ polynomial solution of the differential equation defined by T whose initial jet lies on $H_* = 0$; here regular means that the same jet satisfies $S \neq 0$.

For a fixed guard $C_{r, B_{\text{jet}}}$, if $|E| > C_{r, B_{\text{jet}}}$, the coordinate map and all the displayed polynomials are constructed deterministically in

$$\widetilde{O}_{r, B_{\text{jet}}}(k^{r+1} + D_X)$$

operations in E , with $O_{r, B_{\text{jet}}}(k^{r+1})$ output field coefficients and no extension of E .

Constructing the chart and its coefficient normal forms has the cost stated in [Lemma E.6](#). Given the received pairs, the final loop constructs all agreement polynomials in $\widetilde{O}_{r, B_{\text{jet}}}(n^{r+1})$ additional operations in E .

Proof. Monic chart preparation. We first isolate the regular components and choose coordinates for a monic projection. In characteristic greater than B_{jet} , bounded-degree gcds compute

$$H_{\text{rad}} = \frac{H}{\gcd(H, H_{u_0}, \dots, H_{u_r})}, \quad H_* = \frac{H_{\text{rad}}}{\gcd(H_{\text{rad}}, S)}.$$

Let $(H_*)_b$ be the top homogeneous part. Choose $v \in E^{r+1}$ with $(H_*)_b(v) \neq 0$, make v the last column of M , and scale the equation. A constant grid finds such a v , since b is at most B_{jet} . We obtain

$$h(t, z) = z^b + h_{b-1}(t)z^{b-1} + \dots + h_0(t), \quad \deg h_i \leq b - i. \quad (134)$$

Thus reduction modulo h does not increase total degree: replacing z^b using (134) replaces each monomial only by monomials of no larger total degree.

Making the separant invertible. Write $s(t, z) = S(M(t, z))$. Since H_* contains no component on which S vanishes identically,

$$\Delta(t) := \text{Res}_z(h, s)$$

is a nonzero polynomial of degree bounded in terms of r, B_{jet} . Choose $a \in E^r$ with $\Delta(a) \neq 0$ by testing a constant grid. Set $N = L + 1$ and form

$$C_N = E[\epsilon_0, \dots, \epsilon_{r-1}] / (\epsilon_0^N, \dots, \epsilon_{r-1}^N), \quad R = C_N[z] / (h(a + \epsilon, z)).$$

The ring R is free of rank b over C_N . Moreover, s is a unit in R : a Bézout identity for $h(a, z)$ and $s(a, z)$ gives its inverse in the constant fiber, and Newton inversion lifts that inverse through the nilpotent parameter ideal. The fiber polynomial $h(a, z)$ need not be squarefree. Only the separant s , rather than the projection discriminant, must be a unit.

Taylor recurrence and cost. We shift the X -coefficients of T to $c + Z$ by division-free divide-and-conquer linear composition before entering R , costing $\tilde{O}(D_X)$ even when $D_X \geq p$. We then apply [Lemma E.5](#) over this possibly nonreduced ring. Set $t = a + \epsilon$ and initialize $c_j = M_j(t, z)$ for $0 \leq j \leq r$. The constant residual is zero because $h(t, z) = 0$, while s and every required binomial coefficient are units. Thus we obtain $c_{r+1}, \dots, c_{k-1}$ and only require the equation modulo Z^{k-r} . The inverse of s was computed above; computing the scalar binomial coefficients and their inverses takes $O_r(k)$ operations in E . Every wanted polynomial satisfies the same finite recurrence, so uniqueness proves that the chart recovers it.

Here the cost in [Lemma E.5](#) is evaluated in the specific confluent algebra $R = C_N[z]/h(a + \epsilon, z)$, not from a generic multiplication table. Its dimension is $bN^r = O_{r, B_{\text{jet}}}(k^r)$, and encoding the exponents of Z and the fixed ϵ variables in one variable (Kronecker substitution) gives $M_R(\ell) = \tilde{O}_{r, B_{\text{jet}}}(\ell k^r)$. Since T has only $O_{r, B_{\text{jet}}}(1)$ jet monomials, the recurrence costs $\tilde{O}_{r, B_{\text{jet}}}(k^{r+1})$ field operations.

Recovering the coefficient formulas. For the degree and output interface, apply [Lemma A.4](#) with $K = k$: clearing denominators by S^{2k} gives numerators and a common denominator of degree at most L , and monic reduction modulo h preserves this bound.

Finally compute s^{2k} in R and multiply it by every recovered coefficient c_j . The resulting z -coordinates are the truncated shifts of the degree- $\leq L$ normal forms B_0, N_j . For each coefficient polynomial, the map

$$f(t) \mapsto f(a + \epsilon) \pmod{(\epsilon_0^N, \dots, \epsilon_{r-1}^N)}$$

is injective because $N > L$. Division-free shifts recover the polynomials exactly in another $\tilde{O}_{r, B_{\text{jet}}}(k^{r+1})$ operations. The numerator normal forms B_0, N_j are therefore global polynomials. Consequently $B_0 c_j = N_j$

Algorithm 8 Construct one regular rational Taylor family without enumerating initial jets.

Require: A finite field E of characteristic $p > \max\{k-1, B_{\text{jet}}\}$, $0 \leq r < k$, $|E| > C_{r, B_{\text{jet}}}$, and $T, c, (\alpha_i, y_i)_{i=1}^n$ with the degree bounds of [Lemma E.6](#)

Ensure: A regular chart and its agreement polynomials, or $\emptyset$

- 1: **procedure** FASTREGULARTAYLORFAMILY(E, T, c, α, y, k)
 - 2: $H(u) \leftarrow T(c, u_0, \dots, u_r); S(u) \leftarrow T_{Y_r}(c, u_0, \dots, u_r)$
 - 3: **if** $H = 0$ or H is a nonzero constant **then**
 - 4: **return** $\emptyset$
 - 5: **end if**
 - 6: $H_{\text{rad}} \leftarrow H / \gcd(H, H_{u_0}, \dots, H_{u_r}); H_* \leftarrow H_{\text{rad}} / \gcd(H_{\text{rad}}, S)$
 - 7: **if** H_* is constant **then**
 - 8: **return** $\emptyset$
 - 9: **end if**
 - 10: Construct the linear map M and monic chart $h(t, z) = 0$ as in the proof's monic-chart preparation
 - 11: Set $s(t, z) \leftarrow S(M(t, z)) \bmod h$; choose a constant-grid sample a where $\text{Res}_z(h, s)(a) \neq 0$; put $L \leftarrow 1 + 2k(B_{\text{jet}} - 1)$ and $N \leftarrow L + 1$, construct R , and invert s as in "Making the separant invertible"
 - 12: Shift T from c to $c + Z$ and apply [Lemma E.5](#) over R , starting from $c_j = M_j(a + \epsilon, z)$ for $0 \leq j \leq r$
 - 13: $Y(Z) \leftarrow \sum_{j < k} c_j Z^j$
 - 14: $B_0 \leftarrow s^{2k} \bmod h; N_j \leftarrow B_0[Z^j]Y \bmod h$ for $0 \leq j < k$
 - 15: Recover the global degree- $\leq L$ normal forms from the confluent shifts $a + \epsilon$, as in "Recovering the coefficient formulas"
 - 16: $g_i(t, z) \leftarrow \sum_{j < k} N_j(t, z)(\alpha_i - c)^j - y_i B_0(t, z)$ for $1 \leq i \leq n$
 - 17: **return** $(E, h, s, B_0, N_0, \dots, N_{k-1}, g_1, \dots, g_n, c, r)$
 - 18: **end procedure**
-

holds in $(E[t, z]/(h))[s^{-1}]$, not merely in the chosen fiber, and covers every point of $H_* = 0$ with $S \neq 0$, including ramified projection fibers. $\square$

The same scheduler also covers the rational vector recurrences used for the finite boundary, without adding an inverse coordinate to the state.

Corollary E.7 (Fast rational vector recurrence). *Fix $B \geq 1$. Let E be a finite field of characteristic $p > k - 1$, and let $\mathcal{A}$ be a finite commutative E -algebra of dimension at most B , supplied by a multiplication table. For $d \leq B$, consider*

$$a'_i = \frac{b_i(X, a)}{c(X, a)}, \quad 1 \leq i \leq d, \quad (135)$$

where the circuits for $b_1, \dots, b_d, c$ have total size at most B and their X -coefficient polynomials have total dense length N_X . If $c(x_0, a(0))$ is a unit, then the initial state $a(0) \in \mathcal{A}^d$ extends uniquely to a length- k solution of (135) modulo Z^{k-1} after $X = x_0 + Z$, in

$$\tilde{O}_B(N_X + k)$$

operations in E .

Proof. Shift the X -coefficient polynomials to $x_0 + Z$ in $\tilde{O}_B(N_X)$ operations. The denominator then has unit constant term. At stage m , the known coefficients through degree m determine its inverse and the right-hand side through degree m , hence

$$(a_i)_{m+1} = \frac{[Z^m](b_i/c)}{m+1}, \quad 0 \leq m < k-1. \quad (136)$$

The divisor is a unit. The dyadic scheduler of (133), also applied to unit-series inversion, costs $\tilde{O}_B(k)$ because $\dim_E \mathcal{A} \leq B$. The displayed recurrence proves existence and uniqueness to the asserted precision. $\square$

For a positive-order decoder over a decoder-admissibly presented field, the theorem hypothesis $p > \max\{k-1, B_{\text{jet}}\}$ supplies the recurrence and bounded-degree radical guards. This condition is preserved by every auxiliary field extension. Failure of the fixed field-size grid guard leaves a parameter-bounded length, handled by Section E.8; the characteristic hypothesis itself is not replaced by that fallback. The order-zero recurrence itself has no characteristic restriction; the ordinary decoder uses its characteristic-safe separable preprocessing. Arithmetic in the constructed constant-degree center field costs $\tilde{O}(\log q)$ bits per operation, giving $\tilde{O}_{r, B_{\text{jet}}}((k^{r+1} + D_X) \log q)$ bit complexity.

E.4 Agreement recovery in bounded-fiber towers

We now implement the agreement split from Section 9.3 in the curve's two-level representation. We can afford a bounded amount of work in the fiber variable, but must keep the dependence on the growing base degree nearly linear. The gcd-power identity gives the required partition of points, including fibers with repeated roots. To compute it over the base quotient, we split that quotient whenever a Euclidean division encounters a noninvertible leading coefficient.

Lemma E.8 (Arithmetic for bounded-fiber recovery). *Fix $B \geq 1$. Let $E/\mathbb{F}_q$ be an explicitly represented extension of parameter-bounded degree. Let $G \in E[u]$ be monic and squarefree of degree m , and let*

$$h \in (E[u]/G)[v]$$

be monic of degree $1 \leq d \leq B$, with no reducedness assumption on its fibers. Put

$$\mathcal{A} = (E[u]/G)[v]/h, \quad \Delta = \dim_E \mathcal{A} = md.$$

Algorithm 9 Geometric nilpotent/unit splitting in a bounded-fiber tower

Require: A squarefree monic $g \in E[u]$, a monic bounded-degree $\bar{h} \in (E[u]/g)[v]$, and $e \in (E[u]/g)[v]/\bar{h}$

Ensure: Towers on which e is nilpotent or a unit; total dimension is preserved

```
1: procedure SPLITNILUNIT( $(E[u]/g)[v]/\bar{h}, e$ )
2:    $r \leftarrow \deg_v \bar{h}$ ;  $\bar{e} \leftarrow e^r \bmod \bar{h}$ 
3:   Run the Euclidean algorithm on  $(\bar{h}, \bar{e})$  in  $v$  until it finishes or its next leading coefficient  $a \in E[u]/g$ 
   is a zero divisor
4:   if the Euclidean algorithm stopped at such an  $a$  then
5:     Choose its representative  $\tilde{a} \in E[u]$ ;  $g_0 \leftarrow \gcd(g, \tilde{a})$ ;  $g_1 \leftarrow g/g_0$ 
6:     Continue recursively modulo  $g_0$  after deleting the now-zero leading term
7:     Continue recursively modulo  $g_1$  after inverting  $a$  by extended gcd
8:     return the unions of the tagged nilpotent and unit pieces from both branches
9:   end if
10:   $h_{\text{nil}} \leftarrow \gcd(\bar{h}, \bar{e})$ , made monic
11:  Nil  $\leftarrow \{(E[u]/g)[v]/h_{\text{nil}}\}$  if  $\deg_v h_{\text{nil}} > 0$ , otherwise  $\emptyset$ 
12:  Unit  $\leftarrow \{(E[u]/g)[v]/(\bar{h}/h_{\text{nil}})\}$  if  $\deg_v(\bar{h}/h_{\text{nil}}) > 0$ , otherwise  $\emptyset$ 
13:  return (Nil, Unit)
14: end procedure

15: procedure LOCALIZEFIBER( $(E[u]/G)[v]/h, S$ )
16:   (Bad, Keep)  $\leftarrow$  SPLITNILUNIT( $(E[u]/G)[v]/h, S \bmod h$ )
17:   discard Bad and return Keep
18: end procedure
```

Given $C(X) \in \mathcal{A}[X]_{<k}$ in the tower basis, distinct abscissae $\alpha_i \in \mathbb{F}_q$, received values $y_i \in \mathbb{F}_q$, and $1 \leq k \leq A \leq n$, the splits, coordinate transfers, and recovery operations of [Algorithm 1](#) have deterministic total cost $\tilde{O}_B((n+k)\Delta)$ operations in E , without irreducible factorization, root extraction, or a primitive element for $\mathcal{A}$.

A supplied $S \in (E[u]/G)[v]$ can also be localized in $\tilde{O}_B(md)$ operations. The output towers represent exactly the geometric points with $G = h = 0$ and $S \neq 0$, retain their nilpotents, have total dimension at most md , and make S a unit. No comparison between d and the characteristic is needed.

Proof. We first explain how the algorithm computes a gcd over the reduced base without factoring it. On a current component

$$\mathcal{R} = (E[u]/g)[v]/\bar{h},$$

run the Euclidean algorithm in v on the monic polynomial $\bar{h}$ and $\bar{e} = e^r \bmod \bar{h}$. A leading coefficient $a \in E[u]/g$ can be a zero divisor. Represent it by $\tilde{a} \in E[u]$ and split the squarefree base modulus as

$$g_0 = \gcd(g, \tilde{a}), \quad g_1 = g/g_0.$$

The coefficient is zero modulo g_0 and is a unit modulo g_1 , where an extended gcd computes its inverse. We delete the leading zero coefficient on the first branch and continue the Euclidean step on the second. This is the dynamic-evaluation, or D5, principle [\[DMSX05\]](#).

The v -degrees decrease along both branches. Since they are at most B , the recursion has $O_B(1)$ depth. Consider one residue field K of a leaf and pass to its algebraic closure. There the fiber has the form

$$\bar{h} = \prod_{\beta} (v - \beta)^{m_{\beta}}, \quad m_{\beta} \leq r. \quad (137)$$

If $e(\beta) = 0$, then $(v - \beta)^r$ divides e^r , and hence the full primary factor $(v - \beta)^{m_\beta}$ divides $\bar{e}$. If $e(\beta) \neq 0$, then $(v - \beta)$ does not divide $\bar{e}$. It follows that

$$h_{\text{nil}} = \gcd(\bar{h}, e^r \bmod \bar{h}) = \prod_{e(\beta)=0} (v - \beta)^{m_\beta}. \quad (138)$$

Thus h_{nil} and $\bar{h}/h_{\text{nil}}$ are coprime. On the first factor $e^r = 0$, so e is nilpotent; on the second, e is a unit. The Chinese remainder theorem gives the product decomposition returned by `SPLITNILUNIT`. This proves that a geometric point enters the nilpotent branch exactly when its residual vanishes.

The same calculation with S gives, on every D5 branch,

$$h_{\text{keep}} = h / \gcd(h, S^d \bmod h). \quad (139)$$

It removes the full primary factors above the geometric zeros of S and retains the full primary factors everywhere else. Hence S is a unit in every returned tower, while the nilpotents at the retained points remain. Neither (138) nor (139) differentiates h , so these operations impose no characteristic condition.

Exponentiation to the bounded power r , fast gcds, divisions, and extended gcds cost $\tilde{O}_B(\dim_E \mathcal{R})$ field operations. Every base split preserves the sum of the dimensions. The nilpotent/unit fiber split also preserves it, while localization can only decrease it. This proves the cost statements for one split and for localization.

We next construct all residuals without evaluating C separately in the large algebra. Write

$$C(X) = \sum_{j < k} \sum_{a < m} \sum_{b < d} c_{j,a,b} u^a v^b X^j.$$

For each of the Δ pairs (a, b) , fast multipoint evaluation applies to $\sum_{j < k} c_{j,a,b} X^j$. A shared point tree thus materializes

$$e_i = C(\alpha_i) - y_i \in \mathcal{A}, \quad 1 \leq i \leq n,$$

in $\tilde{O}((n + k)\Delta)$ operations.

The preceding argument supplies the geometric split required by [Algorithm 1](#). It remains to bound coordinate transfers during recovery. For each residual, use a product and remainder tree on its at most B scalar v -coefficients to reduce them into every live base quotient. Repeated base moduli cause no difficulty for remainder evaluation, and

$$\sum_{\mathcal{R} \text{ live}} \deg g_{\mathcal{R}} \leq \sum_{\mathcal{R} \text{ live}} \dim_E \mathcal{R} \leq \Delta.$$

Thus base reductions, bounded-degree fiber reductions, and all D5 splits cost $\tilde{O}_B(\Delta)$ per position. Persistent agreement histories add only one node on an agreeing branch, so all n positions cost $\tilde{O}_B(n\Delta)$.

Finalized components are disjoint and each has positive dimension, giving at most Δ interpolants to check. Their interpolation, full-word evaluation, and deduplication cost $\tilde{O}((n + k)\Delta)$. The same accounting applies to a collection of towers of total dimension Δ , and includes the univariate case $d = 1$. $\square$

E.5 Characteristic-safe radicals

The polynomial H_t from [Lemma 9.9](#) can still contain repeated factors. We need only its geometric zero set, so the following recursion removes repetitions without computing their integer multiplicities.

Lemma E.9 (Fast characteristic-safe radical). *Let K be a perfect field of characteristic $p > 0$, and let $f \in K[U]$ be monic of degree D . Then $\text{rad}(f)$ is computed using $\tilde{O}(D)$ operations in K , excluding inverse Frobenius, together with batched inverse Frobenius on $O(D)$ coefficients across the recursion. If $K = \mathbb{F}_q$ is supplied in a decoder-admissible presentation, the total bit cost is $\tilde{O}(D \log q)$.*

Proof. The constant case returns 1. For $D > 0$, define the factors visible to the derivative and the remaining part by

$$v = \frac{f}{\gcd(f, f')}, \quad R = \frac{f}{\gcd(f, v^D \bmod f)}.$$

If an irreducible factor ϕ occurs in f with multiplicity m_ϕ , then it occurs once in v exactly when $p \nmid m_\phi$. Since $D \geq m_\phi$, the second gcd removes every copy of each such visible factor. The remaining multiplicities are divisible by p , so R is a p -th power over the perfect field K and is coprime to v . We may therefore recurse through the identity

$$\text{rad}(f) = v \text{rad}(R^{1/p}). \quad (140)$$

This proves correctness without labeling any multiplicity.

Fast modular powering, gcds, exact divisions, and multiplication cost $\tilde{O}(D)$ at the current level. If R is nonconstant, then $p \leq D$ and $R^{1/p}$ has at most $D/p + 1$ coefficients. One batched inverse-Frobenius call, including the optional preprocessing in [Definition 9.1](#), costs

$$\tilde{O}(p(D/p + 2) \log q) = \tilde{O}(D \log q)$$

bit operations. The recursive degree is at most D/p , so both the polynomial-arithmetic costs and the inverse-Frobenius charges sum geometrically. This gives the two stated bounds. $\square$

E.6 The polynomial-system algorithm

For higher-order reconstruction, we use Rojas's algorithm to represent isolated roots, not to extract their individual coordinates [[Roj99](#), Main Theorem 1]. For s equations in s variables with degree at most D , it constructs univariate polynomials $h, h_1, \dots, h_s$ whose polynomial map

$$\{(h_1(\theta), \dots, h_s(\theta)) : h(\theta) = 0\}$$

contains all isolated affine roots. The final clause of Rojas's theorem adjoins the origin to each allowed support to obtain this conclusion; our dense degree- D supports already contain it. The theorem permits positive-dimensional components and arbitrary characteristic. Taking dense degree- D supports gives positive mixed volume D^s and degree and deterministic arithmetic bounds $(D + 1)^{O(s)}$, with dimension-dependent prefactors. For this dense specialization we impose the explicit sufficient cardinality condition

$$|E'| > (s + 1)^2 D^{2s}. \quad (141)$$

Indeed, the mixed volume in Rojas's assertion (3) is at most D^s , so this condition dominates its stated field-size requirement.

The arithmetic bound also includes the univariate squarefree-part computations in Rojas's Steps 2 and 3 and in the genericity tests called there. We implement every such computation with the characteristic-safe radical recursion of [Lemma E.9](#). For an input of degree M , this costs $\tilde{O}(M)$ field operations plus batched inverse-Frobenius calls, for a total of $\tilde{O}(M \log |E'|)$ bits. The degrees and aggregate arithmetic size of these internal univariates remain within $(D + 1)^{O(s)}$. The coefficient field E' is a parameter-bounded extension of the supplied field, so $\log |E'| = O_{\text{params}}(\log q)$. Consequently these internal squarefree computations, including those used to certify generic choices, fit the claimed $(D + 1)^{O(s)} \tilde{O}(\log q)$ bit bound.

For the higher-order systems, $s = r + 1$ and $D = O_{B_{\text{jet}}}(n)$. After obtaining the representation, take the radical of its parameter polynomial, filter denominator zeros by gcd, and evaluate the Taylor coefficient formulas in its quotient algebra. This preparation costs $n^{O(r+1)}$; it includes multivariate modular composition. The same radical recursion handles this final parameter polynomial; no multiplicity labels are needed. The isolated-root solver retains its own arithmetic bound. The supplied base field is decoder-admissible, and

each constructed bounded-degree quotient extension provides the same operations by [Lemma E.1](#). Inverse Frobenius is therefore charged in batches rather than treated as a constant-cost field operation. The recovery bound in [Lemma 9.4](#) starts only after these coefficient representations are available.

E.7 Centers and auxiliary fields

At any separant stage, the nonzero specialization $S(X, P, \dots, P^{[r]})$ has degree at most a computable $H = O_{\text{params}}(n)$. Choose $H + 1$ distinct centers. If $q > H$, they lie in $\mathbb{F}_q$. Enumerate a prefix of the supplied $\text{elt}_{\mathbb{F}_q}$ map; repeatedly adding 1 would remain inside the prime subfield. Otherwise $q = O_{\text{params}}(n)$, and we construct a quadratic extension $E/\mathbb{F}_q$. For odd q , find a nonsquare b by applying Euler's criterion to the enumerated field elements and use $U^2 - b$. For $q = 2^e$, find b with absolute trace $\sum_{j < e} b^{2^j} = 1$ and use $U^2 + U + b$. The trace polynomial is nonzero of degree $2^{e-1} < q$, so such b exists. The image of $x \mapsto x^2 + x$ is exactly the trace-zero subspace: both have size $q/2$, and the image is contained in that subspace. Thus the selected quadratic has no root and is irreducible.

In either case, at most q tests using $O(\log q)$ field operations each suffice. This is $\tilde{O}_{\text{params}}(n)$ operations on the only branch where the search occurs. Outside parameter-bounded lengths, $q^2 > H$ gives enough centers; enumerate their pairs of base-field coordinates through the mixed-radix map from [Lemma E.1](#). No primitive-element conversion is required.

For the first-order decoder, all curve-filter and Taylor operations take place in this E . If $E = \mathbb{F}_q$, the supplied presentation applies directly. If $E/\mathbb{F}_q$ is quadratic, [Lemma E.1](#) supplies the required arithmetic, enumeration, and inverse Frobenius. No conversion to an absolute polynomial basis is needed.

For general-order Rojas calls, put $s = r + 1$ and use the system degree bound $D = O_{B_{\text{jet}}}(n)$. The explicit condition $|E'| > (s + 1)^2 D^{2s}$ from (141) is $n^{O(r+1)}$. If the center field is too small, take an extension of the least sufficient relative degree

$$b = \left\lceil \log_{|E|} \left((r + 2)^2 D^{2(r+1)} + 1 \right) \right\rceil.$$

Its cardinality remains $n^{O(r+1)}$, and b is parameter-bounded since $q \geq n$. Enumerating monic polynomials through the supplied element map and testing irreducibility constructs the extension in $n^{O(r+1)}$ operations. This search is never performed in an unrestrictedly large field.

The relative power basis and [Lemma E.1](#) give $\tilde{O}_{\text{params}}(\log q)$ -bit arithmetic and the required batched inverse Frobenius in these extensions. This accounts for the quasilinear dependence on $\log q$ in the positive-order decoder theorems over decoder-admissibly presented fields.

E.8 Bounded lengths and small message degree

Fix one common threshold N_* for the parameter-dependent conditions. When $n < N_*$, enumerate k -subsets of received positions, interpolate, check agreement, and deduplicate. There are at most 2^{N_*} subsets, while arithmetic still costs $\tilde{O}(\log q)$ bits even when q is arbitrarily large. For $k = 1$, frequency counting returns exactly the sufficiently frequent constants in $\tilde{O}(n)$ comparisons.

The same subset algorithm handles $k \leq d$ in the order- d theorem: its cost is $\tilde{O}_d(n^{k+1} \log q)$, within the stated general bound. Otherwise every active order $r \leq d$ satisfies $r < k$ and the Taylor family retains all initial coordinates without tail equations.

If a fixed field-cardinality condition $q > C_{\text{params}}$ fails, then $n \leq q \leq C_{\text{params}}$. Likewise, if the quadratic center field is too small, $q^2 \leq H \leq C_{\text{params}}n$ implies parameter-bounded n . These branches enumerate received positions, not the field. By contrast, the positive-order characteristic bound does not follow from $q \geq n$ over a nonprime field. It is an explicit theorem hypothesis, and failure of that hypothesis is not handled by the bounded-instance procedure.

For an all-rate gap $0 < \delta < 6/25$, use the higher-order construction at δ . For $\delta \geq 6/25$, use the fixed first-order list support of [Corollary 9.13](#) and test the requested threshold. This gives quadratic time in the larger-gap branch and preserves the stated general bound.

F Common-center preparation for first-order curves

We want to run the first-order Taylor construction at one center and still recover every wanted message. The original recurrence covers the states where the separant is nonzero. To recover the remaining states, we enlarge the coordinate ring by integral functions, whose monic equations prevent poles along a wanted message. We then need to find denominators for their derivatives that are nonzero at those states.

We cannot expect these derivatives to have no poles everywhere on the enlarged curve. In characteristic greater than three, for example, take

$$V = Y^2 - (Z - 1)^3, \quad Y = z^3, \quad Z = 1 + z^2,$$

Here D follows the differential equation: $DY = Z$ and $DX = 1$. Differentiating $Y = z^3$ gives $Dz = (1 + z^2)/(3z^2)$, which has a pole at $z = 0$. In this case the denominator ideal below is (z^2) , and

$$1 = z^2(3Dz - 1).$$

For a wanted polynomial, both z and Dz must have no pole at a good center. The displayed identity then rules out $z = 0$. We use an ideal identity of this kind to prove coverage. After restricting to the finite set of boundary states, we split by the denominators one at a time and use each denominator's original differential rule on the factor where it is a unit.

We perform the curve computations over the coefficient field $K = \mathbb{F}_q(X)$. The degrees in the state variables are therefore bounded independently of the block length; only the heights of the coefficients in X grow with the input. We use no factorization of the normal curve and no Gröbner-basis computation.

F.1 A bounded normalization computation

For a reduced rational function $a/b \in K$, put $\text{ht}_X(a/b) = \max\{\deg a, \deg b\}$. For a polynomial, vector, or matrix over K , the height is the maximum height of its entries. Multiplying, differentiating, taking a determinant of bounded size, or applying row reduction to bounded-size matrices raises height by at most a constant factor. Hence a bounded number of such operations on inputs of height N_X costs $\tilde{O}(N_X)$ operations in $\mathbb{F}_q$.

Lemma F.1 (Bounded normalization by three finite spaces). *Let $F \in K[Y, Z]$ be nonconstant and reduced, of total degree at most B , and suppose $\gcd(F, F_Z) = 1$. If*

$$p = \text{char } K > \kappa(B) := 2(B + 1)^3,$$

then one can compute the normalization N of $K[Y, Z]/(F)$, componentwise, using a number of operations in K bounded in terms of B alone. The output consists of a finite separable coordinate u , a free $K[u]$ -basis of N , and its multiplication table. If the coefficients of F have height at most N_X , every output coefficient has height $O_B(N_X)$ and the computation costs $\tilde{O}_B(N_X)$ operations in the constant field.

Proof. Put $R = K[u]$. For $\lambda \in \{0, 1, \dots, 2B\} \subseteq K$, try

$$u = Y + \lambda Z, \quad v = Z.$$

These scalars are distinct under the characteristic hypothesis. The leading v -coefficient of $F(u - \lambda v, v)$ vanishes for at most B values of λ . For an irreducible factor F_0 of F , failure of separability in v means $F_{0,Z} - \lambda F_{0,Y} = 0$. Since $\deg F_0 < p$, its two state partials cannot both vanish unless F_0 is constant, so at most one λ is bad for each factor. There are at most B factors. Direct substitution, leading-coefficient tests, and a subresultant therefore find a λ for which

$$O := K[Y, Z]/(F) = R[v]/(h(u, v)),$$

where h is monic of degree $b \leq B$ in v and is separable over $K(u)$. The substitution and gcd tests use $O_B(1)$ operations in K .

Let $d(u) = \text{disc}_v(h) \neq 0$, made monic. An integral element has integral traces against the power basis $1, v, \dots, v^{b-1}$ of O . These traces belong to the integrally closed ring R . The trace matrix has determinant d up to a nonzero scalar, so its adjugate gives

$$O \subseteq N \subseteq d^{-1}O. \quad (142)$$

Moreover,

$$\deg_u d \leq (2b - 1)B, \quad r := \dim_K(d^{-1}O/O) = b \deg_u d < \kappa(B).$$

If $r = 0$, then d is a unit, O is finite étale over R , and $O = N$; the construction below has no enlargement step. Assume $r > 0$ for the description of the three spaces.

Starting with $O_0 = O$, construct an increasing sequence of intermediate rings. For a ring O_i between the two lattices in (142), let J_i be the inverse image in O_i of the nilradical of O_i/dO_i , and set

$$O_{i+1} = \text{End}_{O_i}(J_i) = \{x \in d^{-1}O : xJ_i \subseteq J_i\}. \quad (143)$$

Each O_i is torsion-free of rank b over the PID R , hence free; therefore O_i/dO_i has K -dimension r . Each of its local lengths and each residue-field degree is therefore less than the characteristic. The residue extensions are separable, because a nontrivial inseparable degree is a positive power of the characteristic. On a local Artin factor, the radical of the trace pairing is then its maximal ideal: modulo that ideal the pairing is the nondegenerate residue-field trace multiplied by the nonzero local length. Thus the nilradical of O_i/dO_i is computed exactly as the kernel of its trace matrix, even though $K = \mathbb{F}_q(X)$ need not be perfect.

For completeness, all products and membership tests in (143) take place in the following three explicit K -spaces:

$$\begin{aligned} E_1 &= d^{-1}O/O, & \dim_K E_1 &= r, \\ E_2 &= d^{-1}O/dO, & \dim_K E_2 &= 2r, \\ E_3 &= d^{-2}O/dO, & \dim_K E_3 &= 3r. \end{aligned}$$

Represent O_i/dO and J_i/dO in E_2 and multiply chosen representatives in E_3 . Quotienting further by the computed image dO_i/dO gives the algebra O_i/dO_i . For an unknown class $x \in E_1$, impose $xJ_i \subseteq J_i$ by linear equations in E_3 . Test a lifted basis of J_i/dO together with $d, dv, \dots, dv^{b-1}$: these at most $2r + b$ elements generate the full ideal over R , including its part invisible modulo dO . Lifting the solution space and adjoining O gives O_{i+1} . All the systems have $O_B(1)$ rows and columns.

The radical ideal J_i contains the nonzerodivisor d , and the nonnormal locus of O_i is contained in $V(d)$: the lattice inclusions give $(O_i)_d = O_d$, which is finite étale over R_d and therefore normal. The Grauert–Remmert criterion and endomorphism-ring step therefore say that O_i is normal exactly when $O_{i+1} = O_i$ [GLS10, Proposition 2.2]. Every strict step enlarges O_i/O inside the r -dimensional space E_1 , so the process stops after at most r strict steps and its terminal order is N .

To obtain a free basis, scale the lattice generators by d and adjoin $d, dv, \dots, dv^{b-1}$. This gives a rectangular matrix of degree at most $\deg d$ whose row span is $dO_i \subseteq R^b$. Cancel the larger leading monomial whenever two rows have the same leading position, using a shifted scalar multiple of the other row. In degree-then-position order, each cancellation lowers that row among at most $b(\deg d + 1)$ leading monomials; its degree never increases. The terminal nonzero rows have distinct leading positions and form a basis, since the preserved row module has rank b . Tracking these operations, dividing by d , and reducing basis products supplies the basis and multiplication table, with u -degrees $O_B(1)$. There are at most r strict enlargement steps and one stabilization test. The bounded dimensions and degrees therefore give the stated height and arithmetic bounds. $\square$

F.2 Denominators that cover every wanted state

The normalized coordinates let us follow a wanted message through a singular point of the original curve. We now need differential rules that determine their later coefficients. We first record all denominators that clear these rules. The main text then proves that they cannot vanish together along a wanted message.

Retain the normalized algebra, derivation, and state generators of (105) and (106). The chain rule gives $D(V) = 0$, so the displayed rule is a derivation extending d/dX on K . The basis reduction records the coordinates of $1, Y, Z$ in the chosen free $R = K[u]$ -basis $e_1, \dots, e_b$. Thus the state generators determine the original message and derivative coordinates even when this basis does not contain 1.

Let

$$\eta(u) = \det(\text{multiplication by } V_Z \text{ on } O),$$

made monic, and choose $g = d\eta$. If a harmless scalar denominator occurs in the chosen linear coordinates, multiply it into g . We now verify (107). Indeed, $dN \subseteq O$ gives the first equality, and the adjugate of the multiplication matrix of V_Z gives the second. Each basis element has the form w_i/d with $w_i \in O$. The two derivatives Du, Dv lie in $\eta^{-1}O$, and the quotient rule gives

$$D(w_i/d) \in d^{-2}\eta^{-1}O \subseteq (d\eta)^{-2}N.$$

This proves the last inclusion, including for $f_0 = Du$. The elementary determinant bounds above give $\deg d, \deg \eta \leq 2B^2$ and hence $\deg g \leq 4B^2$.

We next compute the ideals I, J of (108) and an explicit certificate for (109). From (107),

$$g^2N \subseteq I \subseteq N, \quad \dim_K(N/g^2N) = 2b \deg g = O_B(1). \quad (144)$$

Multiplication by f_i induces a K -linear map

$$N/g^2N \longrightarrow g^{-2}N/N.$$

Compute these matrices in the free basis $e_1, \dots, e_b$. The intersection of their kernels is exactly I/g^2N . Lift a row-reduced K -basis and adjoin $g^2e_1, \dots, g^2e_b$; call the resulting R -module generators $c_1, \dots, c_m$. They generate I also as an N -ideal. Direct multiplication in the module table gives the products $b_{\ell i} = c_\ell f_i \in N$ of (110).

There is also a bounded linear-algebra computation of the coverage identity. Use the R -module generators e_s of N and $e_s, e_s f_i$ of J , where $0 \leq i \leq b$. All products

$$c_\ell e_s, \quad c_\ell e_s f_i = e_s b_{\ell i}$$

belong to N . Reduce them modulo g^2N and close their span under multiplication by u . Because g^2 is monic, the powers $1, u, \dots, u^{2\deg g-1}$ suffice; this is a linear computation in the $2b \deg g$ -dimensional space N/g^2N . Equality (109) says that this span contains 1. Solving for 1 modulo g^2N , and then expressing the remainder in the adjoined generators $g^2e_s \subseteq I$, gives coefficients $d_{\ell,*}, d_{\ell i} \in N$ satisfying the exact identity (111). All coefficients are recorded in the free basis of N , using polynomial reductions of degree $O_B(1)$. The identity will certify coverage after specialization.

Specialization and complexity. We now complete the construction asserted in [Lemma 9.10](#). The preceding steps supply the normalized module, its multiplication table, the denominators and their rational rules, and the coverage identity. We choose H so that the specialized data are well-defined and satisfy the same identities.

First form monic integral equations for the basis elements e_i over $K[u]$, for example from their multiplication matrices. Clear the K -coefficient denominators in these equations, in the module basis and multiplication table, in (107), in the elements $a_i, f_i, c_\ell, b_{\ell i}, d_{\ell,*}, d_{\ell i}$, and in all inverse-chart and coverage identities. Include in H the pivots and leading coefficients used by the finite projection and module reductions, together with a nonzero coefficient of the discriminant in u and the leading coefficient $\text{lc}_u(g)$. There are $O_B(1)$ nonzero factors of degree $O_B(N_X)$. Their product is H . At a center $a \in E$ with $H(a) \neq 0$, let N_a be the finite free $E[u]$ -algebra obtained by specializing the module, basis, and multiplication table recorded over $\mathbb{F}_q[X, H^{-1}][u]$ for N . The same specialization gives $g_a \in E[u]$ and $c_\ell(a), b_{\ell i}(a), d_{\ell,*}(a), d_{\ell i}(a)$. We specialize these recorded certificates; we do not assert that normalization commutes with specialization. Nonvanishing preserves the recorded ranks and multiplication tables, the monic degrees, the reduced open curve, the inverse-chart and coverage identities, and the dimension of $N_a/g_a N_a$.

Successive boundary factors. We already know how to lift every initial state where V_Z is nonzero: apply [Lemma E.6](#) with $r = 1$ to the original equation. Since V_Z is a unit on $g \neq 0$, any wanted state missed by that construction lies in the finite multiplication-table algebra

$$A_0 = N_a/g_a N_a, \quad \dim_E A_0 = \delta = b \deg g = O_B(1).$$

We retain its nilpotents and separate the available differential rules directly.

Set $\mathcal{R}_0 = A_0$. For $\ell = 1, \dots, m$, apply the multiplication-table nilpotent/unit split from the proof of [Lemma 9.4](#) to $c_\ell(a)$ on $\mathcal{R}_{\ell-1}$. Retain its nonzero unit factor $\mathcal{U}_\ell$ and continue only with its nilpotent factor $\mathcal{R}_\ell$. The split also supplies the factor identities, coordinate projections, and inverse of $c_\ell(a)$ on $\mathcal{U}_\ell$. The retained dimensions, including the final unused $\mathcal{R}_m$, sum to δ .

On $\mathcal{U}_\ell$, use the already computed identities

$$c_\ell D a_i = b_{\ell i} \quad (0 \leq i \leq b).$$

The state polynomials come from the recorded normalized-module expressions. Clear each common X -coefficient denominator from numerator and denominator together; it is nonzero at a by the choice of H . The image of $c_\ell(a)$ is a unit, so the rational recurrence of [Corollary E.7](#) determines every coefficient through degree $k - 1$. Extracting the original Y -coordinate gives $C_\ell(X) \in \mathcal{U}_\ell[X]_{<k}$.

All multiplication-table algebras here have B -bounded dimension, and the number of denominators is $O_B(1)$. The Fitting splits therefore cost $O_B(1)$ operations in E . Division-free shifts and all recurrences together cost $\widetilde{O}_B(N_X + k)$ operations because the retained dimensions sum to at most δ . The bit cost is the same bound times $\widetilde{O}(\log q)$ in the supplied presentation. This completes the construction and cost analysis for [Lemma 9.10](#); its proof in the main text establishes that the resulting factors cover every wanted boundary state.

F.3 One-center reconstruction

We can now assemble the three parts at one center: the ordinary tail, the original Taylor curve on the open set where its separant is nonzero, and the finite boundary factors. To use the same center for the ordinary tail, we multiply its leading-coefficient and discriminant guard into H before choosing a . Both guards have degree $O_B(N_X)$, so this does not change the center search.

Algorithm 10 Normalized first-order reconstruction at one common center

Require: Q, α, y, k, A as in [Proposition 9.11](#)**Ensure:** the exact degree- $< k$ agreement list explained by Q

```
1: procedure COMMONCENTERFIRSTORDER( $Q, \alpha, y, k, A$ )
2:   remove the  $X$ -content of  $Q$  and compute its reduced part  $V$ 
3:    $W \leftarrow \gcd(V, V_Z); V_{\text{reg}} \leftarrow V/W$ 
4:   if  $\deg_Z V_{\text{reg}} = 0$  then
5:      $R \leftarrow$  the primitive denominator-cleared polynomial  $W$ 
6:   else
7:      $R \leftarrow$  the primitive denominator-cleared polynomial  $W \text{Res}_Z(V_{\text{reg}}, (V_{\text{reg}})_Z)$ 
8:   end if
9:    $T \leftarrow \text{SEPARABLEPART}(R); D \leftarrow 1$ 
10:  if  $\deg_Y T > 0$  then
11:     $D \leftarrow \text{lc}_Y(T) \text{Res}_Y(T, T_Y)$ 
12:  end if
13:   $H \leftarrow D$ 
14:  if  $\deg_Z V_{\text{reg}} > 0$  then
15:    clear coefficient denominators and take the primitive form of  $V_{\text{reg}}$ 
16:    prepare its normalized module and denominator data by Lemma 9.10; multiply its nonvanishing
    polynomial into  $H$ 
17:  end if
18:  choose the base or affordable quadratic center field  $E$  and  $a \in E$  with  $H(a) \neq 0$ 
19:  FAMILIES  $\leftarrow$  the ordinary family from  $T(a, z)$  and the order-zero lift ▷ empty if  $\deg_Y T = 0$ 
20:  if  $\deg_Z V_{\text{reg}} > 0$  then
21:     $\mathcal{C} \leftarrow \text{FASTREGULARTAYLORFAMILY}(E, V_{\text{reg}}, a, \alpha, y, k)$ 
22:    append FIRSTORDERCURVECANDIDATES( $\mathcal{C}, A, k$ ) to FAMILIES if  $\mathcal{C} \neq \emptyset$ 
23:    split the boundary algebra successively by the  $c_\ell(a)$  and append the nonzero families from
    Lemma 9.10
24:  end if
25:  return RECOVERAGREEMENT(FAMILIES,  $\alpha, y, k, A$ )
26: end procedure
```

Computing the reduced part and W uses bounded-degree gcd, exact division, and recursive subresultant arithmetic in one state variable. No Gröbner basis is required. Correctness, coverage, and the cost of the three parts are proved in [Proposition 9.11](#) beside the main first-order theorem.

G Bounded-input list recovery

List recovery replaces the one received symbol at each position by a small set of permitted symbols. We extend the method to this setting, with bounds that depend on the total input-list size $M = \sum_i |S_i|$. Following Jeronimo’s point–symbol formulation [[Jer26](#)], we extend the rate-dependent support and agreement recovery together: interpolation stacks one local system for every permitted point–symbol pair, while the incidence argument counts at most one satisfied symbol at each position. This section concerns one list-recovery instance; it does not assert a list-recovery version of MCA and is not used elsewhere in the paper.

Fix distinct evaluation points $\alpha_1, \dots, \alpha_n \in F$ and finite sets $S_1, \dots, S_n \subseteq F$. Write

$$M = \sum_{i=1}^n |S_i|, \quad \mathcal{L}_{\text{rec}}(\mathcal{S}, A) = \{P \in F[X] : \deg P < k, |\{i : P(\alpha_i) \in S_i\}| \geq A\}.$$

The next lemma is the labeled analogue of [Lemma 5.1](#).

Lemma G.1 (Dimension-sensitive incidence for input lists). *Assume $k \leq T \leq M$, and work over an algebraically closed field containing the evaluation points and symbols. In the coefficient space of degree- $< k$ polynomials, an algebraic set of dimension at most r and total degree at most B contains at most*

$$BP_r^{\mathcal{S}}(T), \quad P_r^{\mathcal{S}}(T) := \prod_{j=1}^r \frac{M - k + j}{T - k + j}, \quad P_0^{\mathcal{S}}(T) := 1, \quad (145)$$

points belonging to $\mathcal{L}_{\text{rec}}(\mathcal{S}, T)$. If $M < T$, this list is empty.

Proof. For each permitted pair use the hyperplane

$$H_{i,y} = \{P : P(\alpha_i) = y\}, \quad y \in S_i.$$

Hyperplanes with the same position and different symbols are disjoint. Consequently a nonempty irreducible component lies in at most one of them for each position, and a point with T list hits lies on at least T of the M hyperplanes.

We can therefore repeat the induction in [Lemma 5.1](#) with these M hyperplanes. Indeed, if a positive-dimensional irreducible component C of dimension t lies in c of them, their positions are distinct. Vandermonde inversion still gives $c \leq k - t$. The induction step is consequently

$$(T - c)N(C) \leq (M - c) \deg(C) P_{t-1}^{\mathcal{S}}(T),$$

where $N(C)$ counts the retained points. As before, $(M - c)/(T - c) \leq (M - k + t)/(T - k + t)$, giving the next factor in $P_t^{\mathcal{S}}(T)$. Each factor is at least one, so $P_t^{\mathcal{S}}(T) \leq P_r^{\mathcal{S}}(T)$ when $t \leq r$. The zero-dimensional case and the sum over components are unchanged. Thus only the number of hyperplanes changes; agreement still counts distinct positions. $\square$

Theorem G.2 (Rate-dependent bounded-input list recovery). *Fix an integer $L_{\text{in}} \geq 1$, reals $0 < \rho < a < 1$, and an integer $d \geq 500$ such that*

$$\Gamma(\rho, a, d) > L_{\text{in}}. \quad (146)$$

There are explicit finite choices of integers $m, B_{\text{jet}}, N_{\rho, a, d, L_{\text{in}}}$ such that the following holds. Let $n \geq N_{\rho, a, d, L_{\text{in}}}$, $1 \leq k \leq \rho n$, and $an \leq A \leq n$. For arbitrary sets $S_i \subseteq F$ with $|S_i| \leq L_{\text{in}}$,

$$|\mathcal{L}_{\text{rec}}(\mathcal{S}, A)| \leq B_{\text{jet}}^2 \left(\frac{2L_{\text{in}} B_{\text{jet}}}{a - \rho} \right)^d n^d. \quad (147)$$

The bound holds in characteristic zero or in characteristic $p > \max\{k - 1, d, B_{\text{jet}}\}$.

Over a decoder-admissibly presented finite field $\mathbb{F}_q$ with $q \geq n$ and $p > \max\{k - 1, d, B_{\text{jet}}\}$, the complete list can be found deterministically in

$$C_{\rho, a, d, L_{\text{in}}} n^{O(d+1)} \tilde{O}(\log q)$$

bit operations. In particular, the constants regard L_{in} as fixed.

A sufficient explicit order for (146) is

$$d \geq \max \left\{ 500, \frac{1}{6} \left(\frac{40L_{\text{in}}}{9\rho} \right)^{a/(a-\rho)} \right\}. \quad (148)$$

At fixed multiplicative slack $\rho = (1 - \vartheta)a$, this is $O_{\vartheta}((L_{\text{in}}/a)^{1/\vartheta})$.

Proof. If $M < A$, the list is empty and the decoder returns it immediately. Henceforth $M \geq A \geq 1$. Use the support (70). At each permitted pair (α_i, y) , impose the local remainder conditions from Section 3.5. Each local map has rank at most R_{loc} from (72); hence the combined system has rank at most

$$MR_{\text{loc}} \leq L_{\text{in}}nR_{\text{loc}}.$$

The lower bound γ_m on the normalized source-to-rank ratio in Lemma 6.2 tends to $\Gamma(\rho, a, d)$. The strict inequality therefore lets us increase m until

$$N_{\text{coef}} > L_{\text{in}}nR_{\text{loc}} \geq MR_{\text{loc}}.$$

When $R_{\text{loc}} = 0$, the same lemma gives $N_{\text{coef}} > 0$ directly. Choose W, B_{jet} , and the length threshold as in the proof of Theorem 6.7, with this value of m . We obtain a nonzero hidden-derivative explainer Q , computed by Lemma 9.3 on the M permitted pairs.

If P has at least A list hits, then at each such position the local conditions imposed at $(\alpha_i, P(\alpha_i))$ give multiplicity at least m in the specialization $Q(X, P, \dots, P^{[d]})$. Its degree is less than mA_0 , where $A_0 = \lceil an \rceil \leq A$, so the specialization vanishes identically. Thus the interpolation implication used in Theorem 6.7 is unchanged.

Use the same Taylor images and separant stages as in Theorem 6.3, with $K_{\text{rec}} = \max\{k, d+1\} \leq n$. An order- r image in degree- $< k$ coefficient space has dimension $t \leq \min\{r, k\}$. Since $M \leq L_{\text{in}}n$ and $A - k + 1 \geq (a - \rho)n$, Lemma G.1 gives

$$P_t^S(A) \leq \left(\frac{L_{\text{in}}n}{A - k + 1} \right)^t \leq \left(\frac{L_{\text{in}}}{a - \rho} \right)^d.$$

The unchanged image degree is jb_j^r , where $b_j = 1 + 2K_{\text{rec}}(j - 1) \leq 2B_{\text{jet}}n$. Summing the stage degrees, whose j 's sum to at most B_{jet}^2 , gives (147). A jet-degree-zero explainer has no polynomial solutions, so it causes no additional case.

For exact reconstruction, if $k \leq d$, enumerate k distinct positions and one permitted symbol at each, then interpolate and check all positions. This costs $(L_{\text{in}}n)^{O(d)} \tilde{O}(\log q)$. For $k > d$, on a regular Taylor chart, agreement with symbol $y \in S_i$ is the equation

$$g_{i,y} = \sum_{j < k} N_j(\alpha_i - c)^j - yB_0 = 0.$$

At a point satisfying it, its differential on the chart is B_0 times the differential of evaluation at α_i . Hence the proof of Lemma 9.14 still selects r independent equations from distinct hit positions. Enumerate those positions and their permitted symbols, and apply the isolated-root routine of Section 9 to the resulting square systems.

In RECOVERAGREEMENT, process the symbols of one coordinate successively. A nilpotent branch records the labeled pair (i, y) and is not counted again at that coordinate; the branch on which every $C(\alpha_i) - y$ is a unit records no hit. Distinct symbols have disjoint geometric zero loci because B_0 is invertible on the retained chart. Once a branch records k distinctly positioned pairs, interpolate through their base-field values and keep the polynomial exactly when the full test finds at least A list hits. The nilpotent/unit splitting argument in Lemma 9.4 therefore gives completeness and soundness without extracting parameter roots.

There are at most $L_{\text{in}}^r \binom{n}{r}$ labeled square systems per order- r chart and $M \leq L_{\text{in}}n$ residuals to process. With all parameters fixed, these factors are absorbed by the stated $n^{O(d+1)}$ bound; the supplied-presentation arithmetic and characteristic assumption are those of Section 9.

Finally, the sufficient condition (83) shows that (148) gives $\Gamma \geq L_{\text{in}}(1 + 1/d) > L_{\text{in}}$. Substituting $\rho = (1 - \vartheta)a$ gives the stated order dependence. $\square$

H Application details

We record the ProveKit schedules used in [Section 8](#) and the LambdaVM performance diagnostics. The checked artifact supplies expected authentication-size calculations [\[DKT26\]](#); [Table 4](#) reports measured compressed sizes. The LambdaVM algebraic and lookup arguments appear with the application in [Section 8.4](#).

H.1 ProveKit’s round-by-round certificates

[Table 7](#) records every affected WHIR code: (n, k, A) , support $(m, B_\partial, B_{\text{jet}})$, original and revised query counts t_0, t_1 , and original and revised out-of-domain counts u_0, u_1 . All rows satisfy $A^2 < n(k - 1)$, placing them strictly below Johnson. The artifact applies [Proposition 5.10](#) and verifies the local and inter-round error allocations exactly [\[DKT26\]](#).

Code	n	k	A	$(m, B_\partial, B_{\text{jet}})$	t_0	t_1	u_0	u_1
Passport	262144	65536	123208	(384, 168, 688)	127	109	1	1
	131072	8192	29020	(48, 30, 170)	62	55	1	1
	65536	1024	6906	(40, 34, 310)	41	37	1	1
	32768	128	1782	(12, 12, 168)	31	29	1	1
	16384	16	463	(8, 4, 246)	24	23	1	1
	8192	2	85	(4, 4, 339)	20	18	1	1
Passport internal	4096	1024	1933	(96, 42, 174)	127	109	1	1
	2048	128	453	(68, 34, 241)	62	55	1	1
	1024	16	107	(16, 16, 114)	41	37	1	1
	512	2	19	(12, 12, 227)	31	26	1	1
Goldilocks lookup	4096	1024	1933	(96, 42, 174)	127	109	1	2
	2048	128	453	(38, 23, 133)	62	55	1	1
	1024	16	107	(16, 16, 114)	41	37	1	1
	512	2	19	(12, 12, 227)	31	26	1	1
Blinding	32	8	14	(64, 28, 116)	127	99	1	1

Table 7: Certificates for each affected WHIR code. The witness schedules apply separately to both witness parts. Each passport witness has one internal blinding proof; the Goldilocks blinding commitment retains its original 62-query final code.

For the certificates used in the measured configurations, the smallest margin among the changed passport error terms is 128.0015 bits. For the Goldilocks lookup, the first folding and identity term has 131.374 bits, and the first query term has 128.027 bits. These decimal values summarize exact rational inequalities against 2^{-128} ; they do not replace those checks.

H.2 LambdaVM performance diagnostics

The added evaluations nevertheless do real work. Diagnostic runs with grinding disabled record a net increase of 264.4 million retired instructions (0.1845%); single-thread isolation attributes about 305 ms to the early evaluations and modified DEEP computation. At 16 threads, table-level parallelism can hide this work from the critical path. Within verification, the measured CPU components become 0.081 ms faster overall, with interval $[-0.117, -0.048]$ ms for the change: savings from fewer queries exceed the additional evaluation work. The largest new transient allocation is 3 MiB, computed from the buffer sizes and unresolved against the roughly 1.9 GiB process peak. Separate codec measurements differ by less than 0.05 ms; those non-interleaved comparisons are descriptive rather than causal estimates.

```

theorem firstOrderBranch_finiteLength_finiteSlack_bounds
  (rho eta : ℝ) (n k A : ℕ)
  (hrho : 0 < rho) (hrhoOne : rho < 1) (heta : 0 < eta)
  (haOne : firstOrderBranchThreshold rho + eta < 1)
  (hk : 0 < k) (hkRate : (k : ℝ) ≤ rho * n)
  (hA : (firstOrderBranchThreshold rho + eta) * n ≤ A) (hAn : A ≤ n)
  {F : Type*} [Field F] (domain : Fin n → F)
  (hchar : k = 1 ∨ ringChar F = 0 ∨
    max (k - 1) (firstOrderBranchFiniteLengthDerivativeCap rho eta n) < ringChar F) :
  (∀ received : Fin n → F,
    (closePolynomialSet domain received k A).Finite ∧
    ((closePolynomialSet domain received k A).ncard : ℝ) ≤
      7 * firstOrderBranchFiniteLengthMCAConstant rho ^ 3 * n /
        finiteLengthSlack eta n ^ 2) ∧
    ∀ f g : Fin n → F,
      ∃ exceptional : Finset F,
        (exceptional.card : ℝ) ≤
          140 * firstOrderBranchFiniteLengthMCAConstant rho ^ 6 * n ^ 2 /
            finiteLengthSlack eta n ^ 4 ∧
        ∀ z ∉ exceptional,
          ∀ P : F[X], P.degree < k →
            A ≤ (polynomialAgreementSet domain (fun i ↦ f i + z * g i) P).card →
              HasExactCorrelatedPair domain f g (RingHom.id F) k z P
  )

```

Figure 2: The first-order list-decoding and MCA theorem in Lean.

I The main statements in Lean

To connect a paper theorem to its formal statement, start with [ReedSolomon/PaperGuide.lean](#) at the validated mathematics revision [DA26a]. It groups the results by their mathematical role and links to declarations whose comments explain the parameters and hypotheses. We read one joint list-decoding and MCA statement below; the artifact checks this excerpt against the source and supplies the other [literal statements](#) with a [source manifest](#) [DKT26]. The simplified executable decoder is cited separately at its own revision. The artifact’s [implementation-source manifest](#) records the repository, revision, and source paths behind the application parameters. Table 7 gives the ProveKit support ledger; the ZisK and LambdaVM checkers record each batching and fold certificate together with its allocated error.

The RS-specific development spans approximately 125,000 source lines across 587 Lean modules, including the decoder and concrete examples.

I.1 Reading the joint list and MCA statement

Figure 2 records the joint first-order theorem in namespace `ReedSolomon.FirstOrder`. Here `rho` bounds the rate, `eta` is η_1 , and `firstOrderBranchThreshold` is $a_1(\rho)$, including its low-rate branch. The hypotheses require $\eta_1 > 0$, $k \leq \rho n$, $A \geq (a_1(\rho) + \eta_1)n$, and distinct evaluation points; the characteristic restriction is waived for $k = 1$. The first conclusion proves that `closePolynomialSet`, the set of *all* degree- $< k$ polynomials with at least A agreements, is finite and bounds its size. In the second, the exceptional set follows f, g but precedes every challenge z and candidate P . The conclusion `HasExactCorrelatedPair` gives degree- $< k$ witnesses with $P = F_0 + zG_0$ whose common agreement set equals the candidate’s full agreement set. Writing `finiteLengthSlack` as $\eta_1 + 1/n$, the two bounds have the form

$$|\mathcal{L}| \leq C_\rho \frac{n}{(\eta_1 + 1/n)^2}, \quad |\text{Bad}| \leq D_\rho \frac{n^2}{(\eta_1 + 1/n)^4},$$

where C_ρ, D_ρ depend only on the rate bound, not on the field, code length, or received words. Thus the powers 2 and 4 measure dependence on the agreement gap; the dependence on n is linear and quadratic, respectively, when that gap is fixed. For $A > n$, excluded here, `HasCapacityLists` explicitly requires an empty list.

Other quantitative results. The links below lead to the source statements; the source guide records their full declaration names.

Regime and hypotheses	Formalized guarantee
Zeroth order Every characteristic; positive Johnson gap	Complete finite list and Johnson bound; line MCA with full agreement sets
Fixed order $\Gamma(\rho, a, d) > 1$; the stated characteristic assumption	Lists from the derivative-order condition; line MCA from the same parameters
First-order curves Independent ordinary/regular tuning from Lemma 5.6 and Section 5.5.1	Optimized finite curve certificate; squarefree recovery with independently minimized ordinary threshold
Gap from capacity Fixed $\delta > 0$; uniform over the rate	Uniform capacity lists; uniform line, affine, and powers MCA
Interleaving and folds Width-independent transfer and shared-level union bound from Corollary 7.7 and Section 7.4	Arbitrary-width interleaved powers agreement; Reed–Solomon shared-level fold bound

The optimized first-order declarations prove that the independent ordinary minimum over $k \leq L_0 \leq A$ is attained, retains full agreement-set recovery, and is no larger than the choice $L_0 = k$ ([Lemma 5.6](#)).

The bounded-input list-recovery extension in [Theorem G.2](#) is not included in the formalization.

The executable decoder. The simplified coordinate decoder’s [exact-output theorem](#) [DA26b] proves duplicate-free exact membership, the empty-list case, and a bound on recorded primitive operations. It does not formalize the fast reconstruction algorithm of [Section 9](#) or its bit complexity.

Finite-field MCA corollaries give errors E/q for lines and $E/(q - 1)$ for positive-dimensional affine families. The [application guide](#) covers certificates and byte accounting; complete deployed systems are outside scope.